

ZlibValidation

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Chapter 1

ZlibValidation

1.1 Description

Command line tool to validate standard cell libraries in `.lib` format.

ZlibValidation provides a suite of tools for engineers working with digital IC standard cell libraries in the Liberty format (`.lib`). It helps ensure library quality, consistency, and facilitates comparison and conversion tasks. Built with C++ for performance and leveraging robust libraries for parsing and command-line interaction.

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1.3 Motivation

Validating standard cell libraries (.lib) is critical in chip design, but current solutions pose challenges. Commercial tools are expensive, locking out many users (students, startups, researchers), and their closed-source nature prevents customization and hinders innovation. This creates a gap, especially for those needing rapid validation within tight resource constraints.

ZlibValidation aims to fill this gap. It is a **free, open-source command-line tool** designed for comprehensive and efficient .lib file validation. Our goals are to:

- Lower the barrier to entry for library quality assurance.
- Provide **fast, reliable checks** (parsing, consistency, function, parameters, comparison).
- Enable **transparency, customization, and community collaboration** through open source.
- Bridge the divide between academic research and industrial needs in validation technology.

1.4 Installation

ZlibValidation is designed to be easily built and run directly from source **without requiring administrator (sudo) privileges** on Linux systems. This makes it ideal for environments where users have restricted permissions. The recommended installation method is building from source.

1.4.1 Prerequisites

Before you begin, ensure you have the following installed on your **Linux** system:

- **Git:** To clone the repository.
- **C++ Compiler:** C++20 standard support required (e.g., GCC ≥ 12). Developed with GCC 14.2.0.
- **CMake:** Version 3.25 or higher (versions after 4.0.0 may exhibit compatibility issues).
- **Ninja** (recommended) or **Make:** Build systems. Ninja is preferred for its speed.

1.4.2 Optional Pre-requisites

- **ccache:** Optional but recommended for speeding up recompilation. Install it via your package manager (e.g., `conda install conda-forge::ccache`).
- **lld:** Optional but recommended for faster linking. Install it via your package manager (e.g., `conda install conda-forge::lld`).

1.4.3 Building from Source (Recommended Method)

Follow these steps to compile ZlibValidation in your user directory:

1. **Clone the repository:**

```
git clone https://github.com/Cedar17/ZlibValidation.git
cd ZlibValidation
```

2. **Configure the build using CMake:** Create a build directory and run CMake from within it. This keeps build files separate from your source code.

```
mkdir build
cd build
# Use Ninja as the build system (recommended):
cmake .. -G Ninja
# or if you prefer Make:
cmake .. -G "Unix Makefiles"
```

Note: No `sudo` is needed here.

3. **Compile the project:** Use `ninja` to build the executable. If you chose `Make`, use `make` instead, the `-j` flag speeds up compilation by using multiple processor cores.

```
# If you used Ninja:
ninja
# or if you used Make:
make -j$(nproc) # Adjust $(nproc) or use a specific number like -j8 for multi-threading
```

4. **Locate the Executable:** Upon successful compilation, the `zlibvalidation` executable will be located inside the build directory: `./build/zlibvalidation`.

```
# Check the executable version:
./zlibvalidation --version
```

1.4.4 Running ZlibValidation

Since we haven't performed a system-wide install (which would typically require `sudo`), you need to run the executable using its path relative to your current location.

- If you are in the project's root directory (`ZlibValidation/`), run it like this:

```
./build/zlibvalidation --help
```

- If you are inside the build directory, run it like this:

```
./zlibvalidation --help
```

1.4.5 (Optional) Adding to your PATH

For convenience, you can temporarily add the build directory to your shell's `PATH` variable for the current session:

```
export PATH="$PWD/build:$PATH" # Assumes you are in the ZlibValidation root directory
# Now you can run it directly:
zlibvalidation -h
```

To make this change permanent, you can add the above line to your shell's configuration file (e.g., `~/.bashrc`, `~/.zshrc`).

1.5 Help Message / Features

```
ZlibValidation
Usage: ./zlibvalidation [OPTIONS] [SUBCOMMAND]

Options:
  -h,--help            Print this help message and exit
  -v,--version          Display program version information and exit

Subcommands:
  parse                Parse the Liberty file and write JSON (or SQLite DB via --db)
  mono                 Check the monotonicity of timing arc values
  compare              Compare the comparison library against the reference one and report differences
  supercell            Generate supercells for the given Liberty file
  zlibboost            ZlibBoost - Multi-threaded Library Processing Tool
  clear                Clear the log, JSON, map, markdown, Verilog, SPICE files in this directory
  verilog              Generate Verilog file for given Liberty file
  spice                Generate SPICE file for given Liberty file
  func                 Check functional equivalence of two Liberty files or Verilog files
```

1.6 Example Usage

See test directory. There are some shell scripts showing how to use ZlibValidation. You can modify them to validate your own standard cell library pdk.

1.7 Documentation and Reference Manual

See <https://cedar17.github.io/ZlibValidation/> for HTML documentation and PDF reference manual.

1.7.1 Documentation Generation

To generate the documentation, you need to have Doxygen and Graphviz installed. You can install them via your package manager or using conda:

```
conda install -c conda-forge doxygen graphviz
```

Then, run the following command in the root directory of the project:

```
doxygen Doxyfile
```

This will generate the documentation in the doc_doxygen directory. You can open the doc_doxygen/html/index.html file in your web browser to view the documentation.

Alternatively, you can generate the PDF reference manual by LaTeX.

```
cd doc_doxygen/latex
make
```

1.8 Acknowledgements

ZlibValidation leverages the power of several excellent third-party libraries, build tools, and external utilities. We extend our sincere gratitude to the developers and communities behind these projects, which were instrumental in building this tool:

1.8.1 Core Functionality Libraries:

- **Liberty Parser (csguth/LibertyParser):**
 - **Purpose:** Provides the core functionality for parsing the .lib (Liberty) standard cell library format. This open-source implementation forms the basis of our library reading capabilities.
 - **Integration:** The source code was cloned from the repository, manually compiled into a static library (libsi2dr_liberty.a), and included directly within this project's include_3rd_↵party directory.
 - **License:** SYNOPSYS Open Source License Version 1.0.
- **CLI11:**
 - **Purpose:** Enables the robust, feature-rich, and user-friendly command-line interface, handling subcommands and options parsing.
 - **Integration:** Managed via CMake's FetchContent mechanism during the build process.
 - **License:** BSD 3-Clause License.
- **spdlog:**
 - **Purpose:** Provides a highly efficient and flexible library for structured logging throughout the application, critical for debugging and user feedback.
 - **Integration:** Managed via CMake's FetchContent.
 - **License:** MIT License.
- **nlohmann/json:**
 - **Purpose:** Used extensively for representing the parsed Liberty library data internally as JSON objects, facilitating data storage, retrieval, and manipulation (e.g., for monotonicity checks and comparisons).
 - **Integration:** Managed via CMake's FetchContent.
 - **License:** MIT License.
- **ZlibBoost:**
 - **Purpose:** A multi-threaded library processing tool which is the foundation of the zlibboost subcommand. It significantly boost standard cell library characterization with machine learning using Python. See <https://doi.org/10.1145/3658617.3703638> for detailed paper.
 - **Integration:** Need to specify the path to the ZlibBoost main python script and python environment in the ZlibValidation command line.
 - **License:** BSD 3-Clause License with Commercial Use Restriction.
- **tabulate:**
 - **Purpose:** Generates formatted text-based tables, significantly improving the readability of reports generated by the compare subcommand.

- **Integration:** Managed via CMake's FetchContent.
- **License:** MIT License.
- **slang:**
 - **Purpose:** A powerful library for parsing SystemVerilog/Verilog code. It's used here to analyze Verilog netlists, extract Abstract Syntax Trees (AST), generate structural Verilog (`verilog` subcommand), and extract logic for functional equivalence checks (`func` subcommand).
 - **Integration:** Managed via CMake's FetchContent.
 - **License:** MIT License.
- **ExprTk (Expression Toolkit Library):**
 - **Purpose:** A fast mathematical expression parser and evaluation engine. It is used in the `func` subcommand to dynamically evaluate the logical function strings extracted from libraries or Verilog, enabling the functional equivalence check by comparing truth tables.
 - **Integration:** The header file (`exprtk.hpp`) is included directly within this project's `include_3rd_party` directory.
 - **License:** MIT License.
- **SQLiteCpp:**
 - **Purpose:** Provides a C++ wrapper around the SQLite3 C API, enabling the `parse --db` mode to persist Liberty LUT entries directly into a SQLite database for efficient querying and integration with ML data pipelines.
 - **Integration:** Managed via CMake's FetchContent.
 - **License:** MIT License.

1.8.2 Build, Documentation, and External Tools:

- **CMake:** The cross-platform build system generator used to configure and manage the entire compilation process.
- **Doxygen & Graphviz:** Employed for automatically generating source code documentation and visualizing relationships, aiding development and understanding.
- **V2LVS (Calibre® Utility):**
 - **Purpose:** A command-line tool included with the Siemens EDA Calibre® platform, designed to translate structural Verilog netlists into a basic SPICE format, typically for Layout Versus Schematic (LVS) verification.
 - **Integration:** The `spice` subcommand **optionally** utilizes `v2lvs` if it's found in the system's PATH. It serves as the first-pass engine to convert the intermediate Verilog representation into a raw SPICE netlist. ZlibValidation then performs post-processing on this output, and ensure correct formatting.

The seamless integration facilitated by CMake FetchContent for many of these libraries, combined with the direct inclusion of others, significantly streamlined development and enabled the rich feature set of ZlibValidation. We encourage users to consult the individual project licenses for detailed terms of use.

Chapter 2

Development Diary

2.1 2025-01

2.1.1 2025-01-27

- 创建了一个C++空项目，使用CMake管理编译链，并手动添加了外部库 `libsi2dr_liberty.a`。
- 添加了 `CLI11` 库以处理命令行参数解析。
- 添加了 `spdlog` 日志库，实现高效的日志格式化输出。

2.1.2 2025-01-28

- 添加 `nlohmann/json` 库用于 JSON 数据存储和检索。
- 在 `version.h` 中存储项目作者、版本、构建日期等信息，由 CMake 自动维护。
- 简化了 `--version` 参数解析的代码实现，并添加了 `--mode` 选项以选择工作模式。
- 使用 `spdlog` 库的日志记录器将 `debug` 级别及以上的日志输出到文件，将 `info` 级别及以上的日志输出到控制台。
- 将库文件顶层封装为 `LibFile` 对象，提供了读取、解析和修改库文件的方法，包含名称和 PVT 信息作为公共属性。
 - 但仍然使用过程化方法循环迭代读取库文件名称和 PVT 信息。
- 按照 C++ 标准库、第三方库和本地库的顺序重新排列了 `#include` 语句。

2.2 2025-02

2.2.1 2025-02-01

- 使用面向对象编程（OOP）重构了原先用C语言循环遍历的代码，将简单属性的读取封装为 `LibAttribute` 类，
 - 并将函数返回的 `char *` 类型的属性值转换为 `std::string` 类型，便于外部使用。
- 将属性迭代过程封装为 `AttributesIterator` 类，在析构函数中处理迭代退出，实现资源获取即初始化（RAII）。

2.2.2 2025-02-10

- 将组的迭代过程封装为 `GroupsIterator` 类，同样在析构函数中处理迭代退出。
- 封装了 `LibGroup` 类，提供了读取组名、类别、属性、子组的方法。
- 将类的定义和实现分离，将类的实现放在 `src` 目录下，将类的定义放在 `include` 目录下，修改了 `CMakeLists.txt` 自动收集文件，便于模块化编译。

2.2.3 2025-02-11

- 类的头文件改为 `hpp` 后缀，修改了 `CMakeLists.txt` 中的文件收集规则，CMake 编译时间戳更新。
- 为 `LibFile` 类添加了输出同名 json 格式文件的方法 `LibFile::writeJsonToFile`，目前能输出 PVT、`cell_name`、`cell_footprint`、`cell_area` 信息。
- [x] voltage 浮点数误差未解决。

2.2.4 2025-02-14

- 在 `json_utils.cpp` 中，实现了以 JSON 数据结构循环迭代存储库（lib）信息的功能，具体包括 `generateCellJson`、`generatePinJson`、`generatePowerJson`、`generateLutJson` 等函数，并添加了同名的 `hpp` 头文件。
- [x] 时序（timing）信息解析功能待完成。
- 新增辅助函数 `parseStringToVector`，该函数可将以逗号分隔的字符串解析为浮点数向量，以便于读取 Look Up Table (LUT)。
- 为 `LibFile` 对象添加了私有属性 `lib_json_`，用于存储库（lib）的 JSON 对象。
- 封装了用于迭代复杂属性值的 `ValuesIterator` 类。

- 修改了 `LibAttribute::isComplex()` 函数的返回值类型，由原类型变更为 `bool` 类型。
- 重写了 `LibGroup::getName()` 函数，使其直接返回 `std::string` 类型的组名字。
- 重构了迭代器使用代码。移除了中间变量类似 `si2drGroupsIdT sub_groups` 的声明步骤。直接使用 `lib_group.getGroups()` 的返回值进行初始化 `GroupsIterator`，提升了代码的可读性和简洁性。

2.2.5 2025-02-15

- 实现 `generateTimingJson` 函数，用于全面解析时序信息。

2.2.6 2025-02-17

- 简化了迭代器在外部的调用流程，删除所有迭代器的 `begin()` 方法，改为在构造函数内部初始化。

2.2.7 2025-02-18

- 新建 `LibFile::mono()` 方法，准备实现库文件输出引脚的 `values` 单调递增性检查。
- 修改了 `setupLogger()` 函数，将日志文件名可作为参数传入，便于根据不同工作模式切换日志文件。

2.2.8 2025-02-19

- 修改了 `mode == "mono"` 情况下的逻辑，使用 C++17 引入的 `std::filesystem::exists()` 函数检查同名 JSON 文件是否存在，如果 JSON 文件不存在，则先调用库文件解析功能，生成 JSON 文件，然后再执行单调性检查。
- 实现了 `LibFile::mono()` 方法，检查 `cell_rise`、`cell_fall`、`rise_transition`、`fall_transition` 这四种 timing 信息，针对其 LUT 数据结构，检查其 `value` 值在表格的每一行是否保持单调递增。

2.2.9 2025-02-20

- `LibFile::mono()` 方法增加结果统计功能，在程序运行的最后输出单调性检查中 `passed` (通过) 和 `failed` (失败) 的 `cell` 数量，输出格式参考 `liberate_lv` 工具的风格。
- 以 `sl018_ff_3.96_-40.lib` 为例，与 `liberate_lv` 工具对比测试，结果均为 205 out of 559 cells failed。
- 对于非单调信息警告，增加输出 `when` 信息，方便用户查看具体位置。

2.2.10 2025-02-25

- 重构命令行参数解析，使用 CLI11 库的子命令，在每个回调函数中实现 parse、mono 功能。

2.2.11 2025-02-26

- 子命令可自定义输出文件名、日志文件名，并设置了相应的默认值。
- 改为在 Logger 初始化时打印版本信息，避免在 -h, -v 时多余打印。
- `LibFile::mono()` 方法增加对 `input_slew` 的单调性检查，如果在输入时有指定，就检查 value 矩阵的每一列是否单调。经过测试，`tcbn65lpsc.lib` 输出与 `liberate_lv` 工具一致。

2.2.12 2025-02-27

- 多线程并行尝试，采用 GDB 调试：
 - `si2dr_liberty` 库在解析 Liberty 文件时，可能使用了共享的数据结构（例如哈希表、字符串表）来存储解析结果或中间数据。
 - 在多线程并行解析多个文件时，不同的线程同时调用 `si2dr_liberty` 库的函数，并发地访问和操作这些共享数据结构。
 - `si2dr_liberty` 库可能没有采取足够的线程同步措施来保护这些共享数据结构，导致了数据竞争。
 - 数据竞争最终导致了内存损坏，使得 `strcmp` 函数在后续操作中访问了无效内存，触发了段错误。

2.2.13 2025-02-28

- 修改 `LibFile` 的构造函数与析构函数，`read` 方法改为私有，移入 `parse` 方法。将 `si2dr` 初始化与销毁放在 `parse` 方法中，使得 `mono` 方法与 `si2dr` 库解耦，从而实现多线程并行验证单调性。
- [x] 顶层 `si2drPIGetGroups` 未退出的 bug: `WARNING: si2drPIQuit: GetGroups called 1 more times than IterQuit, 内存泄漏?`
- 学到了可以在 `CMakeLists.txt` 中使用 `set(CMAKE_BUILD_TYPE Release)` 减少编译器优化程度，配合 GDB 等调试工具检查堆栈信息。

2.3 2025-03

2.3.1 2025-03-01

- `sub_groups_iter` 是在每次 `for` 循环的迭代中声明的局部变量，在每次迭代结束后其作用域结束，析构函数被调用，释放相关资源。所以 `si2drPIQuit` 未发现内存泄漏。
- 顶层 `group_iter` 的析构函数发生在 `parse` 方法结束时，所以 `parse` 方法末尾的 `si2drPIQuit` 检测到了顶层 `groups` 未退出的情况。
- 综上，`si2drPIQuit` 的警告是由于当时顶层 `groups` 未退出导致的，在 `parse` 方法结束后能正确释放，不会造成实际上的内存泄漏。
- [] 顺序执行 `parse`，`log` 输出仍存在问题：解析第二第三个库的时候会多余打印之前库的 PVT 信息，时间上也比单独解析一个库的总和要慢 (`parse` 6s 左右，`mono` 1s)。并行执行 `mono`，`log` 输出会集中到最后一个库的文件。
- 给 `LibFile` 类的私有属性添加了初始值，避免了未初始化的问题，吗？

2.3.2 2025-03-07

- 实现了 `compare` 子命令的解析，能够对两个库文件依次进行解析，并生成对应的 JSON 文件。
- [] 仍然存在日志重复输出的问题，多文件处理时数据可能会相互干扰。

2.3.3 2025-03-10

- 新增 `supercell` 子命令和 `LibFile::supercell` 方法，用于生成超级单元以供验证。
- 集成了 `zlibboost` 子命令，支持调用开源库特征化工具 `zlibboost`。已与开发者取得联系。
- 调整了子命令输出文件的位置为当前目录，避免文件混淆。
- 添加了 `clear` 子命令，方便清理生成的 JSON 文件和日志文件。

2.3.4 2025-03-14

- 添加了 `tabulate/table` 库，通过 CMake `FetchContent` 进行管理，用于生成格式化的表格输出。
- 新建了 `compare.cpp` 和 `compare.hpp` 文件，创建了 `LibraryComparator` 类，用于比较两个库文件的差异。目前实现了读取参考库 JSON 文件的功能。

2.3.5 2025-03-15

- 使用 `std::filesystem::path` 重构了 `LibFile` 类的文件路径管理，方便文件路径的拼接和处理。`LibFile` 类添加了成员变量 `basename_`、`filename_`、`libname_`、`jsonname_` 和 `loggername_`，用于存储文件名、库名、JSON 文件名和日志文件名。
- 通过作用域 (scope) 管理 `LibFile::parse` 方法中的顶层迭代器的生命周期，提前结束并调用 `si2drPIQuit` 保证资源释放，解决了分析多个库时日志重复输出和数据保存错误的问题。
- `clear` 子命令增加了删除 `.map` 和 `.md` 文件的功能。
- 实现了 `mono` 子命令的多线程并行。首先检查是否是多文件输入，如果是，就顺序检查是否准备好了每个文件的 JSON 文件，否则顺序进行多文件库解析。在所有文件的 JSON 准备就绪后，根据文件数量创建线程池，每个线程负责一个库的单调性检查。
- 重构了 `spdlog` 日志初始化，返回 `logger` 对象而不是设置全局默认 `logger`。保留名为 `APP_NAME` 的全局 `logger`，用于输出总体提示信息。
- 为 `LibFile` 类添加了独立的成员变量 `logger`，用于记录每个库文件的日志信息。
- 完成了 `supercell` 子命令的多线程并行，处理逻辑与 `mono` 子命令类似。
- `LibraryComparator::generateReport()` 方法测试了 `tabulate` 库的 markdown 表格输出功能，完善了报告的头部信息输出，包括参考库、比较库、相对容差、软件信息、报告生成时间和图例说明。

2.3.6 2025-03-16

- 针对 `compare` 子命令，增加了对报告文件名的检查，确保其以 `.md` 或 `.txt` 结尾。若不符合，则给出警告并自动添加 `.md` 后缀。
- 进一步完善了 `LibraryComparator::generateReport()` 方法。现在，该方法能够遍历比较库中的所有 `cell`，并在参考库中查找对应的 `cell`。如果找到，则调用 `LibraryComparator::compareCell()` 方法进行比较；否则，会记录 `cell` 未找到的警告信息。
- 新增了 `LibraryComparator::compareCell()` 方法，用于比较两个库中同名 `cell` 的输出引脚。该方法会遍历比较库 `cell` 的每个输出引脚，在参考库 `cell` 中寻找同名引脚。如果找到，则调用 `comparePin` 函数比较引脚；如果未找到，则记录警告信息。如果比较库 `cell` 没有输出引脚，则会记录一条信息级别的日志。
- 新增了 `LibraryComparator::comparePin()` 方法，用于比较两个库中同名 `cell` 的同名引脚。它会遍历比较库 `pin` 的每个 `timing arc`，并在参考库 `pin` 中查找相同类型的 `timing arc`。如果找到，则调用 `compareTimingArc` 函数进行比较；如果未找到，则记录警告信息。如果比较库 `pin` 没有 `timing arc`，则记录一条信息级别的日志。

- 修改了 `LibraryComparator::compareTimingArc()` 方法，用于比较两个 timing arc JSON 对象。该方法首先记录正在比较的 timing arc 的类型，然后提取 "related_pin" 属性。接着，它遍历预定义的 timing arc 名称列表 ("cell_rise", "cell_fall", "rise_transition", "fall_transition")，并在比较库中查找这些 timing arc。如果找到，则在参考库中查找对应的 timing arc，并调用 `compareLut` 方法进行比较。如果参考库中未找到对应的 timing arc，则记录警告信息。
- 修改了 `LibraryComparator::compareLut()` 方法（原 `compareValue` 方法，已更正命名），用于比较两个 JSON 对象中具有相同名称的 value。它首先提取两个 JSON 对象中的 "index_1" 和 "index_2" 数组，如果这两个数组不相等，则记录错误信息并返回。如果索引匹配，则比较 LUT 中的实际 value 值，并根据相对容差 (reltol) 和绝对容差 (abstol) 标记差异。
- 重新设计了层级比较方法，现在可以逐层级地传入 `cell_name`、`pin_name`、`timing_type`、`related_pin` 和 `arc_name`，以便在最内层制表或提示时能够准确输出层级信息。
- 修复了 `LibraryComparator::compareValue()` 方法的命名错误，已更正为 `LibraryComparator::compareLut`
- 已经可以输出表格，数据数量和商用工具结果一致，但格式还需要进一步调整。
- 新增了 `abstol` 参数，默认值为 0.002ns，用于设置绝对容差，与库验证工具保持一致。

2.3.7 2025-03-17

- 新增 `verilog` 和 `spice` 子命令，用于生成 Verilog 和 SPICE 代码。
- 完善 `LibraryComparator` 报告生成：
 - 为每个 cell 增加表头，仅在存在表格数据时输出表格，避免冗余输出。
 - 增加 cell 结果统计功能，输出结果统计表格，目前仅支持 Timing 中 Delay 的比较。

2.3.8 2025-03-18

- 优化统计表格，包含 | Cell Name | Data Type | Failed Count | Avg Diff | Avg Diff% | Max Diff | Max Diff% | Outliers |。
- 优化报告对比表格，包含 | Pin Name | Reference | Comparison | Diff | Diff % | Type | Arc Name | Row # | Index_1 | Column # | Index_2 | Note |
- 报告表格的 stdout 输出增加表头加粗、黄色的格式化。
- 增加 `si2drSimpleAttrGetBooleanValue` 封装，用于获取布尔类型的属性值，以判断引脚是否是时钟引脚。
- `LibFile::mono()` 增加对 `min_pulse_width` 的单调性检查：
 - 针对包含 "input_pins" 的 cell，遍历每个输入引脚。
 - 如果引脚是时钟引脚，则检查其 "timing_arcs"。

- 针对 "timing_type" 为 "min_pulse_width" 的 timing arc, 对 "rise_constraint" 和 "fall_constraint" 调用 checkTimingArcMonotonicity 进行单调性检查。
- checkTimingArcMonotonicity 函数日志区分有无 when 信息, 输出不同日志。
- checkTimingArcMonotonicity 核心比较功能取消了等于的情况, 相等情况默认为通过。
- checkTimingArcMonotonicity 核心比较功能增加了判断 related_pin 是否等于当前 pin, 不等于则跳过。最终改为: 如果矩阵中当前的值小于前一个值, 并且当前引脚不是相关引脚, 或者当前值和前一个值都为零, 那么就认为这些值不是单调递增的。

2.3.9 2025-03-19

- 完善 supercell 方法, 如果有有时钟信号引脚, 链式长度被设为 1, 再生成超级单元。
- 解决 voltage 浮点数误差问题, `std::round(voltage_ * 100) / 100.0` 保留两位小数。

2.3.10 2025-03-21

- 新增 func 子命令, 用于检查库或者 Verilog 文件的逻辑等价性。

2.3.11 2025-03-24

- 调研了 C++ 操作 Verilog 相关的库, 找到了一个实用的 Verilog 库: `slang`, 它可以解析出抽象语法树 (AST)。已修改 CMakeLists 文件, 将 slang 集成到项目中, 目前正在研究其 API。

2.3.12 2025-03-25

- 新增 `verilog_utils.cpp` 和 `verilog_utils.hpp` 文件, 并创建了 `VerilogVisitor` 类, 用于自定义遍历 Verilog AST。
- 验证了 slang 对不同类型逻辑单元的解析能力:
 - 简单组合逻辑 (如 AND2D0)、复杂时序逻辑 (如 CMPE42D1) 和基本时序逻辑 (如 DFQD1) 均能被正确解析。
 - 用户自定义原语 (UDP), 例如 `tsmc_dff`, 会被错误地识别为模块实例化, 并产生 "Invalid instance declaration" 警告。
 - 成功提取了模块名、端口方向 (input、output) 及名称。
 - 能够解析命名端口连接的层次化实例化, 并获取模块名、实例名称和端口映射关系。
 - 能够解析门级原语实例化, 并获取门类型、实例名称、输出端口和输入端口。

2.3.13 2025-03-26

- 进一步完善了对 UDP（如 tsmc_dff）端口映射关系的解析，但其内部 Entry 尚未处理。
- 尝试了通过继承 slang::syntax::SyntaxRewriter 类创建自定义类 CellExtractor，实现了只保留目标模块、删除其他模块，并将修改后的语法树另存为新文件。使用 slang::syntax::SyntaxPrinter::printFile() 方法可以将 Verilog 代码输出到文件。
- 创建了 CellPrinter 类，继承自 slang::syntax::SyntaxVisitor 类。当访问到目标模块时，使用 module.toString() 方法也能将 Verilog 代码输出到文件，但发现输出结果中空行会被删除。
- 修改了 LibFile::supercell() 方法，该方法可以根据输入的 cell_names 生成指定的 supercell，并在遇到未找到的单元时提示警告信息。
- 优化了时钟引脚的处理方式，现在时钟引脚不再被视为 supercell 的输入引脚，而是仅记录为时序单元，并跳过存入 input_pins 集合的步骤。
- 引入了 Doxygen 文档生成工具，用于可视化分析项目，并生成了 Doxygen HTML 文档和 LaTeX 参考手册。
- [x] 尚未解决手册中文显示不正确的问题，可能需要自定义 LaTeX 头文件。

2.3.14 2025-03-27

- **重大失误！** 误操作 Git 分支，导致部分代码修改丢失。务必牢记：** 及时提交代码！ **
- 实现了 LibFile::verilog 方法的核心流程：
 - 首先调用 this->supercell() 生成超级单元。
 - 读取 .map 文件，提取单元映射关系到 std::pair<std::string, std::string> 中（原始单元名 -> 超级单元名/模块名）。
 - 从 lib_json_ 中提取单元的输入/输出端口信息。
 - 根据是否存在时钟引脚，确定 Verilog 模块中 instance 的生成数量（时序逻辑为 1，组合逻辑等于 chain length）。
 - 通过字符串操作生成 ANSI 风格的端口列表，并与模块名组合成完整的 fullModuleText。
 - 使用 slang::syntax::SyntaxTree::fromText 从 fullModuleText 生成语法树。
 - 将语法树传递给 ModuleRewriter 类，使用其 transform 方法遍历模块成员，并通过 handle() 方法进行处理。
 - 最后，使用 slang::syntax::SyntaxPrinter::printFile(*tree) 将处理后的语法树输出到文件。
- LibFile::verilog 目前能够正确输出模块名和 ANSI 风格的端口声明。

2.3.15 2025-03-28

- 实现了在模块成员列表中插入新节点的功能。具体而言，在 `ModuleRewriter::handle(const slang::syntax::ModuleDeclarationSyntax &module)` 方法中，通过 `insertAtBack(module.members, newDataNode)` 函数，成功地在模块成员的末尾插入了链式单元所需的中间变量声明，例如 `wire OP_i;`。
- 成功地在模块内部添加了关键连接网络变量。通过解析 `CELLNAME__X#__CRITICALPORT__OUTPUTPORT` 格式的字符串，提取出关键输入端口 (critical input port) 和输出端口 (OUTPUTPORT)，用于后续的实例化端口连接。
- 增加了用于处理多输出接口的中间网络变量 `P__UNUSEOUTPUTPORT`，用于连接不需要考察的输出端口。
- 成功地在模块内添加了模块实例化所需的模块名和实例名称，完成了端口映射关系的连接处理。
- 验证了代码对简单组合逻辑单元（如 `INVD0` 反相器）和复杂组合逻辑单元（如 `FA1D0` 全加器）的输出能力，结果均符合预期。
- 为 `ModuleRewriter` 类添加了私有属性 `std::map<std::string, std::string> portInfoMap_`，用于存储端口映射关系，将端口名映射到其方向 (input/output)。
- 完成了实例化端口连接的细节处理：
 - 如果是关键输入端口且为第一个实例，则直接连接到输入端口；否则，连接到 `OP_(i-1)` 这样的中间网络变量。
 - 如果是关键输出端口，且为最后一个实例，则连接到输出端口；否则，连接到 `OP_i` 中间网络变量。
 - 如果是其他输出端口且不是最后一个实例，则连接到 `P_i__portName` 这样的中间网络变量；否则，连接名等于端口名。
 - 其余输入端口，连接名等于端口名。
- 文档方面，GitHub 远程仓库已开放为公开访问，并配置了 GitHub Pages。可以通过 <https://cedar17.github.io/ZlibValidation/> 访问项目文档，该页面包含 Doxygen 生成的 HTML 文档和仓库地址的链接。
- 配置了 GitHub Actions，实现了 Doxygen 文档和 Graphviz 图的自动化构建，以及 LaTeX 参考手册的自动编译（暂不支持中文）。每次在 `dev` 和 `main` 分支的提交都会触发文档和参考手册的生成，并发布到 `gh-pages` 分支。
- 将第三方库放置到 `include_3rd_party` 目录下，方便管理。修改了 `CMakeLists` 文件，将第三方库的头文件路径添加到 `include_directories` 中。修改了 `Doxyfile` 文件，将第三方库的头文件路径添加到 `INPUT` 中，便于文档生成工具理解第三方库函数、类的关系及使用方法。

2.3.16 2025-03-29

- 修复了 Doxygen 生成 LaTeX 参考手册时中文显示不正确的问题。通过修改 Doxyfile 文件，添加 `EXTRA_PACKAGES = ctex` 选项，并指定 `lualatex` 作为编译器，成功解决了中文显示问题。同时，从 `include_3rd_party` 目录中移除了 `Allsyntax.h` 文件，避免了由于 LaTeX 文档过大导致的 `"TeX capacity exceeded, sorry [main memory size=5000000]"` 内存不足问题。
- 实现了 Verilog 文件的顶层模块生成功能，采用非 ANSI 风格的端口声明方式，格式与 `liberate_lv` 工具生成的完全一致。实现过程如下：
 - 首先将各模块的 Verilog 代码存放到临时文件中。
 - 在处理模块的过程中，使用 `std::vector<std::string>` 收集每个模块的 `input_pins` 和 `output_pins`。
 - 遍历所有模块名称列表，拼接模块名和引脚名，生成顶层模块的 `all_input_ports` 和 `all_output_ports`。
 - 使用字符串拼接方式构建 `validate_top` 模块的完整代码。
 - 最后将临时文件内容与顶层模块代码合并，输出到最终的 Verilog 文件中。
- 更改 `deploy-docs.yml`，修复了 GitHub Action 生成的 LaTeX 参考手册中的图像显示问题。具体步骤如下：
 - 设置系统时区为 Asia/Shanghai (东八区)。
 - 安装 miniconda 并更新 conda。
 - 通过 conda 安装 1.9.5 版本的 Doxygen，并安装 graphviz 以解决依赖关系，确保每个 dot 图都能正确编译出 pdf 文件。
 - 使用 `doxygen` 命令生成 HTML 文档和 tex 文件。
 - 采用 action marketplace 提供的 `xu-cheng/texlive-action@v2` 安装全量的 `texlive2024`。
 - 进入 `./doc_doxygen/latex` 目录，运行 `make` 命令使用 `lualatex` 编译 tex 文件，生成 pdf 文件。
 - 使用 `JamesIves/github-pages-deploy-action@v4` 将生成的 pdf 文件上传到 GitHub Pages 发布，并保留 `README.md` 和 `index.html` 文件。

2.3.17 2025-03-30

- 完成了 `LibFile::logic(const std::string &cell_name)` 方法，该方法返回指定 `cell_name` 所有输出引脚对应的逻辑表达式，以 `std::map<std::string, std::string>` 的形式返回，存储输出引脚名称到其逻辑表达式字符串的映射关系。
- 实现了 `func` 子命令的基本框架，用于检查库或 Verilog 文件的逻辑等价性。
 - 完善了 `funcLibFile` 函数，能够处理 Liberty 文件和 Verilog 文件作为参考和比较对象。
 - 增加了文件类型判断，根据文件扩展名选择不同的处理方式。

- 实现了对 .lib 文件的解析，并提取指定 cell 的逻辑表达式。
- 目前仅支持 Liberty 文件的逻辑表达式提取，Verilog 文件的逻辑表达式提取功能尚未完成 (TODO)。
- 增加了对 report 文件名的检查，确保其以 .md 或 .txt 结尾。若不符合，则给出警告并自动添加 .md 后缀。
- 实现了从 Liberty 文件中提取逻辑表达式的功能，使用了 `LibFile::logic` 方法。
- 增加了日志输出，记录每个 cell 的逻辑表达式提取和比较过程。
- 逻辑比较部分目前为空，待实现 (TODO)。

2.3.18 2025-03-31

- 实现了从 Verilog 结构网表中提取逻辑表达式的功能，使用了 `LogicExtractor` 类。
 - 设计了如何使用 Slang 库从 Verilog 结构网表中提取逻辑表达式的方案：
 - * 使用 Slang 解析 Verilog 文件获取语法树 (AST)。
 - * 遍历 AST 定位目标模块。
 - * 在目标模块内构建网表的内部表示，识别输入、输出、线网及门级连接关系。
 - * 对每个输出端口，递归反向追踪门和线网直至到达主输入端口。
 - * 在回溯过程中根据遇到的门类型构建逻辑表达式。
 - * 应用记忆化技术避免对相同线网/信号的重复计算。
 - 创建了 `LogicExtractor` 类，用于从 Verilog 代码中提取逻辑表达式。
 - * `LogicExtractor` 类使用 Slang 库解析 Verilog 文件，并构建目标模块的网表表示。
 - * 实现了 `handle(const slang::syntax::ModuleDeclarationSyntax &module)` 方法，用于定位目标模块，并初始化内部数据结构。
 - * 实现了 `handle(const slang::syntax::PortDeclarationSyntax &portDecl)` 方法，用于提取端口信息，包括端口方向和名称。
 - * 实现了 `handle(const slang::syntax::NetDeclarationSyntax &netDecl)` 方法，用于提取线网信息。
 - * 实现了 `handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst)` 方法，用于提取门级单元信息，包括门类型、输入和输出信号。
 - * 实现了 `getLogicExpressions()` 方法，用于根据提取的网表信息，递归地推导每个输出端口的逻辑表达式。
 - * 实现了 `deriveLogicRecursive(const std::string &signalName)` 方法，用于递归地推导指定信号的逻辑表达式。
 - * 实现了 `formatExpression(const GateInfo &gateInfo, const std::vector<std::string> &inputExprs)` 方法，用于根据门类型和输入表达式，生成逻辑表达式字符串。
 - 实现了 `extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell)` 函数，用于提取并打印网表信息，包括输入、输出、线网和门级单元。

- 实现了 `extractLogicFromVerilog(const std::string &verilog_file, const std::string &cell)` 函数，用于从 Verilog 文件中提取指定 cell 的逻辑表达式，并以 `std::map<std::string, std::string>` 的形式返回。
- 为了更好地组织代码结构，将 `LogicExtractor` 类从 `verilog_utils.hpp` 和 `verilog_utils.cpp` 中分离出来，并创建了独立的 `LogicExtractor.hpp` 和 `LogicExtractor.cpp` 文件。同时，统一了所有 `hpp` 和 `cpp` 文件的命名规范，使其与对应的类名保持一致。
- 新增 `LogicComparator` 类及其头文件、源文件，用于比较两个逻辑表达式的等价性，并生成比较报告。

2.4 2025-04

2.4.1 2025-04-01

- 整理头文件引用，删除了不必要的引用，提高编译效率。
- 引入 `exprtk.cpp` 到第三方头文件目录，为后续的逻辑表达式解析做准备。
- 完善了 `LogicComparator` 类的实现，增加了其示例代码模板 `template <typename T> void logic();`，该模板能够对一个表达式遍历输入变量的所有组合，计算出逻辑值，并打印真值表，为逻辑等价性验证提供基础。
- 实现了 `LogicComparator::preprocessExpression()` 方法，用于预处理逻辑表达式，目的是简化表达式，使其更易于比较。预处理的逻辑如下：
 - **** 统一 AND 处理 ****：将 `*` 和空格隐式 AND 都转换为带空格的 `and`，保证 AND 运算符的显式性。
 - **** 显式符号处理 ****：
 - * 在所有括号 `()` 周围添加空格，方便后续分词。
 - * 将 `+` 替换为 `or`。
 - * 将 `^` 替换为 `xor`。
 - * 将 `!` (非 `!=` 中的 `!`) 替换为 `not`。
 - * 将 `*` 替换为 `and`。
 - **** 分词 ****：基于空格将处理后的字符串分割为 `token` 列表。
 - **** 精确插入隐式 AND ****：
 - * 遍历 `token` 列表。
 - * 检查当前 `token` 和下一个 `token`，如果满足以下条件，则在当前 `token` 之后插入 `and`：
 - 当前 `token` 是一个标识符 (如 `A`, `CIX`) 或右括号 `)`。
 - 下一个 `token` 是一个标识符或左括号 `(`。
 - 当前 `token` 不是一个显式逻辑运算符 (`and`, `or`, `xor`, `not`) 或左括号 `(`。
 - 下一个 `token` 不是一个显式逻辑运算符或右括号 `)`。

- **** 重组与清理 ****: 将处理后的 token 列表用单个空格连接起来, 并进行最终的空格清理 (去除多余空格, 修剪首尾空格), 得到最终的预处理表达式。
- 在 CMakeLists.txt 中, 强制启用了 lld 链接器和 ccache, 以提升链接和编译速度。
- 为了保持 main.cpp 的整洁, 将 LibFile() 函数的相关操作剥离出来, 并放入 LibFileOperations.cpp 文件中。
- 实现了 LogicComparator::extractVariables() 方法, 用于从两个表达式中提取并验证变量名, 确保它们一致。
 - 该方法使用正则表达式初步提取所有可能的标识符。
 - 对每个提取出的标识符, 先用 isIdentifier 函数 (检查大写开头等) 进行验证。
 - 通过 isIdentifier 验证后, 仍然将其转换为小写并与 keywords 集合比较, 以过滤掉可能的关键词 (这是一个保守但安全的操作)。
 - 只有通过这两步验证的标识符才被认为是有效变量, 并以原始大小写形式添加到各自的 std::set 中。
 - 最后比较两个 std::set 是否相等, 如果不等则报告差异并返回 false, 如果相等则将变量列表 (保留原始大小写) 排序后存入 sorted_vars 并返回 true。
- 修复了 LogicComparator::preprocessExpression() 方法中对 not 操作符的处理。exprtk 库要求 not 关键字后必须跟着括号, 如 not(A) 而非 not A。为此增加了专门的处理步骤:
 - 遍历预处理后的 token 列表, 识别所有 not 关键字。
 - 当发现 not 后, 检查其后是否跟随标识符 (通过 isIdentifier() 函数判断)。
 - 如果后接标识符, 自动将 not A 形式转换为 not(A), 具体做法是依次添加 not、(、标识符和), 并跳过已处理的标识符。
 - 如果后接其他元素 (如 (、其他运算符或表达式结尾), 保留原样, 让 ExprTk 处理 not(表达式) 形式。
 - 对其他 token, 直接添加到最终 token 列表, 不做特殊处理。
- 实现了 LogicComparator::compareSingleExpressionPair() 方法, 用于比较两个逻辑表达式的等价性:
 - 接收预处理后的表达式字符串和已排序的变量列表作为输入参数。
 - 为参考表达式和比较表达式分别创建 ExprTk 环境 (符号表、表达式对象和解析器)。
 - 将所有变量添加到两个符号表中, 确保变量一致性和顺序性。
 - 使用 ExprTk 解析器编译两个表达式, 并检查可能的语法错误。
 - 通过二进制计数法, 遍历所有可能的输入变量组合 (2^N 种), 其中 N 为变量数量。
 - 对每种组合:
 - * 设置两个符号表中的变量值 (0 或 1)。
 - * 计算两个表达式的值, 并处理可能的运行时错误。
 - * 将计算结果转换为布尔值并存入结果向量。

- * 同步构建真值表，记录输入组合和对应的输出值。
- 比较两个结果向量判断表达式是否等价。
- 将生成的真值表存入 `PinComparisonResult` 对象的 `std::optional` 成员中，以便后续报告生成。
- 实现了 `template <typename T> std::map<std::string, PinComparisonResult> LogicComparator::compareCellLogic();`
 - 该方法用于比较两个 cell 的逻辑表达式，返回一个 `std::map`，键为 cell 名称，值为 `PinComparisonResult` 结构体。
 - 遍历在类私有属性里的整个独立输出引脚键，提取其值作为逻辑字符串表达式，首先使用 `preprocessExpression` 进行预处理，然后使用 `extractVariables` 提取变量向量。
 - 调用 `LogicComparator::compareSingleExpressionPair()` 方法进行比较，结果存入 `PinComparisonResult` 对象。
 - 最后将 `PinComparisonResult` 对象存入 `std::map` 中，键为 cell 名称，值为 `PinComparisonResult` 对象，返回。
- 实现了 `LogicComparator::generateReport()` 方法，报告格式仍需要调整。

2.4.2 2025-04-02

- 取消所有模板函数的 `template <typename T>` 声明，改为使用 `double` 作为参数类型，加快编译速度。
- 完善了 `LogicComparator::generateReport()` 方法，增加了对逻辑比较结果的详细报告输出，包括：
 - 元数据，例如： `**Performed by ZlibValidation v0.1.0 from Song Zixuan. on: Wed Apr 2 11:11:39 2025**`
 - 图例，包括参考 pin 名称、pin function，参考 pin 名称、pin function，符号解释表。
 - 参考 pin 的真值表
 - 比较 pin 的真值表
 - 比较结果表格，例如：

Property	Value
Status	[OK]
Reference (Raw)	(!(!(A1+A2))+B))
Comparison (Raw)	!((!A1 * !A2) + B)
Reference (Processed)	(not((not(A1 or A2)) or B))
Comparison (Processed)	not((not(A1) and not(A2)) or B)
Ref Expression Compiled	Yes
Comp Expression Compiled	Yes
Logically Equivalent	Yes

2.4.3 2025-04-05

- 新增了 `LibFile::spice()` 方法，为 `spice` 子命令添加了 SPICE 网表生成功能。 - 通过执行 `which v2lvs` 命令检测系统中是否安装了 V2LVS 工具。如果已安装，则调用该工具生成基本的 SPICE 网表。 - 添加了 `--vl` 和 `--sl` 命令行选项，允许用户指定 Verilog 库文件和 SPICE 库文件的路径。 - 目前已能生成基本的 SPICE 网表，后续需要进一步完善输出格式。

Generate SPICE file for given Liberty file

Usage: ./zlibvalidation spice [OPTIONS] library_path...

Positionals:

library_path TEXT:FILE ... REQUIRED

Specify the library file to process

Options:

-h,--help Print this help message and exit

-l,--log TEXT Specify the log file name. Default: <basename>.spice.log

-c,--chain INT Specify the chain length for SPICE generation. Default: 1

--cells TEXT ... Specify the cell names to generate SPICE for

--vl TEXT Specify the location of the Verilog primitive library file

--sl TEXT Specify the location of the SPICE library file to be included in the output

- 实现私有方法 `LibFile::splitString()`，用于将字符串按空格分割成多个子字符串，并返回一个 `std::vector<std::string>`。
- 实现了 `LibFile::generateRCLines()` 方法，用于为给定的 net 生成 RC lines。
 - 该方法根据 `isFinalStage` 参数，生成不同的 RC 结构。
 - 如果 `isFinalStage` 为 `false`，则生成 R1-C1-R2 结构。
 - 如果 `isFinalStage` 为 `true`，则生成 R1-C1 结构。 - RC 的默认值参考目标 SPICE 文件。
- 实现了 `LibFile::modifySpiceNetlist()` 方法，用于修改 `v2lvs` 生成的 SPICE 网表。
 - 该方法添加元数据注释，包括应用名称、版本、作者和时间戳。
 - 在第一个 `.INCLUDE` 指令后插入指定的 global line。
 - 修改 subcircuit，在 `.SUBCKT` 定义的端口列表中添加 VDD VSS。
 - 对于 subcircuit 中的每个 instance line (以 'X' 或 'x' 开头)，修改输出引脚名称，添加 VDD/VSS 连接，并调用 `generateRCLines` 生成 R/C lines。
 - 保留原始注释，跳过空行、`v2lvsheader` line 和原始的 `.GLOBAL` line。
- 完全实现了 `LibFile::spice()` 方法，能够生成 SPICE 网表并输出到指定文件中。SPICE 网表的生成过程如下：
 - 检查是否指定了 Verilog 库文件和 SPICE 库文件，如果没有指定，则使用默认路径。
 - 调用 `v2lvs` 工具生成初步的 SPICE 网表。
 - 调用 `LibFile::generateRCLines()` 方法，为 SPICE 网表中的每个 instance 生成对应的 RC lines。
 - 调用 `LibFile::modifySpiceNetlist()` 方法，修改 SPICE 网表，添加元数据、插入 global line、调整 instance lines，并生成最终的 SPICE 网表。
 - 将最终的 SPICE 网表输出到指定文件中。

2.4.4 2025-04-07

- 将项目研发日记从 [README.md](#) 迁移至独立的 [ChangeLog.md](#) 文件，存放于 `doc/` 目录下，集中记录更新与变更历史。
- 全面完善 [README.md](#)，补充项目描述、目录、动机、安装指南、使用示例、文档与手册、致谢等内容，提升项目信息的完整性。
- 调整 `CMakeLists.txt`，取消强制使用 `ccache` 和 `lld` 的设定，改为检测系统环境，仅在存在相应工具时启用，增强项目兼容性。
- 移除 `FetchContent` 中的 `QUIET` 选项，增加必要提示信息，更清晰地展示依赖库的获取状态。
- 项目版本更新至 `v1.1.0`。
- 明确 `CMake` 版本需求为 `(VERSION 3.21...3.31)`，确保项目在指定版本范围内构建。
- 在 [README.md](#) 中，特别感谢 `ZlibBoost` 项目，并新增编译文档和参考手册的教程，同时将 `ccache` 和 `lld` 列为可选编译准备。

2.4.5 2025-04-10

- 修正了 `printInfo()` 函数中的注释错误，将文件日志级别更正为 `trace`，而非 `debug`。
- 添加了 `mono` 子命令的测试命令到 `test.sh` 脚本中。

2.4.6 2025-04-15

- 修复了 `LibraryComparator::generateReport` 方法中的一个 bug，`max_diff` 现在能基于绝对值比较得出最大差异，并且最大差异百分比保持一致。
- 完善了所有关键子命令的测试命令，确保每个子命令都能在 `test.sh` 脚本中正确执行。

2.5 2026-05

2.5.1 2026-05-13

- 修复作者联系邮件信息在 `cmakelists` 修改后不能自动更新到 `version.h` 的问题。导致这个问题的原因是 `CMakeLists.txt` 中使用了 `CACHE` 变量。`set(VAR "VALUE" CACHE STRING "Doc↵String")` 这种方式只会在这两个变量还未缓存时赋值。如果在 `CMakeLists.txt` 中修改了内容，而且 `CMake` 之前已经生成过并保存了 `Cache` 文件 (`CMakeCache.txt`)，`CMake` 就不会再覆盖它们的值，因此相应的模板替换也就不会更新出新的结果。补充了中文名、备用邮箱、更新了版本号为 `1.1.3`。

- 集成SQLiteCpp 和 GoogleTest。
 - 通过 FetchContent 拉取 SQLiteCpp 3.3.1 和 GoogleTest v1.14.0。
 - 新建 test/ 测试目录及 test/CMakeLists.txt, 通过 file(GLOB_RECURSE) 自动发现测试文件。
 - 新增 test/utlis/common.hpp, 提供 DatabaseTest 测试夹具, 内置 :memory: 内存数据库生命周期管理。
 - 编写 test_sqlite.cpp, 验证 SQLiteCpp 基本 CRUD 读写能力。

2.6 2026-06

2.6.1 2026-06-22

- .gitignore 添加 CLAUDE.md 条目, 避免项目级 AI 辅助文件被跟踪到版本控制中。
- 实现 parse --db SQLite 数据库写入功能, 将 Liberty LUT 条目持久化到 SQLite 数据库:
 - 新建 LibDatabase 类 (RAII 封装), 管理 SQLite 连接生命周期。
 - * initialize() 执行 DDL 创建 lut_entries 表 (仅含此表, 其余表由 Python 构建脚本管理)。
 - * 使用 INSERT OR IGNORE 实现写入幂等性: 相同 UNIQUE 键 (file_path, pvt_corner, aged_year, arc_type, related_pin) 的重复写入自动忽略。
 - * LUT 的 index_1、index_2、values 数组以 float32 BLOB 格式存储, 相比 TEXT/JSON 节省约 60% 空间。
 - * 启用 WAL journal 模式和 busy_timeout, 支持并发读写场景。
 - * scenario_id 字段预留为 NULL, 留待后续 Python 脚本回填。
 - LibFile::writeToDB() 方法实现完整的层次遍历:
 - * 从 lib_json_ 中提取 PVT 头信息 (process, temperature, voltage)。
 - * 逐层遍历 cells → output_pins → timing_arcs → 四种 arc_type (cell_rise, cell_fall, rise_transition, fall_transition)。
 - * 每个 LUT 写为 lut_entries 表的一行, 携带完整的层级上下文。
 - CLI 层新增 --db、--pvt-corner、--aged-year、--lib-json 四个选项。
 - CMakeLists.txt: 将 SQLiteCpp 链接到主目标。
 - 编写数据库写入测试 (8 个 GoogleTest 用例):
 - * test_libdb.cpp: 单元测试 LibDatabase, 覆盖建表、单条目读写、BLOB 逐字节验证、UNIQUE 约束、四种 arc_type。
 - * test_libfile_db.cpp: 集成测试 LibFile::writeToDB() 与真实 .lib 文件 (tcbn65lpsc.ski.lib), 覆盖完整写入流程、BLOB 值合理性、幂等写入、scenario_id IS NULL 不变式。

* 每个测试使用独立的临时 .db 文件，避免并行测试锁冲突。

▪ 重构 parse 子命令的 CLI 选项与 DB 输出行为：

- --pvt-corner 更名为 --pvt，语义更通用（不限于老化场景）。
- --aged-year 从必选项改为可选项（默认 0.0），适用于非老化库。
- 移除 --lib-json 选项：DB 模式下不再同时输出 JSON，JSON 与 DB 输出正交化（纯 parse=JSON，parse -db=仅写 DB）。
- 明确 process 参数的语义：存储 .lib 头部的原始整数值，不映射为 SS/FF/TT，映射由调用者通过 --pvt 负责。

2.6.2 2026-06-23

▪ parse --db SQLite 入库时 when 条件丢失导致去重数据丢失修复：

- **** 问题 ****：when（同一 related_pin 区分不同时序弧的条件布尔表达式）未入库、未进 UNIQUE 约束，导致同一（cell, output_pin, related_pin, arc_type）下不同 when 的时序弧被 INSERT OR IGNORE 静默丢弃。真实生产级 PDK 入库测试：Wrote 35560，DB 仅实存 7936（丢失 77.7）。
- ****DDL****：lut_entries 表新增 "when" TEXT 列（放在 timing_type 与 arc_type 之间），UNIQUE 约束加入 "when"。
- ****INSERT/bind****：INSERT 列和占位符同步加 when（第 11 位）。空 when 绑定空字符串而非 NULL，避免 SQLite UNIQUE 约束视多 NULL 为不冲突导致幂等破坏。
- **** 提取 ****：LibFile::writeToDB() 从 timing arc JSON 中提取 arc.value("when", "") 传入。
- ****TDD 驱动****：新增 test_libdb.cpp:DifferentWhenSameKeyBothPersisted（纯 DB 层验证 when 进 UNIQUE）、test_libfile_db.cpp:WhenConditionPreserved（集成层，ski.lib related_pin=A 有 4 distinct when 全入库）、MultipleWhenArcsAllPersisted(>=12 行)；同步更新现有测试的 SELECT/DISTINCT 加 when 列。
- **** 验证 ****：生产级 PDK 集成验证 DB 条目 7936→35536（恢复 99.9），条件弧 27772 全保留，distinct when=717，二次写入幂等不变。24 条差异（35560→35536）已追溯为 PDK .lib 数据本身的真实重复（某特定单元重复定义），非代码 bug。

2.7 2026-07

2.7.1 2026-07-02

- v1.2.2 file_path 移出 lut_entries UNIQUE 约束。

- ** 问题 **: `file_path` 在 UNIQUE 约束中导致同一份 `.lib` 数据从不同路径 (`aging-ml/↵ DATA/raw/、0516_scripts/LIBRARY/、Aging/ML/DATA/raw/`) 入库时被插入 2~3 次。fresh 数据 249,612 行中 142,360 行冗余。
- ** 修复 **: `file_path` 保留为溯源列 (TEXT NOT NULL), 移出 UNIQUE 约束。改用纯语义键 (`pvt_corner, aged_year, cell_name, output_pin, related_pin, "when", arc_type`) 去重。
- ** 影响 **: 新入库数据按语义键去重, 同一库多路径源不再产生冗余。aging-ml 侧需重建表 + `'INSERT GROUP BY COALESCE(scenario_id,')`` 清理历史冗余。

2.7.2 2026-07-06

- **v1.2.3** `aged_year` → `age_seconds` 全量重构, CLI `--aged-year` 改为 `--age-time`。
 - ** 问题 **: `aged_year` 字段名为“年”, 但 `ZlibValidation` 拿到 `double` 值后不校验单位直接 `bind`。新 `ZlibAging` 管线 fresh 基线从 `0.01y` 改 `0.01h (=36s)`, 继续用“年”存则值为 `1.14e-6`, 浮点比较易腐化。这是命名债务。
 - ** CLI 参数 **: `--aged-year (double)` → `--age-time (string + 单位后缀)`。新增 `parse_age_time()` 解析 `y/d/h/裸秒` → `秒`: `0.01h→36.0s`, `10y→315360000.0s`, `36 (裸秒)→36.0s`, `""→0.0s`。
 - ** DB schema **: `lut_entries.aged_year` → `lut_entries.age_seconds` (DDL + UNIQUE + INSERT + bind 全链路同步)。
 - ** 内部语义 **: 所有变量名/参数名/注释 `aged_year` → `age_seconds` (涉及 `LibFile.hpp`、`LibFile.cpp`、`LibDatabase.hpp`、`LibDatabase.cpp`、`main.cpp`、`test_↵ _libdb.cpp`、`test_libfile_db.cpp` 共 8 个文件)。
 - ** 清理 **: 归档 `doc/sqlite_dedup_analysis.md` (v1.2.1 bug 排查文档, 核心知识点已浓缩至 agent 记忆)。
 - ** 向后兼容 **: CLI 不向后兼容 (唯一调用方 `aging-ml` 已确认同步改); 旧 `aging_v0.db` 归档后不再读; 裸秒值兼容 (`--age-time 36` 等价于 `--age-time 0.01h`)。

Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

AttributesIterator	35
GateInfo	41
GroupsIterator	43
LibAttribute	45
LibDatabase	48
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LibGroup	69
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LogicComparator	78
PinComparisonResult	100
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CellExtractor	37
ModuleRewriter	97
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CellPrinter	39
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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CellExtractor	37
CellPrinter	39
GateInfo	41
GroupsIterator	43
LibAttribute	45
LibDatabase Manages SQLite database connections for writing Liberty LUT data	48
LibFile	53
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LibraryComparator	71
LogicComparator	78
LogicExtractor	87
ModuleRewriter	97
PinComparisonResult	100
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Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

include/ Iterators.hpp	111
include/ json_utils.hpp	112
include/ LibAttribute.hpp	117
include/ LibDatabase.hpp	118
include/ LibFile.hpp	119
include/ LibFileOperations.hpp	121
include/ LibGroup.hpp	129
include/ LibraryComparator.hpp	130
include/ LogicComparator.hpp	131
include/ LogicExtractor.hpp	133
include/ verilog_utils.hpp	136
include/ version.h	139
src/ Iterators.cpp	141
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src/ LibAttribute.cpp	150
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src/ LibraryComparator.cpp	186
src/ LogicComparator.cpp	191
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src/ main.cpp	
This file contains the main function for the ZlibValidation tool	218
src/ verilog_utils.cpp	226

Chapter 6

Class Documentation

6.1 AttributesIterator Class Reference

```
#include <Iterators.hpp>
```

Public Member Functions

- [AttributesIterator](#) (si2drAttrSldT attrs, si2drErrorT &err)
- [~AttributesIterator](#) ()
- void [next](#) ()
- bool [end](#) ()
- [LibAttribute](#) [get](#) ()

Private Attributes

- si2drAttrSldT [attrs_](#)
- si2drAttrIdT [attr_](#)
- si2drErrorT & [err_](#)

6.1.1 Detailed Description

Definition at line 23 of file [lterators.hpp](#).

6.1.2 Constructor & Destructor Documentation

6.1.2.1 AttributesIterator()

```
AttributesIterator::AttributesIterator (
    si2drAttrsIdT attrs,
    si2drErrorT & err )
```

Definition at line 14 of file [lterators.cpp](#).

6.1.2.2 ~AttributesIterator()

```
AttributesIterator::~~AttributesIterator ( )
```

Definition at line 18 of file [lterators.cpp](#).

6.1.3 Member Function Documentation

6.1.3.1 end()

```
bool AttributesIterator::end ( )
```

Definition at line 21 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.1.3.2 get()

```
LibAttribute AttributesIterator::get ( )
```

Definition at line 23 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.1.3.3 next()

```
void AttributesIterator::next ( )
```

Definition at line 20 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.1.4 Member Data Documentation

6.1.4.1 attr_

si2drAttrIdT AttributesIterator::attr_ [private]

Definition at line 33 of file [lterators.hpp](#).

6.1.4.2 attrs_

si2drAttrsIdT AttributesIterator::attrs_ [private]

Definition at line 32 of file [lterators.hpp](#).

6.1.4.3 err_

si2drErrorT& AttributesIterator::err_ [private]

Definition at line 34 of file [lterators.hpp](#).

The documentation for this class was generated from the following files:

- include/[lterators.hpp](#)
- src/[lterators.cpp](#)

6.2 CellExtractor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellExtractor:

Collaboration diagram for CellExtractor:

Public Member Functions

- [CellExtractor](#) (const std::string &targetCell)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)
- bool [foundTargetCell](#) () const

Private Attributes

- const std::string & [targetCell_](#)
- bool [foundTarget_](#)

6.2.1 Detailed Description

Definition at line 33 of file [verilog_utils.hpp](#).

6.2.2 Constructor & Destructor Documentation

6.2.2.1 CellExtractor()

```
CellExtractor::CellExtractor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 35 of file [verilog_utils.hpp](#).

6.2.3 Member Function Documentation

6.2.3.1 foundTargetCell()

```
bool CellExtractor::foundTargetCell ( ) const
```

Definition at line 364 of file [verilog_utils.cpp](#).

Here is the caller graph for this function:

6.2.3.2 handle()

```
void CellExtractor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 348 of file [verilog_utils.cpp](#).

6.2.4 Member Data Documentation

6.2.4.1 foundTarget_

```
bool CellExtractor::foundTarget_ [private]
```

Definition at line 42 of file [verilog_utils.hpp](#).

6.2.4.2 targetCell_

```
const std::string& CellExtractor::targetCell_ [private]
```

Definition at line 41 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

6.3 CellPrinter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellPrinter:

Collaboration diagram for CellPrinter:

Public Member Functions

- [CellPrinter](#) (const std::string &targetCell, std::ostream &out)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Private Attributes

- const std::string & [targetCell_](#)
- std::ostream & [out_](#)
- bool [foundTarget_](#)

6.3.1 Detailed Description

Definition at line [46](#) of file [verilog_utils.hpp](#).

6.3.2 Constructor & Destructor Documentation

6.3.2.1 CellPrinter()

```
CellPrinter::CellPrinter (  
    const std::string & targetCell,  
    std::ostream & out ) [inline], [explicit]
```

Definition at line [48](#) of file [verilog_utils.hpp](#).

6.3.3 Member Function Documentation

6.3.3.1 handle()

```
void CellPrinter::handle (  
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line [367](#) of file [verilog_utils.cpp](#).

6.3.4 Member Data Documentation

6.3.4.1 foundTarget_

```
bool CellPrinter::foundTarget_ [private]
```

Definition at line 55 of file [verilog_utils.hpp](#).

6.3.4.2 out_

```
std::ostream& CellPrinter::out_ [private]
```

Definition at line 54 of file [verilog_utils.hpp](#).

6.3.4.3 targetCell_

```
const std::string& CellPrinter::targetCell_ [private]
```

Definition at line 53 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

6.4 GateInfo Struct Reference

```
#include <LogicExtractor.hpp>
```

Public Attributes

- slang::parsing::TokenKind [kind](#)
- std::string [gateTypeName](#)
- std::vector< std::string > [inputSignals](#)
- std::string [outputSignal](#)

6.4.1 Detailed Description

Definition at line 9 of file [LogicExtractor.hpp](#).

6.4.2 Member Data Documentation

6.4.2.1 gateTypeName

```
std::string GateInfo::gateTypeName
```

Definition at line 11 of file [LogicExtractor.hpp](#).

6.4.2.2 inputSignals

```
std::vector<std::string> GateInfo::inputSignals
```

Definition at line 12 of file [LogicExtractor.hpp](#).

6.4.2.3 kind

```
slang::parsing::TokenKind GateInfo::kind
```

Definition at line 10 of file [LogicExtractor.hpp](#).

6.4.2.4 outputSignal

```
std::string GateInfo::outputSignal
```

Definition at line 13 of file [LogicExtractor.hpp](#).

The documentation for this struct was generated from the following file:

- [include/LogicExtractor.hpp](#)

6.5 GroupsIterator Class Reference

```
#include <Iterators.hpp>
```

Public Member Functions

- [GroupsIterator](#) (si2drGroupsIdT groups, si2drErrorT &err)
- [~GroupsIterator](#) ()
- void [next](#) ()
- bool [end](#) ()
- [LibGroup](#) [get](#) ()

Private Attributes

- si2drGroupsIdT [groups_](#)
- si2drGroupIdT [group_](#)
- si2drErrorT & [err_](#)

6.5.1 Detailed Description

Definition at line 9 of file [l iterators.hpp](#).

6.5.2 Constructor & Destructor Documentation

6.5.2.1 GroupsIterator()

```
GroupsIterator::GroupsIterator (  
    si2drGroupsIdT groups,  
    si2drErrorT & err )
```

Definition at line 3 of file [l iterators.cpp](#).

6.5.2.2 ~GroupsIterator()

```
GroupsIterator::~GroupsIterator ( )
```

Definition at line 7 of file [l iterators.cpp](#).

6.5.3 Member Function Documentation

6.5.3.1 end()

```
bool GroupsIterator::end ( )
```

Definition at line 10 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.5.3.2 get()

```
LibGroup GroupsIterator::get ( )
```

Definition at line 12 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.5.3.3 next()

```
void GroupsIterator::next ( )
```

Definition at line 9 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.5.4 Member Data Documentation

6.5.4.1 err_

```
si2drErrorT& GroupsIterator::err_ [private]
```

Definition at line 20 of file [lterators.hpp](#).

6.5.4.2 group_

si2drGroupIdT GroupsIterator::group_ [private]

Definition at line 19 of file [lterators.hpp](#).

6.5.4.3 groups_

si2drGroupsIdT GroupsIterator::groups_ [private]

Definition at line 18 of file [lterators.hpp](#).

The documentation for this class was generated from the following files:

- include/[lterators.hpp](#)
- src/[lterators.cpp](#)

6.6 LibAttribute Class Reference

```
#include <LibAttribute.hpp>
```

Public Member Functions

- [LibAttribute](#) (si2drAttrIdT attr, si2drErrorT &err)
- [~LibAttribute](#) ()
- std::string [getName](#) ()
- bool [isComplex](#) ()
- si2drValuesIdT [getValues](#) ()
- long int [getInt](#) ()
- double [getFloat](#) ()
- std::string [getString](#) ()
- bool [getBoolean](#) ()

Private Attributes

- si2drAttrIdT [attr_](#)
- si2drErrorT & [err_](#)

6.6.1 Detailed Description

Definition at line 8 of file [LibAttribute.hpp](#).

6.6.2 Constructor & Destructor Documentation

6.6.2.1 LibAttribute()

```
LibAttribute::LibAttribute (
    si2drAttrIdT attr,
    si2drErrorT & err )
```

Definition at line 3 of file [LibAttribute.cpp](#).

6.6.2.2 ~LibAttribute()

```
LibAttribute::~~LibAttribute ( )
```

Definition at line 5 of file [LibAttribute.cpp](#).

6.6.3 Member Function Documentation

6.6.3.1 getBoolean()

```
bool LibAttribute::getBoolean ( )
```

Definition at line 25 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

6.6.3.2 getFloat()

```
double LibAttribute::getFloat ( )
```

Definition at line 18 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

6.6.3.3 getInt()

```
long int LibAttribute::getInt ( )
```

Definition at line 16 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

6.6.3.4 getName()

```
std::string LibAttribute::getName ( )
```

Definition at line 7 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

6.6.3.5 getString()

```
std::string LibAttribute::getString ( )
```

Definition at line 20 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

6.6.3.6 getValues()

```
si2drValuesIdT LibAttribute::getValues ( )
```

Definition at line 14 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

6.6.3.7 isComplex()

```
bool LibAttribute::isComplex ( )
```

Definition at line 12 of file [LibAttribute.cpp](#).

6.6.4 Member Data Documentation

6.6.4.1 attr_

```
si2drAttrIdT LibAttribute::attr_ [private]
```

Definition at line 21 of file [LibAttribute.hpp](#).

6.6.4.2 err_

```
si2drErrorT& LibAttribute::err_ [private]
```

Definition at line 22 of file [LibAttribute.hpp](#).

The documentation for this class was generated from the following files:

- include/[LibAttribute.hpp](#)
- src/[LibAttribute.cpp](#)

6.7 LibDatabase Class Reference

Manages SQLite database connections for writing Liberty LUT data.

```
#include <LibDatabase.hpp>
```

Public Member Functions

- [LibDatabase](#) (const std::string &db_path)
Open (or create) a SQLite database at the given path.
- [~LibDatabase](#) ()
- void [initialize](#) ()
Create the lut_entries table if it does not already exist.
- void [beginTransaction](#) ()
- void [commitTransaction](#) ()
- void [writeLutEntry](#) (const std::string &file_path, const std::string &library_name, const std::string &pvt_corner, double age_seconds, const std::string &cell_name, const std::string &output_pin, const std::string &related_pin, const std::string &timing_sense, const std::string &timing_type, const std::string &when, const std::string &arc_type, int rows_n, int cols_n, const std::vector< double > &index_1, const std::vector< double > &index_2, const std::vector< double > &values, int process, double temperature, double voltage)
Insert one LUT record into lut_entries.

Private Attributes

- std::string [db_path_](#)
- std::unique_ptr< SQLite::Database > [db_](#)
- std::unique_ptr< SQLite::Statement > [insert_stmt_](#)
- bool [initialized_](#) = false

6.7.1 Detailed Description

Manages SQLite database connections for writing Liberty LUT data.

[LibDatabase](#) opens or creates a SQLite database file and provides methods to write timing look-up table (LUT) entries into the `lut_entries` table.

Responsibilities:

- Create the `lut_entries` table (DDL) on first use via [initialize\(\)](#)
- Write one LUT record per call to [writeLutEntry\(\)](#)
- Enforce idempotency via INSERT OR IGNORE

This class is deliberately scoped to Phase 2 of the aging.db migration: it only manages the `lut_entries` table. Other tables (scenarios, `k_values`, etc.) are managed by Python build scripts.

Definition at line 26 of file [LibDatabase.hpp](#).

6.7.2 Constructor & Destructor Documentation

6.7.2.1 LibDatabase()

```
LibDatabase::LibDatabase (
    const std::string & db_path ) [explicit]
```

Open (or create) a SQLite database at the given path.

Parameters

<i>db_path</i>	Filesystem path to the .db file.
----------------	----------------------------------

Definition at line 71 of file [LibDatabase.cpp](#).

6.7.2.2 ~LibDatabase()

```
LibDatabase::~LibDatabase ( ) [default]
```

6.7.3 Member Function Documentation

6.7.3.1 beginTransaction()

```
void LibDatabase::beginTransaction ( )
```

Definition at line 103 of file [LibDatabase.cpp](#).

Here is the caller graph for this function:

6.7.3.2 commitTransaction()

```
void LibDatabase::commitTransaction ( )
```

Definition at line 110 of file [LibDatabase.cpp](#).

Here is the caller graph for this function:

6.7.3.3 initialize()

```
void LibDatabase::initialize ( )
```

Create the `lut_entries` table if it does not already exist.

Safe to call multiple times — uses `CREATE TABLE IF NOT EXISTS`. Sets WAL journal mode and `busy_timeout`.

Definition at line 79 of file [LibDatabase.cpp](#).

Here is the caller graph for this function:

6.7.3.4 writeLutEntry()

```
void LibDatabase::writeLutEntry (
    const std::string & file_path,
    const std::string & library_name,
    const std::string & pvt_corner,
    double age_seconds,
    const std::string & cell_name,
    const std::string & output_pin,
    const std::string & related_pin,
    const std::string & timing_sense,
    const std::string & timing_type,
    const std::string & when,
    const std::string & arc_type,
    int rows_n,
    int cols_n,
    const std::vector< double > & index_1,
    const std::vector< double > & index_2,
    const std::vector< double > & values,
    int process,
    double temperature,
    double voltage )
```

Insert one LUT record into `lut_entries`.

Uses `INSERT OR IGNORE` so repeated calls with the same `UNIQUE` key (`pvt_corner`, `age_seconds`, `cell_name`, `output_pin`, `related_pin`, "when", `arc_type`) are no-ops. `file_path` is deliberately excluded from the `UNIQUE` constraint because the same `.lib` data may exist under multiple paths (symlinks, directory mirrors); it is stored as a provenance column but does not participate in semantic deduplication.

Parameters

<i>file_path</i>	Source .lib file path (for provenance only — not in UNIQUE).
<i>library_name</i>	Library name (basename of .lib file).
<i>pvt_corner</i>	PVT corner string (e.g. "SS_1p08V_125C").
<i>age_seconds</i>	Aging time in seconds (e.g. 36.0 for 0.01h fresh, 315360000.0 for 10y).
<i>cell_name</i>	Cell name extracted from .lib.
<i>output_pin</i>	Output pin name (e.g. "Y").
<i>related_pin</i>	Input related pin name (e.g. "A").
<i>timing_sense</i>	Timing sense (e.g. "negative_unate").
<i>timing_type</i>	Timing type (e.g. "combinational", "rising_edge").
<i>when</i>	Boolean condition expression distinguishing arcs that share the same related_pin (e.g. "A * !B"). Empty string for unconditional arcs — stored as NULL.
<i>arc_type</i>	Arc type: cell_rise, cell_fall, rise_transition, fall_transition.
<i>rows_n</i>	Number of rows in the LUT.
<i>cols_n</i>	Number of columns in the LUT.
<i>index_1</i>	Row index values (size == rows_n).
<i>index_2</i>	Column index values (size == cols_n).
<i>values</i>	Flattened LUT values (size == rows_n × cols_n, row-major).
<i>process</i>	Process number from .lib header (raw integer, e.g. 1). Not mapped to SS/FF/TT — that is the caller's responsibility via the -pvt CLI argument.
<i>temperature</i>	Temperature from .lib header.
<i>voltage</i>	Voltage from .lib header.

Definition at line 121 of file [LibDatabase.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.7.4 Member Data Documentation

6.7.4.1 db_

```
std::unique_ptr<SQLite::Database> LibDatabase::db_ [private]
```

Definition at line 104 of file [LibDatabase.hpp](#).

6.7.4.2 db_path_

```
std::string LibDatabase::db_path_ [private]
```

Definition at line 103 of file [LibDatabase.hpp](#).

6.7.4.3 initialized_

```
bool LibDatabase::initialized_ = false [private]
```

Definition at line 108 of file [LibDatabase.hpp](#).

6.7.4.4 insert_stmt_

```
std::unique_ptr<SQLite::Statement> LibDatabase::insert_stmt_ [private]
```

Definition at line 107 of file [LibDatabase.hpp](#).

The documentation for this class was generated from the following files:

- [include/LibDatabase.hpp](#)
- [src/LibDatabase.cpp](#)

6.8 LibFile Class Reference

```
#include <LibFile.hpp>
```

Public Member Functions

- [LibFile](#) (const std::string &filepath, const std::string &loggername)
Constructs a [LibFile](#) object with specified file path and logger name.
- [~LibFile](#) ()
- void [writeJsonToFile](#) ()
Writes the JSON data stored in the object to a file.
- void [parse](#) ()
Parses the Liberty file and extracts library information into a JSON structure.
- void [modify](#) ()
- void [mono](#) (const bool is_slew)
Validates that lookup tables are monotonically increasing with respect to the output load.
- void [supercell](#) (const int chain_length, const std::vector< std::string > &cell_names)
Creates supercells based on standard cells in the library file.
- void [writeToDB](#) (const std::string &db_path, const std::string &pvt_corner, double age_seconds)
- void [verilog](#) (const int chain_length, const std::vector< std::string > &cell_names)
Generates Verilog files for cell validation based on the library file.
- void [spice](#) (const int chain_length, const std::vector< std::string > &cell_names, const std::string &verilog_lib_file, const std::string &spice_lib_file)
Generates a SPICE netlist for the library based on a temporary Verilog file, using the V2LVS tool.
- std::map< std::string, std::string > [logic](#) (const std::string &cell_name)
Retrieves the logic functions for the output pins of a specified cell within the Liberty file.

Public Attributes

- std::shared_ptr< spdlog::logger > [logger_](#)
- std::filesystem::path [filepath_](#)
- std::string [basename_](#)
- std::string [filename_](#)
- std::string [libname_](#) = ""
- std::string [jsonname_](#) = ""
- std::string [loggername_](#) = ""
- [json lib_json_](#) = json::object()

Private Member Functions

- void [read](#) ()
Reads the Liberty file specified by the filename_ member variable.
- bool [checkTimingArcMonotonicity](#) (const [json](#) &cell, const [json](#) &pin, const [json](#) &arc, const std::string &type, bool is_slew)

Checks if the values in a timing arc are monotonically increasing.

- `std::vector< std::string > splitString (const std::string &s)`

Splits a string into a vector of tokens, using whitespace as the delimiter.

- `void generateRCLines (std::ofstream &outFile, const std::string &netName, int instanceIndex, bool isFinalStage)`

Generates RC lines for a given net in either an intermediate or final stage.

- `bool modifySpiceNetlist (const std::string &v2lvsSpiceFile, const std::string &finalSpiceFile, const std::string &targetGlobalLine)`

Modifies a SPICE netlist generated by v2lvs, adding metadata, inserting a global line, and adjusting instance lines within subcircuits to include VDD/VSS connections and generate corresponding R/C lines.

Private Attributes

- `si2drErrorT err\_`
- `int process\_ = 0`
- `float voltage\_ = 0.0`
- `int temperature\_ = 0`

6.8.1 Detailed Description

Definition at line 23 of file [LibFile.hpp](#).

6.8.2 Constructor & Destructor Documentation

6.8.2.1 LibFile()

```
LibFile::LibFile (
    const std::string & filepath,
    const std::string & loggername )
```

Constructs a [LibFile](#) object with specified file path and logger name.

This constructor initializes a [LibFile](#) object with the given file path and creates a logger with the specified name. It sets up:

- Two logging sinks: a colored console sink and a file sink
- File path information including filename, basename, and a corresponding JSON filename
- Log levels (info for console, debug for file)

Parameters

<i>filepath</i>	The path to the file to be processed
<i>loggername</i>	The name for the logger and the log file

Definition at line 17 of file [LibFile.cpp](#).

6.8.2.2 ~LibFile()

```
LibFile::~LibFile ( )
```

Definition at line 37 of file [LibFile.cpp](#).

6.8.3 Member Function Documentation

6.8.3.1 checkTimingArcMonotonicity()

```
bool LibFile::checkTimingArcMonotonicity (
    const json & cell,
    const json & pin,
    const json & arc,
    const std::string & timing_arc_name,
    bool is_slew ) [private]
```

Checks if the values in a timing arc are monotonically increasing.

This function examines the values in the specified timing arc to ensure they are monotonically increasing. It checks monotonicity in two dimensions:

1. Across rows (by output load capacitance)
2. Across columns (by input slew, only if `is_slew` is true)

The function logs detailed information about any non-monotonic values found, including cell name, pin names, and conditional statements (when clause) if present.

Parameters

<i>cell</i>	The JSON object representing the cell being checked
<i>pin</i>	The JSON object representing the pin being checked
<i>arc</i>	The JSON object representing the timing arc being checked
<i>timing_arc_name</i>	The name of the timing arc to check (e.g., "cell_rise", "cell_fall")
<i>is_slew</i>	Boolean flag indicating whether to check monotonicity across input slew values

Returns

true if all values in the timing arc are monotonic, false otherwise

Exceptions

<i>None, but</i>	logs errors or warnings for invalid data formats or non-monotonic values
------------------	--

Definition at line 312 of file [LibFile.cpp](#).

Here is the caller graph for this function:

6.8.3.2 generateRCLines()

```
void LibFile::generateRCLines (
    std::ofstream & outFile,
    const std::string & netName,
    int instanceIndex,
    bool isFinalStage ) [private]
```

Generates RC lines for a given net in either an intermediate or final stage.

This function generates SPICE-like resistor-capacitor (RC) lines and writes them to the provided output file stream. The generated RC structure differs based on whether it's an intermediate stage or the final stage. In the intermediate stage, an R1-C1-R2 structure is created. In the final stage, a simplified R1-C1 structure is used.

Parameters

<i>outFile</i>	The output file stream to write the RC lines to.
<i>netName</i>	The name of the net to generate RC lines for. This name is used to construct node names.
<i>instanceIndex</i>	A unique index for the instance, used to create unique component names (R1, C1, R2).
<i>isFinalStage</i>	A boolean flag indicating whether this is the final stage. If true, a simplified RC structure is generated. If false, an intermediate RCR structure is generated.

- The function uses predefined default RC values (R1, C1, R2) that are specific to either the intermediate or final stage.
- The CAP_GROUND constant defines the ground node to which capacitors are connected.
- The output is formatted in scientific notation with a precision of 1 before writing the RC lines and then reset to default.
- Node names are constructed by appending ":1" or ":2" to the netName for intermediate nodes.

```
// Example usage:
std::ofstream outFile("rc_lines.sp");
LibFile::generateRCLines(outFile, "net123", 5, false); // Generate intermediate RC lines for net
"net123", instance 5 LibFile::generateRCLines(outFile, "net123", 5, true); // Generate final RC
lines for net "net123", instance 5 outFile.close();
```

Definition at line 1038 of file [LibFile.cpp](#).

Here is the caller graph for this function:

6.8.3.3 logic()

```
std::map< std::string, std::string > LibFile::logic (
    const std::string & cell_name )
```

Retrieves the logic functions for the output pins of a specified cell within the Liberty file.

This method searches for a cell with the given name in the parsed Liberty file (either by parsing the file directly or reading from a JSON representation). It then iterates through the output pins of the cell, extracting the logic function associated with each pin. The logic functions are stored in a map, where the key is the pin name and the value is the logic function string.

If the JSON file does not exist, the Liberty file is parsed, and a JSON file is created. If the JSON file exists but cannot be opened or parsed, an error is logged, and an empty map is returned. If the specified cell is not found or if no logic functions are found for the cell, a warning is logged.

Parameters

<i>cell_name</i>	The name of the cell for which to retrieve the logic functions.
------------------	---

Returns

A map containing the logic functions for the output pins of the specified cell. The keys of the map are the pin names, and the values are the corresponding logic function strings. Returns an empty map if no logic functions are found or if an error occurs during file processing.

Definition at line 1499 of file [LibFile.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.8.3.4 modify()

```
void LibFile::modify ( )
```

Definition at line 289 of file [LibFile.cpp](#).

6.8.3.5 modifySpiceNetlist()

```
bool LibFile::modifySpiceNetlist (
    const std::string & v2lvsSpiceFile,
    const std::string & finalSpiceFile,
    const std::string & targetGlobalLine ) [private]
```

Modifies a SPICE netlist generated by v2lvs, adding metadata, inserting a global line, and adjusting instance lines within subcircuits to include VDD/VSS connections and generate corresponding R/C lines.

This function reads a SPICE netlist from v2lvsSpiceFile, modifies it according to the following rules, and writes the result to finalSpiceFile:

1. **Metadata Comments:** Adds a header with application name, version, author, and timestamp.
2. **Global Line Insertion:** Inserts a specified global line (targetGlobalLine) after the first .INCLUDE directive.
3. **Subcircuit Modifications:**
 - Adds VDD VSS to the port list of .SUBCKT definitions.
 - For each instance line (starting with 'X' or 'x') within a subcircuit:
 - Modifies the output pin name by appending ":1".
 - Inserts VDD VSS before the module name in the instance line.
 - Calls generateRCLines to generate and insert R/C lines for the instance.
4. **Comment Handling:** Preserves original comments, but processes any pending instance line before writing the comment.
5. **Empty/v2lvs Header Line Skipping:** Skips empty lines and lines starting with v2lvs header comments.
6. ****GLOBAL Line Skipping:**** Skips original .GLOBAL lines.

The function handles case-insensitive .SUBCKT and .ENDS directives. It also manages a previousInstanceLine buffer to handle instance lines that might need modification before the next line is processed.

Parameters

<i>v2lvsSpiceFile</i>	The path to the input SPICE netlist file generated by v2lvs.
<i>finalSpiceFile</i>	The path to the output SPICE netlist file after modification. This file will be overwritten if it exists.
<i>targetGlobalLine</i>	The string containing the global line to be inserted after the <code>.INCLUDE</code> directive.

Returns

`true` if the modification was successful, `false` otherwise (e.g., if file opening fails).

Note

The function assumes that the second to last token in an instance line is the output net. It also assumes a specific format for instance lines where VDD/VSS insertion and R/C line generation are applicable. The `generateRCLines` method is expected to handle the actual generation of R/C components.

See also

[generateRCLines](#)

Definition at line 1120 of file [LibFile.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.8.3.6 `mono()`

```
void LibFile::mono (
    const bool is_slew )
```

Validates that lookup tables are monotonically increasing with respect to the output load.

This function checks the following timing data in the current library to ensure monotonicity:

- `cell_rise` and `retaining_rise`
- `cell_fall` and `retaining_fall`
- `rise_transition` and `retain_rise_slew`
- `fall_transition` and `retain_fall_slew`
- `min_pulse_width` constraints (`rise_constraint`, `fall_constraint`)

The function first checks if a parsed JSON representation exists. If not, it parses the Liberty file and creates the JSON file. It then evaluates each timing arc in all cells and reports the pass/fail status at the end.

Parameters

<i>is_slew</i>	Boolean flag indicating whether to check slew values rather than delay values
----------------	---

The function outputs:

- Number of cells that passed/failed the monotonicity check
- List of cells that failed the check (if any)

Definition at line 438 of file [LibFile.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.8.3.7 parse()

```
void LibFile::parse ( )
```

Parses the Liberty file and extracts library information into a JSON structure.

This method handles the parsing of a Liberty file by:

1. Initializing the SI2 parser interface error handler
2. Reading the Liberty file contents
3. Extracting the library name
4. Processing second-level groups including:
 - Cell definitions, which are added to the JSON structure
 - Operating conditions, extracting PVT (Process, Voltage, Temperature) values

The parsed data is stored in the `lib_json_` member variable, with the following structure:

- "library_name": Name of the library
- "cells": Array of cell definitions
- "process": Process value
- "voltage": Voltage value (rounded to 2 decimal places)
- "temperature": Temperature value

Note

This method creates local scopes to ensure proper cleanup of SI2 Iterators.

Definition at line 118 of file [LibFile.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.8.3.8 read()

```
void LibFile::read ( ) [private]
```

Reads the Liberty file specified by the `filename_` member variable.

This function attempts to read a Liberty file and logs the process. It measures the time taken to read the file and logs any errors encountered during the read operation. If an error occurs, the function logs the error and terminates the program.

- Logs the start of the read operation.
- Measures the duration of the read operation.
- Checks for specific errors (invalid name or syntax error) and logs them.
- Terminates the program with specific exit codes if errors are detected.
- Logs the completion of the read operation and the time taken.

Note

This function uses the `spdlog` library for logging and the `si2drReadLibertyFile` function for reading the Liberty file.

Exceptions

<i>This</i>	function will terminate the program with exit codes 301 or 401 if specific errors are encountered.
-------------	--

Definition at line 79 of file [LibFile.cpp](#).

Here is the caller graph for this function:

6.8.3.9 spice()

```
void LibFile::spice (
    const int chain_length,
    const std::vector< std::string > & cell_names,
    const std::string & verilog_lib_file,
    const std::string & spice_lib_file )
```

Generates a SPICE netlist for the library based on a temporary Verilog file, using the V2LVS tool.

This function automates the process of converting a Verilog representation of the library into a SPICE netlist suitable for simulation. It involves the following steps:

1. Generates a temporary Verilog file using the `verilog` method.
2. Checks for the availability of the V2LVS tool in the system's PATH.
3. Constructs and executes the V2LVS command with appropriate options to generate an initial SPICE netlist.
4. Post-processes the generated SPICE netlist to adjust global signals using the `modifySpiceNetlist` method.
5. Logs the progress and any errors encountered during the process.

Parameters

<i>chain_length</i>	The length of the transistor chain for Verilog generation.
<i>cell_names</i>	A vector of cell names to be included in the Verilog generation.
<i>verilog_lib_file</i>	The path to the Verilog library file used by V2LVS for pin order information.
<i>spice_lib_file</i>	The path to the SPICE library file to be included in the generated SPICE netlist.

Note

The function assumes that the `verilog` method is available and correctly generates a temporary Verilog file.

The V2LVS tool must be installed and accessible in the system's PATH for this function to work correctly.

The generated SPICE netlist is post-processed to adjust global signals.

Temporary files (Verilog and intermediate SPICE) are kept for debugging purposes.

Definition at line 1386 of file [LibFile.cpp](#).

Here is the call graph for this function:  Here is the caller graph for this function: 

6.8.3.10 splitString()

```
std::vector< std::string > LibFile::splitString (
    const std::string & s ) [private]
```

Splits a string into a vector of tokens, using whitespace as the delimiter.

This function takes a string as input and splits it into a vector of strings, where each element in the vector represents a token from the original string. Whitespace characters (spaces, tabs, newlines, etc.) are used as delimiters to separate the tokens.

Parameters

<i>s</i>	The string to be split.
----------	-------------------------

Returns

A vector of strings, where each string is a token from the input string.

Definition at line 996 of file [LibFile.cpp](#).

Here is the caller graph for this function:

6.8.3.11 supercell()

```
void LibFile::supercell (
    const int chain_length,
    const std::vector< std::string > & cell_names )
```

Creates supercells based on standard cells in the library file.

This method creates supercells by combining standard cells in chains. For each input-output pin pair of a cell, it creates a supercell entry with naming format: <cellname>**X**<chain_length><input_pin>↔__<output_pin>. The results are written to a .map file.

For sequential cells (with clock pins), the chain length is always set to 1 regardless of the requested chain length.

Parameters

<i>chain_length</i>	The length of the chain for combinational cells
<i>cell_names</i>	Vector of specific cell names to process. If empty, all cells will be processed.

Note

Before creating supercells, this method checks for the existence of a JSON representation of the Liberty file and parses it if not found.

The method will report any requested cells that were not found in the library.

Definition at line 567 of file [LibFile.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.8.3.12 verilog()

```
void LibFile::verilog (
    const int chain_length,
    const std::vector< std::string > & cell_names )
```

Generates Verilog files for cell validation based on the library file.

This function creates Verilog modules for each cell in the specified chain and generates a top-level module that connects them all together. The process involves:

1. Creating supercells based on the specified chain length and cell names
2. Reading the mapping file to identify supercell name relationships
3. Generating individual Verilog modules for each supercell, handling both sequential and combinational cells differently
4. Creating ANSI-style port definitions for each module
5. Using the slang library to build proper syntax trees for Verilog modules
6. Creating a top-level module that instantiates all the individual modules
7. Writing the complete Verilog output to the output file

Sequential cells are instantiated once, while combinational cells are instantiated multiple times based on the specified chain length.

Parameters

<i>chain_length</i>	The number of times to chain combinational cells together
<i>cell_names</i>	Vector of cell names to include in the Verilog generation

Note

The function writes a temporary file during processing that might not be removed when the function completes (commented out cleanup code).

Definition at line 721 of file [LibFile.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.8.3.13 writeJsonToFile()

```
void LibFile::writeJsonToFile ( )
```

Writes the JSON data stored in the object to a file.

This method opens the file specified by `jsonname_` and writes the content of `lib_json_` with an indentation of 2 spaces. If the file cannot be opened, an error message is logged. Upon successful writing, an informational message is logged.

Exceptions

<i>None</i>	directly, but <code>std::ofstream</code> operations may throw exceptions
-------------	--

Definition at line 48 of file [LibFile.cpp](#).

Here is the caller graph for this function:

6.8.3.14 writeToDB()

```
void LibFile::writeToDB (
    const std::string & db_path,
    const std::string & pvt_corner,
    double age_seconds )
```

Definition at line 184 of file [LibFile.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.8.4 Member Data Documentation

6.8.4.1 basename_

```
std::string LibFile::basename_
```

Definition at line 29 of file [LibFile.hpp](#).

6.8.4.2 err_

```
si2drErrorT LibFile::err_ [private]
```

Definition at line 47 of file [LibFile.hpp](#).

6.8.4.3 filename_

```
std::string LibFile::filename_
```

Definition at line 30 of file [LibFile.hpp](#).

6.8.4.4 filepath_

```
std::filesystem::path LibFile::filepath_
```

Definition at line 28 of file [LibFile.hpp](#).

6.8.4.5 jsonname_

```
std::string LibFile::jsonname_ = ""
```

Definition at line 32 of file [LibFile.hpp](#).

6.8.4.6 lib_json_

```
json LibFile::lib_json_ = json::object()
```

Definition at line 34 of file [LibFile.hpp](#).

6.8.4.7 libname_

```
std::string LibFile::libname_ = ""
```

Definition at line 31 of file [LibFile.hpp](#).

6.8.4.8 logger_

```
std::shared_ptr<spdlog::logger> LibFile::logger_
```

Definition at line 27 of file [LibFile.hpp](#).

6.8.4.9 loggername_

```
std::string LibFile::loggername_ = ""
```

Definition at line 33 of file [LibFile.hpp](#).

6.8.4.10 process_

```
int LibFile::process_ = 0 [private]
```

Definition at line 48 of file [LibFile.hpp](#).

6.8.4.11 temperature_

```
int LibFile::temperature_ = 0 [private]
```

Definition at line 50 of file [LibFile.hpp](#).

6.8.4.12 voltage_

```
float LibFile::voltage_ = 0.0 [private]
```

Definition at line 49 of file [LibFile.hpp](#).

The documentation for this class was generated from the following files:

- [include/LibFile.hpp](#)
- [src/LibFile.cpp](#)

6.9 LibGroup Class Reference

```
#include <LibGroup.hpp>
```

Public Member Functions

- [LibGroup](#) (si2drGroupIdT group, si2drErrorT &err)
- [~LibGroup](#) ()
- std::string [getName](#) ()
- std::string [getType](#) ()
- si2drAttrsIdT [getAttrs](#) ()
- si2drGroupsIdT [getGroups](#) ()

Private Attributes

- si2drGroupIdT [group_](#)
- si2drErrorT & [err_](#)

6.9.1 Detailed Description

Definition at line 8 of file [LibGroup.hpp](#).

6.9.2 Constructor & Destructor Documentation

6.9.2.1 LibGroup()

```
LibGroup::LibGroup (  
    si2drGroupIdT group,  
    si2drErrorT & err )
```

Definition at line 3 of file [LibGroup.cpp](#).

6.9.2.2 ~LibGroup()

```
LibGroup::~LibGroup ( )
```

Definition at line 5 of file [LibGroup.cpp](#).

6.9.3 Member Function Documentation

6.9.3.1 getAttrs()

```
si2drAttrsIdT LibGroup::getAttrs ( )
```

Definition at line 19 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

6.9.3.2 getGroups()

```
si2drGroupsIdT LibGroup::getGroups ( )
```

Definition at line 21 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

6.9.3.3 getName()

```
std::string LibGroup::getName ( )
```

Definition at line 7 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

6.9.3.4 getType()

```
std::string LibGroup::getType ( )
```

Definition at line 14 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

6.9.4 Member Data Documentation

6.9.4.1 err_

```
si2drErrorT& LibGroup::err_ [private]
```

Definition at line 19 of file [LibGroup.hpp](#).

6.9.4.2 group_

```
si2drGroupIdT LibGroup::group_ [private]
```

Definition at line 18 of file [LibGroup.hpp](#).

The documentation for this class was generated from the following files:

- include/[LibGroup.hpp](#)
- src/[LibGroup.cpp](#)

6.10 LibraryComparator Class Reference

```
#include <LibraryComparator.hpp>
```

Public Member Functions

- [LibraryComparator](#) ([LibFile](#) &ref_libfile, [LibFile](#) &comp_libfile, double reltol, double abstol)
Constructor for the [LibraryComparator](#) class.
- void [generateReport](#) (const std::string &output_file)
Generates a comparison report between the reference and comparison libraries.

Public Attributes

- std::filesystem::path [ref_lib_path_](#)
- std::filesystem::path [comp_lib_path_](#)

Private Member Functions

- void [compareCell](#) (const std::string &cell_name, const [json](#) &ref_cell, const [json](#) &comp_cell, Table &table)
Compares the output pins of a cell between a reference JSON and a comparison JSON.
- void [comparePin](#) (const std::string &cell_name, const std::string &pin_name, const [json](#) &ref_pin, const [json](#) &comp_pin, Table &table)
Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.
- void [compareTimingArc](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const [json](#) &ref_timing_arc, const [json](#) &comp_timing_arc, Table &table)
Compares a timing arc between two JSON objects.
- void [compareLut](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const std::string &related_pin, const std::string &arc_name, const [json](#) &ref_lut, const [json](#) &comp_lut, Table &table)
Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

Private Attributes

- [json](#) ref_json_
- [json](#) comp_json_
- double reltol_
- double abstol_

6.10.1 Detailed Description

Definition at line 19 of file [LibraryComparator.hpp](#).

6.10.2 Constructor & Destructor Documentation

6.10.2.1 LibraryComparator()

```
LibraryComparator::LibraryComparator (
    LibFile & ref_libfile,
    LibFile & comp_libfile,
    double reltol,
    double abstol )
```

Constructor for the [LibraryComparator](#) class.

Initializes a [LibraryComparator](#) object by loading JSON data from reference and comparison library files. If JSON files don't exist, parses the library files first and creates the JSON files.

The constructor performs the following steps:

1. Looks for reference JSON file, parses library file if not found
2. Loads reference JSON data into `ref_json_` member
3. Looks for comparison JSON file, parses library file if not found
4. Loads comparison JSON data into `comp_json_` member
5. Stores file paths and initializes tolerance values for comparison

Parameters

<i>ref_libfile</i>	Reference library file object
<i>comp_libfile</i>	Comparison library file object
<i>reltol</i>	Relative tolerance for numerical comparisons (default defined elsewhere)
<i>abstol</i>	Absolute tolerance for numerical comparisons (default defined elsewhere)

Definition at line 21 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:

6.10.3 Member Function Documentation

6.10.3.1 compareCell()

```
void LibraryComparator::compareCell (
    const std::string & cell_name,
    const json & ref_cell,
    const json & comp_cell,
    Table & table ) [private]
```

Compares the output pins of a cell between a reference JSON and a comparison JSON.

This function iterates through the output pins of the comparison cell and checks if each pin exists in the reference cell. If a pin is found in both cells, it calls the `comparePin` function to compare the details of the pin. If a pin is not found in the reference cell, a warning is logged. If the comparison cell does not contain any output pins, an informational message is logged.

Parameters

<i>cell_name</i>	The name of the cell being compared.
<i>ref_cell</i>	The reference JSON object containing the cell data.
<i>comp_cell</i>	The comparison JSON object containing the cell data.
<i>table</i>	The table object where the comparison results are stored.

Definition at line 301 of file [LibraryComparator.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.10.3.2 compareLut()

```
void LibraryComparator::compareLut (
    const std::string & cell_name,
    const std::string & pin_name,
    const std::string & timing_type,
    const std::string & related_pin,
    const std::string & arc_name,
    const json & ref_lut,
    const json & comp_lut,
    Table & table ) [private]
```

Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

This function compares the lookup table values between reference and comparison libraries for a given timing arc, checking the consistency of indices and values. It verifies that the indices match between libraries and then compares each value in the LUT against relative and absolute tolerance thresholds. Any discrepancies that exceed the tolerance limits are recorded in the provided table.

Parameters

<i>cell_name</i>	The name of the cell containing the LUT.
<i>pin_name</i>	The name of the pin associated with the LUT.
<i>timing_type</i>	The timing type of the arc (e.g., "rise_transition", "fall_transition").
<i>related_pin</i>	The related pin for the timing arc.
<i>arc_name</i>	The name of the specific timing arc being compared.
<i>ref_lut</i>	The lookup table from the reference library (in JSON format).
<i>comp_lut</i>	The lookup table from the comparison library (in JSON format).
<i>table</i>	Table object to store comparison results for any mismatches found.

Note

The function logs various information/warning/error messages:

- Logs index mismatches as warnings
- Logs value mismatches as debug messages
- Logs format errors in LUT structure as errors
- Records values exceeding tolerance in the provided table

The comparison uses both relative tolerance (`reltol_`) and absolute tolerance (`abstol_`) to determine if values differ significantly.

Definition at line 98 of file [LibraryComparator.cpp](#).

Here is the caller graph for this function:

6.10.3.3 comparePin()

```
void LibraryComparator::comparePin (
    const std::string & cell_name,
    const std::string & pin_name,
    const json & ref_pin,
    const json & comp_pin,
    Table & table ) [private]
```

Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.

This function iterates through the timing arcs of the comparison pin and looks for matching timing types in the reference pin. If a matching timing type is found, it calls the `compareTimingArc` function to compare the timing arcs. If a timing type is not found in the reference JSON, a warning is logged.

Parameters

<i>cell_name</i>	The name of the cell containing the pin.
<i>pin_name</i>	The name of the pin to compare.
<i>ref_pin</i>	The reference JSON object containing the pin's data.
<i>comp_pin</i>	The comparison JSON object containing the pin's data.
<i>table</i>	A reference to a Table object where comparison results are stored.

Definition at line 261 of file [LibraryComparator.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.10.3.4 compareTimingArc()

```
void LibraryComparator::compareTimingArc (
    const std::string & cell_name,
    const std::string & pin_name,
    const std::string & timing_type,
    const json & ref_timing_arc,
```

```
const json & comp_timing_arc,
Table & table ) [private]
```

Compares a timing arc between two JSON objects.

This function compares a specific timing arc (e.g., cell_rise, cell_fall, rise_transition, fall_transition) between a reference JSON object and a comparison JSON object. It extracts the related pin from the reference timing arc and iterates through a predefined list of arc names. If an arc name is found in the comparison timing arc, it checks if the same arc name exists in the reference timing arc. If both exist, it calls the compareLut function to compare the LUT data associated with the arc. If the arc name is found in the comparison JSON but not in the reference JSON, a warning message is logged.

Parameters

<i>cell_name</i>	The name of the cell.
<i>pin_name</i>	The name of the pin.
<i>timing_type</i>	The type of timing (e.g., "combinational", "sequential").
<i>ref_timing_arc</i>	The reference JSON object containing the timing arc information.
<i>comp_timing_arc</i>	The comparison JSON object containing the timing arc information.
<i>table</i>	The table to store comparison results.

Definition at line 220 of file [LibraryComparator.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.10.3.5 generateReport()

```
void LibraryComparator::generateReport (
    const std::string & output_file )
```

Generates a comparison report between the reference and comparison libraries.

This function generates a detailed comparison report between the reference library and the comparison library. The report includes metadata such as the reference and comparison library paths, absolute and relative tolerances, and the application details. It also includes a legend explaining various symbols used in the report.

The function iterates through each cell in the comparison library, compares it with the corresponding cell in the reference library, and generates a table of differences. If there are any differences, it calculates summary statistics such as the average and maximum differences, and the number of outliers. The results are written to the specified output file in either Markdown or plain text format.

Parameters

<i>output_file</i>	The path to the output file where the report will be written.
--------------------	---

Definition at line 343 of file [LibraryComparator.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.10.4 Member Data Documentation

6.10.4.1 abstol_

```
double LibraryComparator::abstol_ [private]
```

Definition at line 30 of file [LibraryComparator.hpp](#).

6.10.4.2 comp_json_

```
json LibraryComparator::comp_json_ [private]
```

Definition at line 28 of file [LibraryComparator.hpp](#).

6.10.4.3 comp_lib_path_

```
std::filesystem::path LibraryComparator::comp_lib_path_
```

Definition at line 23 of file [LibraryComparator.hpp](#).

6.10.4.4 ref_json_

```
json LibraryComparator::ref_json_ [private]
```

Definition at line 27 of file [LibraryComparator.hpp](#).

6.10.4.5 ref_lib_path_

std::filesystem::path LibraryComparator::ref_lib_path_

Definition at line 22 of file [LibraryComparator.hpp](#).

6.10.4.6 reitol_

double LibraryComparator::reitol_ [private]

Definition at line 29 of file [LibraryComparator.hpp](#).

The documentation for this class was generated from the following files:

- include/[LibraryComparator.hpp](#)
- src/[LibraryComparator.cpp](#)

6.11 LogicComparator Class Reference

```
#include <LogicComparator.hpp>
```

Public Member Functions

- [LogicComparator](#) (const std::map< std::string, std::string > &ref_outpin_map, const std::map< std::string, std::string > &comp_outpin_map, const std::string &cell_name)
- void [logic](#) ()

This function demonstrates the usage of the exprtk library to evaluate a boolean logic expression.
- std::string [preprocessExpression](#) (const std::string &input_expr)

Preprocesses a logical expression string to prepare it for evaluation.
- bool [extractVariables](#) (const std::string &expr1, const std::string &expr2, std::vector< std::string > &sorted_vars)

Extracts and validates variables from two expressions, ensuring they match.
- void [compareSingleExpressionPair](#) (const std::string &ref_expression_processed, const std::string &comp_expression_processed, const std::vector< std::string > &sorted_vars, [PinComparisonResult](#) &result)

Compares two preprocessed logic expression strings for equivalence. Internal helper function.
- void [compareCellLogic](#) ()

Compares logic for all output pins defined in the input maps, and stores results in all_pin_results_.
- void [generateReport](#) (const std::string &output_file)

Generates a comparison report file based on cell logic comparison results.

Private Attributes

- `std::map< std::string, std::string > ref\_outpin\_map\_`
- `std::map< std::string, std::string > comp\_outpin\_map\_`
- `std::string cell\_name\_`
- `std::map< std::string, PinComparisonResult > all\_pin\_results\_`

6.11.1 Detailed Description

Definition at line 38 of file [LogicComparator.hpp](#).

6.11.2 Constructor & Destructor Documentation

6.11.2.1 LogicComparator()

```
LogicComparator::LogicComparator (  
    const std::map< std::string, std::string > & ref_outpin_map,  
    const std::map< std::string, std::string > & comp_outpin_map,  
    const std::string & cell_name )
```

Definition at line 3 of file [LogicComparator.cpp](#).

6.11.3 Member Function Documentation

6.11.3.1 compareCellLogic()

```
void LogicComparator::compareCellLogic ( )
```

Compares logic for all output pins defined in the input maps, and stores results in `all_pin_results_`.

Compares the logic expressions for each output pin of a cell between a reference and a comparison design.

This method performs a detailed comparison of the logic expressions associated with each output pin of a cell. It iterates through each unique pin name found in both the reference and comparison designs, retrieves their corresponding logic expressions, preprocesses them, extracts and validates the variables used, and then compares the processed expressions. The results of each pin comparison, including any errors or discrepancies, are stored in the `all_pin_results_` map.

The comparison process includes the following steps:

1. **Pin Collection:** Collects all unique pin names from both the reference (`ref_outpin_map_`) and comparison (`comp_outpin_map_`) maps.
2. **Iteration:** Iterates through each unique pin name.
3. **Expression Retrieval:** Retrieves the raw logic expressions for the current pin from both the reference and comparison maps. If a pin is missing in either map, an error is logged, and the comparison is marked as impossible for that pin.
4. **Preprocessing:** Preprocesses the raw expressions to simplify them and remove any irrelevant characters or formatting. If preprocessing results in an empty expression, an error is logged, and the comparison is marked as impossible.
5. **Variable Extraction and Validation:** Extracts the variables used in both expressions and validates that the sets of variables match. If the variable sets do not match, an error is logged, and the comparison is marked as impossible.
6. **Expression Comparison:** Compares the preprocessed expressions using the extracted variables. This step determines whether the logic functions represented by the expressions are equivalent.
7. **Result Storage:** Stores the detailed results of the pin comparison, including the raw and processed expressions, any errors encountered, and the comparison result, in the `all_pin_results_` map.

Note

The `spdlog` library is used for logging information, warnings, and errors throughout the comparison process.

The `preprocessExpression` method is used to simplify the raw logic expressions before comparison.

The `extractVariables` method is used to extract and validate the variables used in the logic expressions.

The `compareSingleExpressionPair` method is used to compare the preprocessed expressions and determine whether they are equivalent.

See also

[preprocessExpression](#)

[extractVariables](#)

[compareSingleExpressionPair](#)

Definition at line 812 of file [LogicComparator.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.11.3.2 compareSingleExpressionPair()

```
void LogicComparator::compareSingleExpressionPair (
    const std::string & ref_expression_processed,
    const std::string & comp_expression_processed,
    const std::vector< std::string > & sorted_vars,
    PinComparisonResult & result )
```

Compares two preprocessed logic expression strings for equivalence. Internal helper function.

Compares two boolean expressions for logical equivalence using a truth table.

Template Parameters

<i>T</i>	The numeric type used by ExprTk (e.g., double, float).
----------	--

Parameters

<i>ref_expression_processed</i>	The preprocessed reference logic expression.
<i>comp_expression_processed</i>	The preprocessed comparison logic expression.
<i>sorted_vars</i>	A sorted vector of unique variable names common to both expressions.
<i>result</i>	Reference to a PinComparisonResult object to store detailed results.

This function takes two processed boolean expressions (*ref_expression_processed* and *comp_expression_processed*), a sorted list of variable names (*sorted_vars*), and a [PinComparisonResult](#) struct to store the results. It evaluates both expressions for all possible combinations of variable values and compares the resulting truth tables.

The function uses the ExprTk library to parse and evaluate the boolean expressions. It generates a truth table for each expression and compares the results. If the truth tables are identical, the expressions are considered logically equivalent.

Parameters

<i>ref_expression_processed</i>	The processed reference boolean expression.
<i>comp_expression_processed</i>	The processed comparison boolean expression.
<i>sorted_vars</i>	A sorted vector of variable names used in the expressions.
<i>result</i>	A PinComparisonResult struct to store the comparison results, including processed expressions, compilation status, equivalence, error messages, and the generated truth tables.

Note

The function limits the number of variables to 20 to prevent excessively large truth tables. It also performs overflow checking on the number of combinations.

The function uses spdlog for logging debug, warning, and error messages.

The function assumes that the input expressions are valid boolean expressions containing only the variables listed in `sorted_vars`.

Definition at line 553 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:

6.11.3.3 extractVariables()

```
bool LogicComparator::extractVariables (
    const std::string & expr1_raw,
    const std::string & expr2_raw,
    std::vector< std::string > & sorted_vars )
```

Extracts and validates variables from two expressions, ensuring they match.

This function parses two raw expression strings (`expr1_raw` and `expr2_raw`) to identify potential variable names. It uses a regular expression to find identifiers and then validates them against a set of rules:

1. The identifier must be a valid identifier as determined by the `isIdentifier` function.
2. The identifier must not be a keyword or function name defined in the `keywords` set.

The function compares the sets of validated variables from both expressions. If the sets are identical, the variables are extracted, sorted alphabetically, and stored in the `sorted_vars` vector. If the sets differ, an error is reported, and the differences between the sets are logged.

Parameters

<i>expr1_raw</i>	The raw string representation of the first expression.
<i>expr2_raw</i>	The raw string representation of the second expression.
<i>sorted_vars</i>	A vector to store the sorted list of unique variable names if the variable sets from both expressions match. This vector is cleared if the sets do not match.

Returns

`true` if the variable sets from both expressions match, indicating that the `sorted_vars` vector contains the sorted list of unique variable names. `false` if the variable sets do not match, indicating an error.

Note

The `isIdentifier` function is used to validate potential variable names.

The `keywords` set contains a list of reserved words that cannot be used as variable names.

The function uses `spdlog` for logging debug, trace, info, and warning messages.

Definition at line 327 of file [LogicComparator.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.11.3.4 generateReport()

```
void LogicComparator::generateReport (
    const std::string & output_file )
```

Generates a comparison report file based on cell logic comparison results.

Generates a comprehensive report comparing the logic equivalence of a cell's reference and comparison expressions.

Parameters

<code>output_file</code>	Path to the output report file.
--------------------------	---------------------------------

This function performs the following steps:

1. **Initialization:** Sets up logging and determines the output format (Markdown or plain text) based on the file extension.
2. **File Handling:** Opens the specified output file in append mode, creating it if it doesn't exist. Handles potential file opening errors.
3. **Report Header:** Writes a header section to the report, including the cell name, application version, author, and timestamp.
4. **Legend Generation:** Creates a legend explaining the status symbols used in the report (e.g., [OK], [NO], [NA]).

5. **Table Generation:** Creates tables for reference pin functions, comparison pin functions, and the legend.
6. **Pin Result Processing:** Iterates through the results for each pin, generating detailed information including:
 - Pin Name
 - Reference and Comparison Truth Tables (if available)
 - Comparison Summary Table:
 - Status (Logic Equivalence)
 - Raw and Processed Expressions
 - Compilation Status
 - Error Messages (if any)
7. **Output:** Exports the generated tables and pin results to the output file in the determined format (Markdown or plain text).
8. **Closure:** Closes the output file and logs the completion of the report generation.

Parameters

<i>output_file</i>	The path to the output report file. If the file extension is ".md", the report will be generated in Markdown format; otherwise, it will be generated in plain text.
--------------------	---

Note

The function uses `spdlog` for logging and relies on several helper classes/structs:

- `Table`: For creating formatted tables.
- `MarkdownExporter`: For exporting tables to Markdown format.
- `LogicComparator::PinResult`: A struct containing the results of the logic comparison for a single pin.

The function appends to the output file if it already exists.

Definition at line 974 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:

6.11.3.5 `logic()`

```
void LogicComparator::logic ( )
```

This function demonstrates the usage of the `exprtk` library to evaluate a boolean logic expression.

It defines a boolean expression "not(A and B) or C" and evaluates it for all possible combinations of boolean values for the variables A, B, and C. The results are then printed to the console in a tabular format.

The function utilizes the `exprtk` library for:

- Defining a symbol table to hold the variables A, B, and C.
- Creating an expression object and registering the symbol table with it.
- Compiling the boolean expression string into the expression object.
- Iterating through all possible boolean combinations for A, B, and C.
- Assigning the boolean values to the variables in the symbol table.
- Evaluating the expression using `expression.value()`.
- Printing the input values and the result to the console.

Definition at line 24 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:

6.11.3.6 preprocessExpression()

```
std::string LogicComparator::preprocessExpression (  
    const std::string & input_expr )
```

Preprocesses a logical expression string to prepare it for evaluation.

This function performs several steps to transform the input expression:

1. Trims leading/trailing whitespace.
2. Replaces the logical NOT operator '!' with " not " (except for '!=').
3. Adds spaces around operators (+, ^, *) and parentheses.
4. Replaces symbolic operators (+, ^, *) with their keyword equivalents (or, xor, and).
5. Tokenizes the expression based on spaces.
6. Inserts implied 'and' operators between operands where necessary. For example, "A B" becomes "A and B".
7. Handles 'not Identifier' sequences by converting them to 'not(Identifier)'.
8. Reconstructs the final expression string from the processed tokens, adding spaces appropriately.
9. Performs a final cleanup to consolidate multiple spaces and trim the result.

Parameters

<i>input_expr</i>	The input logical expression string.
-------------------	--------------------------------------

Returns

The preprocessed logical expression string, ready for evaluation. Returns an empty string if the input is empty or if no tokens are found after processing.

Note

The function uses spdlog for debugging and tracing.

The function assumes the existence of helper functions `isIdentifier` and `isOperator` to classify tokens.

Definition at line 128 of file [LogicComparator.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.11.4 Member Data Documentation

6.11.4.1 all_pin_results_

```
std::map<std::string, PinComparisonResult> LogicComparator::all_pin_results_ [private]
```

Definition at line 86 of file [LogicComparator.hpp](#).

6.11.4.2 cell_name_

```
std::string LogicComparator::cell_name_ [private]
```

Definition at line 85 of file [LogicComparator.hpp](#).

6.11.4.3 comp_outpin_map_

```
std::map<std::string, std::string> LogicComparator::comp_outpin_map_ [private]
```

Definition at line 84 of file [LogicComparator.hpp](#).

6.11.4.4 ref_outpin_map_

```
std::map<std::string, std::string> LogicComparator::ref_outpin_map_ [private]
```

Definition at line 83 of file [LogicComparator.hpp](#).

The documentation for this class was generated from the following files:

- include/[LogicComparator.hpp](#)
- src/[LogicComparator.cpp](#)

6.12 LogicExtractor Class Reference

```
#include <LogicExtractor.hpp>
```

Inheritance diagram for LogicExtractor:

Collaboration diagram for LogicExtractor:

Public Member Functions

- [LogicExtractor](#) (const std::string &targetCell)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)
Handles a module declaration syntax node.
- void [handle](#) (const slang::syntax::PortDeclarationSyntax &portDecl)
Handles a port declaration syntax node.
- void [handle](#) (const slang::syntax::NonAnsiPortListSyntax &portList)
Handles a non-ANSI port list in a module.
- void [handle](#) (const slang::syntax::NetDeclarationSyntax &netDecl)
Handles a net declaration syntax node.
- void [handle](#) (const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst)
Handles the extraction of logic gate information from a primitive instantiation syntax node.
- const std::unordered_map< std::string, [GateInfo](#) > & [getExtractedGates](#) () const
- const std::unordered_set< std::string > & [getPrimaryInputs](#) () const
- const std::unordered_set< std::string > & [getPrimaryOutputs](#) () const
- const std::unordered_set< std::string > & [getInternalWires](#) () const
- std::map< std::string, std::string > [getLogicExpressions](#) ()
Extracts logic expressions for all primary output ports of the target cell.

Private Member Functions

- `std::string deriveLogicRecursive` (const `std::string` &signalName)
Recursively derives the logic expression for a given signal.
- `std::string formatExpression` (const `GateInfo` &gateInfo, const `std::vector< std::string >` &inputExprs)
Formats a logic expression based on the gate type and input expressions.

Private Attributes

- `const std::string & targetCell_`
- `bool inTargetModule_`
- `bool parsingComplete_`
- `std::unordered_set< std::string > primaryInputs_`
- `std::unordered_set< std::string > primaryOutputs_`
- `std::unordered_set< std::string > internalWires_`
- `std::unordered_map< std::string, std::string > portDirections_`
- `std::unordered_map< std::string, GateInfo > gateOutputDrivers_`
- `std::unordered_map< std::string, std::string > logicCache_`

6.12.1 Detailed Description

Definition at line 17 of file [LogicExtractor.hpp](#).

6.12.2 Constructor & Destructor Documentation

6.12.2.1 LogicExtractor()

```
LogicExtractor::LogicExtractor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 20 of file [LogicExtractor.hpp](#).

6.12.3 Member Function Documentation

6.12.3.1 deriveLogicRecursive()

```
std::string LogicExtractor::deriveLogicRecursive (
    const std::string & signalName ) [private]
```

Recursively derives the logic expression for a given signal.

This function traces back the signal to its driving gates and primary inputs, constructing a logic expression that represents the signal's behavior. It uses memoization to cache previously computed expressions, avoiding redundant calculations.

Parameters

<i>signalName</i>	The name of the signal for which to derive the logic expression.
-------------------	--

Returns

A string representing the logic expression for the signal.

Exceptions

<i>std::runtime_error</i>	if the signal is not a primary input, a known wire, or driven by a recognized gate/assignment. Also thrown if an empty input signal name is encountered or if an internal wire has no identified driver.
---------------------------	--

The function operates in the following steps:

1. **Check Cache (Memoization):** If the logic expression for the signal is already cached in `logicCache_`, it is immediately returned.
2. **Base Case: Is it a primary input?** If the signal is a primary input (present in `primaryInputs_`), its logic expression is simply its own name.
3. **Recursive Step: Is it driven by a gate?** If the signal is driven by a gate (present in `gateOutputDrivers_`), the function recursively derives the logic expressions for all input signals of the gate. These input expressions are then used to format the overall expression for the current signal based on the gate type.
4. **Handle Assign statements:** (Currently not implemented but reserved for future use).
5. **Error Case:** If the signal is not found in any of the above categories, it indicates an error. An exception is thrown, indicating that the signal is either an undriven internal wire or an unknown signal.

Definition at line 510 of file [LogicExtractor.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.12.3.2 formatExpression()

```
std::string LogicExtractor::formatExpression (
    const GateInfo & gateInfo,
    const std::vector< std::string > & inputExprs ) [private]
```

Formats a logic expression based on the gate type and input expressions.

This function takes a [GateInfo](#) structure describing the gate and a vector of input expressions (strings) and constructs a logic expression string representing the gate's operation. It handles AND, NAND, OR, NOR, XOR, XNOR, NOT, and BUF gates. It uses '*', '+', '^', and '!' operators for AND, OR, XOR, and NOT respectively. If the gate type is unsupported, or if the number of inputs is incorrect for NOT/BUF gates, or if a gate requiring inputs receives none, an error string is returned, and a warning is logged.

Parameters

<i>gateInfo</i>	A GateInfo struct containing information about the gate, including its type (slang::parsing::TokenKind) and name.
<i>inputExprs</i>	A vector of strings representing the input expressions to the gate.

Returns

A string representing the formatted logic expression, or an error string if formatting fails due to unsupported gate type or incorrect number of inputs.

Definition at line 584 of file [LogicExtractor.cpp](#).

Here is the caller graph for this function:

6.12.3.3 getExtractedGates()

```
const std::unordered_map< std::string, GateInfo > & LogicExtractor::getExtractedGates ( ) const [inline]
```

Definition at line 37 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

6.12.3.4 getInternalWires()

```
const std::unordered_set< std::string > & LogicExtractor::getInternalWires ( ) const [inline]
```

Definition at line 42 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

6.12.3.5 getLogicExpressions()

```
std::map< std::string, std::string > LogicExtractor::getLogicExpressions ( )
```

Extracts logic expressions for all primary output ports of the target cell.

This method iterates through the primary output ports, derives the logic expression for each, and stores the result in a map. If an error occurs during the derivation of logic for a particular output, an error message is stored in the map instead.

Returns

A map where the key is the output port name (std::string) and the value is the corresponding logic expression (std::string). If AST parsing is not complete or the target module is not found, an empty map is returned. If an error occurs during logic derivation for a specific output, the corresponding value in the map will be an error message.

Definition at line 451 of file [LogicExtractor.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.12.3.6 getPrimaryInputs()

```
const std::unordered_set< std::string > & LogicExtractor::getPrimaryInputs ( ) const [inline]
```

Definition at line 40 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

6.12.3.7 getPrimaryOutputs()

```
const std::unordered_set< std::string > & LogicExtractor::getPrimaryOutputs ( ) const [inline]
```

Definition at line 41 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

6.12.3.8 handle() [1/5]

```
void LogicExtractor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Handles a module declaration syntax node.

This function is the entry point for processing a module declaration. It checks if the module is the target module and, if so, extracts relevant information such as primary inputs, primary outputs, and internal wires. It also visits the children of the module node to process ports, declarations, and instances.

The function ensures that the target module is only processed once and that parsing stops after the target module is found.

Parameters

<i>module</i>	A reference to the module declaration syntax node.
---------------	--

Definition at line 18 of file [LogicExtractor.cpp](#).

6.12.3.9 `handle()` [2/5]

```
void LogicExtractor::handle (
    const slang::syntax::NetDeclarationSyntax & netDecl )
```

Handles a net declaration syntax node.

This function processes a net declaration syntax node, extracting the names of declared wires. It checks if the current scope is within the target module and avoids adding ports again if they were already declared as nets. The extracted wire names are added to the internal wires set.

Parameters

<i>netDecl</i>	A reference to the NetDeclarationSyntax node to handle.
----------------	---

Definition at line 277 of file [LogicExtractor.cpp](#).

6.12.3.10 `handle()` [3/5]

```
void LogicExtractor::handle (
    const slang::syntax::NonAnsiPortListSyntax & portList )
```

Handles a non-ANSI port list in a module.

This method iterates through the ports in a non-ANSI port list, extracts the port name, determines its direction (input, output, or inout), and adds it to the appropriate sets (primaryInputs_, primaryOutputs_, internalWires_). It also handles different types of port declarations, including implicit, explicit, and empty ports.

Parameters

<i>portList</i>	A reference to the NonAnsiPortListSyntax object representing the port list.
-----------------	---

- Skips processing if not in the target module (`inTargetModule_` is false).
- Extracts port names from `ImplicitNonAnsiPortSyntax` and `ExplicitNonAnsiPortSyntax`.
- Handles `PortConcatenationSyntax` by logging a warning and skipping the port.
- Skips `EmptyNonAnsiPortSyntax` (placeholders).
- Uses the `portDirections_` map to determine the direction of each port.
- Adds ports to `primaryInputs_` and `internalWires_` if the direction is "input".
- Adds ports to `primaryOutputs_` and `internalWires_` if the direction is "output".
- Adds ports to both `primaryInputs_` and `primaryOutputs_` if the direction is "inout".
- Logs warnings for ports with unknown or missing directions.
- Logs warnings if the port name cannot be determined.

Note

This method does not call `visitDefault`.

Definition at line 179 of file [LogicExtractor.cpp](#).

6.12.3.11 `handle()` [4/5]

```
void LogicExtractor::handle (
    const slang::syntax::PortDeclarationSyntax & portDecl )
```

Handles a port declaration syntax node.

This method extracts information about port declarations within a SystemVerilog module, including the port's direction (input, output, or inout) and name. It updates internal data structures to track primary inputs, primary outputs, internal wires, and port directions.

Parameters

<i>portDecl</i>	A reference to the <code>PortDeclarationSyntax</code> node being processed.
-----------------	---

- It first checks if the current processing context is within the target module. If not, the method returns early.

- It determines the port's direction by inspecting the port header (`VariablePortHeaderSyntax` or `NetPortHeaderSyntax`). The direction is extracted using `valueText()` to handle keywords represented as text tokens.
- It iterates through the declarators in the port declaration to extract port names.
- If multiple direction declarations are found for the same port, a warning is logged, and the first declared direction is retained.
- Based on the port's direction, the port name is added to the appropriate sets:
 - `primaryInputs_`: For input and inout ports.
 - `primaryOutputs_`: For output and inout ports.
 - `internalWires_`: For all ports (input, output, inout, and ports with unknown directions).
- If a port has an unknown or missing direction, it's treated as an internal wire, and a warning is logged.
- The method avoids calling `visitDefault` to prevent unexpected revisits of syntax nodes.

Note

The `logicCache_` update is commented out, indicating it's part of a later processing step.

Warning

Logs warnings for port declarations without headers, port declarators without names, and multiple direction declarations for the same port.

Definition at line 92 of file [LogicExtractor.cpp](#).

6.12.3.12 `handle()` [5/5]

```
void LogicExtractor::handle (
    const slang::syntax::PrimitiveInstantiationSyntax & primitiveInst )
```

Handles the extraction of logic gate information from a primitive instantiation syntax node.

This method processes a primitive instantiation, extracting the gate type, input signals, and output signal. It populates the `gateOutputDrivers_` map, which stores the driving gate information for each output signal. It also identifies internal wires and checks for multiple drivers on the same signal.

Parameters

<i>primitiveInst</i>	A reference to the PrIMITIVEInstantiationSyntax node representing the gate instance.
----------------------	--

Definition at line 309 of file [LogicExtractor.cpp](#).

6.12.4 Member Data Documentation

6.12.4.1 gateOutputDrivers_

```
std::unordered_map<std::string, GateInfo> LogicExtractor::gateOutputDrivers_ [private]
```

Definition at line 63 of file [LogicExtractor.hpp](#).

6.12.4.2 inTargetModule_

```
bool LogicExtractor::inTargetModule_ [private]
```

Definition at line 50 of file [LogicExtractor.hpp](#).

6.12.4.3 internalWires_

```
std::unordered_set<std::string> LogicExtractor::internalWires_ [private]
```

Definition at line 56 of file [LogicExtractor.hpp](#).

6.12.4.4 logicCache_

```
std::unordered_map<std::string, std::string> LogicExtractor::logicCache_ [private]
```

Definition at line 66 of file [LogicExtractor.hpp](#).

6.12.4.5 parsingComplete_

```
bool LogicExtractor::parsingComplete_ [private]
```

Definition at line 51 of file [LogicExtractor.hpp](#).

6.12.4.6 portDirections_

```
std::unordered_map<std::string, std::string> LogicExtractor::portDirections_ [private]
```

Definition at line 60 of file [LogicExtractor.hpp](#).

6.12.4.7 primaryInputs_

```
std::unordered_set<std::string> LogicExtractor::primaryInputs_ [private]
```

Definition at line 54 of file [LogicExtractor.hpp](#).

6.12.4.8 primaryOutputs_

```
std::unordered_set<std::string> LogicExtractor::primaryOutputs_ [private]
```

Definition at line 55 of file [LogicExtractor.hpp](#).

6.12.4.9 targetCell_

```
const std::string& LogicExtractor::targetCell_ [private]
```

Definition at line 49 of file [LogicExtractor.hpp](#).

The documentation for this class was generated from the following files:

- [include/LogicExtractor.hpp](#)
- [src/LogicExtractor.cpp](#)

6.13 ModuleRewriter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for ModuleRewriter:

Collaboration diagram for ModuleRewriter:

Public Member Functions

- [ModuleRewriter](#) (const std::vector< std::string > &inputPins, const std::vector< std::string > &outputPins, const std::pair< std::string, std::string > &supercell_entry, int instance_count, std::shared_ptr< spdlog::logger > logger)
- void [handle](#) (const slang::syntax::SyntaxNode &node)
Processes a syntax node in the module's AST.
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Public Attributes

- std::shared_ptr< spdlog::logger > [logger_](#)

Private Attributes

- const std::vector< std::string > & [inputPins_](#)
- const std::vector< std::string > & [outputPins_](#)
- const std::string [cellName_](#)
- const std::string [moduleName_](#)
- std::map< std::string, std::string > [portInfoMap_](#)
- int [depth_](#)
- int [instance_count_](#)

6.13.1 Detailed Description

Definition at line 59 of file [verilog_utils.hpp](#).

6.13.2 Constructor & Destructor Documentation

6.13.2.1 ModuleRewriter()

```
ModuleRewriter::ModuleRewriter (
    const std::vector< std::string > & inputPins,
    const std::vector< std::string > & outputPins,
    const std::pair< std::string, std::string > & supercell_entry,
    int instance_count,
    std::shared_ptr< spdlog::logger > logger ) [inline], [explicit]
```

Definition at line 61 of file [verilog_utils.hpp](#).

6.13.3 Member Function Documentation

6.13.3.1 handle() [1/2]

```
void ModuleRewriter::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 406 of file [verilog_utils.cpp](#).

6.13.3.2 handle() [2/2]

```
void ModuleRewriter::handle (
    const slang::syntax::SyntaxNode & node )
```

Processes a syntax node in the module's AST.

This method is called for each syntax node during the traversal of the AST. It logs debug information about the current node and continues processing its child nodes by recursively calling the appropriate visit methods.

Parameters

<i>node</i>	The syntax node to process
-------------	----------------------------

Definition at line 396 of file [verilog_utils.cpp](#).

6.13.4 Member Data Documentation

6.13.4.1 cellName_

```
const std::string ModuleRewriter::cellName_ [private]
```

Definition at line 75 of file [verilog_utils.hpp](#).

6.13.4.2 depth_

```
int ModuleRewriter::depth_ [private]
```

Definition at line 78 of file [verilog_utils.hpp](#).

6.13.4.3 inputPins_

```
const std::vector<std::string>& ModuleRewriter::inputPins_ [private]
```

Definition at line 73 of file [verilog_utils.hpp](#).

6.13.4.4 instance_count_

```
int ModuleRewriter::instance_count_ [private]
```

Definition at line 79 of file [verilog_utils.hpp](#).

6.13.4.5 logger_

```
std::shared_ptr<spdlog::logger> ModuleRewriter::logger_
```

Definition at line 68 of file [verilog_utils.hpp](#).

6.13.4.6 moduleName_

```
const std::string ModuleRewriter::moduleName_ [private]
```

Definition at line 76 of file [verilog_utils.hpp](#).

6.13.4.7 outputPins_

```
const std::vector<std::string>& ModuleRewriter::outputPins_ [private]
```

Definition at line 74 of file [verilog_utils.hpp](#).

6.13.4.8 portInfoMap_

```
std::map<std::string, std::string> ModuleRewriter::portInfoMap_ [private]
```

Definition at line 77 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

6.14 PinComparisonResult Struct Reference

```
#include <LogicComparator.hpp>
```

Public Attributes

- std::string [pin_name](#)
- std::string [ref_expr_raw](#)
- std::string [comp_expr_raw](#)
- std::string [ref_expr_processed](#)
- std::string [comp_expr_processed](#)
- bool [comparison_possible](#) = false
- bool [are_equivalent](#) = false
- bool [ref_compiles](#) = false
- bool [comp_compiles](#) = false
- std::optional< Table > [ref_truth_table](#)
- std::optional< Table > [comp_truth_table](#)
- std::string [error_message](#)

6.14.1 Detailed Description

Definition at line 23 of file [LogicComparator.hpp](#).

6.14.2 Member Data Documentation

6.14.2.1 are_equivalent

```
bool PinComparisonResult::are_equivalent = false
```

Definition at line 30 of file [LogicComparator.hpp](#).

6.14.2.2 comp_compiles

```
bool PinComparisonResult::comp_compiles = false
```

Definition at line 32 of file [LogicComparator.hpp](#).

6.14.2.3 comp_expr_processed

```
std::string PinComparisonResult::comp_expr_processed
```

Definition at line 28 of file [LogicComparator.hpp](#).

6.14.2.4 comp_expr_raw

```
std::string PinComparisonResult::comp_expr_raw
```

Definition at line 26 of file [LogicComparator.hpp](#).

6.14.2.5 comp_truth_table

```
std::optional<Table> PinComparisonResult::comp_truth_table
```

Definition at line 34 of file [LogicComparator.hpp](#).

6.14.2.6 comparison_possible

```
bool PinComparisonResult::comparison_possible = false
```

Definition at line 29 of file [LogicComparator.hpp](#).

6.14.2.7 error_message

```
std::string PinComparisonResult::error_message
```

Definition at line 35 of file [LogicComparator.hpp](#).

6.14.2.8 pin_name

```
std::string PinComparisonResult::pin_name
```

Definition at line 24 of file [LogicComparator.hpp](#).

6.14.2.9 ref_compiles

```
bool PinComparisonResult::ref_compiles = false
```

Definition at line 31 of file [LogicComparator.hpp](#).

6.14.2.10 ref_expr_processed

```
std::string PinComparisonResult::ref_expr_processed
```

Definition at line 27 of file [LogicComparator.hpp](#).

6.14.2.11 ref_expr_raw

```
std::string PinComparisonResult::ref_expr_raw
```

Definition at line 25 of file [LogicComparator.hpp](#).

6.14.2.12 ref_truth_table

```
std::optional<Table> PinComparisonResult::ref_truth_table
```

Definition at line 33 of file [LogicComparator.hpp](#).

The documentation for this struct was generated from the following file:

- [include/LogicComparator.hpp](#)

6.15 ValuesIterator Class Reference

```
#include <Iterators.hpp>
```

Public Member Functions

- [ValuesIterator](#) (si2drValuesIdT values, si2drErrorT &err)
- [~ValuesIterator](#) ()
- void [next](#) ()
- bool [end](#) ()

Public Attributes

- si2drValuesIdT [values_](#)
- si2drValueTypeT [vtype_](#)
- si2drInt32T [int_](#)
- si2drFloat64T [float_](#)
- si2drStringT [str_](#)
- si2drBooleanT [bool_](#)
- si2drExprT * [exprp_](#)
- si2drErrorT & [err_](#)

6.15.1 Detailed Description

Definition at line 37 of file [lterators.hpp](#).

6.15.2 Constructor & Destructor Documentation

6.15.2.1 ValuesIterator()

```
ValuesIterator::ValuesIterator (
    si2drValuesIdT values,
    si2drErrorT & err )
```

Definition at line 25 of file [lterators.cpp](#).

6.15.2.2 ~ValuesIterator()

```
ValuesIterator::~ValuesIterator ( )
```

Definition at line 29 of file [lterators.cpp](#).

6.15.3 Member Function Documentation

6.15.3.1 end()

```
bool ValuesIterator::end ( )
```

Definition at line 34 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.15.3.2 next()

```
void ValuesIterator::next ( )
```

Definition at line 31 of file [lterators.cpp](#).

Here is the caller graph for this function:

6.15.4 Member Data Documentation

6.15.4.1 bool_

```
si2drBooleanT ValuesIterator::bool_
```

Definition at line 49 of file [lterators.hpp](#).

6.15.4.2 err_

```
si2drErrorT& ValuesIterator::err_
```

Definition at line 51 of file [lterators.hpp](#).

6.15.4.3 exprp_

```
si2drExprT* ValuesIterator::exprp_
```

Definition at line 50 of file [lterators.hpp](#).

6.15.4.4 float_

```
si2drFloat64T ValuesIterator::float_
```

Definition at line 47 of file [lterators.hpp](#).

6.15.4.5 int_

```
si2drInt32T ValuesIterator::int_
```

Definition at line 46 of file [lterators.hpp](#).

6.15.4.6 str_

```
si2drStringT ValuesIterator::str_
```

Definition at line 48 of file [lterators.hpp](#).

6.15.4.7 values_

```
si2drValuesIdT ValuesIterator::values_
```

Definition at line 44 of file [lterators.hpp](#).

6.15.4.8 vtype_

```
si2drValueTypeT ValuesIterator::vtype_
```

Definition at line 45 of file [lterators.hpp](#).

The documentation for this class was generated from the following files:

- [include/lterators.hpp](#)
- [src/lterators.cpp](#)

6.16 VerilogVisitor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for VerilogVisitor:

Collaboration diagram for VerilogVisitor:

Public Member Functions

- [VerilogVisitor](#) (const std::string &targetCell)
- void [handle](#) (const slang::syntax::SyntaxNode &node)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)
- void [handle](#) (const slang::syntax::PortDeclarationSyntax &portDecl)
- void [handle](#) (const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst)
- void [handle](#) (const slang::syntax::SpecifyBlockSyntax &specifyBlock)

Private Attributes

- const std::string & [targetCell_](#)
- int [depth_](#)
- bool [inTargetModule_](#)

6.16.1 Detailed Description

Definition at line 15 of file [verilog_utils.hpp](#).

6.16.2 Constructor & Destructor Documentation

6.16.2.1 VerilogVisitor()

```
VerilogVisitor::VerilogVisitor (  
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 17 of file [verilog_utils.hpp](#).

6.16.3 Member Function Documentation

6.16.3.1 `handle()` [1/5]

```
void VerilogVisitor::handle (
    const slang::syntax::HierarchyInstantiationSyntax & hierarchyInst )
```

Definition at line 78 of file [verilog_utils.cpp](#).

6.16.3.2 `handle()` [2/5]

```
void VerilogVisitor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 17 of file [verilog_utils.cpp](#).

6.16.3.3 `handle()` [3/5]

```
void VerilogVisitor::handle (
    const slang::syntax::PortDeclarationSyntax & portDecl )
```

Definition at line 46 of file [verilog_utils.cpp](#).

6.16.3.4 `handle()` [4/5]

```
void VerilogVisitor::handle (
    const slang::syntax::SpecifyBlockSyntax & specifyBlock )
```

Definition at line 258 of file [verilog_utils.cpp](#).

Here is the call graph for this function:

6.16.3.5 handle() [5/5]

```
void VerilogVisitor::handle (  
    const slang::syntax::SyntaxNode & node )
```

Definition at line 6 of file [verilog_utils.cpp](#).

Here is the caller graph for this function:

6.16.4 Member Data Documentation

6.16.4.1 depth_

```
int VerilogVisitor::depth_ [private]
```

Definition at line 28 of file [verilog_utils.hpp](#).

6.16.4.2 inTargetModule_

```
bool VerilogVisitor::inTargetModule_ [private]
```

Definition at line 29 of file [verilog_utils.hpp](#).

6.16.4.3 targetCell_

```
const std::string& VerilogVisitor::targetCell_ [private]
```

Definition at line 27 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

Chapter 7

File Documentation

7.1 doc/ChangeLog.md File Reference

7.2 include/Iterators.hpp File Reference

```
#include "si2dr_liberty.h"  
#include "LibAttribute.hpp"  
#include "LibGroup.hpp"
```

Include dependency graph for Iterators.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [GroupsIterator](#)
- class [AttributesIterator](#)
- class [ValuesIterator](#)

7.3 Iterators.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef ITERATORS_H  
00002 #define ITERATORS_H  
00003  
00004 #include "si2dr_liberty.h"  
00005  
00006 #include "LibAttribute.hpp"  
00007 #include "LibGroup.hpp"  
00008  
00009 class GroupsIterator {  
00010 public:
```

```

00011 GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err);
00012 ~GroupsIterator();
00013 void next();
00014 bool end();
00015 LibGroup get();
00016
00017 private:
00018     si2drGroupsIdT groups_;
00019     si2drGroupIdT group_;
00020     si2drErrorT &err_;
00021 };
00022
00023 class AttributesIterator {
00024 public:
00025     AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err);
00026     ~AttributesIterator();
00027     void next();
00028     bool end();
00029     LibAttribute get();
00030
00031 private:
00032     si2drAttrsIdT attrs_;
00033     si2drAttrIdT attr_;
00034     si2drErrorT &err_;
00035 };
00036
00037 class ValuesIterator {
00038 public:
00039     ValuesIterator(si2drValuesIdT values, si2drErrorT &err);
00040     ~ValuesIterator();
00041     void next();
00042     bool end();
00043
00044     si2drValuesIdT values_;
00045     si2drValueTypeT vtype_;
00046     si2drInt32T int_;
00047     si2drFloat64T float_;
00048     si2drStringT str_;
00049     si2drBooleanT bool_;
00050     si2drExprT *exprp_;
00051     si2drErrorT &err_;
00052 };
00053
00054 #endif // ITERATORS_H

```

7.4 include/json_utils.hpp File Reference

```

#include <string>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"
#include "spdlog/spdlog.h"
#include "Iterators.hpp"
#include "LibGroup.hpp"

```

Include dependency graph for json_utils.hpp: This graph shows which files directly or indirectly include this file:

Typedefs

- using `json` = `nlohmann::json`

Functions

- `json generateLutJson (LibGroup &lib_lut_group, si2drErrorT &err)`
Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.
- `json generatePowerJson (LibGroup &lib_power_group, si2drErrorT &err)`
Converts a Liberty power group into a JSON representation.
- `std::pair< std::string, json > generatePinJson (LibGroup &lib_pin_group, si2drErrorT &err)`
Generates a JSON representation of a Liberty pin group.
- `json generateCellJson (LibGroup &lib_cell_group, si2drErrorT &err)`
Generates a JSON representation of a cell from a [LibGroup](#) object.

7.4.1 Typedef Documentation

7.4.1.1 json

```
using json = nlohmann::json
```

Definition at line 13 of file [json_utils.hpp](#).

7.4.2 Function Documentation

7.4.2.1 generateCellJson()

```
json generateCellJson (  
    LibGroup & lib_cell_group,  
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as area, cell_leakage_power, and cell_footprint. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<i>lib_cell_group</i>	The LibGroup object representing the cell
<i>err</i>	Reference to a <code>si2drErrorT</code> object for error handling

Returns

json A JSON object containing the cell's information with the following structure:

- "cell_name": string - name of the cell
- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 271 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.4.2.2 generateLutJson()

```
json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )
```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an <code>si2drErrorT</code> object to track any errors during processing

Returns

json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 60 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.4.2.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
    LibGroup & lib_pin_group,
    si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 205 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.4.2.4 generatePowerJson()

```
json generatePowerJson (  
    LibGroup & lib_power_group,  
    si2drErrorT & err )
```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

json A JSON object representing the power group data

The function:

- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 152 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.5 json_utils.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef JSON_UTILS_HPP
00002 #define JSON_UTILS_HPP
00003
00004 #include <string>
00005
00006 #include "nlohmann/json.hpp"
00007 #include "si2dr_liberty.h"
00008 #include "spdlog/spdlog.h"
00009
00010 #include "Iterators.hpp"
00011 #include "LibGroup.hpp"
00012
00013 using json = nlohmann::json;
00014
00015 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err);
00016 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err);
00017 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err);
00018 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err);
00019
00020 #endif // JSON_UTILS_HPP
```

7.6 include/LibAttribute.hpp File Reference

```
#include <string>
```

```
#include "si2dr_liberty.h"
```

Include dependency graph for LibAttribute.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibAttribute](#)

7.7 LibAttribute.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef LIB_ATTRIBUTE_H
00002 #define LIB_ATTRIBUTE_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibAttribute {
00009 public:
00010     LibAttribute(si2drAttrIdT attr, si2drErrorT &err);
00011     ~LibAttribute();
00012     std::string getName();
00013     bool isComplex();
00014     si2drValuesIdT getValues();
```

```

00015 long int getInt();
00016 double getFloat();
00017 std::string getString();
00018 bool getBoolean();
00019
00020 private:
00021     si2drAttrIdT attr_;
00022     si2drErrorT &err_;
00023 };
00024
00025 #endif // LIB_ATTRIBUTE_H

```

7.8 include/LibDatabase.hpp File Reference

```

#include <memory>
#include <string>
#include <vector>
#include <SQLiteCpp/Database.h>
#include <SQLiteCpp/Statement.h>

```

Include dependency graph for LibDatabase.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibDatabase](#)

Manages SQLite database connections for writing Liberty LUT data.

7.9 LibDatabase.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIBDATABASE_HPP
00002 #define LIBDATABASE_HPP
00003
00004 #include <memory>
00005 #include <string>
00006 #include <vector>
00007
00008 #include <SQLiteCpp/Database.h>
00009 #include <SQLiteCpp/Statement.h>
00010
00026 class LibDatabase {
00027 public:
00032     explicit LibDatabase(const std::string &db_path);
00033     ~LibDatabase();
00034
00035     // -----
00036     // Schema / lifecycle
00037     // -----
00038
00045     void initialize();
00046

```

```

00047 // -----
00048 // Transaction support (for batched writes)
00049 // -----
00050
00051 void beginTransaction();
00052 void commitTransaction();
00053
00054 // -----
00055 // Write operations
00056 // -----
00057
00092 void writeLutEntry(const std::string &file_path, const std::string &library_name,
00093                  const std::string &pvt_corner, double age_seconds,
00094                  const std::string &cell_name, const std::string &output_pin,
00095                  const std::string &related_pin, const std::string &timing_sense,
00096                  const std::string &timing_type, const std::string &when,
00097                  const std::string &arc_type, int rows_n, int cols_n,
00098                  const std::vector<double> &index_1,
00099                  const std::vector<double> &index_2, const std::vector<double> &values,
00100                  int process, double temperature, double voltage);
00101
00102 private:
00103     std::string db_path_;
00104     std::unique_ptr<SQLite::Database> db_;
00105
00106     // Prepared statements cached across calls
00107     std::unique_ptr<SQLite::Statement> insert_stmt_;
00108     bool initialized_ = false;
00109 };
00110
00111 #endif // LIBDATABASE_HPP

```

7.10 include/LibFile.hpp File Reference

```

#include <chrono>
#include <filesystem>
#include <fstream>
#include <string>
#include <unordered_set>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "Iterators.hpp"
#include "json_utils.hpp"
#include "verilog_utils.hpp"
#include "version.h"

```

Include dependency graph for LibFile.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibFile](#)

Typedefs

- using `json` = `nlohmann::json`

7.10.1 Typedef Documentation

7.10.1.1 json

using `json` = `nlohmann::json`

Definition at line 21 of file [LibFile.hpp](#).

7.11 LibFile.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef LIB_FILE_H
00002 #define LIB_FILE_H
00003
00004 #include <chrono>
00005 #include <filesystem>
00006 #include <fstream>
00007 #include <string>
00008 #include <unordered_set>
00009
00010 #include "nlohmann/json.hpp"
00011 #include "si2dr_liberty.h"
00012 #include "spdlog/sinks/basic_file_sink.h"
00013 #include "spdlog/sinks/stdout_color_sinks.h"
00014 #include "spdlog/spdlog.h"
00015
00016 #include "Iterators.hpp"
00017 #include "json_utils.hpp"
00018 #include "verilog_utils.hpp"
00019 #include "version.h"
00020
00021 using json = nlohmann::json;
00022
00023 class LibFile {
00024 public:
00025     LibFile(const std::string &filepath, const std::string &loggername);
00026     ~LibFile();
00027     std::shared_ptr<spdlog::logger> logger_;
00028     std::filesystem::path filepath_; // full path to the file
00029     std::string basename_;          // file name without extension
00030     std::string filename_;          // full file name with extension
00031     std::string libname_ = "";      // library name obtained through parsing
00032     std::string jsonname_ = "";     // json file name to store parsed data
00033     std::string loggername_ = "";   // log file name
00034     json lib_json_ = json::object();
00035     void writeJsonToFile();
00036     void parse();
00037     void modify();
00038     void mono(const bool is_slew);
```

```

00039 void supercell(const int chain_length, const std::vector<std::string> &cell_names);
00040 void writeToDB(const std::string &db_path, const std::string &pvt_corner, double age_seconds);
00041 void verilog(const int chain_length, const std::vector<std::string> &cell_names);
00042 void spice(const int chain_length, const std::vector<std::string> &cell_names,
00043            const std::string &verilog_lib_file, const std::string &spice_lib_file);
00044 std::map<std::string, std::string> logic(const std::string &cell_name);
00045
00046 private:
00047     si2drErrorT err_;
00048     int process_ = 0;
00049     float voltage_ = 0.0;
00050     int temperature_ = 0;
00051     void read();
00052     bool checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00053                                     const std::string &type, bool is_slew);
00054     // --- Private Helper Methods ---
00055
00056     // Helper function to split a string by whitespace
00057     std::vector<std::string> splitString(const std::string &s);
00058
00059     // Function to generate RC lines based on instance index and stage
00060     void generateRCLines(std::ofstream &outFile, const std::string &netName, int instanceIndex,
00061                         bool isFinalStage);
00062
00063     // Function to modify the SPICE netlist generated by v2lvs
00064     bool modifySpiceNetlist(const std::string &v2lvsSpiceFile, // Input is the v2lvs output
00065                            const std::string &finalSpiceFile, // Output is the final file
00066                            const std::string &targetGlobalLine);
00067 };
00068
00069 #endif // LIB_FILE_H

```

7.12 include/LibFileOperations.hpp File Reference

```

#include <filesystem>
#include <iostream>
#include <thread>
#include "si2dr_liberty.h"
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "LibFile.hpp"
#include "LibraryComparator.hpp"
#include "LogicComparator.hpp"
#include "LogicExtractor.hpp"

```

Include dependency graph for LibFileOperations.hpp: This graph shows which files directly or indirectly include this file:

Functions

- void `printlnInfo` ()

Sets up and configures the global logger and prints application information.

- void `parseLibFile` (const std::string &library_path, const std::string log_file_name)
Parses a library file and generates corresponding output.
- void `monoCheckLibFile` (const std::string &library_path, const std::string log_file_name, bool is↵_slew)
Performs monotonicity check on a library file.
- void `supercellLibFile` (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Creates supercell map structures from a Liberty library file.
- void `verilogLibFile` (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Generates Verilog files from a library file for specified cells.
- void `spiceLibFile` (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names, const std::string &verilog_lib_file, const std::string &spice_lib_file)
Generates a SPICE library file from a given library file, applying a specified chain length and cell names.
- void `compareLibFiles` (const std::string &ref_lib, const std::string &comp_lib, const double reltol, const double abstol, std::string &report_file_name)
Compares two library files and generates a detailed comparison report.
- void `funcLibFile` (const std::string &ref_file, const std::string &comp_file, const std::vector< std↵::string > &cell_names, std::string &report_file_name)
Performs a functional equivalence check between two files (Liberty or Verilog) for a given set of cells.

7.12.1 Function Documentation

7.12.1.1 `compareLibFiles()`

```
void compareLibFiles (
    const std::string & ref_lib,
    const std::string & comp_lib,
    const double reltol,
    const double abstol,
    std::string & report_file_name )
```

Compares two library files and generates a detailed comparison report.

This function compares a reference library file with another library file, performing validation checks based on specified tolerance parameters. The comparison results are written to a report file in markdown or text format.

Parameters

<i>ref_lib</i>	Path to the reference library file to use as the baseline
<i>comp_lib</i>	Path to the library file to compare against the reference
<i>reltol</i>	Relative tolerance for numerical comparisons (must be ≥ 0.0)
<i>abstol</i>	Absolute tolerance for numerical comparisons
<i>report_file_name</i>	[in,out] Name of the file to write the comparison report to. If empty, defaults to [comp_lib_basename].cmp.md. If provided but doesn't end with .txt or .md, .md will be appended.

Note

The function will log an error and return without comparing if reltol is invalid.

Log files will be created with the naming pattern [library_basename].cmp.log

The function uses the [LibraryComparator](#) class to perform the actual comparison.

Definition at line 249 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.12.1.2 funcLibFile()

```
void funcLibFile (
    const std::string & ref_file,
    const std::string & comp_file,
    const std::vector< std::string > & cell_names,
    std::string & report_file_name )
```

Performs a functional equivalence check between two files (Liberty or Verilog) for a given set of cells.

This function compares the logic functions of specified cells in two files, which can be either in Liberty (.lib) or Verilog (.v) format. It extracts the logic functions for each cell from both files, compares them, and generates a report summarizing the comparison results.

Parameters

<i>ref_file</i>	The path to the reference file (Liberty or Verilog).
<i>comp_file</i>	The path to the comparison file (Liberty or Verilog).
<i>cell_names</i>	A vector of cell names to be checked for functional equivalence.
<i>report_file_name</i>	A string to store the name of the report file. If empty, a default name is generated. If the provided name does not end with ".txt" or ".md", ".md" is appended.

The function first checks the file extensions to determine the file format. It then extracts the logic functions for each specified cell from both files. The logic functions are then compared, and a report is generated, which includes the comparison results for each cell. The report is written to the specified report file.

Note

- If no cell names are provided, the function logs an error and returns.
- If the reference or comparison file format is not supported (i.e., not .lib or .v), the function logs an error and returns.
- The report file is cleared before writing the comparison results.
- The function uses spdlog for logging information, warnings, and errors.
- The function utilizes the [LogicComparator](#) class to perform the logic comparison and generate the report.
- Memory allocated for [LibFile](#) objects is managed using raw pointers and must be manually deallocated to prevent memory leaks.

Definition at line 308 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.12.1.3 monoCheckLibFile()

```
void monoCheckLibFile (
    const std::string & library_path,
    const std::string log_file_name,
    bool is_slew )
```

Performs monotonicity check on a library file.

This function validates the monotonicity of timing data in a library file. It creates a log file to record the results of the check and handles any exceptions that occur during the process.

Parameters

<i>library_path</i>	Path to the library file to check
<i>log_file_name</i>	Name of the log file to create (optional). If empty, a default name will be generated from the library file name
<i>is_slew</i>	Flag indicating whether to check input slew monotonicity (true) or output load monotonicity (false)

Exceptions

<i>The</i>	function catches and logs any exceptions but does not rethrow them
------------	--

Definition at line 81 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.12.1.4 parseLibFile()

```
void parseLibFile (
    const std::string & library_path,
    const std::string log_file_name )
```

Parses a library file and generates corresponding output.

This function processes the given library file, parsing its contents and generating a JSON output. It also logs the parsing process to a specified log file or creates a default log file if none is provided.

Parameters

<i>library_path</i>	Path to the library file that needs to be parsed
<i>log_file_name</i>	Optional name for the log file. If empty, a default name is generated based on the library filename with ".parse.log" extension

Exceptions

<i>The</i>	function catches and logs any exceptions but doesn't propagate them
------------	---

Definition at line 49 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.12.1.5 printInfo()

```
void printInfo ( )
```

Sets up and configures the global logger and prints application information.

This function performs the following operations:

1. Creates a console sink for logging with level set to INFO

2. Creates a file sink for logging with level set to TRACE, saving to [APP_NAME].log
3. Configures a logger with both sinks and sets it as the default logger
4. Outputs basic application information:
 - Version and build timestamp
 - Author information
 - Log file location

Note

Uses spdlog library for logging functionality

Depends on APP_NAME, APP_VERSION, BUILD_TIMESTAMP, APP_AUTHOR, and APP_CONTACT macros

Definition at line 18 of file [LibFileOperations.cpp](#).

Here is the caller graph for this function:

7.12.1.6 spiceLibFile()

```
void spiceLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names,
    const std::string & verilog_lib_file,
    const std::string & spice_lib_file )
```

Generates a SPICE library file from a given library file, applying a specified chain length and cell names.

This function takes a library file path, a log file name, a chain length, a vector of cell names, a Verilog library file path, and a SPICE library file path as input. It initializes a [LibFile](#) object, validates the chain length, and then calls the spice method of the [LibFile](#) object to generate the SPICE library file. It logs the start and end of the SPICE generation process, as well as any errors that occur.

Parameters

<i>library_path</i>	The path to the input library file.
<i>log_file_name</i>	The name of the log file. If empty, a default log file name is generated based on the library file name.
<i>chain_length</i>	The chain length to use during SPICE generation. Must be ≥ 1 .
<i>cell_names</i>	A vector of cell names to include in the SPICE generation.
<i>verilog_lib_file</i>	The path to the Verilog library file.
<i>spice_lib_file</i>	The path to the output SPICE library file.

Definition at line 202 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.12.1.7 supercellLibFile()

```
void supercellLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
```

Creates supercell map structures from a Liberty library file.

This function reads a Liberty file, creates supercell structures based on the specified chain length and cell names, and logs the process to a file.

Parameters

<i>library_path</i>	Path to the Liberty file to process
<i>log_file_name</i>	Name of the log file (if empty, defaults to "[library_name].supercell.log")
<i>chain_length</i>	The length of chains to create (must be ≥ 1)
<i>cell_names</i>	Vector of cell names to process for supercell generation

Exceptions

May	pass through exceptions from the LibFile::supercell method
-----	--

Note

The function validates the chain length and logs all activities including errors that might occur during processing

Definition at line 116 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.12.1.8 verilogLibFile()

```
void verilogLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
```

```
int chain_length,
const std::vector< std::string > & cell_names )
```

Generates Verilog files from a library file for specified cells.

This function processes a library file and generates Verilog representation for the specified cell names with a given chain length. The operation results are logged to a specified or default log file.

Parameters

<i>library_path</i>	Path to the library file to process
<i>log_file_name</i>	Name for the log file (if empty, a default name will be generated)
<i>chain_length</i>	Number of cells to chain together, must be ≥ 1
<i>cell_names</i>	Vector of cell names to generate Verilog for

Exceptions

<i>Catches</i>	any exceptions from the verilog generation process and logs them
----------------	--

Definition at line 157 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.13 LibFileOperations.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef LIBFILEOPERATIONS_H
00002 #define LIBFILEOPERATIONS_H
00003
00004 #include <filesystem>
00005 #include <iostream>
00006 #include <thread>
00007
00008 #include "si2dr_liberty.h"
00009 #include "spdlog/sinks/basic_file_sink.h"
00010 #include "spdlog/sinks/stdout_color_sinks.h"
00011 #include "spdlog/spdlog.h"
00012
00013 #include "LibFile.hpp"
00014 #include "LibraryComparator.hpp"
00015 #include "LogicComparator.hpp"
00016 #include "LogicExtractor.hpp"
00017
00018 void printInfo();
00019 void parseLibFile(const std::string &library_path, const std::string log_file_name);
00020 void monoCheckLibFile(const std::string &library_path, const std::string log_file_name,
00021                      bool is_slew);
00022 void supercellLibFile(const std::string &library_path, const std::string &log_file_name,
00023                      int chain_length, const std::vector<std::string> &cell_names);
00024 void verilogLibFile(const std::string &library_path, const std::string &log_file_name,
00025                    int chain_length, const std::vector<std::string> &cell_names);
00026 void spiceLibFile(const std::string &library_path, const std::string &log_file_name,
```

```

00027         int chain_length, const std::vector<std::string> &cell_names,
00028         const std::string &verilog_lib_file, const std::string &spice_lib_file);
00029 void compareLibFiles(const std::string &ref_lib, const std::string &comp_lib, const double reltol,
00030                     const double abstol, std::string &report_file_name);
00031 void funcLibFile(const std::string &ref_file, const std::string &comp_file,
00032                 const std::vector<std::string> &cell_names, std::string &report_file_name);
00033
00034 #endif // LIBFILEOPERATIONS_H

```

7.14 include/LibGroup.hpp File Reference

```
#include <string>
```

```
#include "si2dr_liberty.h"
```

Include dependency graph for LibGroup.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibGroup](#)

7.15 LibGroup.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIB_GROUP_H
00002 #define LIB_GROUP_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibGroup {
00009 public:
00010     LibGroup(si2drGroupIdT group, si2drErrorT &err);
00011     ~LibGroup();
00012     std::string getName();
00013     std::string getType();
00014     si2drAttrIdT getAttrs();
00015     si2drGroupsIdT getGroups();
00016
00017 private:
00018     si2drGroupIdT group_;
00019     si2drErrorT &err_;
00020 };
00021
00022 #endif // LIB_GROUP_H

```

7.16 include/LibraryComparator.hpp File Reference

```
#include <chrono>
#include <filesystem>
#include <fstream>
#include "nlohmann/json.hpp"
#include "spdlog/spdlog.h"
#include "tabulate/table.hpp"
#include <tabulate/markdown_exporter.hpp>
#include "LibFile.hpp"
#include "version.h"
```

Include dependency graph for LibraryComparator.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibraryComparator](#)

Typedefs

- using [json](#) = nlohmann::json

7.16.1 Typedef Documentation

7.16.1.1 json

```
using json = nlohmann::json
```

Definition at line 16 of file [LibraryComparator.hpp](#).

7.17 LibraryComparator.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef COMPARE_HPP
00002 #define COMPARE_HPP
00003
00004 #include <chrono>
00005 #include <filesystem>
00006 #include <fstream>
00007
00008 #include "nlohmann/json.hpp"
00009 #include "spdlog/spdlog.h"
00010 #include "tabulate/table.hpp"
00011 #include <tabulate/markdown_exporter.hpp>
00012
00013 #include "LibFile.hpp"
00014 #include "version.h"
00015
00016 using json = nlohmann::json;
00017 using namespace tabulate;
00018
00019 class LibraryComparator {
00020 public:
00021     LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol, double abstol);
00022     std::filesystem::path ref_lib_path_;
00023     std::filesystem::path comp_lib_path_;
00024     void generateReport(const std::string &output_file);
00025
00026 private:
00027     json ref_json_;
00028     json comp_json_;
00029     double reltol_;
00030     double abstol_;
00031
00032     void compareCell(const std::string &cell_name, const json &ref_cell, const json &comp_cell,
00033                     Table &table);
00034     void comparePin(const std::string &cell_name, const std::string &pin_name, const json &ref_pin,
00035                    const json &comp_pin, Table &table);
00036     void compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00037                           const std::string &timing_type, const json &ref_timing_arc,
00038                           const json &comp_timing_arc, Table &table);
00039     void compareLut(const std::string &cell_name, const std::string &pin_name,
00040                    const std::string &timing_type, const std::string &related_pin,
00041                    const std::string &arc_name, const json &ref_lut, const json &comp_lut,
00042                    Table &table);
00043     // void configureTableFormat(Table &table);
00044 };
00045
00046 #endif // COMPARE_HPP

```

7.18 include/LogicComparator.hpp File Reference

```

#include <algorithm>
#include <chrono>
#include <cmath>
#include <filesystem>
#include <iomanip>
#include <optional>

```

```
#include <regex>
#include <variant>
#include "exprtk.hpp"
#include "tabulate/markdown_exporter.hpp"
#include "tabulate/table.hpp"
#include <spdlog/spdlog.h>
#include "version.h"
```

Include dependency graph for LogicComparator.hpp: This graph shows which files directly or indirectly include this file:

Classes

- struct [PinComparisonResult](#)
- class [LogicComparator](#)

7.19 LogicComparator.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef LOGIC_COMPARATOR_HPP
00002 #define LOGIC_COMPARATOR_HPP
00003
00004 #include <algorithm>
00005 #include <chrono>
00006 #include <cmath>
00007 #include <filesystem>
00008 #include <iomanip>
00009 #include <optional> // To store tables optionally
00010 #include <regex>    // For regular expressions
00011 #include <variant>
00012
00013 #include "exprtk.hpp"
00014 #include "tabulate/markdown_exporter.hpp"
00015 #include "tabulate/table.hpp"
00016 #include <spdlog/spdlog.h>
00017
00018 #include "version.h"
00019
00020 using namespace tabulate;
00021
00022 // Structure to hold results for a single pin comparison
00023 struct PinComparisonResult {
00024     std::string pin_name;
00025     std::string ref_expr_raw;
00026     std::string comp_expr_raw;
00027     std::string ref_expr_processed;
00028     std::string comp_expr_processed;
00029     bool comparison_possible = false; // Was comparison attempted?
00030     bool are_equivalent = false;
00031     bool ref_compiles = false;
00032     bool comp_compiles = false;
00033     std::optional<Table> ref_truth_table; // Store tables only if needed/successful
00034     std::optional<Table> comp_truth_table;
00035     std::string error_message; // Store any error during comparison
00036 };
00037
```

```

00038 class LogicComparator {
00039 public:
00040     LogicComparator(const std::map<std::string, std::string> &ref_outpin_map,
00041                     const std::map<std::string, std::string> &comp_outpin_map,
00042                     const std::string &cell_name);
00043
00044     // Example code from exprtk documentation
00045     void logic();
00046
00047     // Preprocessing function
00048     std::string preprocessExpression(const std::string &input_expr);
00049
00050     // Helper to extract unique sorted variables from TWO expressions
00051     bool extractVariables(const std::string &expr1, const std::string &expr2,
00052                          std::vector<std::string> &sorted_vars);
00053
00064     void compareSingleExpressionPair(const std::string &ref_expression_processed,
00065                                     const std::string &comp_expression_processed,
00066                                     const std::vector<std::string> &sorted_vars,
00067                                     PinComparisonResult &result); // Pass result struct
00068
00073     void compareCellLogic();
00074
00080     void generateReport(const std::string &output_file);
00081
00082 private:
00083     std::map<std::string, std::string> ref_outpin_map_;
00084     std::map<std::string, std::string> comp_outpin_map_;
00085     std::string cell_name_;
00086     std::map<std::string, PinComparisonResult> all_pin_results_;
00087 };
00088
00089 #endif // LOGIC_COMPARATOR_HPP

```

7.20 include/LogicExtractor.hpp File Reference

#include "verilog_utils.hpp"

Include dependency graph for LogicExtractor.hpp: This graph shows which files directly or indirectly include this file:

Classes

- struct [GateInfo](#)
- class [LogicExtractor](#)

Functions

- void [extractAndPrintNetlistInfo](#) (const std::string &verilog_file, const std::string &cell)
Extracts and prints netlist information from a Verilog file for a specified cell.
- std::map< std::string, std::string > [extractLogicFromVerilog](#) (const std::string &verilog_file, const std::string &cell)
Extracts logic expressions from a Verilog file for a specified cell.

7.20.1 Function Documentation

7.20.1.1 `extractAndPrintNetlistInfo()`

```
void extractAndPrintNetlistInfo (
    const std::string & verilog_file,
    const std::string & cell )
```

Extracts and prints netlist information from a Verilog file for a specified cell.

This function parses a Verilog file using the slang library, extracts information about the primary inputs, primary outputs, internal wires, and gate drivers within a specified cell (module). It then prints a summary of the extracted information to the console using `spdlog`.

Parameters

<i>verilog_file</i>	The path to the Verilog file to be parsed.
<i>cell</i>	The name of the cell (module) for which to extract netlist information.

Exceptions

<i>std::exception</i>	If any error occurs during Verilog parsing or info extraction.
-----------------------	--

Note

The function uses the slang library for Verilog parsing and a custom [LogicExtractor](#) class to extract the desired information. Error messages are logged using `spdlog`.

Definition at line [676](#) of file [LogicExtractor.cpp](#).

Here is the call graph for this function:

7.20.1.2 `extractLogicFromVerilog()`

```
std::map< std::string, std::string > extractLogicFromVerilog (
    const std::string & verilog_file,
    const std::string & cell )
```

Extracts logic expressions from a Verilog file for a specified cell.

This function parses a Verilog file using the slang library, identifies the specified cell, and extracts the logic expressions for its outputs. It returns a map where the keys are output signal names and the values are their corresponding logic expressions as strings.

Parameters

<code>verilog_file</code>	The path to the Verilog file to parse.
<code>cell</code>	The name of the cell (module) for which to extract logic expressions.

Returns

A map of output signal names to their logic expressions. Returns an empty map if parsing fails, the cell is not found, or no logic expressions can be derived.

Note

The function uses the slang library for Verilog parsing. Ensure that slang is properly installed and configured before using this function.

The logic extraction process involves traversing the syntax tree of the Verilog code and identifying relevant assignments and expressions within the specified cell.

Error messages and warnings are logged using the spdlog library.

Definition at line 755 of file [LogicExtractor.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.21 LogicExtractor.hpp

[Go to the documentation of this file.](#)

```

00001 // include/LogicExtractor.hpp
00002
00003 #ifndef LOGIC_EXTRACTOR_HPP
00004 #define LOGIC_EXTRACTOR_HPP
00005
00006 #include "verilog_utils.hpp"
00007
00008 // Structure to represent a gate instance's information
00009 struct GateInfo {
00010     slang::parsing::TokenKind kind; // Store the TokenKind for reliable checks
00011     std::string gateTypeName;       // Store the string name like "and", "xor"
00012     std::vector<std::string> inputSignals;
00013     std::string outputSignal;
00014 };
00015
00016 // Visitor class to extract netlist information and derive logic expressions
00017 class LogicExtractor : public slang::syntax::SyntaxVisitor<LogicExtractor> {
00018 public:
00019     // Constructor takes the target cell name
00020     explicit LogicExtractor(const std::string &targetCell)
00021         : targetCell_(targetCell), inTargetModule_(false), parsingComplete_(false) {}
00022
00023     // --- Visitor Handlers ---
00024     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00025     void handle(
00026         const slang::syntax::PortDeclarationSyntax &portDecl); // Handles direction/type declaration

```

```

00027 void
00028 handle(const slang::syntax::NonAnsiPortListSyntax &portList); // Handles port names list in header
00029 void
00030 handle(const slang::syntax::NetDeclarationSyntax &netDecl); // To find explicitly declared wires
00031 void handle(const slang::syntax::PrimitiveInstantiationSyntax
00032             &primitiveInst); // Handles gate instantiation
00033 // Potentially handle ContinuousAssignSyntax if needed later:
00034 // void handle(const slang::syntax::ContinuousAssignSyntax& assign);
00035
00036 // --- Access Extracted Info (for Step 1 debugging) ---
00037 const std::unordered_map<std::string, GateInfo> &getExtractedGates()const {
00038     return gateOutputDrivers_;
00039 }
00040 const std::unordered_set<std::string> &getPrimaryInputs()const { return primaryInputs_; }
00041 const std::unordered_set<std::string> &getPrimaryOutputs()const { return primaryOutputs_; }
00042 const std::unordered_set<std::string> &getInternalWires()const { return internalWires_; }
00043
00044 // --- Logic Derivation (Commented out for Step 1) ---
00045 std::map<std::string, std::string> getLogicExpressions();
00046
00047 private:
00048 // --- Internal State ---
00049 const std::string &targetCell_;
00050 bool inTargetModule_;
00051 bool parsingComplete_; // Flag to indicate AST traversal is done
00052
00053 // Netlist Information
00054 std::unordered_set<std::string> primaryInputs_;
00055 std::unordered_set<std::string> primaryOutputs_;
00056 std::unordered_set<std::string> internalWires_; // Includes gate outputs
00057
00058 // Temporary map to store directions found in PortDeclarationSyntax
00059 // Key: port name, Value: direction ("input", "output", "inout", "unknown")
00060 std::unordered_map<std::string, std::string> portDirections_;
00061
00062 // Map: output signal name -> GateInfo driving it
00063 std::unordered_map<std::string, GateInfo> gateOutputDrivers_;
00064
00065 // Map: signal name -> Logic expression string (Memoization Cache) - (Used in Step 2)
00066 std::unordered_map<std::string, std::string> logicCache_;
00067
00068 // --- Helper Methods ---
00069 // Recursive function to derive logic for a given signal (Used in Step 2)
00070 std::string deriveLogicRecursive(const std::string &signalName);
00071 // Helper to format expressions based on gate type (Used in Step 2)
00072 std::string formatExpression(const GateInfo &gateInfo,
00073                             const std::vector<std::string> &inputExprs);
00074 };
00075
00076 // --- Function Declaration ---
00077 // For Step 1, this function will just run the visitor and maybe print summary
00078 void extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell);
00079
00080 // Function to be implemented in Step 2
00081 std::map<std::string, std::string> extractLogicFromVerilog(const std::string &verilog_file,
00082                                                           const std::string &cell);
00083
00084 #endif // LOGIC_EXTRACTOR_HPP

```

7.22 include/verilog_utils.hpp File Reference

```

#include <fstream>
#include <unordered_map>

```

```
#include <unordered_set>
#include "slang/syntax/SyntaxPrinter.h"
#include "slang/syntax/SyntaxVisitor.h"
#include "spdlog/spdlog.h"
```

Include dependency graph for verilog_utils.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [VerilogVisitor](#)
- class [CellExtractor](#)
- class [CellPrinter](#)
- class [ModuleRewriter](#)

Functions

- void [getAST](#) (const std::string &verilog_file, const std::string &cell)

7.22.1 Function Documentation

7.22.1.1 getAST()

```
void getAST (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line [544](#) of file [verilog_utils.cpp](#).

Here is the call graph for this function:

7.23 verilog_utils.hpp

[Go to the documentation of this file.](#)

```
00001 // verilog_utils.hpp
00002
00003 #ifndef VERILOG_UTILS_H
00004 #define VERILOG_UTILS_H
00005
00006 #include <fstream>
00007 #include <unordered_map>
00008 #include <unordered_set>
00009
00010 #include "slang/syntax/SyntaxPrinter.h"
00011 #include "slang/syntax/SyntaxVisitor.h"
00012 #include "spdlog/spdlog.h" // Include spdlog
00013
00014 // Creating custom visitor class
00015 class VerilogVisitor : public slang::syntax::SyntaxVisitor<VerilogVisitor> {
00016 public:
00017     explicit VerilogVisitor(const std::string &targetCell)
00018         : targetCell_(targetCell), depth_(0), inTargetModule_(false) {}
00019     void handle(const slang::syntax::SyntaxNode &node);
00020     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00021     void handle(const slang::syntax::PortDeclarationSyntax &portDecl);
00022     void handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst);
00023     // void handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst);
00024     void handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock);
00025
00026 private:
00027     const std::string &targetCell_;
00028     int depth_;
00029     bool inTargetModule_;
00030 };
00031
00032 // Creating a Rewriter to extract a specific cell
00033 class CellExtractor : public slang::syntax::SyntaxRewriter<CellExtractor> {
00034 public:
00035     explicit CellExtractor(const std::string &targetCell)
00036         : targetCell_(targetCell), foundTarget_(false) {}
00037     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00038     bool foundTargetCell() const;
00039
00040 private:
00041     const std::string &targetCell_;
00042     bool foundTarget_;
00043 };
00044
00045 // Print specific cell when visiting SyntaxTree
00046 class CellPrinter : public slang::syntax::SyntaxVisitor<CellPrinter> {
00047 public:
00048     explicit CellPrinter(const std::string &targetCell, std::ostream &out)
00049         : targetCell_(targetCell), out_(out), foundTarget_(false) {}
00050     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00051
00052 private:
00053     const std::string &targetCell_;
00054     std::ostream &out_;
00055     bool foundTarget_;
00056 };
00057
00058 // Comprehensive module rewriter for adding ports, instances, and connections
00059 class ModuleRewriter : public slang::syntax::SyntaxRewriter<ModuleRewriter> {
00060 public:
00061     explicit ModuleRewriter(const std::vector<std::string> &inputPins,
00062                             const std::vector<std::string> &outputPins,
00063                             const std::pair<std::string, std::string> &superCellEntry,
00064                             int instanceCount, std::shared_ptr<spdlog::logger> logger)
```

```

00065     : inputPins_(inputPins), outputPins_(outputPins), cellName_(supercell_entry.first),
00066     moduleName_(supercell_entry.second), logger_(logger), depth_(0),
00067     instance_count_(instance_count) {}
00068 std::shared_ptr<spdlog::logger> logger_;
00069 void handle(const slang::syntax::SyntaxNode &node);
00070 void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00071
00072 private:
00073     const std::vector<std::string> &inputPins_;
00074     const std::vector<std::string> &outputPins_;
00075     const std::string cellName_;
00076     const std::string moduleName_;
00077     std::map<std::string, std::string> portInfoMap_; // Map from port name to direction
00078     int depth_;
00079     int instance_count_;
00080 };
00081
00082 void getAST(const std::string &verilog_file, const std::string &cell);
00083
00084 #endif // VERILOG_UTILS_H

```

7.24 include/version.h File Reference

This graph shows which files directly or indirectly include this file:

Macros

- #define APP_NAME "ZlibValidation"
- #define APP_VERSION_MAJOR 1
- #define APP_VERSION_MINOR 2
- #define APP_VERSION_PATCH 3
- #define APP_VERSION "1.2.3"
- #define APP_AUTHOR "Song Zixuan (宋子轩)"
- #define APP_CONTACT "zx.song@zju.edu.cn, songcedar@outlook.com"
- #define BUILD_TIMESTAMP "2026-07-06 17:15:44"

7.24.1 Macro Definition Documentation

7.24.1.1 APP_AUTHOR

```
#define APP_AUTHOR "Song Zixuan (宋子轩)"
```

Definition at line 10 of file [version.h](#).

7.24.1.2 APP_CONTACT

```
#define APP_CONTACT "zx.song@zju.edu.cn, songcedar@outlook.com"
```

Definition at line 11 of file [version.h](#).

7.24.1.3 APP_NAME

```
#define APP_NAME "ZlibValidation"
```

Definition at line 5 of file [version.h](#).

7.24.1.4 APP_VERSION

```
#define APP_VERSION "1.2.3"
```

Definition at line 9 of file [version.h](#).

7.24.1.5 APP_VERSION_MAJOR

```
#define APP_VERSION_MAJOR 1
```

Definition at line 6 of file [version.h](#).

7.24.1.6 APP_VERSION_MINOR

```
#define APP_VERSION_MINOR 2
```

Definition at line 7 of file [version.h](#).

7.24.1.7 APP_VERSION_PATCH

```
#define APP_VERSION_PATCH 3
```

Definition at line 8 of file [version.h](#).

7.24.1.8 BUILD_TIMESTAMP

```
#define BUILD_TIMESTAMP "2026-07-06 17:15:44"
```

Definition at line 12 of file [version.h](#).

7.25 version.h

[Go to the documentation of this file.](#)

```
00001 // version.h.in
00002 #pragma once
00003
00004 // Auto-generated by CMake - DO NOT EDIT
00005 #define APP_NAME          "ZlibValidation"
00006 #define APP_VERSION_MAJOR 1
00007 #define APP_VERSION_MINOR 2
00008 #define APP_VERSION_PATCH 3
00009 #define APP_VERSION       "1.2.3"
00010 #define APP_AUTHOR        "Song Zixuan (宋子轩)"
00011 #define APP_CONTACT       "zx.song@zju.edu.cn, songcedar@outlook.com"
00012 #define BUILD_TIMESTAMP   "2026-07-06 17:15:44"
```

7.26 README.md File Reference

7.27 src/Iterators.cpp File Reference

```
#include "Iterators.hpp"
```

Include dependency graph for Iterators.cpp:

7.28 Iterators.cpp

[Go to the documentation of this file.](#)

```
00001 #include "Iterators.hpp"
00002
00003 GroupsIterator::GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err)
00004     : groups_(groups), err_(err) {
00005     group_ = si2drIterNextGroup(groups_, &err_);
00006 }
00007 GroupsIterator::~GroupsIterator() { si2drIterQuit(groups_, &err_); }
00008
00009 void GroupsIterator::next() { group_ = si2drIterNextGroup(groups_, &err_); }
00010 bool GroupsIterator::end() { return si2drObjectIsNull(group_, &err_); }
00011
00012 LibGroup GroupsIterator::get() { return LibGroup(group_, err_); }
00013
00014 AttributesIterator::AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err)
00015     : attrs_(attrs), err_(err) {
00016     attr_ = si2drIterNextAttr(attrs_, &err_);
00017 }
00018 AttributesIterator::~AttributesIterator() { si2drIterQuit(attrs_, &err_); }
00019
00020 void AttributesIterator::next() { attr_ = si2drIterNextAttr(attrs_, &err_); }
00021 bool AttributesIterator::end() { return si2drObjectIsNull(attr_, &err_); }
00022
00023 LibAttribute AttributesIterator::get() { return LibAttribute(attr_, err_); }
00024
00025 ValuesIterator::ValuesIterator(si2drValuesIdT values, si2drErrorT &err)
00026     : values_(values), err_(err) {
00027     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00028 }
00029 ValuesIterator::~ValuesIterator() { si2drIterQuit(values_, &err_); }
00030
00031 void ValuesIterator::next() {
00032     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00033 }
00034 bool ValuesIterator::end() { return vtype_ == SI2DR_UNDEFINED_VALUETYPE; }
```

7.29 src/json_utils.cpp File Reference

```
#include "json_utils.hpp"
```

Include dependency graph for json_utils.cpp:

Functions

- `std::vector< double > parseStringToVector (const std::string &str)`
Parses a string representation of a vector of doubles into a vector of doubles.
- `json generateLutJson (LibGroup &lib_lut_group, si2drErrorT &err)`
Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.
- `json generateTimingJson (LibGroup &lib_timing_group, si2drErrorT &err)`
Generates a JSON representation of timing information from a library timing group.
- `json generatePowerJson (LibGroup &lib_power_group, si2drErrorT &err)`
Converts a Liberty power group into a JSON representation.

- `std::pair< std::string, json > generatePinJson (LibGroup &lib_pin_group, si2drErrorT &err)`
Generates a JSON representation of a Liberty pin group.
- `json generateCellJson (LibGroup &lib_cell_group, si2drErrorT &err)`
Generates a JSON representation of a cell from a [LibGroup](#) object.

7.29.1 Function Documentation

7.29.1.1 generateCellJson()

```
json generateCellJson (
    LibGroup & lib_cell_group,
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as area, cell_leakage_power, and cell_footprint. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<i>lib_cell_group</i>	The LibGroup object representing the cell
<i>err</i>	Reference to a si2drErrorT object for error handling

Returns

json A JSON object containing the cell's information with the following structure:

- "cell_name": string - name of the cell
- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 271 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.29.1.2 generateLutJson()

```
json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )
```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an <code>si2drErrorT</code> object to track any errors during processing

Returns

json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 60 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.29.1.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
    LibGroup & lib_pin_group,
    si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 205 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.29.1.4 generatePowerJson()

```
json generatePowerJson (
    LibGroup & lib_power_group,
    si2drErrorT & err )
```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

json A JSON object representing the power group data

The function:

- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 152 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.29.1.5 generateTimingJson()

```
json generateTimingJson (  
    LibGroup & lib_timing_group,  
    si2drErrorT & err )
```

Generates a JSON representation of timing information from a library timing group.

This function extracts timing attributes and sub-groups from a Liberty timing group and organizes them into a JSON object. It processes standard timing attributes like 'related_pin', 'timing_type', 'timing_sense', and 'when', as well as timing tables such as 'cell_fall', 'cell_rise', 'fall_transition', 'rise_transition', 'fall_constraint', and 'rise_constraint'.

Parameters

<i>lib_timing_group</i>	The Liberty timing group to process
<i>err</i>	Reference to an si2drErrorT object for error reporting

Returns

json A JSON object containing the extracted timing information

Definition at line 104 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.29.1.6 parseStringToVector()

```
std::vector< double > parseStringToVector (
    const std::string & str )
```

Parses a string representation of a vector of doubles into a vector of doubles.

This function takes a string containing comma-separated numbers, cleans it by removing backslashes and newline characters, and then converts each comma-separated value into a double that is added to the resulting vector.

Parameters

<i>str</i>	The input string to be parsed, containing comma-separated numbers
------------	---

Returns

`std::vector<double>` A vector of doubles parsed from the input string

Exceptions

<i>std::invalid_argument</i>	If the string contains values that cannot be converted to double
<i>std::out_of_range</i>	If the string contains values that are out of range for double

Definition at line 25 of file `json_utils.cpp`.

Here is the caller graph for this function:

7.30 json_utils.cpp

[Go to the documentation of this file.](#)

```
00001 #include "json_utils.hpp"
00002
00003 /*
00004  json 类型的值可以是以下几种类型之一:
00005  方括号 []: 用于包含一组有序的值 (数组)。
00006  数组中的每个值可以是任意类型的 JSON 值, 包括对象、数组、字符串、数字、布尔值或 null。
00007  花括号 {}: 用于包含一组键值对 (对象)。
00008  对象中的每个键都是一个字符串, 值可以是任意类型的 JSON 值, 包括对象、数组、字符串、数字、布尔值或
00009  null。
00010 */
00011
00025 std::vector<double> parseStringToVector(const std::string &str) {
00026     std::vector<double> result;
00027     std::string cleaned_str;
00028
00029     // Remove backslashes and newline characters
00030     for (char c : str) {
```

```

00031     if (c != '\\\' && c != '\n') {
00032         cleaned_str += c;
00033     }
00034 }
00035
00036 std::stringstream ss(cleaned_str);
00037 std::string item;
00038 while (std::getline(ss, item, ',')) {
00039     result.push_back(std::stod(item));
00040 }
00041 return result;
00042 }
00043
00060 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err) {
00061     json lut_json;
00062
00063     AttributesIterator attr_iter(lib_lut_group.getAttrs(), err);
00064     for (; !attr_iter.end(); attr_iter.next()) {
00065         LibAttribute lib_attr = attr_iter.get();
00066         std::string lut_attr_name = lib_attr.getName();
00067
00068         // spdlog::debug("LUT Attribute Name: {}", lib_attr.getName());
00069         // spdlog::debug("Is Complex? {}", lib_attr.isComplex());
00070
00071         if (lut_attr_name == "index_1" || lut_attr_name == "index_2") {
00072             ValuesIterator values_iter(lib_attr.getValues(), err);
00073             for (; !values_iter.end(); values_iter.next()) {
00074                 // spdlog::debug("Type: {}", int(values_iter.vtype_)); // 5 is string
00075                 // spdlog::debug("Str: {}", values_iter.str_);
00076                 lut_json[lut_attr_name] = parseStringToVector(std::string(values_iter.str_));
00077             }
00078         } else if (lut_attr_name == "values") {
00079             ValuesIterator values_iter(lib_attr.getValues(), err);
00080             for (; !values_iter.end(); values_iter.next()) {
00081                 // spdlog::debug("{} ", values_iter.str_);
00082                 lut_json["values"].push_back(parseStringToVector(std::string(values_iter.str_)));
00083             }
00084         } else {
00085             spdlog::warn("Unknown LUT attribute name: {}", lut_attr_name);
00086         }
00087     }
00088     return lut_json;
00089 }
00090
00104 json generateTimingJson(LibGroup &lib_timing_group, si2drErrorT &err) {
00105     json timing_json;
00106
00107     AttributesIterator attr_iter(lib_timing_group.getAttrs(), err);
00108     for (; !attr_iter.end(); attr_iter.next()) {
00109         LibAttribute lib_attr = attr_iter.get();
00110
00111         std::string attr_name = lib_attr.getName();
00112         if (attr_name == "related_pin" || attr_name == "timing_type" || attr_name == "timing_sense" ||
00113             attr_name == "when") {
00114             timing_json[attr_name] = lib_attr.getString();
00115         }
00116         // More timing attributes can be added here
00117     }
00118
00119     GroupsIterator timing_sub_group_iter(lib_timing_group.getGroups(), err);
00120     for (; !timing_sub_group_iter.end(); timing_sub_group_iter.next()) {
00121         LibGroup lib_timing_sub_group = timing_sub_group_iter.get();
00122
00123         std::string timing_sub_group_type = lib_timing_sub_group.getType();
00124         // std::string timing_sub_group_name = lib_timing_sub_group.getName();
00125         if (timing_sub_group_type == "cell_fall" || timing_sub_group_type == "cell_rise" ||
00126             timing_sub_group_type == "fall_transition" || timing_sub_group_type == "rise_transition" ||
00127             timing_sub_group_type == "fall_constraint" || timing_sub_group_type == "rise_constraint") {
00128             timing_json[timing_sub_group_type] = generateLutJson(lib_timing_sub_group, err);

```

```

00129     } else {
00130         spdlog::warn("Unknown timing sub group type: {}", timing_sub_group_type);
00131     }
00132 }
00133 return timing_json;
00134 }
00135
00152 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err) {
00153     json power_json;
00154
00155     AttributesIterator attr_iter(lib_power_group.getAttrs(), err);
00156     for (; !attr_iter.end(); attr_iter.next()) {
00157         LibAttribute lib_attr = attr_iter.get();
00158
00159         std::string attr_name = lib_attr.getName();
00160         if (attr_name == "when" || attr_name == "related_pin" || attr_name == "related_pg_pin") {
00161             power_json[attr_name] = lib_attr.getString();
00162         }
00163         // More power attributes can be added here
00164     }
00165
00166     GroupsIterator power_sub_group_iter(lib_power_group.getGroups(), err);
00167     for (; !power_sub_group_iter.end(); power_sub_group_iter.next()) {
00168         LibGroup lib_power_sub_group = power_sub_group_iter.get();
00169
00170         std::string power_sub_group_type = lib_power_sub_group.getType();
00171         // std::string power_sub_group_name = lib_power_sub_group.getName();
00172         if (power_sub_group_type == "rise_power") {
00173             // spdlog::debug("Rise Power: {}", power_sub_group_name);
00174             power_json["rise_power"] = generateLutJson(lib_power_sub_group, err);
00175         } else if (power_sub_group_type == "fall_power") {
00176             power_json["fall_power"] = generateLutJson(lib_power_sub_group, err);
00177         } else {
00178             spdlog::warn("Unknown power sub group type: {}", power_sub_group_type);
00179         }
00180     }
00181     return power_json;
00182 }
00183
00205 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err) {
00206     json pin_json;
00207     std::string direction;
00208     pin_json["pin_name"] = lib_pin_group.getName();
00209
00210     AttributesIterator attr_iter(lib_pin_group.getAttrs(), err);
00211     for (; !attr_iter.end(); attr_iter.next()) {
00212         LibAttribute lib_attr = attr_iter.get();
00213
00214         std::string attr_name = lib_attr.getName();
00215         if (attr_name == "direction") {
00216             direction = lib_attr.getString();
00217         } else if (attr_name == "max_transition" || attr_name == "capacitance" ||
00218             attr_name == "rise_capacitance" || attr_name == "fall_capacitance" ||
00219             attr_name == "max_capacitance") {
00220             pin_json[attr_name] = lib_attr.getFloat();
00221         } else if (attr_name == "function" || attr_name == "power_down_function" ||
00222             attr_name == "related_ground_pin" || attr_name == "related_power_pin" ||
00223             attr_name == "three_state") {
00224             pin_json[attr_name] = lib_attr.getString();
00225         } else if (attr_name == "clock") {
00226             pin_json[attr_name] = lib_attr.getBoolean();
00227         }
00228         // More pin attributes can be added here
00229     }
00230
00231     GroupsIterator pin_sub_group_iter(lib_pin_group.getGroups(), err);
00232     for (; !pin_sub_group_iter.end(); pin_sub_group_iter.next()) {
00233         LibGroup lib_pin_sub_group = pin_sub_group_iter.get();
00234

```

```

00235     std::string pin_sub_group_type = lib_pin_sub_group.getType();
00236     // std::string pin_sub_group_name = lib_pin_sub_group.getName();
00237     if (pin_sub_group_type == "internal_power") {
00238         json power_json = generatePowerJson(lib_pin_sub_group, err);
00239         pin_json["power_arcs"].push_back(power_json);
00240     } else if (pin_sub_group_type == "timing") {
00241         // spdlog::debug("Has Timing: {}", lib_pin_group.getName());
00242         json timing_json = generateTimingJson(lib_pin_sub_group, err);
00243         pin_json["timing_arcs"].push_back(timing_json);
00244     } else {
00245         spdlog::warn("Unknown pin sub group type: {}", pin_sub_group_type);
00246     }
00247 }
00248 return std::make_pair(direction, pin_json);
00249 }
00250
00271 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err) {
00272     json cell_json;
00273     cell_json["cell_name"] = lib_cell_group.getName();
00274
00275     AttributesIterator attr_iter(lib_cell_group.getAttrs(), err);
00276     for (; !attr_iter.end(); attr_iter.next()) {
00277         LibAttribute lib_attr = attr_iter.get();
00278
00279         std::string attr_name = lib_attr.getName();
00280         if (attr_name == "area" || attr_name == "cell_leakage_power") {
00281             cell_json[attr_name] = lib_attr.getFloat();
00282         } else if (attr_name == "cell_footprint") {
00283             cell_json[attr_name] = lib_attr.getString();
00284         }
00285         // More cell attributes can be added here
00286     }
00287
00288     GroupsIterator cell_sub_group_iter(lib_cell_group.getGroups(), err);
00289     for (; !cell_sub_group_iter.end(); cell_sub_group_iter.next()) {
00290         LibGroup lib_cell_sub_group = cell_sub_group_iter.get();
00291
00292         std::string cell_sub_group_type = lib_cell_sub_group.getType();
00293         // std::string cell_sub_group_name = lib_cell_sub_group.getName();
00294
00295         // spdlog::debug("Cell Sub Group Type: {}", cell_sub_group_type);
00296         // spdlog::debug("Cell Sub Group Name: {}", cell_sub_group_name);
00297
00298         if (cell_sub_group_type == "pin") {
00299             auto [direction, pin_json] = generatePinJson(lib_cell_sub_group, err);
00300             if (direction == "input") {
00301                 cell_json["input_pins"].push_back(pin_json);
00302             } else if (direction == "output") {
00303                 cell_json["output_pins"].push_back(pin_json);
00304             } else if (direction == "internal") {
00305                 cell_json["internal_pins"].push_back(pin_json);
00306             } else if (direction == "inout") {
00307                 cell_json["inout_pins"].push_back(pin_json);
00308             } else {
00309                 spdlog::warn("Unknown direction: {}", direction);
00310             }
00311         }
00312     }
00313     return cell_json;
00314 }

```

7.31 src/LibAttribute.cpp File Reference

```
#include "LibAttribute.hpp"
```

Include dependency graph for LibAttribute.cpp:

7.32 LibAttribute.cpp

[Go to the documentation of this file.](#)

```
00001 #include "LibAttribute.hpp"
00002
00003 LibAttribute::LibAttribute(si2drAttrIdT attr, si2drErrorT &err) : attr_(attr), err_(err) {}
00004
00005 LibAttribute::~LibAttribute() {}
00006
00007 std::string LibAttribute::getName() {
00008     si2drStringT name = si2drAttrGetName(attr_, &err_);
00009     return name ? std::string(name) : std::string();
00010 }
00011
00012 bool LibAttribute::isComplex() { return si2drAttrGetAttrType(attr_, &err_) ? 1 : 0; }
00013
00014 si2drValuesIdT LibAttribute::getValues() { return si2drComplexAttrGetValues(attr_, &err_); }
00015
00016 long int LibAttribute::getInt() { return si2drSimpleAttrGetInt32Value(attr_, &err_); }
00017
00018 double LibAttribute::getFloat() { return si2drSimpleAttrGetFloat64Value(attr_, &err_); }
00019
00020 std::string LibAttribute::getString() {
00021     si2drStringT str = si2drSimpleAttrGetStringValue(attr_, &err_);
00022     return str ? std::string(str) : std::string();
00023 }
00024
00025 bool LibAttribute::getBoolean() { return si2drSimpleAttrGetBooleanValue(attr_, &err_) ? 1 : 0; }
```

7.33 src/LibDatabase.cpp File Reference

```
#include "LibDatabase.hpp"
#include <SQLiteCpp/Database.h>
#include <SQLiteCpp/Statement.h>
#include <cstring>
#include <vector>
```

Include dependency graph for LibDatabase.cpp:

Functions

- static std::vector< float > [toFloatBlob](#) (const std::vector< double > &src)

Variables

- static const char * [DDL_LUT_ENTRIES](#)
- static const char * [INSERT_SQL](#)

7.33.1 Function Documentation

7.33.1.1 toFloatBlob()

```
static std::vector< float > toFloatBlob (  
    const std::vector< double > & src ) [static]
```

Definition at line 12 of file [LibDatabase.cpp](#).

Here is the caller graph for this function:

7.33.2 Variable Documentation

7.33.2.1 DDL_LUT_ENTRIES

```
const char* DDL_LUT_ENTRIES [static]
```

Initial value:

```
= R"sql(  
CREATE TABLE IF NOT EXISTS lut_entries (  
entry_id      INTEGER PRIMARY KEY AUTOINCREMENT,  
file_path     TEXT NOT NULL,  
library_name  TEXT NOT NULL,  
scenario_id   TEXT,  
pvt_corner    TEXT NOT NULL,  
age_seconds   REAL NOT NULL,  
cell_name     TEXT NOT NULL,  
output_pin    TEXT NOT NULL DEFAULT 'Y',  
related_pin   TEXT NOT NULL,  
timing_sense  TEXT,  
timing_type   TEXT,  
"when"       TEXT,  
arc_type      TEXT NOT NULL,  
rows_n        INTEGER NOT NULL,  
cols_n        INTEGER NOT NULL,  
index_1_blob  BLOB NOT NULL,  
index_2_blob  BLOB NOT NULL,  
values_blob   BLOB NOT NULL,  
process       INTEGER,  
temperature   REAL,  
voltage       REAL,  
ingested_at   TEXT NOT NULL DEFAULT (datetime('now')),  
UNIQUE(pvt_corner, age_seconds, cell_name, output_pin,  
related_pin, "when", arc_type)  
);  
)sql"
```

Definition at line 26 of file [LibDatabase.cpp](#).

7.33.2.2 INSERT_SQL

```
const char* INSERT_SQL [static]
```

Initial value:

```
= R"sql(
INSERT OR IGNORE INTO lut_entries
(file_path, library_name, scenario_id, pvt_corner, age_seconds,
cell_name, output_pin, related_pin, timing_sense, timing_type, "when", arc_type,
rows_n, cols_n, index_1_blob, index_2_blob, values_blob,
process, temperature, voltage)
VALUES (?, ?, ?, ?, ?,
?, ?, ?, ?, ?, ?, ?,
?, ?, ?, ?, ?,
?, ?, ?)
)sql"
```

Definition at line 55 of file [LibDatabase.cpp](#).

7.34 LibDatabase.cpp

[Go to the documentation of this file.](#)

```
00001 #include "LibDatabase.hpp"
00002
00003 #include <SQLiteCpp/Database.h>
00004 #include <SQLiteCpp/Statement.h>
00005 #include <cstring>
00006 #include <vector>
00007
00008 // =====
00009 // Helpers: convert vector<double> -> blob (vector<float>)
00010 // =====
00011
00012 static std::vector<float> toFloatBlob(const std::vector<double> &src) {
00013     std::vector<float> out;
00014     out.reserve(src.size());
00015     for (double v : src) out.push_back(static_cast<float>(v));
00016     return out;
00017 }
00018
00019 // =====
00020 // DDL — only the lut_entries table (the rest is managed by Python)
00021 // NOTE: file_path is deliberately excluded from the UNIQUE constraint.
00022 // The same .lib data may exist under multiple paths (symlinks, directory mirrors),
00023 // so UNIQUE uses semantic keys only. file_path remains as a provenance column.
00024 // =====
00025
00026 static const char *DDL_LUT_ENTRIES = R"sql(
00027 CREATE TABLE IF NOT EXISTS lut_entries (
00028     entry_id      INTEGER PRIMARY KEY AUTOINCREMENT,
00029     file_path     TEXT NOT NULL,
00030     library_name  TEXT NOT NULL,
00031     scenario_id   TEXT,
00032     pvt_corner    TEXT NOT NULL,
00033     age_seconds   REAL NOT NULL,
00034     cell_name     TEXT NOT NULL,
00035     output_pin    TEXT NOT NULL DEFAULT 'Y',
00036     related_pin   TEXT NOT NULL,
00037     timing_sense  TEXT,
00038     timing_type   TEXT,
00039     "when"        TEXT,
```

```

00040 arc_type      TEXT NOT NULL,
00041 rows_n          INTEGER NOT NULL,
00042 cols_n          INTEGER NOT NULL,
00043 index_1_blob    BLOB NOT NULL,
00044 index_2_blob    BLOB NOT NULL,
00045 values_blob     BLOB NOT NULL,
00046 process         INTEGER,
00047 temperature     REAL,
00048 voltage         REAL,
00049 ingested_at     TEXT NOT NULL DEFAULT (datetime('now')),
00050 UNIQUE(pvt_corner, age_seconds, cell_name, output_pin,
00051 related_pin, "when", arc_type)
00052 );
00053 )sql";
00054
00055 static const char *INSERT_SQL = R"sql(
00056 INSERT OR IGNORE INTO lut_entries
00057 (file_path, library_name, scenario_id, pvt_corner, age_seconds,
00058 cell_name, output_pin, related_pin, timing_sense, timing_type, "when", arc_type,
00059 rows_n, cols_n, index_1_blob, index_2_blob, values_blob,
00060 process, temperature, voltage)
00061 VALUES (?, ?, ?, ?, ?,
00062 ?, ?, ?, ?, ?, ?, ?,
00063 ?, ?, ?, ?, ?,
00064 ?, ?, ?)
00065 )sql";
00066
00067 // =====
00068 // Constructor / Destructor
00069 // =====
00070
00071 LibDatabase::LibDatabase(const std::string &db_path) : db_path_(db_path) {}
00072
00073 LibDatabase::~LibDatabase() = default;
00074
00075 // =====
00076 // initialize()
00077 // =====
00078
00079 void LibDatabase::initialize() {
00080     if (initialized_) return;
00081
00082     db_ = std::make_unique<SQLite::Database>(
00083         db_path_,
00084         SQLite::OPEN_READWRITE | SQLite::OPEN_CREATE);
00085
00086     // WAL mode + busy timeout for concurrent access
00087     db_->exec("PRAGMA journal_mode = WAL");
00088     db_->exec("PRAGMA busy_timeout = 30000");
00089
00090     // Create table
00091     db_->exec(DDL_LUT_ENTRIES);
00092
00093     // Cache the INSERT statement
00094     insert_stmt_ = std::make_unique<SQLite::Statement>(*db_, INSERT_SQL);
00095
00096     initialized_ = true;
00097 }
00098
00099 // =====
00100 // Transaction support
00101 // =====
00102
00103 void LibDatabase::beginTransaction() {
00104     if (!initialized_) {
00105         throw std::runtime_error("LibDatabase::beginTransaction() called before initialize()");
00106     }
00107     db_->exec("BEGIN");
00108 }

```

```

00109
00110 void LibDatabase::commitTransaction() {
00111     if (!initialized_) {
00112         throw std::runtime_error("LibDatabase::commitTransaction() called before initialize()");
00113     }
00114     db_>exec("COMMIT");
00115 }
00116
00117 // =====
00118 // writeLutEntry()
00119 // =====
00120
00121 void LibDatabase::writeLutEntry(const std::string &file_path,
00122                                const std::string &library_name,
00123                                const std::string &pvt_corner, double age_seconds,
00124                                const std::string &cell_name, const std::string &output_pin,
00125                                const std::string &related_pin, const std::string &timing_sense,
00126                                const std::string &timing_type, const std::string &when,
00127                                const std::string &arc_type, int rows_n, int cols_n,
00128                                const std::vector<double> &index_1,
00129                                const std::vector<double> &index_2,
00130                                const std::vector<double> &values, int process,
00131                                double temperature, double voltage) {
00132     if (!initialized_) {
00133         throw std::runtime_error("LibDatabase::writeLutEntry() called before initialize()");
00134     }
00135
00136     auto &stmt = *insert_stmt_;
00137
00138     // Reset before binding — safe even after a prior exec() failure
00139     stmt.reset();
00140
00141     // Convert doubles to float32 blobs
00142     auto idx1_f = toFloatBlob(index_1);
00143     auto idx2_f = toFloatBlob(index_2);
00144     auto vals_f = toFloatBlob(values);
00145
00146     // Empty vectors would bind NULL to BLOB NOT NULL columns — refuse early
00147     if (idx1_f.empty()) {
00148         throw std::runtime_error("Empty index_1 in LUT entry for " + cell_name + "/" + arc_type);
00149     }
00150     if (vals_f.empty()) {
00151         throw std::runtime_error("Empty values in LUT entry for " + cell_name + "/" + arc_type);
00152     }
00153     // index_2 may be zero-filled for 1D LUTs — accept empty as 1-column placeholder
00154     if (idx2_f.empty()) {
00155         // BIND NULL is rejected by NOT NULL, so write a single zero
00156         idx2_f.push_back(0.0f);
00157     }
00158
00159     // Bind parameters
00160     stmt.bind(1, file_path);
00161     stmt.bind(2, library_name);
00162     stmt.bind(3); // scenario_id → NULL (Python fills later)
00163     stmt.bind(4, pvt_corner);
00164     stmt.bind(5, age_seconds);
00165     stmt.bind(6, cell_name);
00166     stmt.bind(7, output_pin);
00167     stmt.bind(8, related_pin);
00168     stmt.bind(9, timing_sense);
00169     stmt.bind(10, timing_type);
00170     // Empty when → "" (NOT NULL string), so unconditional arcs still honor the
00171     // UNIQUE constraint for idempotent re-writes. Binding NULL would let SQLite
00172     // treat multiple NULLs as non-conflicting and break INSERT OR IGNORE dedup.
00173     stmt.bind(11, when);
00174     stmt.bind(12, arc_type);
00175     stmt.bind(13, rows_n);
00176     stmt.bind(14, cols_n);
00177     stmt.bind(15, idx1_f.data(), static_cast<int>(idx1_f.size() * sizeof(float)));

```

```

00178 stmt.bind(16, idx2_f.data(), static_cast<int>(idx2_f.size() * sizeof(float)));
00179 stmt.bind(17, vals_f.data(), static_cast<int>(vals_f.size() * sizeof(float)));
00180 stmt.bind(18, process);
00181 stmt.bind(19, temperature);
00182 stmt.bind(20, voltage);
00183
00184 // Execute
00185 stmt.exec();
00186 }

```

7.35 src/LibFile.cpp File Reference

```
#include "LibFile.hpp"
```

```
#include "LibDatabase.hpp"
```

Include dependency graph for LibFile.cpp:

7.36 LibFile.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LibFile.hpp"
00002
00003 #include "LibDatabase.hpp"
00004
00017 LibFile::LibFile(const std::string &filepath, const std::string &loggername)
00018     : filepath_(filepath), loggername_(loggername) {
00019     filename_ = filepath_.string();
00020     basename_ = filepath_.stem().string();
00021     jsonname_ = basename_ + ".json";
00022
00023     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00024     console_sink->set_level(spdlog::level::info);
00025
00026     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(loggername_, true);
00027     file_sink->set_level(spdlog::level::debug);
00028
00029     std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00030     logger_ = std::make_shared<spdlog::logger>(loggername_, sinks.begin(), sinks.end());
00031     logger_->set_level(spdlog::level::debug);
00032
00033     logger_->info("Created LibFile object for '{}'", filename_);
00034     logger_->info("Debug log in: '{}'", loggername_);
00035 }
00036
00037 LibFile::~LibFile() { logger_->info("Closing file: '{}'", filename_); }
00038
00048 void LibFile::writeJsonToFile() {
00049     std::ofstream out(jsonname_);
00050     if (!out.is_open()) {
00051         logger_->error("Could not open file '{}' for writing", jsonname_);
00052         return;
00053     }
00054     out << lib_json_.dump(2);
00055     out.close();
00056     logger_->info("JSON data written to '{}'", jsonname_);
00057 }
00058
00079 void LibFile::read() {
00080     logger_->info("Reading '{} ...'", filename_);

```

```

00081
00082  auto start = std::chrono::high_resolution_clock::now();
00083  si2drReadLibertyFile(const_cast<char *>(filepath_.c_str()), &err_);
00084  auto end = std::chrono::high_resolution_clock::now();
00085  std::chrono::duration<double> duration = end - start;
00086
00087  if (err_ == SI2DR_INVALID_NAME) {
00088      logger_>error("Could not open file '{}' for parsing, quitting...", filename_);
00089      exit(301);
00090  } else if (err_ == SI2DR_SYNTAX_ERROR) {
00091      logger_>error("Syntax Errors were detected in the input file!");
00092      exit(401);
00093  } else {
00094      logger_>info("Done. Read time: {:.2f} seconds", duration.count());
00095  }
00096 }
00097
00118 void LibFile::parse() {
00119     si2drPIInit(&err_); // Initialize private error handler
00120     this->read();        // Read the Liberty file
00121
00122     logger_>info("Parsing '{}' ...", filename_);
00123     // Create a scope for the top-level groups iteration
00124     {
00125         GroupsIterator group_iter(si2drPIGetGroups(&err_), err_);
00126         for (; !group_iter.end(); group_iter.next()) {
00127             LibGroup lib_group = group_iter.get();
00128
00129             if (lib_group.getName() != "") {
00130                 libname_ = lib_group.getName();
00131                 lib_json["library_name"] = this->libname_;
00132                 logger_>info("Library Name: {}", libname_);
00133             } else {
00134                 logger_>warn("Library Name: <NONAME>");
00135             }
00136             // Second level groups
00137             GroupsIterator sub_group_iter(lib_group.getGroups(), err_);
00138             for (; !sub_group_iter.end(); sub_group_iter.next()) {
00139                 LibGroup lib_sub_group = sub_group_iter.get();
00140
00141                 std::string sub_group_type = lib_sub_group.getType();
00142                 std::string sub_group_name = lib_sub_group.getName();
00143
00144                 if (sub_group_type == "cell") {
00145                     logger_>debug("Cell Name: {}", sub_group_name);
00146                     // if (sub_group_name != "AN2D0") {
00147                     //     continue;
00148                     // }
00149                     // Handle each cell
00150                     json cell_json = generateCellJson(lib_sub_group, err_);
00151                     lib_json["cells"].push_back(cell_json);
00152                 } else if (sub_group_type == "operating_conditions") {
00153                     logger_>info("Operating Conditions: {}", sub_group_name);
00154                     // Get PVT from Operating Conditions
00155                     AttributesIterator attr_iter(lib_sub_group.getAttrs(), err_);
00156                     for (; !attr_iter.end(); attr_iter.next()) {
00157                         LibAttribute lib_attr = attr_iter.get();
00158                         logger_>debug("Attribute Name: {}", lib_attr.getName());
00159                         logger_>debug("Int Value: {}", lib_attr.getInt());
00160                         logger_>debug("Float Value: {}", lib_attr.getFloat());
00161                         logger_>debug("String Value: {}", lib_attr.getString());
00162                         if (lib_attr.getName() == "process") {
00163                             process_ = lib_attr.getInt();
00164                             lib_json["process"] = process_;
00165                         } else if (lib_attr.getName() == "voltage") {
00166                             voltage_ = lib_attr.getFloat();
00167                             // Round to 2 decimal places to avoid floating-point precision issues
00168                             lib_json["voltage"] = std::round(voltage_ * 100) / 100.0;
00169                         } else if (lib_attr.getName() == "temperature") {

```

```

00170         temperature_ = lib_attr.getInt();
00171         lib_json["temperature"] = temperature_;
00172     }
00173 }
00174     logger_>info("P: {}, V: {}, T: {}", process_, voltage_, temperature_);
00175 }
00176     // sub_group_iter's lifetime ends here, and the destructor is called
00177 }
00178 }
00179     // group_iter's lifetime ends here, and the destructor is called
00180 }
00181 si2drPIQuit(&err_);
00182 }
00183
00184 void LibFile::writeToDB(const std::string &db_path, const std::string &pvt_corner,
00185     double age_seconds) {
00186     logger_>info("Writing LUT entries to DB: '{}'", db_path);
00187
00188     if (!lib_json_.contains("cells") || !lib_json_["cells"].is_array()) {
00189         logger_>warn("No cells found in parsed JSON, skipping DB write");
00190         return;
00191     }
00192
00193     LibDatabase database(db_path);
00194     database.initialize();
00195
00196     int total_entries = 0;
00197
00198     // --- Extract header-level PVT ---
00199     int process = lib_json_.value("process", 0);
00200     double temperature = lib_json_.value("temperature", 0.0);
00201     double voltage = lib_json_.value("voltage", 0.0);
00202
00203     // library_name = .lib basename (scenario_id form)
00204     std::string library_name = basename_;
00205
00206     try {
00207         // Walk: cells → output_pins → timing_arcs → arc_types
00208         database.beginTransaction();
00209
00210         for (const auto &cell : lib_json_["cells"]) {
00211             if (!cell.contains("cell_name") || !cell["cell_name"].is_string()) continue;
00212             std::string cell_name = cell["cell_name"].get<std::string>();
00213
00214             if (!cell.contains("output_pins") || !cell["output_pins"].is_array()) continue;
00215
00216             for (const auto &pin : cell["output_pins"]) {
00217                 if (!pin.contains("pin_name") || !pin["pin_name"].is_string()) continue;
00218                 std::string output_pin = pin["pin_name"].get<std::string>();
00219
00220                 if (!pin.contains("timing_arcs") || !pin["timing_arcs"].is_array()) continue;
00221
00222                 for (const auto &arc : pin["timing_arcs"]) {
00223                     std::string related_pin = arc.value("related_pin", "");
00224                     std::string timing_sense = arc.value("timing_sense", "");
00225                     std::string timing_type = arc.value("timing_type", "");
00226                     std::string when = arc.value("when", "");
00227
00228                     // Four arc types to check
00229                     static const char *ARC_NAMES[] = {"cell_rise", "cell_fall", "rise_transition",
00230                         "fall_transition"};
00231                     for (const char *arc_type : ARC_NAMES) {
00232                         if (!arc.contains(arc_type)) continue;
00233                         const auto &lut = arc[arc_type];
00234                         if (!lut.is_object()) continue;
00235                         if (!lut.contains("index_1") || !lut["index_1"].is_array() ||
00236                             !lut.contains("values") || !lut["values"].is_array()) continue;
00237
00238                         // index_1 → rows_n

```

```

00239         std::vector<double> index_1 = lut["index_1"].get<std::vector<double>>();
00240         int rows_n = static_cast<int>(index_1.size());
00241
00242         // index_2 (optional, derive cols_n from values shape)
00243         std::vector<double> index_2;
00244         if (lut.contains("index_2") && lut["index_2"].is_array()) {
00245             index_2 = lut["index_2"].get<std::vector<double>>();
00246         }
00247
00248         // values → flatten
00249         const auto &vals_json = lut["values"];
00250         std::vector<double> values;
00251         int cols_n = 0;
00252         if (vals_json.is_array() && !vals_json.empty()) {
00253             cols_n = static_cast<int>(vals_json[0].size());
00254             for (const auto &row : vals_json) {
00255                 for (const auto &v : row) {
00256                     values.push_back(v.get<double>());
00257                 }
00258             }
00259         }
00260
00261         // If index_2 is missing, default to empty vector of size cols_n
00262         if (index_2.empty()) {
00263             index_2.resize(cols_n, 0.0);
00264         }
00265
00266         database.writeLutEntry(
00267             filename_, // file_path (original path for provenance)
00268             library_name, pvt_corner, age_seconds, cell_name, output_pin,
00269             related_pin, timing_sense, timing_type, when, arc_type, rows_n,
00270             cols_n, index_1, index_2, values, process, temperature, voltage);
00271
00272         total_entries++;
00273     }
00274 }
00275 }
00276 }
00277
00278 database.commitTransaction();
00279
00280 logger_>info("Wrote {} LUT entries to DB: '{}', total_entries, db_path);
00281 } catch (const std::exception &e) {
00282     // Rollback on any error, then rethrow
00283     database.commitTransaction();
00284     logger_>error("DB write failed, rolled back: {}", e.what());
00285     throw;
00286 }
00287 }
00288
00289 void LibFile::modify() { logger_>info("Modifying the file..."); }
00290
00312 bool LibFile::checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00313                                           const std::string &timing_arc_name, const bool is_slew) {
00314     if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00315         logger_>debug(
00316             "Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}', when: '{}'",
00317             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00318             arc["related_pin"].get<std::string>(), timing_arc_name, arc["when"].get<std::string>());
00319     } else {
00320         logger_>debug("Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}'",
00321             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00322             arc["related_pin"].get<std::string>(), timing_arc_name);
00323     }
00324     bool is_monotonic = true; // Assume the values are monotonic initially
00325     if (arc.contains(timing_arc_name) && arc[timing_arc_name].contains("values")) {
00326         std::vector<std::vector<double>> value_matrix;
00327         // Generate a matrix of values from the JSON array
00328         for (const auto &row_val : arc[timing_arc_name]["values"]) {

```

```

00329     std::vector<double> row_data;
00330     // Check if the value is an array
00331     if (!row_val.is_array()) {
00332         logger_>error("Invalid format: '{}' values should be an array of arrays in cell '{}', pin "
00333             "'{}', related_pin '{}'",
00334             timing_arc_name, cell["cell_name"].get<std::string>(),
00335             pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00336         return false;
00337     }
00338     // Check for non-numeric values
00339     for (const auto &val : row_val) {
00340         if (val.is_number()) {
00341             row_data.push_back(val.get<double>());
00342         } else {
00343             logger_>warn("Non-numeric value found in '{}'.values, skipping value: {} in cell '{}', "
00344                 "pin '{}', related_pin '{}'",
00345                 timing_arc_name, val.dump(), cell["cell_name"].get<std::string>(),
00346                 pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00347         }
00348     }
00349     value_matrix.push_back(row_data);
00350 }
00351 // Check the matrix for monotonicity
00352 if (!value_matrix.empty() && !value_matrix[0].empty()) {
00353     // Check the monotonic incrementality of rows (by output load capacitance, index 2)
00354     for (size_t i = 0; i < value_matrix.size(); ++i) {
00355         for (size_t j = 1; j < value_matrix[i].size(); ++j) {
00356             if ((value_matrix[i][j] < value_matrix[i][j - 1] &&
00357                 pin["pin_name"] != arc["related_pin"]) ||
00358                 (value_matrix[i][j] == 0 && value_matrix[i][j - 1] == 0)) {
00359                 // If contains "when" key, log the when value
00360                 if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00361                     logger_>warn("Non-monotonic (by load) '{}' values: ({} , {}) {} < ({} , {}) {} "
00362                         "for Cell: {} Pin: {}->{} when: \"{}\\\"",
00363                         timing_arc_name, i, j, value_matrix[i][j], i, j - 1,
00364                         value_matrix[i][j - 1], cell["cell_name"].get<std::string>(),
00365                         arc["related_pin"].get<std::string>(),
00366                         pin["pin_name"].get<std::string>(), arc["when"].get<std::string>());
00367                 } else {
00368                     logger_>warn("Non-monotonic (by load) '{}' values: ({} , {}) {} < ({} , {}) {} "
00369                         "for Cell: {} Pin: {}->{}",
00370                         timing_arc_name, i, j, value_matrix[i][j], i, j - 1,
00371                         value_matrix[i][j - 1], cell["cell_name"].get<std::string>(),
00372                         arc["related_pin"].get<std::string>(),
00373                         pin["pin_name"].get<std::string>());
00374                 }
00375                 is_monotonic = false;
00376             }
00377         }
00378     }
00379     if (is_slew) {
00380         // Check the monotonic incrementality of columns (by input slew, index 1)
00381         for (size_t j = 0; j < value_matrix[0].size(); ++j) {
00382             for (size_t i = 1; i < value_matrix.size(); ++i) {
00383                 if ((value_matrix[i][j] < value_matrix[i - 1][j] &&
00384                     pin["pin_name"] != arc["related_pin"]) ||
00385                     (value_matrix[i][j] == 0 && value_matrix[i - 1][j] == 0)) {
00386                     // If contains "when" key, log the when value
00387                     if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00388                         logger_>warn("Non-monotonic (by slew) '{}' values: ({} , {}) {} < ({} , {}) {} "
00389                             "for Cell: {} Pin: {}->{} when: \"{}\\\"",
00390                             timing_arc_name, i, j, value_matrix[i][j], i - 1, j,
00391                             value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00392                             arc["related_pin"].get<std::string>(),
00393                             pin["pin_name"].get<std::string>(), arc["when"].get<std::string>());
00394                     } else {
00395                         logger_>warn("Non-monotonic (by slew) '{}' values: ({} , {}) {} < ({} , {}) {} "
00396                             "for Cell: {} Pin: {}->{}",
00397                             timing_arc_name, i, j, value_matrix[i][j], i - 1, j,

```

```

00398         value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00399         arc["related_pin"].get<std::string>(),
00400         pin["pin_name"].get<std::string>());
00401     }
00402     is_monotonic = false;
00403 }
00404 }
00405 }
00406 }
00407 } else {
00408     logger_>warn(
00409         "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "
00410         "timing_arc: '{}'",
00411         cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00412         arc["related_pin"].get<std::string>(), timing_arc_name);
00413     }
00414 }
00415 return is_monotonic;
00416 }
00417
00438 void LibFile::mono(const bool is_slew) {
00439     /*
00440     * Use this command to check the following data in the current library to ensure
00441     * the tables are monotonically increasing with respect to output load:
00442     * cell_rise retaining_rise
00443     * cell_fall retaining_fall
00444     * rise_transition retain_rise_slew
00445     * fall_transition retain_fall_slew
00446     * mpw
00447     */
00448     logger_>info("Monotonicity check of '{}' starting ...", filename_);
00449
00450     if (!std::filesystem::exists(jsonname_)) {
00451         logger_>info("JSON file not found. Parsing Liberty file first.");
00452         this->parse();
00453         this->writeJsonToFile();
00454     } else {
00455         // Read the JSON file into json object
00456         std::ifstream in(jsonname_);
00457         if (!in.is_open()) {
00458             logger_>error("Could not open file '{}' for reading", jsonname_);
00459             return;
00460         }
00461         try {
00462             lib_json_ = json::parse(in);
00463         } catch (const json::parse_error &e) {
00464             logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00465             in.close();
00466             return;
00467         }
00468         in.close();
00469     }
00470
00471     std::map<std::string, bool> cell_monotonicity_status; // Track pass/fail status for each cell
00472     std::vector<std::string> failed_cells;               // List of failed cell names
00473     int total_cells = 0;
00474     int passed_cells = 0;
00475
00476     // Check the monotonicity of delay values
00477     // The index 1 (time values) must be monotonically increasing and >= 0
00478     for (const auto &cell : lib_json_["cells"]) {
00479         total_cells++;
00480         bool cell_is_monotonic = true; // Assume cell is monotonic initially
00481         std::string cell_name = cell["cell_name"].get<std::string>();
00482         cell_monotonicity_status[cell_name] =
00483             true; // Initialize to pass, will be set to false if any check fails
00484
00485         if (cell.contains("output_pins")) {
00486             for (const auto &pin : cell["output_pins"]) {

```

```

00487         if (pin.contains("timing_arcs")) {
00488             for (const auto &arc : pin["timing_arcs"]) {
00489                 // Check 4 timing arc names and accumulate monotonicity status
00490                 for (const auto &name :
00491                     {"cell_rise", "cell_fall", "rise_transition", "fall_transition"}) {
00492                     if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00493                         cell_is_monotonic = false;
00494                     }
00495                 }
00496             }
00497         }
00498     }
00499 }
00500 if (cell.contains("input_pins")) {
00501     for (const auto &pin : cell["input_pins"]) {
00502         if (!pin.contains("clock")) {
00503             continue; // Skip non-clock pins
00504         }
00505         // logger->debug("Clock pin: {}", pin["pin_name"].get<std::string>());
00506         if (pin.contains("timing_arcs")) {
00507             for (const auto &arc : pin["timing_arcs"]) {
00508                 if (arc.contains("timing_type")) {
00509                     // logger->debug("Timing type: {}", arc["timing_type"].get<std::string>());
00510                     if (arc["timing_type"].get<std::string>() == "min_pulse_width") {
00511                         for (const auto &name : {"rise_constraint", "fall_constraint"}) {
00512                             if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00513                                 cell_is_monotonic = false;
00514                             }
00515                         }
00516                     }
00517                 }
00518             }
00519         }
00520     }
00521 }
00522 // Update cell status based on all checks
00523 cell_monotonicity_status[cell_name] = cell_is_monotonic;
00524 if (cell_is_monotonic) {
00525     passed_cells++;
00526 } else {
00527     failed_cells.push_back(cell_name);
00528 }
00529 }
00530 logger->info("Monotonicity check of '{}' completed.", filename_);
00531
00532 // Output summary statistics
00533 logger->info("validate_monotonicity : {} out of {} cells passed", passed_cells, total_cells);
00534 logger->info("validate_monotonicity : {} out of {} cells failed", total_cells - passed_cells,
00535             total_cells);
00536 if (!failed_cells.empty()) {
00537     std::stringstream failed_cells_ss;
00538     for (size_t i = 0; i < failed_cells.size(); ++i) {
00539         failed_cells_ss << failed_cells[i];
00540         if (i < failed_cells.size() - 1) {
00541             failed_cells_ss << ", "; // Add comma if not the last element
00542         }
00543     }
00544     logger->info("Failed cell list : {}", failed_cells_ss.str());
00545 }
00546 }
00547
00567 void LibFile::supercell(const int chain_length, const std::vector<std::string> &cell_names) {
00568     /*
00569     * Supercells are named as follows:
00570     * <cellname>_X<chain_length>_<input_pin>_<output_pin>
00571     */
00572     logger->info("Creating supercells for '{}'", filename_);
00573
00574     if (!std::filesystem::exists(jsonname_)) {

```

```

00575     logger_>info("JSON file not found. Parsing Liberty file first.");
00576     this->parse();
00577     this->writeJsonToFile();
00578 } else {
00579     // Read the JSON file into json object
00580     std::ifstream in(jsonname_);
00581     if (!in.is_open()) {
00582         logger_>error("Could not open file '{}' for reading", jsonname_);
00583         return;
00584     }
00585     try {
00586         lib_json_ = json::parse(in);
00587     } catch (const json::parse_error &e) {
00588         logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00589         in.close();
00590         return;
00591     }
00592     in.close();
00593 }
00594
00595 // write to .map file
00596 std::ofstream out(basename_ + ".map");
00597 if (!out.is_open()) {
00598     logger_>error("Could not open file '{}' for writing", basename_ + ".map");
00599     return;
00600 }
00601 // if cell_names is empty, create supercells for all cells
00602 // else create supercells for only the specified cells
00603
00604 // Check if we're processing specific cells or all cells
00605 bool process_all_cells = cell_names.empty();
00606
00607 if (process_all_cells) {
00608     logger_>info("Creating supercells for ALL cells in '{}'", filename_);
00609 } else {
00610     logger_>info("Creating supercells for {} specified cells in '{}'", cell_names.size(),
00611                 filename_);
00612 }
00613
00614 // Create a set for faster lookups if we have specific cell names
00615 std::unordered_set<std::string> cell_set;
00616 std::unordered_set<std::string> found_cells;
00617 if (!process_all_cells) {
00618     cell_set.insert(cell_names.begin(), cell_names.end());
00619 }
00620
00621 for (const auto &cell : lib_json_["cells"]) {
00622     bool is_sequential = false;
00623     std::string cell_name = cell["cell_name"].get<std::string>();
00624
00625     // Skip cells not in the specified list
00626     if (!process_all_cells && cell_set.find(cell_name) == cell_set.end()) {
00627         continue;
00628     }
00629
00630     // Mark this cell as found
00631     if (!process_all_cells) {
00632         found_cells.insert(cell_name);
00633     }
00634
00635     logger_>debug("Creating supercells for cell: '{}'", cell_name);
00636
00637     std::vector<std::string> output_pins;
00638     std::vector<std::string> input_pins;
00639     // Extract input and output pins
00640     if (cell.contains("output_pins")) {
00641         for (const auto &pin : cell["output_pins"]) {
00642             output_pins.push_back(pin["pin_name"].get<std::string>());
00643         }

```

```

00644     }
00645     if (cell.contains("input_pins")) {
00646         for (const auto &pin : cell["input_pins"]) {
00647             if (pin.contains("clock")) {
00648                 is_sequential = true;
00649                 // clock pin not use for supercell
00650                 continue;
00651             }
00652             input_pins.push_back(pin["pin_name"].get<std::string>());
00653         }
00654     }
00655
00656     // create supercells for all combinations of input and output pins
00657     for (const auto &output_pin : output_pins) {
00658         for (const auto &input_pin : input_pins) {
00659             if (is_sequential) {
00660                 const int chain_len = 1;
00661                 std::string supercell_name =
00662                     cell_name + "__X" + std::to_string(chain_len) + "__" + input_pin + "__" + output_pin;
00663                 out << cell_name << " " << supercell_name << std::endl;
00664             } else {
00665                 std::string supercell_name = cell_name + "__X" + std::to_string(chain_length) + "__" +
00666                     input_pin + "__" + output_pin;
00667                 out << cell_name << " " << supercell_name << std::endl;
00668             }
00669         }
00670     }
00671 }
00672
00673 // Check for cells that were specified but not found
00674 if (!process_all_cells) {
00675     std::vector<std::string> not_found_cells;
00676     for (const auto &requested_cell : cell_names) {
00677         if (found_cells.find(requested_cell) == found_cells.end()) {
00678             not_found_cells.push_back(requested_cell);
00679         }
00680     }
00681
00682     // Output warning for cells that weren't found
00683     if (!not_found_cells.empty()) {
00684         std::string missing_cells = not_found_cells[0];
00685         for (size_t i = 1; i < not_found_cells.size(); ++i) {
00686             missing_cells += ", " + not_found_cells[i];
00687         }
00688         logger_>warn("{} specified cell{} not found in the library: {}", not_found_cells.size(),
00689             not_found_cells.size() > 1 ? "s were" : " was", missing_cells);
00690     }
00691 }
00692
00693 out.close();
00694 logger_>info("Supercell creation complete in '{}', basename_ + ".map");
00695 }
00696
00721 void LibFile::verilog(const int chain_length, const std::vector<std::string> &cell_names) {
00722     logger_>info("Creating Verilog for '{}', filename_");
00723
00724     this->supercell(chain_length, cell_names);
00725
00726     // Create a temporary file for individual modules
00727     std::string temp_file = basename_ + "_temp.v";
00728     std::ofstream out(temp_file);
00729     if (!out.is_open()) {
00730         logger_>error("Could not open temp file '{}' for writing", temp_file);
00731         return;
00732     }
00733
00734     // Read the .map file into a vector to preserve all entries and their order
00735     std::vector<std::pair<std::string, std::string>> supercell_entries;
00736     std::ifstream in(basename_ + ".map");

```

```

00737 if (!in.is_open()) {
00738     logger->error("Could not open file '{}' for reading", basename_ + ".map");
00739     return;
00740 }
00741
00742 std::string line;
00743 while (std::getline(in, line)) {
00744     std::istringstream iss(line);
00745     std::string cell_name, supercell_name;
00746     if (!(iss >> cell_name >> supercell_name)) {
00747         logger->warn("Invalid line format in map file: '{}'", line);
00748         continue;
00749     }
00750     supercell_entries.push_back({cell_name, supercell_name});
00751 }
00752 in.close();
00753 logger->info("Read {} supercells from '{}'", supercell_entries.size(), basename_ + ".map");
00754
00755 // Store module information for top-level creation
00756 std::vector<std::string> module_names;
00757 std::map<std::string, std::vector<std::string>> module_inputs;
00758 std::map<std::string, std::vector<std::string>> module_outputs;
00759
00760 // Generate verilog module for each supercell
00761 for (const auto &supercell_entry : supercell_entries) {
00762     // Check if the cell is sequential
00763     bool is_sequential = false;
00764     // Count the number of instances will be created in verilog
00765     int instance_count = 0;
00766     // Get original cell name from pair
00767     const std::string &cell_name = supercell_entry.first;
00768     // Get supercell name(as well as module name) from pair
00769     const std::string &module_name = supercell_entry.second;
00770
00771     // Store module name for top-level creation
00772     module_names.push_back(module_name);
00773
00774     logger->info("Creating Module: '{}' from Cell '{}'", module_name, cell_name);
00775
00776     // Get the cell JSON object
00777     json cell_json;
00778     for (const auto &cell : lib_json_["cells"]) {
00779         if (cell["cell_name"].get<std::string>() == cell_name) {
00780             cell_json = cell;
00781             break;
00782         }
00783     }
00784
00785     // Get input/output pins from the cell JSON object
00786     std::vector<std::string> input_pins;
00787     std::vector<std::string> output_pins;
00788     std::stringstream input_pins_ss;
00789     std::stringstream output_pins_ss;
00790     if (cell_json.contains("input_pins")) {
00791         for (const auto &pin : cell_json["input_pins"]) {
00792             if (pin.contains("clock")) {
00793                 is_sequential = true;
00794             }
00795             std::string pin_name = pin["pin_name"].get<std::string>();
00796             input_pins.push_back(pin_name);
00797             input_pins_ss << pin_name << ", ";
00798
00799             // Store input pin name for top-level creation
00800             module_inputs[module_name].push_back(pin_name);
00801         }
00802     }
00803     if (cell_json.contains("output_pins")) {
00804         for (const auto &pin : cell_json["output_pins"]) {
00805             std::string pin_name = pin["pin_name"].get<std::string>();

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00806         output_pins.push_back(pin_name);
00807         output_pins_ss << pin_name << ", ";
00808
00809         // Store output pin name for top-level creation
00810         module_outputs[module_name].push_back(pin_name);
00811     }
00812 }
00813 logger_>debug("Input pins set: {}", input_pins_ss.str());
00814 logger_>debug("Output pins set: {}", output_pins_ss.str());
00815
00816 // Sequential cells will have only one instance
00817 // Combinational cells will have instances equal to chain length
00818 if (is_sequential) {
00819     instance_count = 1;
00820 } else {
00821     instance_count = chain_length;
00822 }
00823
00824 // Create ANSI port list
00825 std::string port_list_text = "(";
00826 // Add input ports
00827 bool isFirstPort = true;
00828 for (const auto &input_pin : input_pins) {
00829     if (!isFirstPort) {
00830         port_list_text += ", ";
00831     }
00832     port_list_text += "input " + input_pin;
00833     isFirstPort = false;
00834 }
00835 // Add output ports
00836 for (const auto &output_pin : output_pins) {
00837     if (!isFirstPort) {
00838         port_list_text += ", ";
00839     }
00840     port_list_text += "output " + output_pin;
00841     isFirstPort = false;
00842 }
00843 port_list_text += ")";
00844 logger_>debug("ANSI Port list: {}", port_list_text);
00845
00846 // Create full module header text
00847 std::string fullModuleText = "module " + module_name + port_list_text + ";\nendmodule";
00848 logger_>debug("Full module text: {}", fullModuleText);
00849
00850 // Generate the syntax tree for the module using slang library
00851 auto tree = slang::syntax::SyntaxTree::fromText(fullModuleText);
00852 if (tree) {
00853     // Add ports to the syntax tree
00854     ModuleRewriter rewriter(input_pins, output_pins, supercell_entry, instance_count, logger_);
00855
00856     tree = rewriter.transform(tree);
00857     // Revisit the syntax tree to check architecture
00858     // rewriter.visit(tree->root());
00859
00860     // Output the transformed syntax tree
00861     out << "timescale 1ns/10ps" << std::endl;
00862     out << slang::syntax::SyntaxPrinter::printFile(*tree) << std::endl << std::endl;
00863 } else {
00864     logger_>error("Failed to create syntax tree for module '{}'", module_name);
00865     continue;
00866 }
00867 }
00868
00869 out.close();
00870
00871 // Create the top-level module
00872 std::ofstream final_out(basename_ + ".v");
00873 if (!final_out.is_open()) {
00874     logger_>error("Could not open file '{}' for writing", basename_ + ".v");

```

```

00875     return;
00876 }
00877
00878 // Collect all port names separately for inputs and outputs
00879 std::vector<std::string> all_input_ports;
00880 std::vector<std::string> all_output_ports;
00881 std::stringstream top_instances;
00882
00883 // First pass: collect all port names
00884 for (const auto &module_name : module_names) {
00885     // Collect input port names
00886     for (const auto &pin : module_inputs[module_name]) {
00887         all_input_ports.push_back(module_name + "__" + pin);
00888     }
00889
00890     // Collect output port names
00891     for (const auto &pin : module_outputs[module_name]) {
00892         all_output_ports.push_back(module_name + "__" + pin);
00893     }
00894
00895     // Create instance
00896     std::string instance_name = "I_" + module_name.substr(0, module_name.find("__X"));
00897     for (size_t i = module_name.find("__X") + 3; i < module_name.length(); i++) {
00898         if (module_name[i] == '_' && module_name[i + 1] == '_') {
00899             instance_name += "__" + module_name.substr(i + 2);
00900             break;
00901         }
00902     }
00903
00904     // Generate port connections for this instance
00905     std::stringstream instance_ports;
00906     for (const auto &pin : module_inputs[module_name]) {
00907         instance_ports << "." << pin << "(" << module_name << "__" << pin << ")", ";";
00908     }
00909     for (const auto &pin : module_outputs[module_name]) {
00910         instance_ports << "." << pin << "(" << module_name << "__" << pin << ")", ";";
00911     }
00912
00913     // Remove last comma and space
00914     std::string instance_ports_str = instance_ports.str();
00915     if (!instance_ports_str.empty()) {
00916         instance_ports_str = instance_ports_str.substr(0, instance_ports_str.length() - 2);
00917     }
00918
00919     top_instances << " " << module_name << " " << instance_name << " (" << instance_ports_str
00920         << ");\n";
00921 }
00922
00923 // Now generate the port list with all inputs first, then all outputs
00924 std::stringstream top_ports;
00925
00926 // Add all input ports first
00927 for (size_t i = 0; i < all_input_ports.size(); ++i) {
00928     top_ports << all_input_ports[i];
00929     if (i < all_input_ports.size() - 1 || !all_output_ports.empty()) {
00930         top_ports << ", ";
00931     }
00932 }
00933
00934 // Then add all output ports
00935 for (size_t i = 0; i < all_output_ports.size(); ++i) {
00936     top_ports << all_output_ports[i];
00937     if (i < all_output_ports.size() - 1) {
00938         top_ports << ", ";
00939     }
00940 }
00941
00942 // Generate input and output declarations
00943 std::stringstream top_inputs;

```

```

00944 std::stringstream top_outputs;
00945
00946 // Generate input declarations
00947 for (const auto &port : all_input_ports) {
00948     top_inputs << "    input " << port << ";\n";
00949 }
00950
00951 // Generate output declarations
00952 for (const auto &port : all_output_ports) {
00953     top_outputs << "    output " << port << ";\n";
00954 }
00955
00956 // Write the top module
00957 final_out << "timescale 1ns/10ps" << std::endl;
00958 final_out << "module validate_top ( " << top_ports.str() << " );" << std::endl;
00959 final_out << std::endl; // Add newline before port declarations
00960 final_out << top_inputs.str();
00961 final_out << top_outputs.str();
00962 final_out << std::endl;
00963 final_out << top_instances.str();
00964 final_out << "endmodule" << std::endl << std::endl;
00965
00966 // Append the module definitions from the temporary file
00967 std::ifstream temp_in(temp_file);
00968 if (temp_in.is_open()) {
00969     final_out << temp_in.rdbuf();
00970     temp_in.close();
00971 } else {
00972     logger_>error("Could not open temp file '{}' for reading", temp_file);
00973 }
00974
00975 final_out.close();
00976
00977 // Remove temporary file
00978 // if (std::filesystem::exists(temp_file)) {
00979 //     std::filesystem::remove(temp_file);
00980 // }
00981
00982 logger_>info("Verilog creation complete in '{}'", basename_ + ".v");
00983 }
00984
00996 std::vector<std::string> LibFile::splitString(const std::string &s) {
00997     std::vector<std::string> tokens;
00998     std::stringstream ss(s);
00999     std::string token;
01000     while (ss >> token) {
01001         tokens.push_back(token);
01002     }
01003     return tokens;
01004 }
01005
01038 void LibFile::generateRCLines(std::ofstream &outFile, const std::string &netName, int instanceIndex,
01039                             bool isFinalStage) {
01040     // --- Default RC Values (as observed in the target SPICE) ---
01041     const double R1_INTERMEDIATE = 0.01;
01042     const double C1_INTERMEDIATE = 5.3e-16;
01043     const double R2_INTERMEDIATE = 0.01;
01044     const double R1_FINAL = 1e-2;
01045     const double C1_FINAL = 1e-18;
01046     const std::string CAP_GROUND = "COREGND1"; // Assumed ground for capacitors
01047
01048     std::string r1Name = "R1_" + std::to_string(instanceIndex);
01049     std::string c1Name = "C1_" + std::to_string(instanceIndex);
01050     std::string r2Name = "R2_" + std::to_string(instanceIndex);
01051     std::string node1 = netName + ":1"; // Intermediate node 1
01052     std::string node2 = netName + ":2"; // Intermediate node 2
01053
01054     // Set precision for scientific notation output
01055     outFile << std::scientific << std::setprecision(1);

```

```

01056
01057 if (!isFinalStage) {
01058     // Intermediate RCR structure (R1-C1-R2)
01059     outFile << r1Name << " " << node1 << " " << node2 << " " << R1_INTERMEDIATE << std::endl;
01060     outFile << c1Name << " " << node2 << " " << CAP_GROUND << " " << C1_INTERMEDIATE << std::endl;
01061     outFile << r2Name << " " << node2 << " " << netName << " " << R2_INTERMEDIATE << std::endl;
01062 } else {
01063     // Final RC structure (R1-C1 connected to the instance output node :1)
01064     // Note: The target SPICE connects R1 between node1 and the final port (netName).
01065     outFile << r1Name << " " << node1 << " " << netName << " " << R1_FINAL << std::endl;
01066     outFile << c1Name << " " << node1 << " " << CAP_GROUND << " " << C1_FINAL << std::endl;
01067 }
01068 // Reset precision/format to default
01069 outFile << std::defaultfloat << std::setprecision(6);
01070 }
01071
01120 bool LibFile::modifySpiceNetlist(const std::string &v2lvsSpiceFile, // Input is the v2lvs output
01121                                 const std::string &finalSpiceFile, // Output is the final file
01122                                 const std::string &targetGlobalLine) {
01123     logger_>info("Post-processing SPICE netlist: {} -> {}", v2lvsSpiceFile, finalSpiceFile);
01124
01125     std::ifstream inFile(v2lvsSpiceFile);
01126     // Write directly to the final output file, overwriting if it exists
01127     std::ofstream outFile(finalSpiceFile, std::ios::trunc);
01128
01129     if (!inFile) {
01130         logger_>error("Could not open input v2lvs SPICE file for modification: {}", v2lvsSpiceFile);
01131         return false;
01132     }
01133     if (!outFile) {
01134         logger_>error("Could not open final output SPICE file for writing: {}", finalSpiceFile);
01135         inFile.close(); // Close input file if output fails
01136         return false;
01137     }
01138
01139     // --- Add Metadata Comments ---
01140     outFile << "** Generated by " << APP_NAME << " v" << APP_VERSION << " from " << APP_AUTHOR;
01141     auto now = std::chrono::system_clock::now();
01142     std::time_t now_time_t = std::chrono::system_clock::to_time_t(now);
01143     // Use localtime_s on Windows, localtime_r on POSIX, or std::localtime (less safe)
01144     // std::tm now_tm;
01145     // localtime_s(&now_tm, &now_time_t); // Example for Windows
01146     std::tm *now_tm_ptr = std::localtime(&now_time_t); // Standard C++, potentially less thread-safe
01147     if (now_tm_ptr) {
01148         outFile << ". On: " << std::put_time(now_tm_ptr, "%c %Z") << " **\n" << std::endl;
01149     } else {
01150         outFile << ". Timestamp unavailable **\n" << std::endl;
01151     }
01152     // --- End Metadata ---
01153
01154     std::string line;
01155     bool includeFound = false;
01156     bool globalInserted = false;
01157     bool inSubckt = false;
01158     std::string previousInstanceLine = "";
01159     int instanceIndex = 0; // Counter for R/C naming within a subckt
01160
01161     while (std::getline(inFile, line)) {
01162         // Trim leading/trailing whitespace
01163         line.erase(0, line.find_first_not_of(" \t\n\r\f\v"));
01164         line.erase(line.find_last_not_of(" \t\n\r\f\v") + 1);
01165
01166         // Skip empty lines and v2lvs header comments
01167         if (line.empty() || line.find("$ Spice netlist generated by v2lvs") == 0 ||
01168             line.find("$ v2") == 0 || // Catch version line too
01169             line.find("*.BUSDELIMITER") == 0) {
01170             continue; // Skip these lines entirely
01171         }
01172

```

```

01173 // Write other comments directly after processing any pending instance
01174 if (line[0] == '*') {
01175     // Process previous instance before writing comment if needed
01176     if (!previousInstanceLine.empty()) {
01177         std::vector<std::string> tokens = this->splitString(previousInstanceLine);
01178         if (tokens.size() > 2 && (tokens[0][0] == 'X' || tokens[0][0] == 'x')) {
01179             std::string moduleName = tokens.back();
01180             std::string outputNet = tokens[tokens.size() - 2]; // Assumption
01181             bool isFinal = (outputNet == "C0" || outputNet == "S" || outputNet == "ZN");
01182             tokens[tokens.size() - 2] = outputNet + ":1"; // Modify output pin
01183
01184             // Reconstruct line with correct VDD/VSS order
01185             for (size_t i = 0; i < tokens.size() - 1; ++i) { // Up to module name
01186                 outFile << tokens[i] << " ";
01187             }
01188             outFile << "VDD VSS " << moduleName << std::endl; // Insert VDD VSS before module name
01189
01190             this->generateRCLines(outFile, outputNet, instanceIndex - 1, isFinal);
01191         } else {
01192             logger_>warn("Previous line stored but not recognized as instance: {}",
01193                 previousInstanceLine);
01194             outFile << previousInstanceLine << std::endl;
01195         }
01196         previousInstanceLine = ""; // Clear after processing
01197     }
01198     outFile << line << std::endl; // Write the comment line
01199     continue;
01200 }
01201
01202 // Handle .INCLUDE
01203 if (line.find(".INCLUDE") == 0) {
01204     outFile << line << std::endl;
01205     includeFound = true;
01206     continue; // Process next line to insert global
01207 }
01208
01209 // Insert the target global line after .INCLUDE
01210 if (includeFound && !globalInserted) {
01211     outFile << targetGlobalLine << std::endl << std::endl; // Add extra newline for spacing
01212     globalInserted = true;
01213     // Fall through to process the current line
01214 }
01215
01216 // Handle .SUBCKT
01217 if (line.find(".SUBCKT") == 0 || line.find(".subckt") == 0) { // Handle case-insensitivity
01218     if (!previousInstanceLine.empty()) {
01219         std::vector<std::string> tokens = this->splitString(previousInstanceLine);
01220         if (tokens.size() > 2 && (tokens[0][0] == 'X' || tokens[0][0] == 'x')) {
01221             std::string moduleName = tokens.back();
01222             std::string outputNet = tokens[tokens.size() - 2];
01223             bool isFinal = (outputNet == "C0" || outputNet == "S" || outputNet == "ZN");
01224             tokens[tokens.size() - 2] = outputNet + ":1";
01225
01226             for (size_t i = 0; i < tokens.size() - 1; ++i) {
01227                 outFile << tokens[i] << " ";
01228             }
01229             outFile << "VDD VSS " << moduleName << std::endl;
01230
01231             this->generateRCLines(outFile, outputNet, instanceIndex - 1, isFinal);
01232         } else {
01233             logger_>warn("Previous line stored but not recognized as instance: {}",
01234                 previousInstanceLine);
01235             outFile << previousInstanceLine << std::endl;
01236         }
01237         previousInstanceLine = "";
01238     }
01239
01240     inSubckt = true;
01241     instanceIndex = 0; // Reset R/C counter for new subcircuit

```

```

01242     outFile << line << " VDD VSS" << std::endl; // Add VDD VSS to ports
01243 }
01244 // Handle .ENDS
01245 else if (line.find(".ENDS") == 0 || line.find(".ends") == 0) { // Handle case-insensitivity
01246     inSubckt = false;
01247     // Process the last instance line *before* writing .ENDS
01248     if (!previousInstanceLine.empty()) {
01249         std::vector<std::string> tokens = this->splitString(previousInstanceLine);
01250         if (tokens.size() > 2 && (tokens[0][0] == 'X' || tokens[0][0] == 'x')) {
01251             std::string moduleName = tokens.back();
01252             std::string outputNet = tokens[tokens.size() - 2];
01253             bool isFinal = (outputNet == "C0" || outputNet == "S" || outputNet == "ZN");
01254             tokens[tokens.size() - 2] = outputNet + ":1";
01255
01256             for (size_t i = 0; i < tokens.size() - 1; ++i) {
01257                 outFile << tokens[i] << " ";
01258             }
01259             outFile << "VDD VSS " << moduleName << std::endl;
01260
01261             this->generateRCLines(outFile, outputNet, instanceIndex - 1, isFinal);
01262         } else {
01263             logger_>warn("Previous line stored but not recognized as instance: {}",
01264                 previousInstanceLine);
01265             outFile << previousInstanceLine << std::endl;
01266         }
01267         previousInstanceLine = ""; // Clear after processing
01268     }
01269     outFile << line << std::endl << std::endl; // Add extra newline after .ENDS
01270 }
01271 // Skip original .GLOBAL lines
01272 else if (line.find(".GLOBAL") == 0 || line.find(".global") == 0) {
01273     continue;
01274 }
01275 // Handle Instance lines
01276 else if (inSubckt && (line[0] == 'X' || line[0] == 'x')) {
01277     if (!previousInstanceLine.empty()) {
01278         std::vector<std::string> tokens = this->splitString(previousInstanceLine);
01279         if (tokens.size() > 2 && (tokens[0][0] == 'X' || tokens[0][0] == 'x')) {
01280             std::string moduleName = tokens.back();
01281             std::string outputNet = tokens[tokens.size() - 2]; // Assumption
01282             bool isFinal = (outputNet == "C0" || outputNet == "S" || outputNet == "ZN");
01283             tokens[tokens.size() - 2] = outputNet + ":1"; // Modify output pin
01284
01285             // Reconstruct line with correct VDD/VSS order
01286             for (size_t i = 0; i < tokens.size() - 1; ++i) { // Up to module name
01287                 outFile << tokens[i] << " ";
01288             }
01289             outFile << "VDD VSS " << moduleName << std::endl; // Insert VDD VSS before module name
01290
01291             this->generateRCLines(outFile, outputNet, instanceIndex - 1,
01292                 isFinal); // Use previous index
01293         } else {
01294             logger_>warn("Previous line stored but not recognized as instance: {}",
01295                 previousInstanceLine);
01296             outFile << previousInstanceLine << std::endl;
01297         }
01298     }
01299     previousInstanceLine = line;
01300     instanceIndex++;
01301 }
01302 // Handle other lines
01303 else {
01304     if (!previousInstanceLine.empty()) {
01305         std::vector<std::string> tokens = this->splitString(previousInstanceLine);
01306         if (tokens.size() > 2 && (tokens[0][0] == 'X' || tokens[0][0] == 'x')) {
01307             std::string moduleName = tokens.back();
01308             std::string outputNet = tokens[tokens.size() - 2];
01309             bool isFinal = (outputNet == "C0" || outputNet == "S" || outputNet == "ZN");
01310             tokens[tokens.size() - 2] = outputNet + ":1";

```

```

01311
01312         for (size_t i = 0; i < tokens.size() - 1; ++i) {
01313             outFile << tokens[i] << " ";
01314         }
01315         outFile << "VDD VSS " << moduleName << std::endl;
01316
01317         this->generateRCLines(outFile, outputNet, instanceIndex - 1, isFinal);
01318     } else {
01319         logger->warn("Previous line stored but not recognized as instance: {}",
01320             previousInstanceLine);
01321         outFile << previousInstanceLine << std::endl;
01322     }
01323     previousInstanceLine = ""; // Clear after processing
01324 }
01325 outFile << line << std::endl;
01326 }
01327 }
01328
01329 // Process the very last instance line
01330 if (!previousInstanceLine.empty()) {
01331     std::vector<std::string> tokens = this->splitString(previousInstanceLine);
01332     if (tokens.size() > 2 && (tokens[0][0] == 'X' || tokens[0][0] == 'x')) {
01333         std::string moduleName = tokens.back();
01334         std::string outputNet = tokens[tokens.size() - 2];
01335         bool isFinal = (outputNet == "CO" || outputNet == "S" || outputNet == "ZN");
01336         tokens[tokens.size() - 2] = outputNet + ":1";
01337
01338         for (size_t i = 0; i < tokens.size() - 1; ++i) {
01339             outFile << tokens[i] << " ";
01340         }
01341         outFile << "VDD VSS " << moduleName << std::endl;
01342
01343         this->generateRCLines(outFile, outputNet, instanceIndex - 1, isFinal);
01344     } else {
01345         logger->warn("Last stored line not recognized as instance: {}", previousInstanceLine);
01346         outFile << previousInstanceLine << std::endl;
01347     }
01348 }
01349
01350 inFile.close();
01351 outFile.close();
01352
01353 // No renaming needed, we wrote directly to the final file
01354 logger->info("SPICE netlist post-processing successful for: {}", finalSpiceFile);
01355 return true;
01356 }
01357
01386 void LibFile::spice(const int chain_length, const std::vector<std::string> &cell_names,
01387     const std::string &verilog_lib_file, const std::string &spice_lib_file) {
01388     logger->info("Creating SPICE for '{}'", filename_);
01389
01390     // 1. Generate the temporary Verilog file first
01391     this->verilog(chain_length, cell_names);
01392
01393     // Define input and output filenames
01394     std::string temp_verilog_file = basename_ + "_temp.v";
01395     std::string v2lvs_output_spice_file = basename_ + ".v2lvs.spi"; // v2lvs output
01396     std::string final_output_spice_file = basename_ + ".spi"; // Final processed output
01397
01398     // Check if temporary Verilog file exists
01399     std::ifstream check_v_in(temp_verilog_file);
01400     if (!check_v_in.is_open()) {
01401         logger->error("Temporary Verilog file {} not found or could not be opened. Verilog generation "
01402             "might have failed.",
01403             temp_verilog_file);
01404         return;
01405     }
01406     check_v_in.close(); // Close after checking
01407

```

```

01408 // 2. Check if V2LVS tool is available
01409 logger_>debug("Checking for v2lvs tool...");
01410 std::string v2lvs_path_cmd = "which v2lvs";
01411 std::string v2lvs_path_result;
01412 std::array<char, 128> buffer;
01413 std::unique_ptr<FILE, decltype(&pclose)> pipe(popen(v2lvs_path_cmd.c_str(), "r"), pclose);
01414 if (!pipe) {
01415     logger_>error("Failed to run command to find v2lvs: {}", v2lvs_path_cmd);
01416     // Do NOT delete temp_verilog_file here if popen fails
01417     return;
01418 }
01419 while (fgets(buffer.data(), buffer.size(), pipe.get()) != nullptr) {
01420     v2lvs_path_result += buffer.data();
01421 }
01422
01423 // Trim potential newline from the result
01424 if (!v2lvs_path_result.empty() && v2lvs_path_result.back() == '\n') {
01425     v2lvs_path_result.pop_back();
01426 }
01427
01428 if (v2lvs_path_result.empty()) {
01429     logger_>error(
01430         "V2LVS tool not found in PATH. Please install Calibre or ensure v2lvs is accessible.");
01431     // Do NOT delete temp_verilog_file
01432     return;
01433 } else {
01434     logger_>info("V2LVS tool found at: {}", v2lvs_path_result);
01435 }
01436
01437 // 3. Construct and Run V2LVS command
01438 logger_>info("Running V2LVS to generate initial SPICE netlist...");
01439 std::string v2lvs_cmd_full = "v2lvs"; // Start with the command name
01440 v2lvs_cmd_full += " -v " + temp_verilog_file; // Input Verilog
01441 v2lvs_cmd_full += " -o " + v2lvs_output_spice_file; // Output SPICE
01442 v2lvs_cmd_full += " -s0 VSS"; // Default ground net, generates .GLOBAL VSS
01443 v2lvs_cmd_full += " -s1 VDD"; // Default power net, generates .GLOBAL VDD
01444 v2lvs_cmd_full += " -i"; // Use positional pins (traditional SPICE)
01445 v2lvs_cmd_full += " -l " + verilog_lib_file; // Verilog library for pin order
01446 v2lvs_cmd_full += " -s " + spice_lib_file; // SPICE library to include
01447
01448 logger_>debug("Executing V2LVS command: {}", v2lvs_cmd_full);
01449 int ret = system(v2lvs_cmd_full.c_str());
01450
01451 // Do NOT delete temp_verilog_file
01452
01453 if (ret != 0) {
01454     logger_>error("V2LVS command failed with exit code: {}. Command: {}", ret, v2lvs_cmd_full);
01455     // Attempt to remove potentially incomplete v2lvs output SPICE file
01456     std::remove(v2lvs_output_spice_file.c_str());
01457     return;
01458 }
01459 logger_>info("V2LVS completed successfully, output at '{}'.", v2lvs_output_spice_file);
01460
01461 // 4. Post-process the generated SPICE file
01462 std::string target_global = ".global VSS GND COREGND1 VDD COREVDD1";
01463
01464 // Call the modification function, reading from .v2lvs.spi and writing to .spi
01465 if (!this->modifySpiceNetlist(v2lvs_output_spice_file, final_output_spice_file, target_global)) {
01466     logger_>error("Failed to post-process the generated SPICE netlist: {}",
01467         v2lvs_output_spice_file);
01468     // Keep both files for debugging if modification fails
01469     return;
01470 }
01471
01472 // 5. Final success message
01473 logger_>info("Intermediate v2lvs output kept in '{}'", v2lvs_output_spice_file);
01474 logger_>info("Temporary Verilog input kept in '{}'", temp_verilog_file);
01475 logger_>info("SPICE generation and post-processing complete. Final output in '{}'",
01476     final_output_spice_file);

```

```

01477 }
01478
01499 std::map<std::string, std::string> LibFile::logic(const std::string &cell_name) {
01500     std::map<std::string, std::string> logic_map = {};
01501
01502     logger_>info("Getting output pin logic function for '{}', filename_");
01503
01504     if (!std::filesystem::exists(jsonname_)) {
01505         logger_>info("JSON file not found. Parsing Liberty file first.");
01506         this->parse();
01507         this->writeJsonToFile();
01508     } else {
01509         // Read the JSON file into json object
01510         std::ifstream in(jsonname_);
01511         if (!in.is_open()) {
01512             logger_>error("Could not open file '{}' for reading", jsonname_);
01513             return logic_map;
01514         }
01515         try {
01516             lib_json_ = json::parse(in);
01517         } catch (const json::parse_error &e) {
01518             logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
01519             in.close();
01520             return logic_map;
01521         }
01522         in.close();
01523     }
01524
01525     // Check if the cell name exists in the JSON object
01526     for (const auto &cell : lib_json_["cells"]) {
01527         if (cell["cell_name"].get<std::string>() == cell_name) {
01528             logger_>debug("Found cell: '{}', cell_name");
01529
01530             // Tranverse all output pins
01531             if (cell.contains("output_pins")) {
01532                 for (const auto &pin : cell["output_pins"]) {
01533                     std::string pin_name = pin["pin_name"].get<std::string>();
01534                     // Check if the pin has a "timing_arcs" attribute
01535                     if (pin.contains("function")) {
01536                         std::string function = pin["function"].get<std::string>();
01537                         logic_map[pin_name] = function;
01538                         logger_>debug("Pin: '{}' Logic Function: '{}', pin_name, function);
01539                     } else {
01540                         logger_>warn("Pin: '{}' does not have a 'function' attribute", pin_name);
01541                     }
01542                 }
01543             }
01544         }
01545     }
01546     if (logic_map.empty()) {
01547         logger_>warn("No logic functions found for cell: '{}', cell_name);
01548     } else {
01549         logger_>info("Found {} Logic functions for cell: '{}', logic_map.size(), cell_name);
01550         // Print the logic functions
01551         for (const auto &pair : logic_map) {
01552             logger_>info("Pin -> Logic Function: {} -> {}", pair.first, pair.second);
01553         }
01554     }
01555     return logic_map;
01556 }

```

7.37 src/LibFileOperations.cpp File Reference

#include "LibFileOperations.hpp"

Include dependency graph for LibFileOperations.cpp:

Functions

- void `printInfo` ()
Sets up and configures the global logger and prints application information.
- void `parseLibFile` (const std::string &library_path, const std::string log_file_name)
Parses a library file and generates corresponding output.
- void `monoCheckLibFile` (const std::string &library_path, const std::string log_file_name, bool is↵_slew)
Performs monotonicity check on a library file.
- void `supercellLibFile` (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Creates supercell map structures from a Liberty library file.
- void `verilogLibFile` (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Generates Verilog files from a library file for specified cells.
- void `spiceLibFile` (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names, const std::string &verilog_lib_file, const std::string &spice_lib_file)
Generates a SPICE library file from a given library file, applying a specified chain length and cell names.
- void `compareLibFiles` (const std::string &ref_lib, const std::string &comp_lib, const double reltol, const double abstol, std::string &report_file_name)
Compares two library files and generates a detailed comparison report.
- void `funcLibFile` (const std::string &ref_file, const std::string &comp_file, const std::vector< std↵::string > &cell_names, std::string &report_file_name)
Performs a functional equivalence check between two files (Liberty or Verilog) for a given set of cells.

7.37.1 Function Documentation

7.37.1.1 `compareLibFiles()`

```
void compareLibFiles (
    const std::string & ref_lib,
    const std::string & comp_lib,
    const double reltol,
    const double abstol,
    std::string & report_file_name )
```

Compares two library files and generates a detailed comparison report.

This function compares a reference library file with another library file, performing validation checks based on specified tolerance parameters. The comparison results are written to a report file in markdown or text format.

Parameters

<i>ref_lib</i>	Path to the reference library file to use as the baseline
<i>comp_lib</i>	Path to the library file to compare against the reference
<i>reltol</i>	Relative tolerance for numerical comparisons (must be ≥ 0.0)
<i>abstol</i>	Absolute tolerance for numerical comparisons
<i>report_file_name</i>	[in,out] Name of the file to write the comparison report to. If empty, defaults to [comp_lib_basename].cmp.md. If provided but doesn't end with .txt or .md, .md will be appended.

Note

The function will log an error and return without comparing if reltol is invalid.

Log files will be created with the naming pattern [library_basename].cmp.log

The function uses the [LibraryComparator](#) class to perform the actual comparison.

Definition at line 249 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.37.1.2 funcLibFile()

```
void funcLibFile (
    const std::string & ref_file,
    const std::string & comp_file,
    const std::vector< std::string > & cell_names,
    std::string & report_file_name )
```

Performs a functional equivalence check between two files (Liberty or Verilog) for a given set of cells.

This function compares the logic functions of specified cells in two files, which can be either in Liberty (.lib) or Verilog (.v) format. It extracts the logic functions for each cell from both files, compares them, and generates a report summarizing the comparison results.

Parameters

<i>ref_file</i>	The path to the reference file (Liberty or Verilog).
<i>comp_file</i>	The path to the comparison file (Liberty or Verilog).
<i>cell_names</i>	A vector of cell names to be checked for functional equivalence.
<i>report_file_name</i>	A string to store the name of the report file. If empty, a default name is generated. If the provided name does not end with ".txt" or ".md", ".md" is appended.

The function first checks the file extensions to determine the file format. It then extracts the logic functions for each specified cell from both files. The logic functions are then compared, and a report is generated, which includes the comparison results for each cell. The report is written to the specified report file.

Note

- If no cell names are provided, the function logs an error and returns.
- If the reference or comparison file format is not supported (i.e., not .lib or .v), the function logs an error and returns.
- The report file is cleared before writing the comparison results.
- The function uses spdlog for logging information, warnings, and errors.
- The function utilizes the [LogicComparator](#) class to perform the logic comparison and generate the report.
- Memory allocated for [LibFile](#) objects is managed using raw pointers and must be manually deallocated to prevent memory leaks.

Definition at line 308 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.37.1.3 monoCheckLibFile()

```
void monoCheckLibFile (
    const std::string & library_path,
    const std::string log_file_name,
    bool is_slew )
```

Performs monotonicity check on a library file.

This function validates the monotonicity of timing data in a library file. It creates a log file to record the results of the check and handles any exceptions that occur during the process.

Parameters

<i>library_path</i>	Path to the library file to check
<i>log_file_name</i>	Name of the log file to create (optional). If empty, a default name will be generated from the library file name
<i>is_slew</i>	Flag indicating whether to check input slew monotonicity (true) or output load monotonicity (false)

Exceptions

<i>The</i>	function catches and logs any exceptions but does not rethrow them
------------	--

Definition at line 81 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.37.1.4 parseLibFile()

```
void parseLibFile (
    const std::string & library_path,
    const std::string log_file_name )
```

Parses a library file and generates corresponding output.

This function processes the given library file, parsing its contents and generating a JSON output. It also logs the parsing process to a specified log file or creates a default log file if none is provided.

Parameters

<i>library_path</i>	Path to the library file that needs to be parsed
<i>log_file_name</i>	Optional name for the log file. If empty, a default name is generated based on the library filename with ".parse.log" extension

Exceptions

<i>The</i>	function catches and logs any exceptions but doesn't propagate them
------------	---

Definition at line 49 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.37.1.5 printInfo()

```
void printInfo ( )
```

Sets up and configures the global logger and prints application information.

This function performs the following operations:

1. Creates a console sink for logging with level set to INFO

2. Creates a file sink for logging with level set to TRACE, saving to [APP_NAME].log
3. Configures a logger with both sinks and sets it as the default logger
4. Outputs basic application information:
 - Version and build timestamp
 - Author information
 - Log file location

Note

Uses spdlog library for logging functionality

Depends on APP_NAME, APP_VERSION, BUILD_TIMESTAMP, APP_AUTHOR, and APP_CONTACT macros

Definition at line 18 of file [LibFileOperations.cpp](#).

Here is the caller graph for this function:

7.37.1.6 spiceLibFile()

```
void spiceLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names,
    const std::string & verilog_lib_file,
    const std::string & spice_lib_file )
```

Generates a SPICE library file from a given library file, applying a specified chain length and cell names.

This function takes a library file path, a log file name, a chain length, a vector of cell names, a Verilog library file path, and a SPICE library file path as input. It initializes a [LibFile](#) object, validates the chain length, and then calls the spice method of the [LibFile](#) object to generate the SPICE library file. It logs the start and end of the SPICE generation process, as well as any errors that occur.

Parameters

<i>library_path</i>	The path to the input library file.
<i>log_file_name</i>	The name of the log file. If empty, a default log file name is generated based on the library file name.
<i>chain_length</i>	The chain length to use during SPICE generation. Must be ≥ 1 .
<i>cell_names</i>	A vector of cell names to include in the SPICE generation.
<i>verilog_lib_file</i>	The path to the Verilog library file.
<i>spice_lib_file</i>	The path to the output SPICE library file.

Definition at line 202 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.37.1.7 supercellLibFile()

```
void supercellLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
```

Creates supercell map structures from a Liberty library file.

This function reads a Liberty file, creates supercell structures based on the specified chain length and cell names, and logs the process to a file.

Parameters

<i>library_path</i>	Path to the Liberty file to process
<i>log_file_name</i>	Name of the log file (if empty, defaults to "[library_name].supercell.log")
<i>chain_length</i>	The length of chains to create (must be ≥ 1)
<i>cell_names</i>	Vector of cell names to process for supercell generation

Exceptions

May	pass through exceptions from the LibFile::supercell method
-----	--

Note

The function validates the chain length and logs all activities including errors that might occur during processing

Definition at line 116 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.37.1.8 verilogLibFile()

```
void verilogLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
```

```
int chain_length,
const std::vector< std::string > & cell_names )
```

Generates Verilog files from a library file for specified cells.

This function processes a library file and generates Verilog representation for the specified cell names with a given chain length. The operation results are logged to a specified or default log file.

Parameters

<i>library_path</i>	Path to the library file to process
<i>log_file_name</i>	Name for the log file (if empty, a default name will be generated)
<i>chain_length</i>	Number of cells to chain together, must be ≥ 1
<i>cell_names</i>	Vector of cell names to generate Verilog for

Exceptions

<i>Catches</i>	any exceptions from the verilog generation process and logs them
----------------	--

Definition at line 157 of file [LibFileOperations.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.38 LibFileOperations.cpp

[Go to the documentation of this file.](#)

```
00001 #include "LibFileOperations.hpp"
00002
00018 void printInfo() {
00019     // Set Global Logger
00020     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00021     console_sink->set_level(spdlog::level::info);
00022     std::string app_name = APP_NAME;
00023     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(app_name + ".log", true);
00024     file_sink->set_level(spdlog::level::trace);
00025
00026     std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00027     auto logger = std::make_shared<spdlog::logger>(APP_NAME, sinks.begin(), sinks.end());
00028     logger->set_level(spdlog::level::trace);
00029     spdlog::set_default_logger(logger);
00030
00031     spdlog::info("Version {}, Built: {}", APP_VERSION, BUILD_TIMESTAMP);
00032     spdlog::info("Author: {}, Email: {}", APP_AUTHOR, APP_CONTACT);
00033     spdlog::info("Global log file in: '{}'", app_name + ".log");
00034 }
00035
00049 void parseLibFile(const std::string &library_path, const std::string log_file_name) {
00050     std::string logname = log_file_name.empty()
00051         ? std::filesystem::path(library_path).stem().string() + ".parse.log"
00052         : log_file_name;
00053     LibFile libfile(library_path, logname);
00054 }
```

```

00055 libfile.logger_>info("Starting parse for file: '{}'. ...", library_path);
00056
00057 try {
00058     libfile.parse();
00059     libfile.writeJsonToFile();
00060     libfile.logger_>info("Successfully parsed file: '{}'", library_path);
00061 } catch (const std::exception &e) {
00062     libfile.logger_>error("Error parsing file '{}': {}", library_path, e.what());
00063 }
00064 }
00065
00081 void monoCheckLibFile(const std::string &library_path, const std::string log_file_name,
00082                      bool is_slew) {
00083     std::string logname = log_file_name.empty()
00084         ? std::filesystem::path(library_path).stem().string() + ".mono.log"
00085         : log_file_name;
00086     LibFile libfile(library_path, logname);
00087
00088     libfile.logger_>info("Starting monotonicity check for file: '{}', input slew: {}", library_path,
00089                         is_slew);
00090
00091     try {
00092         libfile.mono(is_slew);
00093         libfile.logger_>info("Successfully completed monotonicity check for file: '{}'", library_path);
00094     } catch (const std::exception &e) {
00095         libfile.logger_>error("Error during monotonicity check for file '{}': {}", library_path,
00096                             e.what());
00097     }
00098 }
00099
00116 void supercellLibFile(const std::string &library_path, const std::string &log_file_name,
00117                      int chain_length, const std::vector<std::string> &cell_names) {
00118     std::string logname = log_file_name.empty()
00119         ? std::filesystem::path(library_path).stem().string() + ".supercell.log"
00120         : log_file_name;
00121     LibFile libfile(library_path, logname);
00122
00123     // Check chain length validity
00124     if (chain_length < 1) {
00125         libfile.logger_>error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00126         return;
00127     }
00128     std::stringstream ss;
00129     for (const auto &cell : cell_names) {
00130         ss << cell << ", ";
00131     }
00132     libfile.logger_>info("Starting supercell generation for file: '{}', chain length: {}, cells: {}",
00133                         library_path, chain_length, ss.str());
00134     try {
00135         libfile.supercell(chain_length, cell_names);
00136         libfile.logger_>info("Successfully generated supercells for file: '{}'", library_path);
00137     } catch (const std::exception &e) {
00138         libfile.logger_>error("Error during supercell generation for file '{}': {}", library_path,
00139                             e.what());
00140     }
00141 }
00142
00157 void verilogLibFile(const std::string &library_path, const std::string &log_file_name,
00158                    int chain_length, const std::vector<std::string> &cell_names) {
00159     std::string logname = log_file_name.empty()
00160         ? std::filesystem::path(library_path).stem().string() + ".verilog.log"
00161         : log_file_name;
00162     LibFile libfile(library_path, logname);
00163
00164     // Check chain length validity
00165     if (chain_length < 1) {
00166         libfile.logger_>error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00167         return;
00168     }

```

```

00169     std::stringstream ss;
00170     for (const auto &cell : cell_names) {
00171         ss << cell << ", ";
00172     }
00173     libfile.logger_>info("Starting Verilog generation for file: '{}', chain length: {}, cells: {}",
00174         library_path, chain_length, ss.str());
00175     try {
00176         libfile.verilog(chain_length, cell_names);
00177         libfile.logger_>info("Successfully generated Verilog for file: '{}', library_path);
00178     } catch (const std::exception &e) {
00179         libfile.logger_>error("Error during Verilog generation for file '{}': {}", library_path,
00180             e.what());
00181     }
00182 }
00183
00202 void spiceLibFile(const std::string &library_path, const std::string &log_file_name,
00203     int chain_length, const std::vector<std::string> &cell_names,
00204     const std::string &verilog_lib_file, const std::string &spice_lib_file) {
00205     std::string logname = log_file_name.empty()
00206         ? std::filesystem::path(library_path).stem().string() + ".spice.log"
00207         : log_file_name;
00208     LibFile libfile(library_path, logname);
00209
00210     // Check chain length validity
00211     if (chain_length < 1) {
00212         libfile.logger_>error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00213         return;
00214     }
00215     std::stringstream ss;
00216     for (const auto &cell : cell_names) {
00217         ss << cell << ", ";
00218     }
00219     libfile.logger_>info("Starting SPICE generation for file: '{}', chain length: {}, cells: {}",
00220         library_path, chain_length, ss.str());
00221     try {
00222         libfile.spice(chain_length, cell_names, verilog_lib_file, spice_lib_file);
00223         libfile.logger_>info("Successfully generated SPICE for file: '{}', library_path);
00224     } catch (const std::exception &e) {
00225         libfile.logger_>error("Error during SPICE generation for file '{}': {}", library_path,
00226             e.what());
00227     }
00228 }
00229
00249 void compareLibFiles(const std::string &ref_lib, const std::string &comp_lib, const double reltol,
00250     const double abstol, std::string &report_file_name) {
00251     std::string ref_logname = std::filesystem::path(ref_lib).stem().string() + ".cmp.log";
00252     std::string comp_logname = std::filesystem::path(comp_lib).stem().string() + ".cmp.log";
00253     LibFile ref_libfile(ref_lib, ref_logname), comp_libfile(comp_lib, comp_logname);
00254
00255     // Check relative tolerance validity
00256     if (reltol < 0.0) {
00257         spdlog::error("Invalid relative tolerance: {}. Relative tolerance must be >= 0.0.", reltol);
00258         return;
00259     }
00260
00261     // Check the report file name
00262     if (report_file_name.empty()) {
00263         report_file_name = comp_libfile.basename_ + ".cmp.md";
00264     } else if (std::filesystem::path(report_file_name).extension() != ".txt" &&
00265         std::filesystem::path(report_file_name).extension() != ".md") {
00266         // report file name not end in .txt or .md, warn user and append .md
00267         report_file_name = report_file_name + ".md";
00268         spdlog::warn("Report file name does not end in .txt or .md. Report will be written to '{}'",
00269             report_file_name);
00270     }
00271
00272     LibraryComparator comparator(ref_libfile, comp_libfile, reltol, abstol);
00273     comparator.generateReport(report_file_name);
00274     spdlog::info("Comparison completed. Report written to: '{}', report_file_name);

```

```

00275 }
00276
00308 void funcLibFile(const std::string &ref_file, const std::string &comp_file,
00309                 const std::vector<std::string> &cell_names, std::string &report_file_name) {
00310     // Check if the files are Liberty or Verilog
00311     std::string ref_ext = std::filesystem::path(ref_file).extension();
00312     std::string comp_ext = std::filesystem::path(comp_file).extension();
00313     std::string ref_logname = std::filesystem::path(ref_file).stem().string() + ".cmp.log";
00314     std::string comp_logname = std::filesystem::path(comp_file).stem().string() + ".cmp.log";
00315
00316     // Pre cell names check
00317     if (cell_names.empty()) {
00318         spdlog::error("No cell names provided for functional equivalence check.");
00319         return;
00320     }
00321
00322     // Check the report file name
00323     if (report_file_name.empty()) {
00324         report_file_name = std::filesystem::path(comp_file).stem().string() + ".logic.cmp.md";
00325     } else if (std::filesystem::path(report_file_name).extension() != ".txt" &&
00326               std::filesystem::path(report_file_name).extension() != ".md") {
00327         // report file name not end in .txt or .md, warn user and append .md
00328         report_file_name = report_file_name + ".md";
00329         spdlog::warn("Report file name does not end in .txt or .md. Report will be written to '{}'",
00330                     report_file_name);
00331     }
00332
00333     // Clear the report file
00334     std::ofstream report_file(report_file_name);
00335     if (!report_file.is_open()) {
00336         spdlog::error("Failed to open report file: '{}'", report_file_name);
00337         return;
00338     }
00339     report_file.close();
00340     spdlog::info("Report file cleared: '{}'", report_file_name);
00341
00342     // Declare LibFile objects as pointers, initialized to nullptr
00343     LibFile *ref_libfile = nullptr;
00344     LibFile *comp_libfile = nullptr;
00345
00346     // Check the reference file extension
00347     if (ref_ext == ".v") {
00348         spdlog::info("Reference file in Verilog format.");
00349     } else if (ref_ext == ".lib") {
00350         spdlog::info("Reference file in Liberty format.");
00351         ref_libfile = new LibFile(ref_file, ref_logname);
00352     } else {
00353         spdlog::error("Unsupported reference file format: '{}'", ref_ext);
00354         return;
00355     }
00356
00357     // Check the comparison file extension
00358     if (comp_ext == ".v") {
00359         spdlog::info("Comparison file in Verilog format.");
00360     } else if (comp_ext == ".lib") {
00361         spdlog::info("Comparison file in Liberty format.");
00362         comp_libfile = new LibFile(comp_file, comp_logname);
00363     } else {
00364         spdlog::error("Unsupported comparison file format: '{}'", comp_ext);
00365         return;
00366     }
00367
00368     // Tranverse the cell names and check functional equivalence
00369     for (const auto &cell : cell_names) {
00370         spdlog::info("Checking functional equivalence for cell: '{}'", cell);
00371         std::map<std::string, std::string> ref_outpin_map;
00372         std::map<std::string, std::string> comp_outpin_map;
00373
00374         if (ref_ext == ".v") {
00375             ref_outpin_map = extractLogicFromVerilog(ref_file, cell);

```

```

00375     } else if (ref_ext == ".lib") {
00376         ref_outpin_map = ref_libfile->logic(cell);
00377     }
00378
00379     if (comp_ext == ".v") {
00380         // getAST(comp_file, cell);
00381         // extractAndPrintNetlistInfo(comp_file, cell);
00382         comp_outpin_map = extractLogicFromVerilog(comp_file, cell);
00383     } else if (comp_ext == ".lib") {
00384         comp_outpin_map = comp_libfile->logic(cell);
00385     }
00386
00387     spdlog::info("Logic function expressions collected for cell: '{}'", cell);
00388     spdlog::info("Starting logic comparison ...");
00389
00390     LogicComparator comparator(ref_outpin_map, comp_outpin_map, cell);
00391     // Easter egg
00392     if (cell == "easteregg") {
00393         spdlog::info("Easter egg found! Running ExprTk Eample 07...");
00394         comparator.logic();
00395         return;
00396     }
00397
00398     // // Test for preprocessing
00399     // std::string ref_expr;
00400     // std::string comp_expr;
00401     // for (const auto &pin : ref_outpin_map) {
00402     //     spdlog::info("Pin -> Expression: {} -> {}", pin.first, pin.second);
00403     //     ref_expr = comparator.preprocessExpression(pin.second);
00404     // }
00405     // for (const auto &pin : comp_outpin_map) {
00406     //     spdlog::info("Pin => Expression: {} => {}", pin.first, pin.second);
00407     //     comp_expr = comparator.preprocessExpression(pin.second);
00408     // }
00409     // std::vector<std::string> sorted_vars;
00410     // comparator.extractVariables(ref_expr, comp_expr, sorted_vars);
00411     // struct PinComparisonResult pin_comparison_result;
00412     // // comp_expr = "not ((not(A1) and not(A2)) or B)";
00413     // comparator.compareSingleExpressionPair(ref_expr, comp_expr, sorted_vars,
00414     //                                         pin_comparison_result);
00415
00416     // comparator.compareCellLogic();
00417     comparator.compareCellLogic();
00418     comparator.generateReport(report_file_name);
00419
00420     spdlog::info("Functional equivalence check completed for cell: '{}'", cell);
00421 }
00422 spdlog::info("Functional equivalence check completed for all cells.");
00423 spdlog::info("Report written to: '{}'", report_file_name);
00424 }

```

7.39 src/LibGroup.cpp File Reference

```
#include "LibGroup.hpp"
```

Include dependency graph for LibGroup.cpp:

7.40 LibGroup.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LibGroup.hpp"
00002
00003 LibGroup::LibGroup(si2drGroupIdT group, si2drErrorT &err) : group_(group), err_(err) {}
00004
00005 LibGroup::~LibGroup() {}
00006
00007 std::string LibGroup::getName() {
00008     si2drNamesIdT names = si2drGroupGetNames(group_, &err_);
00009     si2drStringT name = si2drIterNextName(names, &err_);
00010     si2drIterQuit(names, &err_);
00011     return name ? std::string(name) : std::string();
00012 }
00013
00014 std::string LibGroup::getType() {
00015     si2drStringT type = si2drGroupGetGroupType(group_, &err_);
00016     return type ? std::string(type) : std::string();
00017 }
00018
00019 si2drAttrsIdT LibGroup::getAttrs() { return si2drGroupGetAttrs(group_, &err_); }
00020
00021 si2drGroupsIdT LibGroup::getGroups() { return si2drGroupGetGroups(group_, &err_); }

```

7.41 src/LibraryComparator.cpp File Reference

#include "LibraryComparator.hpp"

Include dependency graph for LibraryComparator.cpp:

7.42 LibraryComparator.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LibraryComparator.hpp"
00002
00003 LibraryComparator::LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol,
00004                                     double abstol)
00005     : reltol_(reltol), abstol_(abstol) {
00006     std::string ref_json_name = ref_libfile.basename_ + ".json";
00007     if (!std::filesystem::exists(ref_json_name)) {
00008         spdlog::info("Reference JSON file {} not found. Parsing first.", ref_json_name);
00009         ref_libfile.parse();
00010         ref_libfile.writeJsonToFile();
00011         ref_json_ = ref_libfile.lib_json_;
00012     } else {
00013         // Read the JSON file into json object
00014         std::ifstream ref_in(ref_json_name);
00015         if (!ref_in.is_open()) {
00016             spdlog::error("Could not open file '{}' for reading", ref_json_name);
00017             return;
00018         }
00019         try {
00020             ref_json_ = json::parse(ref_in);
00021         } catch (const json::parse_error &e) {
00022             spdlog::error("Error parsing reference JSON file '{}': {}", ref_json_name, e.what());
00023             return;
00024         }
00025     }
00026     std::string comp_json_name = comp_libfile.basename_ + ".json";
00027     if (!std::filesystem::exists(comp_json_name)) {
00028         spdlog::info("Comparison JSON file {} not found. Parsing first.", comp_json_name);

```

```

00047     comp_libfile.parse();
00048     comp_libfile.writeJsonToFile();
00049     comp_json_ = comp_libfile.lib_json_;
00050 } else {
00051     // Read the JSON file into json object
00052     std::ifstream comp_in(comp_json_name);
00053     if (!comp_in.is_open()) {
00054         spdlog::error("Could not open file '{}' for reading", comp_json_name);
00055         return;
00056     }
00057     try {
00058         comp_json_ = json::parse(comp_in);
00059     } catch (const json::parse_error &e) {
00060         spdlog::error("Error parsing comparison JSON file '{}': {}", comp_json_name, e.what());
00061         return;
00062     }
00063 }
00064
00065 ref_lib_path_ = ref_libfile.filepath_;
00066 comp_lib_path_ = comp_libfile.filepath_;
00067 spdlog::info("Successfully loaded JSON files for comparison");
00068 }
00069
00098 void LibraryComparator::compareLut(const std::string &cell_name, const std::string &pin_name,
00099                                     const std::string &timing_type, const std::string &related_pin,
00100                                     const std::string &arc_name, const json &ref_lut,
00101                                     const json &comp_lut, Table &table) {
00102     spdlog::info("Comparing LUT of '{}' ...", arc_name);
00103     std::vector<double> ref_index_1 = ref_lut["index_1"].get<std::vector<double>>();
00104     std::vector<double> ref_index_2 = ref_lut["index_2"].get<std::vector<double>>();
00105     std::vector<double> comp_index_1 = comp_lut["index_1"].get<std::vector<double>>();
00106     std::vector<double> comp_index_2 = comp_lut["index_2"].get<std::vector<double>>();
00107     if (ref_index_1 != comp_index_1 || ref_index_2 != comp_index_2) {
00108         spdlog::warn("Mismatch in LUT indices for '{}' in cell: '{}', pin: '{}', related_pin: '{}', "
00109                     "timing_type: '{}'",
00110                     arc_name, cell_name, pin_name, related_pin, timing_type);
00111         return;
00112     } else {
00113         spdlog::debug("LUT indices match for '{}'", arc_name);
00114
00115         std::vector<std::vector<double>> comp_value_matrix;
00116         // Generate a matrix of values from the JSON array for comparison library
00117         for (const auto &row_val : comp_lut["values"]) {
00118             std::vector<double> row_data;
00119             // Check if the row value is an array
00120             if (!row_val.is_array()) {
00121                 spdlog::error(
00122                     "Invalid LUT format: '{}' values should be an array of arrays in comp_lib '{}', "
00123                     "cell '{}', pin '{}', "
00124                     "related_pin '{}', timing_type '{}'",
00125                     arc_name, this->comp_lib_path_.string(), cell_name, pin_name, related_pin, timing_type);
00126                 return;
00127             }
00128             // Check if non-numeric values
00129             for (const auto &val : row_val) {
00130                 if (val.is_number()) {
00131                     row_data.push_back(val.get<double>());
00132                 } else {
00133                     spdlog::error("Non-numeric value found in '{}'.values, skipping value: {} in comp_lib "
00134                                 "'{}', cell '{}', "
00135                                 "pin '{}', related_pin '{}', timing_type '{}'",
00136                                 arc_name, val.dump(), this->comp_lib_path_.string(), cell_name, pin_name,
00137                                 related_pin, timing_type);
00138                     return;
00139                 }
00140             }
00141             comp_value_matrix.push_back(row_data);
00142         }
00143         // Generate a matrix of values from the JSON array for reference library

```

```

00144     std::vector<std::vector<double>> ref_value_matrix;
00145     for (const auto &row_val : ref_lut["values"]) {
00146         std::vector<double> row_data;
00147         // Check if the row value is an array
00148         if (!row_val.is_array()) {
00149             spdlog::error(
00150                 "Invalid LUT format: '{}' values should be an array of arrays in ref_lib '{}', "
00151                 "cell '{}', pin '{}', "
00152                 "related_pin '{}', timing_type '{}'",
00153                 arc_name, this->ref_lib_path_.string(), cell_name, pin_name, related_pin, timing_type);
00154             return;
00155         }
00156         // Check if non-numeric values
00157         for (const auto &val : row_val) {
00158             if (val.is_number()) {
00159                 row_data.push_back(val.get<double>());
00160             } else {
00161                 spdlog::error("Non-numeric value found in '{}'.values, skipping value: {} in ref_lib "
00162                     "'{}', cell '{}', "
00163                     "pin '{}', related_pin '{}', timing_type '{}'",
00164                     arc_name, val.dump(), this->ref_lib_path_.string(), cell_name, pin_name,
00165                     related_pin, timing_type);
00166                 return;
00167             }
00168         }
00169         ref_value_matrix.push_back(row_data);
00170     }
00171     // Compare the matrix for relative tolerance
00172     if (!comp_value_matrix.empty() && !comp_value_matrix[0].empty()) {
00173         for (size_t i = 0; i < comp_value_matrix.size(); ++i) {
00174             for (size_t j = 0; j < comp_value_matrix[i].size(); ++j) {
00175                 double ref_val = ref_value_matrix[i][j];
00176                 double comp_val = comp_value_matrix[i][j];
00177                 if (std::abs(ref_val - comp_val) > reltol_ * std::abs(ref_val) &&
00178                     std::abs(ref_val - comp_val) > abstol_) {
00179                     spdlog::debug("LUT value mismatch for '{}', index_1: {}, index_2: {}", arc_name,
00180                         ref_index_1[i], ref_index_2[j]);
00181                     table.add_row({related_pin + "->" + pin_name, std::to_string(ref_val),
00182                         std::to_string(comp_val), std::to_string(comp_val - ref_val),
00183                         std::to_string((comp_val - ref_val) / ref_val * 100), timing_type,
00184                         arc_name, std::to_string(i + 1), std::to_string(ref_index_1[i]),
00185                         std::to_string(j + 1), std::to_string(ref_index_2[j]), "<"});
00186                     spdlog::debug("table size: {}", table.size());
00187                 } else {
00188                     spdlog::debug("LUT value match for '{}', index_1: {}, index_2: {}", arc_name,
00189                         ref_index_1[i], ref_index_2[j]);
00190                 }
00191             }
00192         }
00193     } else {
00194         spdlog::error(
00195             "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "
00196             "timing_type: '{}', arc: '{}'",
00197             cell_name, pin_name, related_pin, timing_type, arc_name);
00198     }
00199 }
00200 }
00201
00220 void LibraryComparator::compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00221     const std::string &timing_type, const json &ref_timing_arc,
00222     const json &comp_timing_arc, Table &table) {
00223     spdlog::info("Comparing timing type '{}' ...", timing_type);
00224
00225     std::string related_pin = ref_timing_arc["related_pin"].get<std::string>();
00226     spdlog::debug("Related pin: '{}'", related_pin);
00227
00228     std::vector<std::string> arc_names = {"cell_rise", "cell_fall", "rise_transition",
00229         "fall_transition"};
00230     for (auto arc_name : arc_names) {

```

```

00231     if (comp_timing_arc.contains(arc_name)) {
00232         spdlog::debug("Comparing timing arc: '{}'", arc_name);
00233         // Look for the arc in the reference JSON
00234         if (ref_timing_arc.contains(arc_name)) {
00235             // Found this arc
00236             spdlog::debug("Found timing arc: '{}'", arc_name);
00237             compareLut(cell_name, pin_name, timing_type, related_pin, arc_name,
00238                 ref_timing_arc[arc_name], comp_timing_arc[arc_name], table);
00239         } else {
00240             // arc not found in reference JSON
00241             spdlog::warn("Timing arc: '{}'" not found in reference JSON", arc_name);
00242         }
00243     }
00244 }
00245 }
00246
00261 void LibraryComparator::comparePin(const std::string &cell_name, const std::string &pin_name,
00262     const json &ref_pin, const json &comp_pin, Table &table) {
00263     spdlog::info("Comparing pin '{}'" ..., pin_name);
00264
00265     if (comp_pin.contains("timing_arcs")) {
00266         for (const auto &comp_arc : comp_pin["timing_arcs"]) {
00267             std::string timing_type = comp_arc["timing_type"].get<std::string>();
00268             spdlog::debug("Comparing timing type: '{}'", timing_type);
00269             // Look for the timing type in the reference JSON
00270             auto ref_arc_it = std::find_if(
00271                 ref_pin["timing_arcs"].begin(), ref_pin["timing_arcs"].end(),
00272                 [&timing_type](const json &ref_arc) { return ref_arc["timing_type"] == timing_type; });
00273             if (ref_arc_it != ref_pin["timing_arcs"].end()) {
00274                 // Found this timing type
00275                 spdlog::debug("Found timing type: '{}'", timing_type);
00276                 compareTimingArc(cell_name, pin_name, timing_type, *ref_arc_it, comp_arc, table);
00277             } else {
00278                 // timing arc not found in reference JSON
00279                 spdlog::warn("Timing arc: '{}'" not found in reference JSON", timing_type);
00280             }
00281         }
00282     } else {
00283         spdlog::info("No timing arcs found for pin: '{}'", comp_pin["pin_name"].get<std::string>());
00284     }
00285 }
00286
00301 void LibraryComparator::compareCell(const std::string &cell_name, const json &ref_cell,
00302     const json &comp_cell, Table &table) {
00303     spdlog::info("Comparing cell '{}'" ..., cell_name);
00304
00305     if (comp_cell.contains("output_pins")) {
00306         for (const auto &comp_pin : comp_cell["output_pins"]) {
00307             std::string pin_name = comp_pin["pin_name"].get<std::string>();
00308             spdlog::debug("Comparing pin: '{}'", pin_name);
00309             // Look for the pin in the reference JSON
00310             auto ref_pin_it = std::find_if(
00311                 ref_cell["output_pins"].begin(), ref_cell["output_pins"].end(),
00312                 [&pin_name](const json &ref_pin) { return ref_pin["pin_name"] == pin_name; });
00313             if (ref_pin_it != ref_cell["output_pins"].end()) {
00314                 // Found this pin
00315                 spdlog::debug("Found pin: '{}'", pin_name);
00316                 comparePin(cell_name, pin_name, *ref_pin_it, comp_pin, table);
00317             } else {
00318                 // pin not found in reference JSON
00319                 spdlog::warn("Pin: '{}'" not found in reference JSON", pin_name);
00320             }
00321         }
00322     } else {
00323         spdlog::info("No output pins found for cell: '{}'", cell_name);
00324     }
00325 }
00326
00343 void LibraryComparator::generateReport(const std::string &output_file) {

```

```

00344 std::ofstream outfile(output_file);
00345 outfile << "# LIBRARY comparison\n" << std::endl;
00346 outfile << "**Reference library: " << ref_lib_path_ << "**\n" << std::endl;
00347 outfile << "**Comparison library: " << comp_lib_path_ << "**\n" << std::endl;
00348 outfile << "**Absolute tolerance: " << abstol_ << "**\n" << std::endl;
00349 outfile << "**Relative tolerance: " << reltol_ << "**\n" << std::endl;
00350 outfile << "**Performed by " << APP_NAME << " v" << APP_VERSION << " from " << APP_AUTHOR;
00351 auto now = std::chrono::system_clock::now();
00352 std::time_t now_time_t = std::chrono::system_clock::to_time_t(now);
00353 std::tm *now_tm = std::localtime(&now_time_t);
00354 outfile << ". on: " << std::put_time(now_tm, "%c") << "**\n" << std::endl;
00355 outfile << "> Legend: < outlier, * scaled, ! indices switched, ^ slews extrapolated, - loads "
00356         "extrapolated,\n";
00357 outfile << "> \n";
00358 outfile << "> Legend: + padding added, /0 divide by zero, / slews interpolated, # loads "
00359         "interpolated\n";
00360 outfile << "> \n";
00361 outfile << "> Legend: << value is less but unknown, >> value is more but unknown\n\n";
00362
00363 spdlog::info("Starting comparison report generation ...");
00364 for (const auto &comp_cell : comp_json_["cells"]) {
00365     std::string cell_name = comp_cell["cell_name"].get<std::string>();
00366     spdlog::debug("Comparing cell: '{}'", cell_name);
00367     // Look for the cell in the reference JSON
00368     auto ref_cell_it = std::find_if(
00369         ref_json_["cells"].begin(), ref_json_["cells"].end(),
00370         [&cell_name](const json &ref_cell) { return ref_cell["cell_name"] == cell_name; });
00371     if (ref_cell_it != ref_json_["cells"].end()) {
00372         // Found this cell
00373         spdlog::debug("Found cell: '{}'", cell_name);
00374         Table table;
00375         table.add_row({"Pin Name", "Reference", "Comparison", "Diff", "Diff %", "Type", "Arc Name",
00376             "Row #", "Index_1", "Col #", "Index_2", "Note"});
00377         // Header formatting
00378         for (size_t i = 0; i < table[0].size(); ++i) {
00379             table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00380         }
00381         compareCell(cell_name, *ref_cell_it, comp_cell, table);
00382
00383         if (table.size() > 1) {
00384             // Data starts from row 1, row 0 is header
00385             size_t failed_count = table.size() - 1;
00386             double sum_diff = 0.0;
00387             double sum_diff_percent = 0.0;
00388             size_t outlier_count = 0;
00389             double max_diff = 0;
00390             double max_diff_percent = 0;
00391             std::vector<double> diff_values; // Store diff values for std deviation calculation
00392
00393             // Index of columns
00394             const int diff_index = 3;
00395             const int diff_percent_index = 4;
00396             const int note_index = 11;
00397
00398             for (size_t i = 1; i < table.size(); ++i) {
00399                 try {
00400                     double diff = std::stod(table[i][diff_index].get_text());
00401                     double diff_percent = std::stod(table[i][diff_percent_index].get_text());
00402
00403                     sum_diff += diff;
00404                     sum_diff_percent += diff_percent;
00405                     diff_values.push_back(diff_percent);
00406
00407                     if (table[i][note_index].get_text() == "<") {
00408                         outlier_count++;
00409                     }
00410                     // Change to absolute value for max diff
00411                     if (std::abs(diff) > std::abs(max_diff)) {
00412                         max_diff = diff;

```

```

00413         max_diff_percent = diff_percent;
00414     }
00415
00416     } catch (const std::invalid_argument &e) {
00417         spdlog::error("Could not convert value to double: {}", e.what());
00418         continue; // Skip this row if there's an error
00419     }
00420 }
00421
00422 double avg_diff = sum_diff / failed_count;
00423 double avg_diff_percent = sum_diff_percent / failed_count;
00424
00425 outfile << "## " << cell_name << "\n\n";
00426 outfile << "Delay Comparison in ns\n\n";
00427 if (output_file.substr(output_file.size() - 3) == ".md") {
00428     MarkdownExporter exporter;
00429     auto markdown = exporter.dump(table);
00430     outfile << markdown << std::endl;
00431 } else {
00432     outfile << table << std::endl;
00433 }
00434 outfile << std::endl;
00435 std::cout << table << std::endl;
00436 outfile << cell_name << " delay SUMMARY (abstol: " << abstol_ << ", reltol: " << reltol_
00437     << ") \n";
00438
00439 // Create summary table
00440 Table summary_table;
00441 summary_table.add_row({"Cell Name", "Data Type", "Failed Count", "Avg Diff", "Avg Diff%",
00442     "Max Diff", "Max Diff%", "Outliers"});
00443 summary_table.add_row({cell_name, "delay(ns)", std::to_string(failed_count),
00444     std::to_string(avg_diff), std::to_string(avg_diff_percent) + "%",
00445     std::to_string(max_diff), std::to_string(max_diff_percent) + "%",
00446     std::to_string(outlier_count)});
00447
00448 // Output summary table
00449 // Use std::filesystem to reliably get the extension
00450 std::filesystem::path file_path(output_file);
00451 if (file_path.has_extension() && file_path.extension() == ".md") {
00452     MarkdownExporter exporter;
00453     auto markdown = exporter.dump(summary_table);
00454     outfile << markdown << std::endl;
00455 } else {
00456     outfile << summary_table << std::endl;
00457 }
00458 outfile << "Worst delay outlier: Max Abs: " << max_diff << ", Max Rel: " << max_diff_percent
00459     << "%, Outliers: " << outlier_count << "\n\n";
00460 }
00461 } else {
00462     // cell not found in reference JSON
00463     spdlog::warn("Cell: '{} ' not found in reference JSON", cell_name);
00464 }
00465 }
00466
00467 outfile.close();
00468 }

```

7.43 src/LogicComparator.cpp File Reference

```
#include "LogicComparator.hpp"
```

Include dependency graph for LogicComparator.cpp:

Functions

- `bool isIdentifier (const std::string &token)`
Checks if a given string is a valid identifier.
- `bool isOperator (const std::string &token)`
Checks if a given token is a logical operator.

7.43.1 Function Documentation

7.43.1.1 isIdentifier()

```
bool isIdentifier (  
    const std::string & token )
```

Checks if a given string is a valid identifier.

A valid identifier must:

- Not be empty.
- Consist of alphanumeric characters and underscores.
- Start with an uppercase alphabetic character.
- Contain at least one alphabetic character.

Parameters

<i>token</i>	The string to check.
--------------	----------------------

Returns

True if the string is a valid identifier, false otherwise.

Definition at line 74 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:

7.43.1.2 isOperator()

```
bool isOperator (  
    const std::string & token )
```

Checks if a given token is a logical operator.

This function determines whether the input string token is one of the supported logical operators: "and", "or", "xor", or "not".

Parameters

<i>token</i>	The string to check.
--------------	----------------------

Returns

True if the token is a logical operator, false otherwise.

Definition at line 99 of file [LogicComparator.cpp](#).

Here is the caller graph for this function:

7.44 LogicComparator.cpp

[Go to the documentation of this file.](#)

```
00001 #include "LogicComparator.hpp"
00002
00003 LogicComparator::LogicComparator(const std::map<std::string, std::string> &ref_outpin_map,
00004                                 const std::map<std::string, std::string> &comp_outpin_map,
00005                                 const std::string &cell_name)
00006     : ref_outpin_map(ref_outpin_map), comp_outpin_map(comp_outpin_map), cell_name(cell_name) {}
00024 void LogicComparator::logic() {
00025     typedef exprtk::symbol_table<double> symbol_table_t;
00026     typedef exprtk::expression<double> expression_t;
00027     typedef exprtk::parser<double> parser_t;
00028
00029     const std::string expression_string = "not(A and B) or C";
00030
00031     symbol_table_t symbol_table;
00032     symbol_table.create_variable("A");
00033     symbol_table.create_variable("B");
00034     symbol_table.create_variable("C");
00035
00036     expression_t expression;
00037     expression.register_symbol_table(symbol_table);
00038
00039     parser_t parser;
00040     parser.compile(expression_string, expression);
00041
00042     printf(" # | A | B | C | %s\n"
00043           "-----+-----+-%s\n",
00044           expression_string.c_str(), std::string(expression_string.size(), '-').c_str());
00045
00046     for (int i = 0; i < 8; ++i) {
00047         symbol_table.get_variable("A")->ref() = double(i & 0x01 ? 1 : 0);
00048         symbol_table.get_variable("B")->ref() = double(i & 0x02 ? 1 : 0);
00049         symbol_table.get_variable("C")->ref() = double(i & 0x04 ? 1 : 0);
00050
00051         const int result = static_cast<int>(expression.value());
00052
00053         printf(" %d | %d | %d | %d | %d \n", i,
```

```

00054         static_cast<int>(symbol_table.get_variable("A")->value()),
00055         static_cast<int>(symbol_table.get_variable("B")->value()),
00056         static_cast<int>(symbol_table.get_variable("C")->value()), result);
00057     }
00058 }
00059
00060 // --- Preprocessing Function ---
00061
00074 bool isIdentifier(const std::string &token) {
00075     if (token.empty())
00076         return false;
00077     // Check if the token is a valid identifier (upper alpha+ numeric + underscore)
00078     bool has_alpha = false;
00079     for (char c : token) {
00080         if (std::isalpha(c)) {
00081             has_alpha = true;
00082         } else if (!std::isalnum(c) && c != '_' ) {
00083             return false; // Contains invalid char
00084         }
00085     }
00086     // Must start with an uppercase alphabetic character
00087     return has_alpha && std::isupper(token[0]);
00088 }
00089
00099 bool isOperator(const std::string &token) {
00100     static const std::set<std::string> operators = {"and", "or", "xor", "not"};
00101     return operators.count(token);
00102 }
00103
00128 std::string LogicComparator::preprocessExpression(const std::string &input_expr) {
00129     std::string processed = input_expr;
00130     spdlog::debug("Preprocessing raw expression: {}", processed);
00131
00132     // 0. Trim leading/trailing whitespace first
00133     processed = std::regex_replace(processed, std::regex("^\\s+|\\s+$"), "");
00134     if (processed.empty()) {
00135         spdlog::debug("Input expression is empty.");
00136         return ""; // Handle empty input early
00137     }
00138
00139     // --- Helper function: build log string ---
00140     auto build_log_string = [](const auto &container) {
00141         std::ostringstream oss;
00142         for (auto it = container.begin(); it != container.end(); ++it) {
00143             oss << *it;
00144             if (std::next(it) != container.end()) {
00145                 oss << " ";
00146             }
00147         }
00148         return oss.str();
00149     };
00150
00151     // 1. Handle '!' for NOT carefully BEFORE adding general spaces.
00152     // Replace logical NOT '!' with " not ", ensuring not to replace '!='.
00153     std::string temp_processed;
00154     temp_processed.reserve(processed.length() * 1.2); // Pre-allocate a bit more space
00155     for (size_t i = 0; i < processed.length(); ++i) {
00156         if (processed[i] == '!') {
00157             if (i + 1 < processed.length() && processed[i + 1] == '=') {
00158                 temp_processed += "!="; // Keep inequality operator
00159                 i++; // Skip the '='
00160             } else {
00161                 temp_processed += " not "; // Replace logical NOT with spaced operator
00162             }
00163         } else {
00164             temp_processed += processed[i];
00165         }
00166     }
00167     processed = temp_processed;

```

```

00168     spdlog::trace("After '!' to 'not' conversion: {}", processed);
00169
00170     // 2. Add spaces around other operators and parentheses that need them
00171     processed = std::regex_replace(processed, std::regex("\\("), " ( ");
00172     processed = std::regex_replace(processed, std::regex("\\)"), " ) ");
00173     processed = std::regex_replace(processed, std::regex("\\+"), " + "); // OR
00174     processed = std::regex_replace(processed, std::regex("\\^"), " ^ "); // XOR
00175     processed = std::regex_replace(processed, std::regex("\\*"), " * "); // AND
00176     // Consolidate multiple spaces into one and trim again
00177     processed = std::regex_replace(processed, std::regex("\\s+"), " ");
00178     processed = std::regex_replace(processed, std::regex("^\\s+|\\s+$"), "");
00179     spdlog::trace("After adding spaces: {}", processed);
00180
00181     // 3. Replace symbolic operators with their keyword equivalents
00182     // Note: 'not' is already handled.
00183     processed = std::regex_replace(processed, std::regex("\\+"), " or ");
00184     processed = std::regex_replace(processed, std::regex("\\^"), " xor ");
00185     processed = std::regex_replace(processed, std::regex("\\*"), " and ");
00186     spdlog::trace("After operator keyword replacement: {}", processed);
00187
00188     // 4. Tokenize based on spaces
00189     std::stringstream ss(processed);
00190     // Read tokens skipping whitespace
00191     std::vector<std::string> tokens{std::istream_iterator<std::string>(ss),
00192                                     std::istream_iterator<std::string>()};
00193
00194     if (tokens.empty()) {
00195         spdlog::debug("No tokens found after tokenization.");
00196         return "";
00197     }
00198     spdlog::trace("Tokens after initial processing: [{}]", build_log_string(tokens));
00199
00200     // 5. Insert implied 'and'
00201     std::vector<std::string> tokens_with_implied_and;
00202     if (!tokens.empty()) {
00203         tokens_with_implied_and.push_back(tokens[0]);
00204         for (size_t i = 0; i < tokens.size() - 1; ++i) {
00205             const std::string &current_token = tokens[i];
00206             const std::string &next_token = tokens[i + 1];
00207
00208             // An operand ends if it's an identifier or a closing parenthesis.
00209             bool current_ends_operand = isIdentifier(current_token) || current_token == ")";
00210             // An operand starts if it's an identifier, an opening parenthesis, or 'not'.
00211             bool next_starts_operand =
00212                 isIdentifier(next_token) || next_token == "(" || next_token == "not";
00213
00214             // We should insert 'and' if the current token ends an operand AND the next token starts one,
00215             // UNLESS the current token is already an operator/opening bracket OR the next token is an
00216             // operator/closing bracket. Explicit check: Don't insert 'and' if an operator already exists
00217             // between them.
00218             bool current_is_op_or_open =
00219                 isOperator(current_token) || current_token == "not" || current_token == "(";
00220             bool next_is_op_or_close = isOperator(next_token) || next_token == "not" || next_token == ")";
00221
00222             if (current_ends_operand && next_starts_operand && !current_is_op_or_open &&
00223                 !next_is_op_or_close) {
00224                 spdlog::trace("Inserting implied 'and' between '{}' and '{}'", current_token, next_token);
00225                 tokens_with_implied_and.push_back("and");
00226             }
00227
00228             tokens_with_implied_and.push_back(next_token); // Add the next token regardless
00229         }
00230     } else {
00231         // Handle case where initial tokenization yielded no tokens
00232         return "";
00233     }
00234
00235     spdlog::trace("Tokens after implied 'and': [{}]", build_log_string(tokens_with_implied_and));
00236

```

```

00237 // 6. Handle 'not Identifier' ONLY during reconstruction (NEW LOGIC)
00238 std::vector<std::string> final_tokens;
00239 for (size_t i = 0; i < tokens_with_implied_and.size(); ++i) {
00240     const std::string &current_token = tokens_with_implied_and[i];
00241
00242     if (current_token == "not") {
00243         // Check if the *next* token is an Identifier
00244         if (i + 1 < tokens_with_implied_and.size() && isIdentifier(tokens_with_implied_and[i + 1])) {
00245             // Found 'not Identifier', transform to 'not(Identifier)'
00246             final_tokens.push_back("not");
00247             final_tokens.push_back("(");
00248             final_tokens.push_back(tokens_with_implied_and[i + 1]); // The identifier
00249             final_tokens.push_back(")");
00250             i++; // Increment i to skip the identifier we just processed
00251         } else {
00252             // 'not' is followed by '(', another operator, or is at the end.
00253             // Add 'not' as is, let ExprTk parse 'not(...)' or handle errors.
00254             final_tokens.push_back("not");
00255         }
00256     } else {
00257         // Token is not 'not', just add it to the final list
00258         final_tokens.push_back(current_token);
00259     }
00260 }
00261 spdlog::trace("Tokens after 'not' parenthesis handling: [{}]", build_log_string(final_tokens));
00262
00263 // 7. Reconstruct the final expression string from tokens
00264 std::string final_expr;
00265 if (!final_tokens.empty()) {
00266     final_expr = final_tokens[0];
00267     for (size_t i = 1; i < final_tokens.size(); ++i) {
00268         const std::string &prev = final_tokens[i - 1];
00269         const std::string &curr = final_tokens[i];
00270
00271         // Add space smartly: No space after '(', no space before ')',
00272         // no space after 'not' if followed by '(', no space before ',' (in function args, if
00273         // applicable)
00274         bool add_space = true;
00275         if (prev == "(" || curr == ")" || curr == ",") {
00276             add_space = false;
00277         }
00278         if (prev == "not" && curr == "(") {
00279             add_space = false; // Already handled by 'not(...)' structure
00280         }
00281         // Potentially add more rules if needed for other operators/functions
00282
00283         if (add_space) {
00284             final_expr += " ";
00285         }
00286         final_expr += curr;
00287     }
00288 }
00289
00290 // 8. Final cleanup (consolidate any remaining multiple spaces and trim)
00291 final_expr = std::regex_replace(final_expr, std::regex("\\s+"), " ");
00292 final_expr = std::regex_replace(final_expr, std::regex("^\\s+|\\s+$"), "");
00293
00294 spdlog::debug("Preprocessed expression result: {}", final_expr);
00295 return final_expr;
00296 }
00297
00298 // --- Variable Extraction Helper ---
00327 bool LogicComparator::extractVariables(const std::string &expr1_raw, const std::string &expr2_raw,
00328                                         std::vector<std::string> &sorted_vars) {
00329     // Define the set of ExprTk keywords/function names to filter out (lowercase)
00330     // Even though isIdentifier checks for uppercase, keep this filter as a safety measure
00331     static const std::set<std::string> keywords = {"abs",
00332                                                     "acos",
00333                                                     "acosh",

```

```
00334         "and",
00335         "asin",
00336         "asinh",
00337         "assert",
00338         "atan",
00339         "atan2",
00340         "atanh",
00341         "avg",
00342         "break",
00343         "case",
00344         "ceil",
00345         "clamp",
00346         "continue",
00347         "cosh",
00348         "cos",
00349         "cot",
00350         "csc",
00351         "default",
00352         "deg2grad",
00353         "deg2rad",
00354         "else",
00355         "equal",
00356         "erfc",
00357         "erf",
00358         "exp",
00359         "expm1",
00360         "false",
00361         "floor",
00362         "for",
00363         "frac",
00364         "grad2deg",
00365         "hypot",
00366         "iclamp",
00367         "if",
00368         "ilike",
00369         "in",
00370         "inrange",
00371         /*"in",*/ "like",
00372         "log",
00373         "log10",
00374         "log1p",
00375         "log2",
00376         "logn",
00377         "mand",
00378         "max",
00379         "min",
00380         "mod",
00381         "mor",
00382         "mul",
00383         "nand",
00384         "ncdf",
00385         "nor",
00386         "not",
00387         "not_equal",
00388         /*"not",*/ "null",
00389         "or",
00390         "pow",
00391         "rad2deg",
00392         "repeat",
00393         "return",
00394         "root",
00395         "roundn",
00396         "round",
00397         "sec",
00398         "sgn",
00399         "shl",
00400         "shr",
00401         "sinc",
00402         "sinh",
```

```

00403         "sin",
00404         "sqrt",
00405         "sum",
00406         "swap",
00407         "switch",
00408         "tanh",
00409         "tan",
00410         "true",
00411         "trunc",
00412         "until",
00413         "var",
00414         "while",
00415         "xnor",
00416         "xor"};
00417
00418 // Regular expression to find potential identifiers (unchanged)
00419 const std::regex identifier_regex("[a-zA-Z_][a-zA-Z0-9_]*");
00420
00421 // Store validated and filtered variable names
00422 std::set<std::string> actual_vars1_set;
00423 std::set<std::string> actual_vars2_set;
00424
00425 // --- Helper function: build log string ---
00426 auto build_log_string = [](const auto &container) {
00427     std::ostringstream oss;
00428     oss << "[";
00429     for (auto it = container.begin(); it != container.end(); ++it) {
00430         oss << *it;
00431         if (std::next(it) != container.end()) {
00432             oss << ", ";
00433         }
00434     }
00435     oss << "]";
00436     return oss.str();
00437 };
00438
00439 // --- Process the first expression ---
00440 spdlog::debug("Extracting and validating identifiers from raw expr1: {}", expr1_raw);
00441 auto words_begin1 = std::sregex_iterator(expr1_raw.begin(), expr1_raw.end(), identifier_regex);
00442 auto words_end1 = std::sregex_iterator();
00443 for (std::sregex_iterator i = words_begin1; i != words_end1; ++i) {
00444     std::string potential_var = i->str();
00445     spdlog::trace("Regex found in expr1: {}", potential_var);
00446
00447     // 1. Validate using isIdentifier (now checks for uppercase etc.)
00448     if (isIdentifier(potential_var)) {
00449         // 2. Check if it's a keyword (still use lowercase comparison as a precaution)
00450         std::string lower_var = potential_var;
00451         std::transform(lower_var.begin(), lower_var.end(), lower_var.begin(),
00452             [](unsigned char c) { return std::tolower(c); });
00453
00454         if (keywords.find(lower_var) == keywords.end()) {
00455             actual_vars1_set.insert(potential_var); // Keep original case
00456             spdlog::trace("Kept variable from expr1: {}", potential_var);
00457         } else {
00458             spdlog::trace("Filtered keyword from expr1: {}", potential_var);
00459         }
00460     } else {
00461         spdlog::trace("Filtered invalid identifier from expr1: {}", potential_var);
00462     }
00463 }
00464 spdlog::debug("Validated variables found in expr1: {}", build_log_string(actual_vars1_set));
00465
00466 // --- Process the second expression ---
00467 spdlog::debug("Extracting and validating identifiers from raw expr2: {}", expr2_raw);
00468 auto words_begin2 = std::sregex_iterator(expr2_raw.begin(), expr2_raw.end(), identifier_regex);
00469 auto words_end2 = std::sregex_iterator();
00470 for (std::sregex_iterator i = words_begin2; i != words_end2; ++i) {
00471     std::string potential_var = i->str();

```

```

00472     spdlog::trace("Regex found in expr2: {}", potential_var);
00473
00474     // 1. Validate using isIdentifier
00475     if (isIdentifier(potential_var)) {
00476         // 2. Check if it's a keyword (still use lowercase comparison)
00477         std::string lower_var = potential_var;
00478         std::transform(lower_var.begin(), lower_var.end(), lower_var.begin(),
00479             [](unsigned char c) { return std::tolower(c); });
00480
00481         if (keywords.find(lower_var) == keywords.end()) {
00482             actual_vars2_set.insert(potential_var); // Keep original case
00483             spdlog::trace("Kept variable from expr2: {}", potential_var);
00484         } else {
00485             spdlog::trace("Filtered keyword from expr2: {}", potential_var);
00486         }
00487     } else {
00488         spdlog::trace("Filtered invalid identifier from expr2: {}", potential_var);
00489     }
00490 }
00491 spdlog::debug("Validated variables found in expr2: {}", build_log_string(actual_vars2_set));
00492
00493 // --- Compare the two variable sets ---
00494 if (actual_vars1_set == actual_vars2_set) {
00495     // Sets are equal, extract the variable list
00496     sorted_vars.assign(actual_vars1_set.begin(), actual_vars1_set.end());
00497     std::sort(sorted_vars.begin(), sorted_vars.end()); // Sort alphabetically
00498     spdlog::info("Variable sets match. Found unique variables for comparison: {}",
00499         build_log_string(sorted_vars));
00500     spdlog::info("Extracted variables done !");
00501     return true; // Variable sets are consistent
00502 } else {
00503     // Sets are not equal, report an error and return false
00504     spdlog::warn("Variable sets do not match between the two expressions!");
00505     // Calculate the differences for more detailed logging
00506     std::vector<std::string> diff1, diff2;
00507     std::set_difference(actual_vars1_set.begin(), actual_vars1_set.end(), actual_vars2_set.begin(),
00508         actual_vars2_set.end(),
00509         std::back_inserter(diff1)); // Vars only in expr1
00510     std::set_difference(actual_vars2_set.begin(), actual_vars2_set.end(), actual_vars1_set.begin(),
00511         actual_vars1_set.end(),
00512         std::back_inserter(diff2)); // Vars only in expr2
00513
00514     if (!diff1.empty()) {
00515         spdlog::warn("Variables only in reference expression: {}", build_log_string(diff1));
00516     }
00517     if (!diff2.empty()) {
00518         spdlog::warn("Variables only in comparison expression: {}", build_log_string(diff2));
00519     }
00520     sorted_vars.clear(); // Clear the output variable list
00521     return false; // Variable sets are inconsistent
00522 }
00523 }
00524
00525 // --- Implementation of compareSingleExpressionPair ---
00553 void LogicComparator::compareSingleExpressionPair(const std::string &ref_expression_processed,
00554     const std::string &comp_expression_processed,
00555     const std::vector<std::string> &sorted_vars,
00556     PinComparisonResult &result) {
00557     // Store processed expressions in the result struct
00558     result.ref_expr_processed = ref_expression_processed;
00559     result.comp_expr_processed = comp_expression_processed;
00560     result.comparison_possible = true; // Assume comparison is possible initially
00561
00562     spdlog::debug("Comparing processed expressions for pin '{}':", result.pin_name);
00563     spdlog::debug("  Ref : {}", ref_expression_processed);
00564     spdlog::debug("  Comp: {}", comp_expression_processed);
00565
00566     // --- ExprTk Setup ---
00567     typedef exprtk::symbol_table<double> symbol_table_t;

```

```

00568 typedef exprtk::expression<double> expression_t;
00569 typedef exprtk::parser<double> parser_t;
00570
00571 // Setup for Reference Expression
00572 symbol_table_t ref_symbol_table;
00573 expression_t ref_expression;
00574 parser_t ref_parser;
00575
00576 for (const auto &var_name : sorted_vars) {
00577     // Create variable, ExprTk manages its memory within the table
00578     if (!ref_symbol_table.create_variable(var_name)) {
00579         result.error_message = "Failed to create variable '" + var_name + "' in ref symbol table.";
00580         spdlog::error(result.error_message);
00581         result.comparison_possible = false;
00582         return; // Cannot proceed
00583     }
00584 }
00585 ref_expression.register_symbol_table(ref_symbol_table);
00586
00587 // Compile Reference Expression
00588 result.ref_compiles = ref_parser.compile(ref_expression_processed, ref_expression);
00589 if (!result.ref_compiles) {
00590     result.error_message = "Reference expression compilation failed: " + ref_parser.error();
00591     spdlog::warn("Pin '{}': {}", result.pin_name, result.error_message);
00592     result.comparison_possible = false;
00593     // Don't return yet, maybe the comparison one also fails
00594 } else {
00595     spdlog::debug("Pin '{}': Reference expression compiled successfully.", result.pin_name);
00596 }
00597
00598 // Setup for Comparison Expression
00599 symbol_table_t comp_symbol_table;
00600 expression_t comp_expression;
00601 parser_t comp_parser;
00602
00603 for (const auto &var_name : sorted_vars) {
00604     if (!comp_symbol_table.create_variable(var_name)) {
00605         // Append error if ref also failed, otherwise set it
00606         std::string comp_err = "Failed to create variable '" + var_name + "' in comp symbol table.";
00607         result.error_message += (result.error_message.empty() ? "" : "\n") + comp_err;
00608         spdlog::error(comp_err);
00609         result.comparison_possible = false; // Mark as impossible now
00610         return; // Cannot proceed if variable creation fails
00611     }
00612 }
00613 comp_expression.register_symbol_table(comp_symbol_table);
00614
00615 // Compile Comparison Expression
00616 result.comp_compiles = comp_parser.compile(comp_expression_processed, comp_expression);
00617 if (!result.comp_compiles) {
00618     std::string comp_err = "Comparison expression compilation failed: " + comp_parser.error();
00619     result.error_message += (result.error_message.empty() ? "" : "\n") + comp_err;
00620     spdlog::warn("Pin '{}': {}", result.pin_name, comp_err);
00621     result.comparison_possible = false; // Mark as impossible if not already
00622 } else {
00623     spdlog::debug("Pin '{}': Comparison expression compiled successfully.", result.pin_name);
00624 }
00625
00626 // If either failed to compile, comparison is not possible
00627 if (!result.ref_compiles || !result.comp_compiles) {
00628     result.comparison_possible = false;
00629     return;
00630 }
00631
00632 // --- Truth Table Generation and Comparison ---
00633 size_t num_vars = sorted_vars.size();
00634 // Use unsigned long long for potentially large number of combinations
00635 unsigned long long num_combinations = 1ULL << num_vars;
00636 // Prevent excessively large tables (e.g., > 20 variables is 1M+ rows)

```

```

00637  const unsigned long long MAX_COMBINATIONS = 1ULL << 20; // Limit to 2^20 combinations
00638  if (num_vars > 20) { // Check against var count for clarity
00639      result.error_message = "Too many variables (" + std::to_string(num_vars) +
00640          "). Max supported for truth table is 20.";
00641      spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00642      result.comparison_possible = false;
00643      return;
00644  }
00645  if (num_combinations == 0 && num_vars > 0) { // Overflow check
00646      result.error_message = "Number of combinations calculation overflowed for " +
00647          std::to_string(num_vars) + " variables.";
00648      spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00649      result.comparison_possible = false;
00650      return;
00651  }
00652
00653  std::vector<bool> ref_results;
00654  std::vector<bool> comp_results;
00655  ref_results.reserve(static_cast<size_t>(num_combinations)); // Avoid reallocations
00656  comp_results.reserve(static_cast<size_t>(num_combinations));
00657
00658  // --- Prepare Tabulate Tables ---
00659  Table ref_table;
00660  Table comp_table;
00661  Table::Row_t header_row;
00662  header_row.push_back("#"); // Row number column
00663  for (const auto &var_name : sorted_vars) {
00664      header_row.push_back(var_name);
00665  }
00666  header_row.push_back(ref_expression_processed); // Reference expression as last column header
00667  ref_table.add_row(header_row);
00668
00669  // Adjust header for comparison table
00670  header_row.back() = comp_expression_processed; // Change last header to comparison expression
00671  comp_table.add_row(header_row);
00672
00673  // Format header rows
00674  for (size_t i = 0; i < ref_table[0].size(); ++i) {
00675      ref_table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00676      comp_table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00677  }
00678
00679  spdlog::debug("Pin '{}': Generating truth table with {} variables ({} combinations)...",
00680      result.pin_name, num_vars, num_combinations);
00681
00682  bool evaluation_error = false;
00683  for (unsigned long long i = 0; i < num_combinations; ++i) {
00684      Table::Row_t ref_data_row;
00685      Table::Row_t comp_data_row;
00686      ref_data_row.push_back(std::to_string(i));
00687      comp_data_row.push_back(std::to_string(i));
00688
00689      // Set variable values for this combination
00690      for (size_t j = 0; j < num_vars; ++j) {
00691          // The j-th bit of i determines the value of the j-th variable
00692          bool bit_value = ((i >> j) & 1ULL);
00693          double var_value = bit_value ? double(1.0) : double(0.0); // ExprTk uses floating point
00694
00695          // Get variable reference and assign value
00696          // Need error checking here in theory, but create_variable should have caught issues
00697          ref_symbol_table.get_variable(sorted_vars[j])->ref() = var_value;
00698          comp_symbol_table.get_variable(sorted_vars[j])->ref() = var_value;
00699
00700          // Add input value to table rows (as string "0" or "1")
00701          std::string bit_str = bit_value ? "1" : "0";
00702          ref_data_row.push_back(bit_str);
00703          comp_data_row.push_back(bit_str);
00704      }
00705  }

```

```

00706 // Evaluate expressions
00707 double ref_val, comp_val;
00708 try {
00709     ref_val = ref_expression.value();
00710 } catch (const std::exception &e) {
00711     result.error_message +=
00712         "\nReference evaluation failed at combination " + std::to_string(i) + ": " + e.what();
00713     spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00714     result.comparison_possible = false;
00715     evaluation_error = true;
00716     break; // Stop evaluation
00717 }
00718 try {
00719     comp_val = comp_expression.value();
00720 } catch (const std::exception &e) {
00721     result.error_message +=
00722         "\nComparison evaluation failed at combination " + std::to_string(i) + ": " + e.what();
00723     spdlog::error("Pin '{}': {}", result.pin_name, result.error_message);
00724     result.comparison_possible = false;
00725     evaluation_error = true;
00726     break; // Stop evaluation
00727 }
00728
00729 // Store boolean results (commonly, non-zero is true in logic contexts)
00730 bool ref_bool_result = (ref_val != double(0.0));
00731 bool comp_bool_result = (comp_val != double(0.0));
00732
00733 ref_results.push_back(ref_bool_result);
00734 comp_results.push_back(comp_bool_result);
00735
00736 // Add output results to table rows (as string "0" or "1")
00737 ref_data_row.push_back(ref_bool_result ? "1" : "0");
00738 comp_data_row.push_back(comp_bool_result ? "1" : "0");
00739
00740 // Add rows to tables
00741 ref_table.add_row(ref_data_row);
00742 comp_table.add_row(comp_data_row);
00743
00744 } // End of combination loop
00745
00746 if (evaluation_error) {
00747     return; // Don't proceed if evaluation failed
00748 }
00749
00750 // --- Final Comparison ---
00751 if (result.comparison_possible) {
00752     result.are_equivalent = (ref_results == comp_results);
00753     if (result.are_equivalent) {
00754         spdlog::info("Pin '{}': Expressions ARE logically equivalent.", result.pin_name);
00755     } else {
00756         spdlog::warn("Pin '{}': Expressions ARE NOT logically equivalent.", result.pin_name);
00757         result.error_message +=
00758             "\nTruth table outputs differ."; // Add specific reason for non-equivalence
00759     }
00760
00761 // Store the generated tables in the result struct
00762 result.ref_truth_table = ref_table;
00763 result.comp_truth_table = comp_table;
00764 }
00765 }
00766
00812 void LogicComparator::compareCellLogic() {
00813
00814 // 1. Collect all unique pin names from both maps
00815 std::set<std::string> unique_pin_names;
00816 for (const auto &pair : ref_outpin_map_) {
00817     unique_pin_names.insert(pair.first);
00818 }
00819 for (const auto &pair : comp_outpin_map_) {

```

```

00820     unique_pin_names.insert(pair.first);
00821 }
00822
00823 spdlog::info("Starting logic comparison for cell '{}' with {} unique output pins...", cell_name_,
00824             unique_pin_names.size());
00825
00826 // 2. Iterate through each unique pin name
00827 for (const std::string &pin_name : unique_pin_names) {
00828     PinComparisonResult pin_result;
00829     pin_result.pin_name = pin_name;
00830     pin_result.comparison_possible = true; // Assume possible initially
00831
00832 // 3. Get raw expressions
00833 auto ref_it = ref_outpin_map_.find(pin_name);
00834 auto comp_it = comp_outpin_map_.find(pin_name);
00835
00836 if (ref_it == ref_outpin_map_.end()) {
00837     pin_result.error_message = "Pin not found in reference map.";
00838     spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name, pin_result.error_message);
00839     pin_result.comparison_possible = false;
00840     // Still try to get the comparison expression for reporting
00841     if (comp_it != comp_outpin_map_.end()) {
00842         pin_result.comp_expr_raw = comp_it->second;
00843     }
00844 } else {
00845     pin_result.ref_expr_raw = ref_it->second;
00846     spdlog::debug("Pin -> Expression: {} -> {}", pin_name, pin_result.ref_expr_raw);
00847 }
00848
00849 if (comp_it == comp_outpin_map_.end()) {
00850     std::string comp_err = "Pin not found in comparison map.";
00851     pin_result.error_message += (pin_result.error_message.empty() ? "" : "\n") + comp_err;
00852     spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name, comp_err);
00853     pin_result.comparison_possible = false;
00854     // If ref existed, store it
00855     if (ref_it != ref_outpin_map_.end()) {
00856         pin_result.ref_expr_raw = ref_it->second; // Already stored if ref exists
00857     }
00858 } else {
00859     pin_result.comp_expr_raw = comp_it->second;
00860     spdlog::debug("Pin => Expression: {} => {}", pin_name, pin_result.comp_expr_raw);
00861 }
00862
00863 // If pin missing in either, we can't compare, but store result and continue
00864 if (!pin_result.comparison_possible) {
00865     all_pin_results_[pin_name] = pin_result;
00866     continue;
00867 }
00868
00869 // 4. Preprocess expressions
00870 std::string ref_processed = preprocessExpression(pin_result.ref_expr_raw);
00871 std::string comp_processed = preprocessExpression(pin_result.comp_expr_raw);
00872
00873 if (ref_processed.empty()) {
00874     pin_result.error_message += "\nReference expression became empty after preprocessing.";
00875     spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name,
00876                 "Reference expression became empty after preprocessing.");
00877     pin_result.comparison_possible = false;
00878 }
00879 if (comp_processed.empty()) {
00880     std::string comp_err = "\nComparison expression became empty after preprocessing.";
00881     pin_result.error_message += (pin_result.error_message.empty() ? "" : "\n") + comp_err;
00882     spdlog::warn("Cell '{}', Pin '{}': {}", cell_name_, pin_name,
00883                 "Comparison expression became empty after preprocessing.");
00884     pin_result.comparison_possible = false; // Mark as impossible if not already
00885 }
00886 // If preprocessing failed for either, store and continue
00887 if (ref_processed.empty() || comp_processed.empty()) {
00888     pin_result.ref_expr_processed = ref_processed; // Store possibly empty strings

```

```

00889     pin_result.comp_expr_processed = comp_processed;
00890     all_pin_results_[pin_name] = pin_result;
00891     continue;
00892 }
00893
00894 // 5. Extract and validate variables
00895 std::vector<std::string> sorted_vars;
00896 // Pass RAW expressions to extractVariables as it uses regex on original format
00897 bool variables_match =
00898     extractVariables(pin_result.ref_expr_raw, pin_result.comp_expr_raw, sorted_vars);
00899
00900 if (!variables_match) {
00901     pin_result.error_message = "Variable sets do not match between expressions.";
00902     // extractVariables already logs details
00903     pin_result.comparison_possible = false;
00904     pin_result.ref_expr_processed = ref_processed; // Store processed even if vars mismatch
00905     pin_result.comp_expr_processed = comp_processed;
00906     all_pin_results_[pin_name] = pin_result;
00907     continue;
00908 }
00909 // --- Helper function: build log string ---
00910 auto build_log_string = [](const auto &container) {
00911     std::ostringstream oss;
00912     for (auto it = container.begin(); it != container.end(); ++it) {
00913         oss << *it;
00914         if (std::next(it) != container.end()) {
00915             oss << ", ";
00916         }
00917     }
00918     return oss.str();
00919 };
00920 spdlog::info("Pin '{}': Variable sets match. Found unique variables: {}", pin_name,
00921     build_log_string(sorted_vars));
00922
00923 // 6. Compare the single pair using the processed expressions
00924 // Pass pin_result by reference - it will be populated by the function
00925 compareSingleExpressionPair(ref_processed, comp_processed, sorted_vars, pin_result);
00926
00927 // 7. Store the detailed result for this pin
00928 all_pin_results_[pin_name] = pin_result;
00929
00930 } // End of pin loop
00931
00932 spdlog::info("Logic comparison finished for cell '{}'.", cell_name_);
00933 }
00934
00974 void LogicComparator::generateReport(const std::string &output_file) {
00975     // --- 0. Start Info ---
00976     spdlog::info("Generating report for cell '{}' to '{}'....", cell_name_, output_file);
00977
00978     // --- 1. Determine Output Format ---
00979     bool output_markdown = false;
00980     try {
00981         // Use std::filesystem to reliably get the extension
00982         std::filesystem::path file_path(output_file);
00983         if (file_path.has_extension() && file_path.extension() == ".md") {
00984             output_markdown = true;
00985         }
00986     } catch (const std::exception &e) {
00987         spdlog::warn("Could not determine file extension for '{}': {}. Defaulting to plain text.",
00988             output_file, e.what());
00989         output_markdown = false; // Default to plain text on error
00990     }
00991
00992     // --- 2. Open Output File ---
00993     std::ofstream outfile(output_file,
00994         std::ios::app); // Use app to append if file exists, or create new
00995     if (!outfile.is_open()) {
00996         spdlog::error("Failed to open output report file: {}", output_file);

```

```

00997     return;
00998 }
00999 spdlog::info("Generating report for cell '{0}' to '{0}' (Format: {})...", cell_name_, output_file,
01000             output_markdown ? "Markdown" : "Plain Text");
01001
01002 // --- 3. Report Header ---
01003 outfile << "# Logic Equivalence Comparison Report\n\n";
01004 outfile << "**Cell Name: " << cell_name_ << "**\n\n";
01005 // Reference Pin Functions Table
01006 Table ref_pin_table;
01007 ref_pin_table.add_row({"Reference Pin", "Function"});
01008 ref_pin_table[0][0].format().font_style({FontStyle::bold});
01009 ref_pin_table[0][1].format().font_style({FontStyle::bold});
01010 for (const auto &[pin, function] : ref_outpin_map_) {
01011     ref_pin_table.add_row({pin, "`" + function + "`"}); // Add backticks to function
01012 }
01013 // Comparison Pin Functions Table
01014 Table comp_pin_table;
01015 comp_pin_table.add_row({"Comparison Pin", "Function"});
01016 comp_pin_table[0][0].format().font_style({FontStyle::bold});
01017 comp_pin_table[0][1].format().font_style({FontStyle::bold});
01018 for (const auto &[pin, function] : comp_outpin_map_) {
01019     comp_pin_table.add_row({pin, "`" + function + "`"}); // Add backticks to function
01020 }
01021
01022 // --- 4. Report Metadata ---
01023 outfile << "**Performed by " << APP_NAME << " v" << APP_VERSION << " from " << APP_AUTHOR;
01024 auto now = std::chrono::system_clock::now();
01025 std::time_t now_time_t = std::chrono::system_clock::to_time_t(now);
01026 std::tm *now_tm = std::localtime(&now_time_t);
01027 outfile << ". on: " << std::put_time(now_tm, "%c") << "**\n" << std::endl;
01028
01029 // --- 5. Legend Generation ---
01030 outfile << "## Legend\n\n";
01031 Table legend_table;
01032 legend_table.add_row({"Symbol", "Meaning"});
01033 legend_table[0][0].format().font_style({FontStyle::bold});
01034 legend_table[0][1].format().font_style({FontStyle::bold});
01035 legend_table.add_row({"OK", "Logically Equivalent"});
01036 legend_table.add_row({"NO", "Not Logically Equivalent"});
01037 legend_table.add_row({"NA", "Comparison Not Applicable (General)"}); // Changed from possible
01038 legend_table.add_row({"VM", "Variable Mismatch (Cannot Compare)"});
01039 legend_table.add_row({"CE", "Compilation Error (Cannot Compare)"});
01040 legend_table.add_row({"EE", "Evaluation Error (Cannot Compare)"});
01041 legend_table.add_row({"PE", "Preprocessing/Pin/Setup Error (Cannot Compare)"});
01042
01043 // --- 6. Export Legend Table ---
01044 if (output_markdown) {
01045     MarkdownExporter exporter;
01046     outfile << exporter.dump(ref_pin_table) << "\n"; // Export ref pin table
01047     outfile << exporter.dump(comp_pin_table) << "\n"; // Export comp pin table
01048     outfile << exporter.dump(legend_table) << "\n"; // Export legend table
01049 } else {
01050     outfile << ref_pin_table << "\n"; // Export ref pin table
01051     outfile << comp_pin_table << "\n"; // Export comp pin table
01052     outfile << legend_table << "\n"; // Export legend table (plain text)
01053 }
01054
01055 // --- 7. Process Pin Results ---
01056 for (const auto &[pin_name, result] : all_pin_results_) { // Use passed argument
01057     outfile << "## Pin: `" << pin_name << "`\n\n"; // Use backticks for pin name
01058
01059     // --- Output Truth Tables (Always if available) ---
01060     if (result.ref_truth_table.has_value()) {
01061         outfile << "### Reference Truth Table\n\n";
01062         Table ref_table = result.ref_truth_table.value();
01063         if (output_markdown) {
01064             MarkdownExporter exporter;
01065             outfile << exporter.dump(ref_table) << "\n";

```

```

01066     } else {
01067         outfile << ref_table << "\n";
01068     }
01069 } else {
01070     outfile << "*Reference truth table not generated (e.g., due to compilation error).*\n\n";
01071 }
01072
01073 if (result.comp_truth_table.has_value()) {
01074     outfile << "### Comparison Truth Table\n\n";
01075     Table comp_table = result.comp_truth_table.value();
01076     if (output_markdown) {
01077         MarkdownExporter exporter;
01078         outfile << exporter.dump(comp_table) << "\n";
01079     } else {
01080         outfile << comp_table << "\n";
01081     }
01082 } else {
01083     outfile << "*Comparison truth table not generated (e.g., due to compilation error).*\n\n";
01084 }
01085
01086 // --- Generate Summary Table ---
01087 outfile << "### Comparison Summary\n\n";
01088 Table summary_table;
01089 summary_table.add_row({"Property", "Value"});
01090 summary_table[0][0].format().font_style({FontStyle::bold});
01091 summary_table[0][1].format().font_style({FontStyle::bold});
01092 summary_table[0][0].format().font_color(Color::yellow);
01093 summary_table[0][1].format().font_color(Color::yellow);
01094
01095 // Determine Status Symbol using ASCII codes
01096 std::string status_symbol;
01097 if (!result.comparison_possible) {
01098     status_symbol = "[NA]"; // Default Not Applicable
01099     if (!result.error_message.empty()) {
01100         if (result.error_message.find("Variable sets do not match") != std::string::npos) {
01101             status_symbol = "[VM]";
01102         } else if (!result.ref_compiles || !result.comp_compiles ||
01103             result.error_message.find("compilation failed") != std::string::npos) {
01104             // Check flags first, then error message as fallback
01105             status_symbol = "[CE]";
01106         } else if (result.error_message.find("evaluation failed") != std::string::npos) {
01107             status_symbol = "[EE]";
01108         } else {
01109             // If comparison_possible is false for other reasons (e.g., pin missing, setup errors)
01110             status_symbol = "[PE]";
01111         }
01112     } else if (!result.ref_compiles || !result.comp_compiles) {
01113         // Handle cases where comparison_possible might be true but compilation failed (should
01114         // ideally not happen if logic is sound)
01115         status_symbol = "[CE]";
01116     } else {
01117         status_symbol = "[PE]"; // Fallback if no error message but still not possible
01118     }
01119 } else if (result.are_equivalent) {
01120     status_symbol = "[OK]";
01121 } else {
01122     status_symbol = "[NO]";
01123 }
01124
01125 summary_table.add_row({"Status", status_symbol});
01126 summary_table.add_row({"Reference (Raw)", "" + result.ref_expr_raw + ""});
01127 summary_table.add_row({"Comparison (Raw)", "" + result.comp_expr_raw + ""});
01128 summary_table.add_row({"Reference (Processed)", "" + result.ref_expr_processed + ""});
01129 summary_table.add_row({"Comparison (Processed)", "" + result.comp_expr_processed + ""});
01130 summary_table.add_row({"Ref Expression Compiled", result.ref_compiles ? "Yes" : "No"});
01131 summary_table.add_row({"Comp Expression Compiled", result.comp_compiles ? "Yes" : "No"});
01132 summary_table.add_row({"Logically Equivalent", result.comparison_possible
01133     ? (result.are_equivalent ? "Yes" : "No")
01134     : "N/A"});

```

```

01135     summary_table[1][0].format().font_style({FontStyle::bold});
01136     summary_table[2][0].format().font_style({FontStyle::bold});
01137     summary_table[3][0].format().font_style({FontStyle::bold});
01138     summary_table[4][0].format().font_style({FontStyle::bold});
01139     summary_table[5][0].format().font_style({FontStyle::bold});
01140     summary_table[6][0].format().font_style({FontStyle::bold});
01141     summary_table[7][0].format().font_style({FontStyle::bold});
01142     summary_table[8][0].format().font_style({FontStyle::bold});
01143
01144     if (!result.error_message.empty()) {
01145         std::string formatted_error = result.error_message;
01146         if (output_markdown) {
01147             // Basic newline replacement for Markdown
01148             size_t pos = 0;
01149             while ((pos = formatted_error.find('\n', pos)) != std::string::npos) {
01150                 formatted_error.replace(pos, 1, "<br>");
01151                 pos += 4; // Length of "<br>"
01152             }
01153         }
01154         summary_table.add_row({"Details/Error", formatted_error});
01155     }
01156
01157     // Output Summary Table
01158     if (output_markdown) {
01159         MarkdownExporter exporter;
01160         outfile << exporter.dump(summary_table) << "\n";
01161     } else {
01162         outfile << summary_table << "\n";
01163     }
01164     std::cout << summary_table << "\n"; // Print to console for immediate feedback
01165
01166     outfile << "---\n\n"; // Separator between pins
01167
01168 } // End of pin results loop
01169
01170 outfile.close();
01171 spdlog::info("Report generation complete for cell '{}'.", cell_name_);
01172 }

```

7.45 src/LogicExtractor.cpp File Reference

#include "LogicExtractor.hpp"

Include dependency graph for LogicExtractor.cpp:

Functions

- void [extractAndPrintNetlistInfo](#) (const std::string &verilog_file, const std::string &cell)

Extracts and prints netlist information from a Verilog file for a specified cell.
- std::map< std::string, std::string > [extractLogicFromVerilog](#) (const std::string &verilog_file, const std::string &cell)

Extracts logic expressions from a Verilog file for a specified cell.

7.45.1 Function Documentation

7.45.1.1 extractAndPrintNetlistInfo()

```
void extractAndPrintNetlistInfo (
    const std::string & verilog_file,
    const std::string & cell )
```

Extracts and prints netlist information from a Verilog file for a specified cell.

This function parses a Verilog file using the slang library, extracts information about the primary inputs, primary outputs, internal wires, and gate drivers within a specified cell (module). It then prints a summary of the extracted information to the console using spdlog.

Parameters

<i>verilog_file</i>	The path to the Verilog file to be parsed.
<i>cell</i>	The name of the cell (module) for which to extract netlist information.

Exceptions

<i>std::exception</i>	If any error occurs during Verilog parsing or info extraction.
-----------------------	--

Note

The function uses the slang library for Verilog parsing and a custom [LogicExtractor](#) class to extract the desired information. Error messages are logged using spdlog.

Definition at line 676 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:

7.45.1.2 extractLogicFromVerilog()

```
std::map< std::string, std::string > extractLogicFromVerilog (
    const std::string & verilog_file,
    const std::string & cell )
```

Extracts logic expressions from a Verilog file for a specified cell.

This function parses a Verilog file using the slang library, identifies the specified cell, and extracts the logic expressions for its outputs. It returns a map where the keys are output signal names and the values are their corresponding logic expressions as strings.

Parameters

<code>verilog_file</code>	The path to the Verilog file to parse.
<code>cell</code>	The name of the cell (module) for which to extract logic expressions.

Returns

A map of output signal names to their logic expressions. Returns an empty map if parsing fails, the cell is not found, or no logic expressions can be derived.

Note

The function uses the slang library for Verilog parsing. Ensure that slang is properly installed and configured before using this function.

The logic extraction process involves traversing the syntax tree of the Verilog code and identifying relevant assignments and expressions within the specified cell.

Error messages and warnings are logged using the spdlog library.

Definition at line 755 of file [LogicExtractor.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

7.46 LogicExtractor.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LogicExtractor.hpp"
00002
00003 // --- Implementation for LogicExtractor ---
00004
00018 void LogicExtractor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00019     if (parsingComplete_)
00020         return; // Don't re-process if called again
00021
00022     if (module.header && module.header->name.valid()) {
00023         std::string_view moduleName = module.header->name.valueText();
00024         if (!targetCell_.empty() && moduleName == targetCell_) {
00025             spdlog::info("LogicExtractor: Found target module: {}", targetCell_);
00026             inTargetModule_ = true;
00027
00028             // Reset state for the target module
00029             primaryInputs_.clear();
00030             primaryOutputs_.clear();
00031             internalWires_.clear();
00032             portDirections_.clear(); // Clear temporary direction map
00033             gateOutputDrivers_.clear();
00034             // logicCache_.clear(); // For Step 2
00035
00036             // Visit children (ports, declarations, instances) IN ORDER
00037             // It's often better to visit declarations first, then the port list
00038             // Slang's default visitDefault might handle this, but be aware.
00039             visitDefault(module);

```

```

00040
00041     // --- Finalize Ports after visiting declarations and port list ---
00042     // This part might be better placed *after* visiting NonAnsiPortList,
00043     // assuming the list visit confirms the names from the header.
00044     // Let's move the finalization logic to handle(NonAnsiPortListSyntax)
00045
00046     // Mark that we finished processing the target module
00047     inTargetModule_ = false; // Important to prevent processing other modules
00048     parsingComplete_ = true; // Stop processing after finding the target
00049     spdlog::info("LogicExtractor: Finished visiting target module '{}'.", targetCell_);
00050
00051 } else if (!parsingComplete_) {
00052     // If not the target module yet, continue searching
00053     visitDefault(module);
00054 }
00055 } else if (!parsingComplete_) {
00056     // Handle modules without valid headers if necessary, or just traverse
00057     visitDefault(module);
00058 }
00059 }
00060
00061 // Handle Port Declarations (defines direction and type, usually inside module body)
00092 void LogicExtractor::handle(const slang::syntax::PortDeclarationSyntax &portDecl) {
00093     if (!inTargetModule_)
00094         return;
00095     spdlog::debug("LogicExtractor: Handling Port Declaration");
00096
00097     std::string direction = "unknown";
00098     if (portDecl.header) {
00099         // Determine direction (input/output/inout)
00100         // Important: Use valueText() as Slang typically represents keywords as text tokens
00101         if (auto varHeader = portDecl.header->as_if<slang::syntax::VariablePortHeaderSyntax>()) {
00102             if (varHeader->direction.valid()) {
00103                 direction = std::string(varHeader->direction.valueText());
00104             }
00105         } else if (auto netHeader = portDecl.header->as_if<slang::syntax::NetPortHeaderSyntax>()) {
00106             if (netHeader->direction.valid()) {
00107                 direction = std::string(netHeader->direction.valueText());
00108             }
00109         }
00110         // Add InterfacePortHeaderSyntax if needed
00111     } else {
00112         spdlog::warn("LogicExtractor: Port declaration without header found.");
00113     }
00114
00115     for (const auto &declarator : portDecl.declarators) {
00116         if (declarator->name.valid()) {
00117             std::string portName = std::string(declarator->name.valueText());
00118             if (portDirections_.count(portName) && portDirections_[portName] != "unknown") {
00119                 spdlog::warn("LogicExtractor: Multiple direction declarations for port '{}'. Keeping first "
00120                     "one '{}'. New direction: '{}'",
00121                     portName, portDirections_[portName], direction);
00122             } else {
00123                 spdlog::info("LogicExtractor: Storing direction '{}' for port '{}'", direction, portName);
00124                 portDirections_[portName] = direction;
00125                 if (direction == "input") {
00126                     primaryInputs_.insert(portName);
00127                     internalWires_.insert(portName); // Inputs are also technically 'wires' usable internally
00128                 } else if (direction == "output") {
00129                     primaryOutputs_.insert(portName);
00130                     internalWires_.insert(portName); // Outputs are also wires driven by something
00131                 } else if (direction == "inout") {
00132                     spdlog::info("LogicExtractor: Inout port '{}' found. Adding to both inputs and outputs.",
00133                         portName);
00134                     primaryInputs_.insert(portName);
00135                     primaryOutputs_.insert(portName);
00136                     internalWires_.insert(portName);
00137                 } else {
00138                     spdlog::warn("LogicExtractor: Port '{}' found in declaration but has unknown or missing "

```

```

00139         "direction '{}'. Treating as internal wire.",
00140         portName, direction);
00141         internalWires_.insert(portName);
00142     }
00143     // logicCache_[portName] = portName; // For Step 2
00144 }
00145 } else {
00146     spdlog::warn(
00147         "LogicExtractor: Port declarator without a name found in PortDeclarationSyntax.");
00148 }
00149 }
00150 // Don't call visitDefault here, as it might revisit things unexpectedly.
00151 // Let the main module visitor handle traversing into children if necessary.
00152 }
00153
00154 // Handle Non-ANSI Port List (defines names, usually in module header)
00179 void LogicExtractor::handle(const slang::syntax::NonAnsiPortListSyntax &portList) {
00180     if (!inTargetModule_)
00181         return;
00182     spdlog::debug("LogicExtractor: Handling NonAnsi Port List");
00183
00184     for (const auto portSyntax : portList.ports) {
00185         if (!portSyntax)
00186             continue;
00187
00188         std::string portName = "";
00189         // Most common case: ImplicitNonAnsiPortSyntax contains the expression (usually just the name)
00190         if (auto implicitPort = portSyntax->as_if<slang::syntax::ImplicitNonAnsiPortSyntax>()) {
00191             if (implicitPort->expr) {
00192                 // The expression itself might be complex, try to get the simple name
00193                 if (auto portRef = implicitPort->expr->as_if<slang::syntax::PortReferenceSyntax>()) {
00194                     if (portRef->name.valid()) {
00195                         portName = std::string(portRef->name.valueText());
00196                     }
00197                 } else if (auto portConcat =
00198                     implicitPort->expr->as_if<slang::syntax::PortConcatenationSyntax>()) {
00199                     // Handle concatenations if necessary - complex
00200                     spdlog::warn("LogicExtractor: Port concatenation found in NonAnsi port list - currently "
00201                         "not fully handled for logic extraction. Port: {}",
00202                         implicitPort->toString());
00203                     continue; // Skip complex ports for now
00204                 } else {
00205                     // Fallback: try getting name from the expression directly (might be just identifier)
00206                     portName = implicitPort->toString();
00207                 }
00208             }
00209         }
00210         // Handle ExplicitNonAnsiPortSyntax (e.g., .A(A)) if needed
00211         else if (auto explicitPort = portSyntax->as_if<slang::syntax::ExplicitNonAnsiPortSyntax>()) {
00212             if (explicitPort->name.valid()) {
00213                 portName = std::string(explicitPort->name.valueText());
00214                 // Note: explicitPort->expr is the internal signal it connects to.
00215                 // We are primarily interested in the port's own name (explicitPort->name) here.
00216             }
00217         }
00218         // Handle EmptyNonAnsiPortSyntax (commas for placeholders) if necessary
00219         else if (portSyntax->kind == slang::syntax::SyntaxKind::EmptyNonAnsiPort) {
00220             spdlog::debug("LogicExtractor: Skipping empty non-ANSI port placeholder.");
00221             continue;
00222         } else {
00223             spdlog::warn("LogicExtractor: Unhandled NonAnsi port syntax kind: {}",
00224                 slang::syntax::toString(portSyntax->kind));
00225             continue;
00226         }
00227
00228         // Now we (hopefully) have the portName.
00229         if (!portName.empty()) {
00230             if (portDirections_.count(portName)) {
00231                 const std::string &direction = portDirections_[portName];

```

```

00232     spdlog::debug("LogicExtractor: Finalizing port '{}' with direction '{}'", portName,
00233                   direction);
00234
00235     if (direction == "input") {
00236         primaryInputs_.insert(portName);
00237         internalWires_.insert(portName); // Inputs are also technically 'wires' usable internally
00238                                         // logicCache_[portName] = portName; // For Step 2
00239     } else if (direction == "output") {
00240         primaryOutputs_.insert(portName);
00241         internalWires_.insert(portName); // Outputs are also wires driven by something
00242     } else if (direction == "inout") {
00243         spdlog::info("LogicExtractor: Inout port '{}' found. Adding to both inputs and outputs.",
00244                     portName);
00245         primaryInputs_.insert(portName);
00246         primaryOutputs_.insert(portName);
00247         internalWires_.insert(portName);
00248         // logicCache_[portName] = portName; // For Step 2
00249     } else {
00250         spdlog::warn("LogicExtractor: Port '{}' found in list but has unknown or missing "
00251                     "direction '{}'. Treating as internal wire.",
00252                     portName, direction);
00253         internalWires_.insert(portName);
00254     }
00255 } else {
00256     spdlog::debug("LogicExtractor: Port '{}' found in NonAnsi list, no direction declaration ",
00257                 portName);
00258 }
00259 } else {
00260     spdlog::warn("LogicExtractor: Could not determine port name from NonAnsi port list item: {}",
00261                 portSyntax->toString());
00262 }
00263 }
00264 // Don't call visitDefault here
00265 }
00266
00267 // Handle explicit wire declarations
00277 void LogicExtractor::handle(const slang::syntax::NetDeclarationSyntax &netDecl) {
00278     if (!inTargetModule_)
00279         return;
00280
00281     for (const auto &declarator : netDecl.declarators) {
00282         if (declarator->name.valid()) {
00283             std::string wireName = std::string(declarator->name.valueText());
00284             spdlog::debug("LogicExtractor: Found wire declaration: {}", wireName);
00285             // Avoid adding ports again if they were also declared as nets (common)
00286             if (!primaryInputs_.count(wireName) && !primaryOutputs_.count(wireName)) {
00287                 internalWires_.insert(wireName);
00288             } else {
00289                 spdlog::trace("LogicExtractor: Wire '{}' is already known as a port.", wireName);
00290             }
00291         }
00292     }
00293     // Don't call visitDefault here
00294 }
00295
00296 // Handle primitive gate instantiations (MOST IMPORTANT PART)
00309 void LogicExtractor::handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst) {
00310     if (!inTargetModule_)
00311         return;
00312
00313     if (!primitiveInst.type.valid()) {
00314         spdlog::warn("LogicExtractor: Primitive instance without a type token found.");
00315         return;
00316     }
00317
00318     // Use token kind for reliable checking, text for name storage
00319     slang::parsing::TokenKind gateKind = primitiveInst.type.kind;
00320     std::string gateTypeName = std::string(primitiveInst.type.valueText());
00321

```

```

00322 spdlog::debug("LogicExtractor: Found Primitive Instance of Type: {} (Kind: {})", gateTypeName,
00323             slang::parsing::toString(gateKind));
00324
00325 for (const auto &instance : primitiveInst.instances) {
00326     // if (!instance || !instance->connections.isInitialized()) { // Check if connections are valid
00327     //     spdlog::warn("LogicExtractor: Skipping primitive instance of type {} due to null pointer or
00328     //         "
00329     //             "uninitialized connections.",
00330     //             gateTypeName);
00331     //     continue;
00332     // }
00333
00334     GateInfo currentGateInfo;
00335     currentGateInfo.gateTypeName = gateTypeName;
00336     currentGateInfo.kind = gateKind;
00337
00338     // Primitives usually have ordered connections.
00339     // The FIRST connection is typically the OUTPUT.
00340     // The REST are INPUTS.
00341     if (instance->connections.empty()) {
00342         spdlog::warn("LogicExtractor: Gate instance of type {} has no connections.", gateTypeName);
00343         continue;
00344     }
00345
00346     // Extract Output Signal
00347     // Ensure the connection itself is not null
00348     if (!instance->connections[0]) {
00349         spdlog::error("LogicExtractor: First connection (output) is null for primitive {}",
00350             gateTypeName);
00351         continue;
00352     }
00353     if (auto firstConn =
00354         instance->connections[0]->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00355         currentGateInfo.outputSignal =
00356             firstConn->expr->toString(); // Get the output signal name from the connection
00357             // remove extra spaces in the signal name
00358         currentGateInfo.outputSignal.erase(
00359             std::remove_if(currentGateInfo.outputSignal.begin(), currentGateInfo.outputSignal.end(),
00360                 [](unsigned char x) { return std::isspace(x); }),
00361             currentGateInfo.outputSignal.end());
00362         if (!currentGateInfo.outputSignal.empty()) {
00363             spdlog::debug(" Output Signal: {}", currentGateInfo.outputSignal);
00364             // Gate outputs are internal signals (unless they are module outputs)
00365             // Add to internal wires if not already a primary output.
00366             if (!primaryOutputs_.count(currentGateInfo.outputSignal)) {
00367                 internalWires_.insert(currentGateInfo.outputSignal);
00368             }
00369         } else {
00370             spdlog::error(
00371                 "LogicExtractor: Could not determine output signal name for gate instance of type {}",
00372                 gateTypeName);
00373             continue; // Skip this instance if output is unknown
00374         }
00375     } else {
00376         spdlog::error("LogicExtractor: Expected OrderedPortConnectionSyntax for output of primitive "
00377             "{}", but got kind: {}. Conn: {}",
00378             gateTypeName, slang::syntax::toString(instance->connections[0]->kind),
00379             instance->connections[0]->toString());
00380         continue; // Skip this instance
00381     }
00382
00383     // Extract Input Signals
00384     for (size_t i = 1; i < instance->connections.size(); ++i) {
00385         // Ensure the connection itself is not null
00386         if (!instance->connections[i]) {
00387             spdlog::warn("LogicExtractor: Input connection {} is null for primitive {}", i,
00388                 gateTypeName);
00389             continue;
00390         }

```

```

00391     if (auto conn =
00392         instance->connections[i]->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00393         std::string inputSig =
00394             conn->expr->toString(); // Get the input signal name from the connection
00395         // remove extra spaces in the signal name
00396         inputSig.erase(std::remove_if(inputSig.begin(), inputSig.end(),
00397             [](unsigned char x) { return std::isspace(x); }),
00398             inputSig.end());
00399         if (!inputSig.empty()) {
00400             currentGateInfo.inputSignals.push_back(inputSig);
00401             spdlog::debug(" Input Signal {}: {}", i, inputSig);
00402         } else {
00403             spdlog::warn("LogicExtractor: Could not determine input signal name for input {} of gate "
00404                 "instance type {}. Conn: {}",
00405                 i, gateTypeName, conn->toString());
00406             // Decide whether to skip or continue with partial inputs
00407         }
00408     } else {
00409         spdlog::warn("LogicExtractor: Expected OrderedPortConnectionSyntax for input {} of "
00410             "primitive {}, but got kind: {}. Conn: {}",
00411             i, gateTypeName, slang::syntax::toString(instance->connections[i]->kind),
00412             instance->connections[i]->toString());
00413     }
00414 }
00415
00416 // Store the gate information, mapping the output signal to its driving gate
00417 if (!currentGateInfo.outputSignal.empty()) {
00418     if (gateOutputDrivers_.count(currentGateInfo.outputSignal)) {
00419         // This is a critical warning - indicates multiple drivers for the same net!
00420         spdlog::error("LogicExtractor: Multiple drivers found for signal '{}'. Previous driver: "
00421             "{}. New driver: {}. Netlist is likely invalid.",
00422             currentGateInfo.outputSignal,
00423             gateOutputDrivers_[currentGateInfo.outputSignal].gateTypeName, gateTypeName);
00424         // Keep the first one found for now, or decide on error handling
00425     } else {
00426         spdlog::info(" Storing driver for '{}': Gate Type '{}'", currentGateInfo.outputSignal,
00427             gateTypeName);
00428         gateOutputDrivers_[currentGateInfo.outputSignal] = currentGateInfo;
00429     }
00430 }
00431 }
00432 // Don't call visitDefault here
00433 }
00434
00435 // --- Logic Derivation Implementation ---
00436
00437 // Public method called after visiting the tree
00438 std::map<std::string, std::string> LogicExtractor::getLogicExpressions() {
00439     std::map<std::string, std::string> result_map;
00440     if (!parsingComplete_) {
00441         spdlog::error("LogicExtractor: AST parsing did not complete or target module '{}' not found. "
00442             "Cannot extract logic.",
00443             targetCell_);
00444         return result_map; // Return empty map
00445     }
00446
00447     spdlog::info("LogicExtractor: Deriving logic expressions for {} output ports...",
00448         primaryOutputs_.size());
00449
00450     for (const std::string &outputPort : primaryOutputs_) {
00451         spdlog::debug("LogicExtractor: Deriving logic for output: {}", outputPort);
00452         try {
00453             result_map[outputPort] = deriveLogicRecursive(outputPort);
00454             spdlog::info(" Output: {} => {}", outputPort, result_map[outputPort]);
00455         } catch (const std::runtime_error &e) {
00456             spdlog::error("LogicExtractor: Error deriving logic for output '{}': {}", outputPort,
00457                 e.what());
00458             result_map[outputPort] = "/* Error deriving logic */";
00459         } catch (...) {

```

```

00473     spdlog::error("LogicExtractor: Unknown error deriving logic for output '{}'", outputPort);
00474     result_map[outputPort] = "/* Unknown error deriving logic */";
00475 }
00476 }
00477 return result_map;
00478 }
00479
00480 // Recursive function with memoization
00510 std::string LogicExtractor::deriveLogicRecursive(const std::string &signalName) {
00511     // 1. Check Cache (Memoization)
00512     if (logicCache_.count(signalName)) {
00513         return logicCache_.at(signalName);
00514     }
00515
00516     // 2. Base Case: Is it a primary input?
00517     if (primaryInputs_.count(signalName)) {
00518         // Already cached during port handling, but double-check
00519         if (!logicCache_.count(signalName)) {
00520             logicCache_[signalName] = signalName;
00521         }
00522         return signalName;
00523     }
00524
00525     // 3. Recursive Step: Is it driven by a gate?
00526     if (gateOutputDrivers_.count(signalName)) {
00527         const GateInfo &driverGate = gateOutputDrivers_.at(signalName);
00528         std::vector<std::string> inputExpressions;
00529         spdlog::debug("    Tracing signal '{}', driven by {} gate", signalName,
00530             driverGate.gateTypeName);
00531
00532         // Recursively find expressions for all inputs of this gate
00533         for (const std::string &inputSig : driverGate.inputSignals) {
00534             if (inputSig.empty()) {
00535                 throw std::runtime_error("Empty input signal name encountered for gate driving " +
00536                     signalName);
00537             }
00538             spdlog::debug("        Recursing for input: {}", inputSig);
00539             inputExpressions.push_back(deriveLogicRecursive(inputSig));
00540         }
00541
00542         // Format the expression based on gate type and input expressions
00543         std::string currentExpr = formatExpression(driverGate, inputExpressions);
00544
00545         // Cache the result
00546         logicCache_[signalName] = currentExpr;
00547         return currentExpr;
00548     }
00549
00550     // 4. Handle Assign statements (if implemented)
00551     // if (assignDrivers_.count(signalName)) { ... }
00552
00553     // 5. Error Case: Signal not found or not driven by known element
00554     // Check if it's just an internal wire that wasn't driven?
00555     if (internalWires_.count(signalName)) {
00556         throw std::runtime_error(
00557             "Signal '" + signalName +
00558             "' is an internal wire but has no identified driver (gate or assign).");
00559     } else {
00560         throw std::runtime_error(
00561             "Signal '" + signalName +
00562             "' is not a primary input, known wire, or driven by a recognized gate/assignment.");
00563     }
00564 }
00565
00566 // Helper to format the expression string based on gate type
00584 std::string LogicExtractor::formatExpression(const GateInfo &gateInfo,
00585     const std::vector<std::string> &inputExprs) {
00586     if (inputExprs.empty() && gateInfo.kind != slang::parsing::TokenKind::NotKeyword &&
00587         gateInfo.kind != slang::parsing::TokenKind::BufKeyword) {

```

```

00588 // Gates like AND/OR/XOR need inputs
00589 spdlog::warn("Gate type {} requires inputs, but none were provided/derived for output {}",
00590             gateInfo.gateTypeName, gateInfo.outputSignal);
00591 return "/*<Error: Missing Inputs for " + gateInfo.gateTypeName + ">*/";
00592 }
00593
00594 std::string result = "";
00595
00596 // Use gateInfo.type (enum) for reliable checking
00597 switch (gateInfo.kind) {
00598 case slang::parsing::TokenKind::AndKeyword:
00599     result = "(" + inputExprs[0];
00600     for (size_t i = 1; i < inputExprs.size(); ++i)
00601         result += " * " + inputExprs[i]; // Use * for AND
00602     result += ")";
00603     break;
00604 case slang::parsing::TokenKind::NandKeyword:
00605     result = "!(" + inputExprs[0];
00606     for (size_t i = 1; i < inputExprs.size(); ++i)
00607         result += " * " + inputExprs[i];
00608     result += ")";
00609     break;
00610 case slang::parsing::TokenKind::OrKeyword:
00611     result = "(" + inputExprs[0];
00612     for (size_t i = 1; i < inputExprs.size(); ++i)
00613         result += " + " + inputExprs[i]; // Use + for OR
00614     result += ")";
00615     break;
00616 case slang::parsing::TokenKind::NorKeyword:
00617     result = "!(" + inputExprs[0];
00618     for (size_t i = 1; i < inputExprs.size(); ++i)
00619         result += " + " + inputExprs[i];
00620     result += ")";
00621     break;
00622 case slang::parsing::TokenKind::XorKeyword:
00623     result = "(" + inputExprs[0];
00624     for (size_t i = 1; i < inputExprs.size(); ++i)
00625         result += " ^ " + inputExprs[i]; // Use ^ for XOR
00626     result += ")";
00627     break;
00628 case slang::parsing::TokenKind::XnorKeyword:
00629     result = "!(" + inputExprs[0];
00630     for (size_t i = 1; i < inputExprs.size(); ++i)
00631         result += " ^ " + inputExprs[i];
00632     result += ")";
00633     break;
00634 case slang::parsing::TokenKind::NotKeyword: // NOT gate
00635     if (inputExprs.size() != 1) {
00636         spdlog::warn("NOT gate expects 1 input, got {}", inputExprs.size());
00637         return "/*<Error: Incorrect Inputs for NOT>*/";
00638     }
00639     result = "!" + inputExprs[0]; // Use ! for NOT
00640     break;
00641 case slang::parsing::TokenKind::BufKeyword: // BUF gate
00642     if (inputExprs.size() != 1) {
00643         spdlog::warn("BUF gate expects 1 input, got {}", inputExprs.size());
00644         return "/*<Error: Incorrect Inputs for BUF>*/";
00645     }
00646     result = inputExprs[0]; // Output is the same as input
00647     break;
00648 // Add cases for bufif0, bufif1, notif0, notif1, pullup, pulldown, cmos, nmos, pmos, tran etc. if
00649 // needed
00650 default:
00651     spdlog::warn("Unsupported primitive gate type for logic expression generation: {}",
00652                 gateInfo.gateTypeName);
00653     result = "/*<Unsupported Gate: " + gateInfo.gateTypeName + ">*/";
00654     break;
00655 }
00656 return result;

```

```

00657 }
00658
00659 // --- New Function Implementation (Step 1: Visit and Print) ---
00660
00676 void extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell) {
00677     spdlog::info("--- Step 1: Starting Netlist Info Extraction ---");
00678     spdlog::info("Verilog file: '{}', Target cell: '{}'", verilog_file, cell);
00679
00680     try {
00681         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00682         if (!result) { // Check if result is valid
00683             spdlog::error("Error parsing Verilog file '{}'. Cannot extract info.", verilog_file);
00684             return;
00685         }
00686
00687         spdlog::info("Successfully parsed Verilog file.");
00688         std::shared_ptr<slang::syntax::SyntaxTree> tree = result.value();
00689
00690         LogicExtractor extractor(cell);
00691         tree->root().visit(extractor); // Populate extractor's internal state
00692
00693         // --- Print Summary (Debug for Step 1) ---
00694         spdlog::info("--- Extraction Summary for Cell '{}': ---", cell);
00695
00696         const auto &inputs = extractor.getPrimaryInputs();
00697         spdlog::info("Found {} Primary Inputs:", inputs.size());
00698         for (const auto &name : inputs) {
00699             spdlog::info("  - {}", name);
00700         }
00701
00702         const auto &outputs = extractor.getPrimaryOutputs();
00703         spdlog::info("Found {} Primary Outputs:", outputs.size());
00704         for (const auto &name : outputs) {
00705             spdlog::info("  - {}", name);
00706         }
00707
00708         const auto &wires = extractor.getInternalWires();
00709         spdlog::info("Found {} Internal Wires:", wires.size());
00710         for (const auto &name : wires) {
00711             spdlog::info("  - {}", name);
00712         }
00713
00714         const auto &gates = extractor.getExtractedGates();
00715         spdlog::info("Found {} Gate Drivers:", gates.size());
00716         for (const auto &pair : gates) {
00717             const std::string &outputNet = pair.first;
00718             const GateInfo &info = pair.second;
00719             std::string inputsStr = "";
00720             for (size_t i = 0; i < info.inputSignals.size(); ++i) {
00721                 inputsStr += info.inputSignals[i] + (i == info.inputSignals.size() - 1 ? "" : ", ");
00722             }
00723             spdlog::info("  - Output: {} <= Driven by: {} ({} Inputs: [{}]", outputNet,
00724                 info.gateTypeName, slang::parsing::toString(info.kind), inputsStr);
00725         }
00726         spdlog::info("--- End Netlist Info Extraction ---");
00727
00728     } catch (const std::exception &e) {
00729         spdlog::error("Exception during Verilog parsing or info extraction: {}", e.what());
00730     } catch (...) {
00731         spdlog::error("Unknown exception during Verilog parsing or info extraction.");
00732     }
00733 }
00734
00735 // --- (Step 2: Extract Logic Expressions) ---
00736
00755 std::map<std::string, std::string> extractLogicFromVerilog(const std::string &verilog_file,
00756     const std::string &cell) {
00757     spdlog::info("--- Starting Logic Expression Extraction for cell: '{}' ---", cell);
00758     std::map<std::string, std::string> logicMap; // Default empty map

```

```

00759
00760 try {
00761     auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00762     if (!result) {
00763         spdlog::error("Error parsing Verilog file '{}'. Cannot extract logic.", verilog_file);
00764         return logicMap;
00765     }
00766
00767     spdlog::info("Successfully parsed Verilog file.");
00768     std::shared_ptr<slang::syntax::SyntaxTree> tree = result.value();
00769
00770     LogicExtractor extractor(cell);
00771     tree->root().visit(extractor); // Populate extractor's internal state
00772
00773     // Now, call the method to derive and get the expressions
00774     logicMap = extractor.getLogicExpressions();
00775
00776 } catch (const std::exception &e) {
00777     spdlog::error("Exception during Verilog parsing or logic extraction: {}", e.what());
00778 } catch (...) {
00779     spdlog::error("Unknown exception during Verilog parsing or logic extraction.");
00780 }
00781
00782 if (logicMap.empty()) {
00783     spdlog::warn("Logic extraction finished, but no expressions were derived for cell '{}'. Check "
00784                 "if cell exists and is correctly defined.",
00785                 cell);
00786 } else {
00787     spdlog::info("Logic extraction completed for cell '{}'. Found {} output expression strings",
00788                 cell, logicMap.size());
00789 }
00790
00791 spdlog::info("--- End Logic Expression Extraction ---");
00792 return logicMap;
00793 }

```

7.47 src/main.cpp File Reference

This file contains the main function for the ZlibValidation tool.

```

#include "CLI/CLI.hpp"
#include "LibFileOperations.hpp"
#include "version.h"

```

Include dependency graph for main.cpp:

Functions

- static double [parse_age_time](#) (const std::string &s)
- int [main](#) (int argc, char *argv[])

7.47.1 Detailed Description

This file contains the main function for the ZlibValidation tool.

The ZlibValidation tool is a command-line application that provides several functionalities for processing and validating Liberty files, including parsing, monotonicity checking, comparison, supercell generation, Verilog/SPICE netlist generation, functional equivalence check, and a utility to clear generated files. It uses the CLI11 library for command-line argument parsing and spdlog for logging.

The main function parses command-line arguments using CLI11 to determine the desired operation and its parameters. It then calls the appropriate functions to perform the requested task. The tool supports processing multiple Liberty files in parallel using multi-threading for monotonicity checking, supercell generation, Verilog generation, and SPICE generation.

Parameters

<i>argc</i>	The number of command-line arguments.
<i>argv</i>	An array of command-line argument strings.

Returns

0 if the program executes successfully, or an error code otherwise.

The main function performs the following steps:

1. Initializes command-line argument parsing using CLI11.
2. Defines subcommands for each supported operation:
 - `parse`: Parses a Liberty file and writes JSON to a file.
 - `mono`: Checks the monotonicity of timing arc values in a Liberty file.
 - `compare`: Compares two Liberty files and reports differences.
 - `supercell`: Generates supercells for a given Liberty file.
 - `zlibboost`: Runs the ZlibBoost tool for multi-threaded library processing.
 - `clear`: Clears log, JSON, map, markdown, Verilog, and SPICE files in the current directory.
 - `verilog`: Generates Verilog netlist for a given Liberty file.
 - `spice`: Generates SPICE netlist for a given Liberty file.
 - `func`: Checks functional equivalence of two Liberty or Verilog files.
3. Parses the command-line arguments using CLI11.
4. Calls the appropriate function based on the selected subcommand.
5. Logs the program's exit status and timestamp.

Note

The tool relies on external libraries such as CLI11, spdlog, and potentially others depending on the specific operation being performed. Ensure that these libraries are properly installed and configured before running the tool.

Definition in file [main.cpp](#).

7.47.2 Function Documentation

7.47.2.1 main()

```
int main (
    int argc,
    char * argv[] )
```

Definition at line 63 of file [main.cpp](#).

Here is the call graph for this function:

7.47.2.2 parse_age_time()

```
static double parse_age_time (
    const std::string & s ) [static]
```

Definition at line 10 of file [main.cpp](#).

Here is the caller graph for this function:

7.48 main.cpp

[Go to the documentation of this file.](#)

```
00001 // ./src/main.cpp
00002
00003 #include "CLI/CLI.hpp"
00004
00005 #include "LibFileOperations.hpp"
00006 #include "version.h"
00007
00008 // Parse age-time string with optional unit suffix (y/d/h) into seconds.
00009 // Empty string -> 0.0 (non-aging). Bare number (no suffix) -> seconds.
00010 static double parse_age_time(const std::string &s) {
00011     if (s.empty()) return 0.0;
00012     char unit = s.back();
00013     double num = std::stod(s); // stod 解析前导数字，忽略尾部非数字字符
00014     switch (unit) {
00015         case 'y': return num * 365.0 * 24.0 * 3600.0; // 31536000
00016         case 'd': return num * 24.0 * 3600.0; // 86400
00017         case 'h': return num * 3600.0; // 3600
00018         default: return num; // 无后缀 → 秒
00019     }
00020 }
00021
00063 int main(int argc, char *argv[]) {
00064     // Parse command line arguments
00065     std::vector<std::string> library_paths; // Support multiple files
```

```

00066 std::string log_file_name = "";
00067
00068 // Command line parameter parsing using CLI11
00069 CLI::App app{APP_NAME};
00070 app.set_version_flag("-v,--version", APP_VERSION);
00071
00072 // Add subcommand for parse mode
00073 CLI::App *parse_cmd =
00074     app.add_subcommand("parse", "Parse the Liberty file and write JSON (or SQLite DB via --db)");
00075 parse_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00076     ->check(CLI::ExistingFile)
00077     ->required();
00078 parse_cmd->add_option("-l,--log", log_file_name,
00079     "Specify the log file name. Default: <basename>.parse.log");
00080
00081 // DB writing options (SQLite database output)
00082 std::string db_path = "";
00083 std::string pvt_corner = "";
00084 std::string age_time_str;
00085 double age_seconds = 0.0;
00086 parse_cmd->add_option("--db", db_path,
00087     "Path to SQLite database for storing LUT entries (enables DB write)");
00088 parse_cmd->add_option("--pvt", pvt_corner,
00089     "PVT corner string (e.g. SS_1p08V_125C). Recommended when writing to DB.");
00090 parse_cmd->add_option("--age-time", age_time_str,
00091     "Age time with unit suffix (e.g. 0.01h, 10y, 100d). Converted to seconds. Bare number = seconds.");
00092
00093 parse_cmd->callback([&] {
00094     printInfo();
00095     bool has_db = !db_path.empty();
00096
00097     // Parse --age-time string -> seconds (empty/absent -> 0.0)
00098     age_seconds = parse_age_time(age_time_str);
00099
00100     if (!has_db) {
00101         // ---- Existing behaviour: no DB ----
00102         if (library_paths.size() > 1) {
00103             spdlog::info("Running sequential parsing for {} files.", library_paths.size());
00104             spdlog::info("Each library will write to its own log file.");
00105             for (const auto &library_path : library_paths) {
00106                 parseLibFile(library_path, log_file_name = "");
00107             }
00108         } else {
00109             parseLibFile(library_paths[0], log_file_name);
00110         }
00111     } else {
00112         // ---- DB mode: parse each file and write LUT entries to DB ----
00113         spdlog::info("DB write enabled: '{}' (corner={}, age_time='{}' -> {}s)",
00114             db_path, pvt_corner, age_time_str, age_seconds);
00115         for (const auto &library_path : library_paths) {
00116             std::string logname = log_file_name.empty()
00117                 ? std::filesystem::path(library_path).stem().string() + ".parse.log"
00118                 : log_file_name;
00119             LibFile libfile(library_path, logname);
00120             libfile.parse();
00121             libfile.writeToDB(db_path, pvt_corner, age_seconds);
00122         }
00123     }
00124 });
00125
00126 // Add subcommand for mono check mode
00127 bool is_slew = false;
00128 CLI::App *mono_cmd = app.add_subcommand("mono", "Check the monotonicity of timing arc values");
00129 mono_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00130     ->check(CLI::ExistingFile)
00131     ->required();
00132 mono_cmd->add_option("-l,--log", log_file_name,
00133     "Specify the log file name. Default: <basename>.mono.log");
00134 mono_cmd->add_flag("-s,--slew", is_slew,

```

```

00135         "Specify that monotonicity checks also include input slew.");
00136 mono_cmd->callback([&] {
00137     printInfo();
00138     // Check if multi files
00139     if (library_paths.size() > 1) {
00140         spdlog::info("Running multi-threaded monotonicity check for {} files.", library_paths.size());
00141         spdlog::info("Each thread will write to its own log file.");
00142
00143         // Sequential json file check
00144         for (const auto &library_path : library_paths) {
00145             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00146                 ".json")) {
00147                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00148                 parseLibFile(library_path, log_file_name = "");
00149             }
00150         }
00151         spdlog::info("All JSON files prepared.");
00152         // Parallel monotonicity check
00153         std::vector<std::thread> threads;
00154         for (const auto &library_path : library_paths) {
00155             threads.emplace_back(monoCheckLibFile, library_path, log_file_name = "", is_slew);
00156         }
00157         for (auto &thread : threads) {
00158             thread.join();
00159         }
00160         spdlog::info("All threads completed.");
00161     } else {
00162         monoCheckLibFile(library_paths[0], log_file_name, is_slew);
00163     }
00164 });
00165
00166 // Add subcommand for compare mode
00167 std::string ref_lib, comp_lib, report_file_name;
00168 CLI::App *compare_cmd = app.add_subcommand(
00169     "compare", "Compare the comparison library against the reference one and report differences");
00170 compare_cmd->add_option("--ref", ref_lib, "Specify the reference library file")
00171     ->check(CLI::ExistingFile)
00172     ->required();
00173 compare_cmd->add_option("--comp", comp_lib, "Specify the comparison library file")
00174     ->check(CLI::ExistingFile)
00175     ->required();
00176 double abstol = 0.002;
00177 compare_cmd->add_option("--abstol", abstol,
00178     "Specify the absolute tolerance for comparison. Default: 0.002ns");
00179 double reltol = 0.02;
00180 compare_cmd->add_option("--reltol", reltol,
00181     "Specify the relative tolerance for comparison. Default: 0.02/2.0%");
00182 compare_cmd->add_option("--report", report_file_name,
00183     "Specify the report file name. Default: <comp_lib>.cmp.md");
00184 compare_cmd->callback([&] {
00185     printInfo();
00186     compareLibFiles(ref_lib, comp_lib, reltol, abstol, report_file_name);
00187 });
00188
00189 // Add subcommand for supercell generation
00190 int chain_length = 1;
00191 CLI::App *supercell_cmd =
00192     app.add_subcommand("supercell", "Generate supercells for the given Liberty file");
00193 supercell_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00194     ->check(CLI::ExistingFile)
00195     ->required();
00196 supercell_cmd->add_option("-l,--log", log_file_name,
00197     "Specify the log file name. Default: <basename>.supercell.log");
00198 supercell_cmd->add_option("-c,--chain", chain_length,
00199     "Specify the chain length for supercell generation. Default: 1");
00200 std::vector<std::string> cell_names = {}; // "CMPE42D1" "AN2D0", "DFQD1"
00201 supercell_cmd->add_option("--cells", cell_names,
00202     "Specify the cell names to generate supercells for");
00203 supercell_cmd->callback([&] {

```

```

00204     printInfo();
00205     // Check if multi files
00206     if (library_paths.size() > 1) {
00207         spdlog::info("Running multi-threaded supercell generation for {} files.",
00208             library_paths.size());
00209         spdlog::info("Each library will write to its own log file.");
00210
00211         // Sequential json file check
00212         for (const auto &library_path : library_paths) {
00213             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00214                 ".json")) {
00215                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00216                 parseLibFile(library_path, log_file_name = "");
00217             }
00218         }
00219         spdlog::info("All JSON files prepared.");
00220         // Parallel supercell generation
00221         std::vector<std::thread> threads;
00222         for (const auto &library_path : library_paths) {
00223             threads.emplace_back(supercellLibFile, library_path, log_file_name = "", chain_length,
00224                 cell_names);
00225         }
00226         for (auto &thread : threads) {
00227             thread.join();
00228         }
00229         spdlog::info("All threads completed.");
00230     } else {
00231         supercellLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00232     }
00233 });
00234
00235 // Add subcommand for zlibboost
00236 CLI::App *zlibboost_cmd =
00237     app.add_subcommand("zlibboost", "ZlibBoost - Multi-threaded Library Processing Tool");
00238 std::string config_dir = "/home/songzx/Projects/zlibboost/config.tcl";
00239 std::string python_dir = "/home/guocj/anaconda3/envs/myenv/bin/python";
00240 std::string main_py_dir = "/home/songzx/Projects/zlibboost/zlibboost.py";
00241 zlibboost_cmd->add_option(
00242     "-c, --config", config_dir,
00243     "Specify the configuration TCL file. Default: /home/songzx/Projects/zlibboost/config.tcl");
00244 zlibboost_cmd->add_option(
00245     "--python", python_dir,
00246     "Specify the python directory. Default: /home/guocj/anaconda3/envs/myenv/bin/python");
00247 zlibboost_cmd->add_option("--main", main_py_dir,
00248     "Specify the main python script directory. Default: "
00249     "/home/songzx/Projects/zlibboost/zlibboost.py");
00250 zlibboost_cmd->callback([&] {
00251     printInfo();
00252     // Run the ZlibBoost tool
00253     std::string command = python_dir + " " + main_py_dir + " -c " + config_dir;
00254     spdlog::info("Running ZlibBoost with command: '{}'", command);
00255     int ret = std::system(command.c_str());
00256     if (ret == 0) {
00257         spdlog::info("ZlibBoost completed successfully.");
00258     } else {
00259         spdlog::error("ZlibBoost failed with return code: {}", ret);
00260     }
00261 });
00262
00263 // Add subcommand for clear
00264 CLI::App *clear_cmd = app.add_subcommand(
00265     "clear", "Clear the log, JSON, map, markdown, Verilog, SPICE files in this directory");
00266 clear_cmd->callback([&] {
00267     printInfo();
00268     std::filesystem::path current_dir = std::filesystem::current_path();
00269     for (const auto &entry : std::filesystem::directory_iterator(current_dir)) {
00270         if (entry.path().extension() == ".log" || entry.path().extension() == ".json" ||
00271             entry.path().extension() == ".map" || entry.path().extension() == ".md" ||
00272             entry.path().extension() == ".v" || entry.path().extension() == ".spi") {

```

```

00273         spdlog::info("Removing file: '{}'", entry.path().string());
00274         std::filesystem::remove(entry.path());
00275     }
00276 }
00277 spdlog::info("All log, JSON, map, markdown files cleared.");
00278 });
00279
00280 // Add subcommand for Verilog generation
00281 CLI::App *verilog_cmd =
00282     app.add_subcommand("verilog", "Generate Verilog file for given Liberty file");
00283 verilog_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00284     ->check(CLI::ExistingFile)
00285     ->required();
00286 verilog_cmd->add_option("-l,--log", log_file_name,
00287     "Specify the log file name. Default: <basename>.verilog.log");
00288 verilog_cmd->add_option("-c,--chain", chain_length,
00289     "Specify the chain length for verilog generation. Default: 1");
00290 verilog_cmd->add_option("--cells", cell_names, "Specify the cell names to generate Verilog for");
00291 verilog_cmd->callback([&] {
00292     printInfo();
00293     // Check if multi files
00294     if (library_paths.size() > 1) {
00295         spdlog::info("Running multi-threaded Verilog generation for {} files.", library_paths.size());
00296         spdlog::info("Each library will write to its own log file.");
00297
00298         // Sequential json file check
00299         for (const auto &library_path : library_paths) {
00300             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00301                 ".json")) {
00302                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00303                 parseLibFile(library_path, log_file_name = "");
00304             }
00305         }
00306         spdlog::info("All JSON files prepared.");
00307         // Parallel Verilog generation
00308         std::vector<std::thread> threads;
00309         for (const auto &library_path : library_paths) {
00310             threads.emplace_back(verilogLibFile, library_path, log_file_name = "", chain_length,
00311                 cell_names);
00312         }
00313         for (auto &thread : threads) {
00314             thread.join();
00315         }
00316         spdlog::info("All threads completed.");
00317     } else {
00318         verilogLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00319     }
00320 });
00321
00322 // Add subcommand for SPICE generation
00323 CLI::App *spice_cmd = app.add_subcommand("spice", "Generate SPICE file for given Liberty file");
00324 spice_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00325     ->check(CLI::ExistingFile)
00326     ->required();
00327 spice_cmd->add_option("-l,--log", log_file_name,
00328     "Specify the log file name. Default: <basename>.spice.log");
00329 spice_cmd->add_option("-c,--chain", chain_length,
00330     "Specify the chain length for SPICE generation. Default: 1");
00331 spice_cmd->add_option("--cells", cell_names, "Specify the cell names to generate SPICE for");
00332 std::string verilog_lib_file =
00333     "/home/songzx/examples/mydpk/TSMC65/TSMC65NM_CLN65LP_STDI0_STDCELL/tcbn65lp_220a/"
00334     "0396011_20170308/TSMCHOME/digital/Front_End/verilog/tcbn65lp.v";
00335 std::string spice_lib_file =
00336     "/home/songzx/examples/mydpk/TSMC65/TSMC65NM_CLN65LP_STDI0_STDCELL/tcbn65lp_220a/"
00337     "0396011_20170308/TSMCHOME/digital/Back_End/lpe_spice/tcbn65lp_200a/tcbn65lp_200a_lpe.spi";
00338 spice_cmd->add_option("--v1", verilog_lib_file,
00339     "Specify the location of the Verilog primitive library file");
00340 spice_cmd->add_option(
00341     "--s1", spice_lib_file,

```

```

00342     "Specify the location of the SPICE library file to be included in the output");
00343 spice_cmd->callback([&] {
00344     printInfo();
00345     // Check if multi files
00346     if (library_paths.size() > 1) {
00347         spdlog::info("Running multi-threaded SPICE generation for {} files.", library_paths.size());
00348         spdlog::info("Each library will write to its own log file.");
00349
00350         // Sequential json file check
00351         for (const auto &library_path : library_paths) {
00352             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00353                                     ".json")) {
00354                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00355                 parseLibFile(library_path, log_file_name = "");
00356             }
00357         }
00358         spdlog::info("All JSON files prepared.");
00359         // Parallel SPICE generation
00360         std::vector<std::thread> threads;
00361         for (const auto &library_path : library_paths) {
00362             threads.emplace_back(spiceLibFile, library_path, log_file_name = "", chain_length,
00363                                 cell_names, verilog_lib_file, spice_lib_file);
00364         }
00365         for (auto &thread : threads) {
00366             thread.join();
00367         }
00368         spdlog::info("All threads completed.");
00369     } else {
00370         spiceLibFile(library_paths[0], log_file_name, chain_length, cell_names, verilog_lib_file,
00371                     spice_lib_file);
00372     }
00373 });
00374
00375 // Add subcommand for funtional equivalence check
00376 CLI::App *func_cmd = app.add_subcommand(
00377     "func", "Check functional equivalence of two Liberty files or Verilog files");
00378 std::string ref_file, comp_file;
00379 func_cmd->add_option("--ref", ref_file, "Specify the reference Liberty or Verilog file")
00380     ->check(CLI::ExistingFile)
00381     ->required();
00382 func_cmd->add_option("--comp", comp_file, "Specify the comparison Liberty or Verilog file")
00383     ->check(CLI::ExistingFile)
00384     ->required();
00385 func_cmd->add_option("--cells", cell_names,
00386                     "Specify the cell names to check functional equivalence for");
00387 func_cmd->add_option("--report", report_file_name,
00388                     "Specify the report file name. Default: <comp_lib>.cmp.md");
00389
00390 func_cmd->callback([&] {
00391     printInfo();
00392     funcLibFile(ref_file, comp_file, cell_names, report_file_name);
00393 });
00394
00395 CLI11_PARSE(app, argc, argv);
00396
00397 // End of program
00398 char hostname[256];
00399 gethostname(hostname, sizeof(hostname));
00400
00401 auto now = std::chrono::system_clock::now();
00402 auto time_now = std::chrono::system_clock::to_time_t(now);
00403 std::stringstream ss;
00404 ss << std::put_time(std::localtime(&time_now), "%c");
00405
00406 spdlog::info("ZlibValidation exited on '{}' at {}", hostname, ss.str());
00407 return 0;
00408 }

```

7.49 src/verilog_utils.cpp File Reference

```
#include "verilog_utils.hpp"
```

Include dependency graph for verilog_utils.cpp:

Functions

- void [getAST](#) (const std::string &verilog_file, const std::string &cell)

7.49.1 Function Documentation

7.49.1.1 getAST()

```
void getAST (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 544 of file [verilog_utils.cpp](#).

Here is the call graph for this function:

7.50 verilog_utils.cpp

[Go to the documentation of this file.](#)

```
00001 // verilog_utils.cpp
00002
00003 #include "verilog_utils.hpp"
00004
00005 // Adding a generic handler method to record all visited nodes and increment the depth counter
00006 void VerilogVisitor::handle(const slang::syntax::SyntaxNode &node) {
00007     std::string indent(depth_ * 2, ' ');
00008     spdlog::debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00009
00010     // Increase depth, visit child nodes, then decrease depth
00011     depth_++;
00012     visitDefault(node);
00013     depth_--;
00014 }
00015
00016 // Handle module declarations
00017 void VerilogVisitor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00018     std::string indent(depth_ * 2, ' ');
00019
00020     if (module.header && module.header->name.valid()) {
00021         std::string_view moduleName = module.header->name.valueText();
00022         spdlog::debug("{}Module Name:{}", indent, moduleName);
```

```

00023
00024     inTargetModule_ = !targetCell_.empty() && moduleName == targetCell_;
00025     if (!inTargetModule_) {
00026         return;
00027     }
00028
00029     // If a specific module name is specified, check if it matches
00030     if (inTargetModule_) {
00031         spdlog::info("{}Found target module: {}", indent, targetCell_);
00032         // Print the module ports
00033         if (module.header->ports) {
00034             spdlog::info("{}Ports: {}", indent, module.header->ports->toString());
00035         }
00036     }
00037 }
00038
00039 // Continue processing child nodes
00040 depth_++;
00041 visitDefault(module);
00042 depth--;
00043 }
00044
00045 // Handle port declarations
00046 void VerilogVisitor::handle(const slang::syntax::PortDeclarationSyntax &portDecl) {
00047     if (!inTargetModule_) {
00048         return;
00049     }
00050
00051     std::string indent = std::string(depth_ * 2, ' ');
00052
00053     // Get port direction
00054     std::string direction = "unknown";
00055     if (portDecl.header) {
00056         // Try to convert the header to different types of port headers
00057         if (auto *varPort = portDecl.header->as_if<slang::syntax::VariablePortHeaderSyntax>()) {
00058             if (varPort->direction) {
00059                 direction = varPort->direction.valueText();
00060             }
00061         } else if (auto *netPort = portDecl.header->as_if<slang::syntax::NetPortHeaderSyntax>()) {
00062             if (netPort->direction) {
00063                 direction = netPort->direction.valueText();
00064             }
00065         }
00066
00067         // Get port name
00068         for (const auto &declarator : portDecl.declarators) {
00069             if (declarator->name.valid()) {
00070                 std::string_view portName = declarator->name.valueText();
00071                 spdlog::info("{}Port Name: {} ({})", indent, portName, direction);
00072             }
00073         }
00074     }
00075 }
00076
00077 // Handle hierarchical instantiation (module instance)
00078 void VerilogVisitor::handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst) {
00079     if (!inTargetModule_) {
00080         return;
00081     }
00082
00083     std::string indent(depth_ * 2, ' ');
00084
00085     if (hierarchyInst.type.valid()) {
00086         std::string_view instType = hierarchyInst.type.valueText();
00087         spdlog::info("{}Hierarchy Instance Type: {}", indent, instType);
00088
00089         // Safely print parameter values (if any)
00090         try {
00091             if (hierarchyInst.parameters) {

```

```

00092     spdlog::info("{}Parameters:", indent);
00093     for (const auto &param : hierarchyInst.parameters->parameters) {
00094         if (!param)
00095             continue; // Null pointer check
00096
00097         if (auto *orderedParam = param->as_if<slang::syntax::OrderedParamAssignmentSyntax>()) {
00098             if (orderedParam->expr) {
00099                 spdlog::info("{} Parameter: {}", indent, orderedParam->expr->toString());
00100             }
00101         } else if (auto *namedParam = param->as_if<slang::syntax::NamedParamAssignmentSyntax>()) {
00102             if (namedParam->name.valid() && namedParam->expr) {
00103                 spdlog::info("{} Parameter {}: {}", indent, namedParam->name.valueText(),
00104                     namedParam->expr->toString());
00105             }
00106         }
00107     }
00108 }
00109 } catch (const std::exception &e) {
00110     spdlog::warn("{}Exception while processing parameters: {}", indent, e.what());
00111 }
00112
00113 // Safely handle instances
00114 try {
00115     for (const auto &instance : hierarchyInst.instances) {
00116         if (!instance) {
00117             spdlog::warn("{}Invalid instance", indent);
00118             continue;
00119         } else if (!instance->decl) {
00120             spdlog::warn("{}Invalid instance declaration", indent);
00121             // continue;
00122         } else {
00123             try {
00124                 if (instance->decl->name.valid()) {
00125                     std::string_view instName = instance->decl->name.valueText();
00126                     spdlog::info("{}Instance Name: {}", indent, instName);
00127                 }
00128             } catch (...) {
00129                 spdlog::debug("{}Error printing name", indent);
00130             }
00131
00132             // Safely print dimensions
00133             try {
00134                 if (!instance->decl->dimensions.empty()) {
00135                     spdlog::info("{} Dimensions: {}", indent, instance->decl->dimensions.toString());
00136                 }
00137             } catch (...) {
00138                 spdlog::debug("{}Error printing dimensions", indent);
00139             }
00140         }
00141
00142         // Safely print port connections
00143         spdlog::info("{} Port connections:", indent);
00144         for (const auto &conn : instance->connections) {
00145             if (!conn) {
00146                 spdlog::warn("{} Null connection", indent);
00147                 continue; // Null pointer check
00148             }
00149
00150             try {
00151                 // Use as_if method for safe type conversion
00152                 if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00153                     if (ordered->expr) {
00154                         spdlog::info("{} Ordered connection: {}", indent, ordered->expr->toString());
00155                     } else {
00156                         spdlog::info("{} Ordered connection: <empty>", indent);
00157                     }
00158                 } else if (auto *named = conn->as_if<slang::syntax::NamedPortConnectionSyntax>()) {
00159                     if (named->name.valid()) {
00160                         std::string exprStr = named->expr ? named->expr->toString() : "<empty>";

```

```

00161         spdlog::info("{}    .{}({})", indent, named->name.valueText(), exprStr);
00162     }
00163     } else if (conn->as_if<slang::syntax::EmptyPortConnectionSyntax>()) {
00164         spdlog::info("{}    <empty connection>", indent);
00165     } else if (conn->as_if<slang::syntax::WildcardPortConnectionSyntax>()) {
00166         spdlog::info("{}    .* (wildcard connection)", indent);
00167     } else {
00168         spdlog::info("{}    Unknown connection type: {}", indent,
00169             slang::syntax::toString(conn->kind));
00170     }
00171     } catch (const std::exception &e) {
00172         spdlog::warn("{}Exception processing connection: {}", indent, e.what());
00173     } catch (...) {
00174         spdlog::warn("{}Unknown exception processing connection", indent);
00175     }
00176     }
00177     }
00178     } catch (const std::exception &e) {
00179         spdlog::warn("{}Exception while processing instances: {}", indent, e.what());
00180     }
00181
00182     // Continue processing child nodes
00183     depth++;
00184     visitDefault(hierarchyInst);
00185     depth--;
00186 }
00187 }
00188 /*
00189 // Handle primitive gate instantiation
00190 void VerilogVisitor::handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst) {
00191     if (!inTargetModule_) {
00192         return;
00193     }
00194
00195     std::string indent(depth_ * 2, ' ');
00196
00197     // Get primitive gate type
00198     if (primitiveInst.type.valid()) {
00199         std::string_view gateType = primitiveInst.type.valueText();
00200         spdlog::info("{}Primitive Gate: {}", indent, gateType);
00201
00202         // Print delay information (if any)
00203         if (primitiveInst.delay) {
00204             spdlog::info("{}Delay: {}", indent, primitiveInst.delay->toString());
00205         }
00206
00207         // Print strength information (if any)
00208         if (primitiveInst.strength) {
00209             spdlog::info("{}Strength: {}", indent, primitiveInst.strength->toString());
00210         }
00211
00212         // Print each instance
00213         for (const auto &instance : primitiveInst.instances) {
00214             // Primitive gates usually don't have names, but if they do, print them
00215             if (instance->decl && instance->decl->name.valid()) {
00216                 std::string_view instName = instance->decl->name.valueText();
00217                 spdlog::info("{} Gate instance: {}", indent, instName);
00218             }
00219
00220             // Print connections
00221             spdlog::info("{} Gate connections:", indent);
00222             for (size_t i = 0; i < instance->connections.size(); ++i) {
00223                 const auto &conn = instance->connections[i];
00224                 // For gate-level instantiation, connections are usually ordered
00225                 if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00226                     if (ordered->expr) {
00227                         // The first is usually the output, the rest are inputs
00228                         std::string portType = (i == 0) ? "output" : "input";
00229                         spdlog::info("{}    {}: {}", indent, portType, i, ordered->expr->toString());

```

```

00230 }
00231 }
00232 }
00233 }
00234
00235 // Continue to traverse deeper when specific standard gate type
00236 // Create a set of safe standard gate types
00237 static const std::unordered_set<std::string_view> safeGateTypes = {
00238     "and",    "or",    "nand",    "nor",    "xor",    "xnor",    "not",
00239     "buf",    "bufif0", "bufif1", "notif0", "notif1", "pullup", "pulldown",
00240     "cmos",   "rcmos",  "nmos",   "pmos",  "rnmos",  "rpmos",  "tran",
00241     "rtran",  "tranif0", "tranif1", "rtranif0", "rtranif1";
00242
00243 if (safeGateTypes.find(gateType) != safeGateTypes.end()) {
00244     // Continue processing child nodes
00245     depth_++;
00246     visitDefault(primitiveInst);
00247     depth--;
00248 } else {
00249     spdlog::warn("{}Skipping deeper traversal of primitive: {}", indent, gateType);
00250 }
00251
00252 } else {
00253     spdlog::warn("{}Primitive instantiation missing gate type", indent);
00254 }
00255 }
00256 */
00257 // Handle specify block
00258 void VerilogVisitor::handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock) {
00259     if (!inTargetModule_) {
00260         return;
00261     }
00262
00263     std::string indent(depth_ * 2, ' ');
00264     spdlog::info("{}Specify Block:", indent);
00265
00266     // Iterate through all path declarations
00267     for (const auto &item : specifyBlock.items) {
00268         if (auto *pathDecl = item->as_if<slang::syntax::PathDeclarationSyntax>()) {
00269             if (pathDecl->desc) {
00270                 std::string pathSrc;
00271                 std::string pathDst;
00272
00273                 // Get path source
00274                 if (!pathDecl->desc->inputs.empty()) {
00275                     // Get the first input
00276                     if (auto *identifier =
00277                         pathDecl->desc->inputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00278                         if (identifier->identifier.valid()) {
00279                             pathSrc = identifier->identifier.valueText();
00280                         }
00281                     }
00282                 }
00283
00284                 // Get path destination
00285                 if (pathDecl->desc->suffix) {
00286                     if (auto *simpleSuffix =
00287                         pathDecl->desc->suffix->as_if<slang::syntax::SimplePathSuffixSyntax>()) {
00288                         if (!simpleSuffix->outputs.empty()) {
00289                             if (auto *identifier =
00290                                 simpleSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00291                                 if (identifier->identifier.valid()) {
00292                                     pathDst = identifier->identifier.valueText();
00293                                 }
00294                             }
00295                         } else if (auto *edgeSuffix =
00296                             pathDecl->desc->suffix
00297                                 ->as_if<slang::syntax::EdgeSensitivePathSuffixSyntax>()) {
00298                             if (!edgeSuffix->outputs.empty()) {

```

```

00299         if (auto *identifier =
00300             edgeSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00301             if (identifier->identifier.valid()) {
00302                 pathDst = identifier->identifier.valueText();
00303             }
00304         }
00305     }
00306 }
00307 }
00308
00309 // Construct path string
00310 std::string pathStr = pathSrc + " => " + pathDst;
00311
00312 // Get path delay
00313 std::string delayStr = "(";
00314 for (size_t i = 0; i < pathDecl->delays.size(); ++i) {
00315     if (i > 0)
00316         delayStr += ", ";
00317     delayStr += pathDecl->delays[i]->toString();
00318 }
00319 delayStr += ")";
00320
00321 spdlog::info("{} Path: {} = {}", indent, pathStr, delayStr);
00322 }
00323 }
00324 // Can add handling for ConditionalPathDeclarationSyntax and IfNonePathDeclarationSyntax
00325 else if (auto *condPath = item->as_if<slang::syntax::ConditionalPathDeclarationSyntax>()) {
00326     if (condPath->path && condPath->predicate) {
00327         spdlog::info("{} Conditional Path (if {})", indent, condPath->predicate->toString());
00328         // Recursively handle internal path
00329         handle(*condPath->path);
00330     }
00331 } else if (auto *ifNonePath = item->as_if<slang::syntax::IfNonePathDeclarationSyntax>()) {
00332     if (ifNonePath->path) {
00333         spdlog::info("{} If-None Path", indent);
00334         // Recursively handle internal path
00335         handle(*ifNonePath->path);
00336     }
00337 }
00338 }
00339
00340 // Continue processing child nodes
00341 depth++;
00342 visitDefault(specifyBlock);
00343 depth--;
00344 }
00345 }
00346
00347 // Handle module declarations, only keep the target module
00348 void CellExtractor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00349     if (module.header && module.header->name.valid()) {
00350         std::string_view moduleName = module.header->name.valueText();
00351
00352         // If it is the target module, mark as found, otherwise remove it
00353         if (!targetCell_.empty() && moduleName == targetCell_) {
00354             foundTarget_ = true;
00355             // Do not modify, keep this module
00356         } else {
00357             // Remove non-target modules
00358             remove(module);
00359         }
00360     }
00361 }
00362
00363 // Get result
00364 bool CellExtractor::foundTargetCell()const { return foundTarget_; }
00365
00366 // Handle module declarations
00367 void CellPrinter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {

```

```

00368 if (module.header && module.header->name.valid()) {
00369     std::string_view moduleName = module.header->name.valueText();
00370
00371     // Check if it is the target module
00372     if (!targetCell_.empty() && moduleName == targetCell_) {
00373         foundTarget_ = true;
00374
00375         // Print module definition
00376         out_ << "timescale 1ns/10ps\n";
00377         out_ << module.toString();
00378
00379         return; // Do not traverse further
00380     }
00381 }
00382
00383 // Continue traversing other modules
00384 visitDefault(module);
00385 }
00386
00396 void ModuleRewriter::handle(const slang::syntax::SyntaxNode &node) {
00397     std::string indent(depth_ * 2, ' ');
00398     logger_>debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00399
00400     // Continue processing child nodes
00401     depth_++;
00402     visitDefault(node);
00403     depth_--;
00404 }
00405
00406 void ModuleRewriter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00407     logger_>debug("Processing module: {}", module.header->name.valueText());
00408
00409     // Create intermediate wires for connections between instances
00410     for (int i = 0; i < instance_count_ - 1; i++) {
00411         auto &newNetNode = parse("\n wire OP_" + std::to_string(i) + ";");
00412         insertAtBack(module.members, newNetNode);
00413         logger_>debug("Added intermediate wire: {}", newNetNode.toString());
00414     }
00415
00416     // Get critical input port & output port from
00417     // the module name pattern: CELLNAME_#_#CRITICALPORT__OUTPUTPORT
00418     std::string criticalInputPort = "";
00419     std::string criticalOutputPort = "";
00420     if (!this->moduleName_.empty()) {
00421         size_t firstSep = this->moduleName_.find("__");
00422         if (firstSep != std::string::npos) {
00423             size_t secondSep = this->moduleName_.find("__", firstSep + 2);
00424             if (secondSep != std::string::npos) {
00425                 size_t thirdSep = this->moduleName_.find("__", secondSep + 2);
00426                 if (thirdSep != std::string::npos) {
00427                     criticalInputPort = this->moduleName_.substr(secondSep + 2, thirdSep - (secondSep + 2));
00428                     criticalOutputPort = this->moduleName_.substr(thirdSep + 2);
00429                     logger_>debug("Extracted critical input port: {}", criticalInputPort);
00430                     logger_>debug("Extracted output port: {}", criticalOutputPort);
00431                 } else {
00432                     logger_>warn("Can't find thirdSep. Invalid module name pattern: {}", this->moduleName_);
00433                     return;
00434                 }
00435             } else {
00436                 logger_>warn("Can't find secondSep. Invalid module name pattern: {}", this->moduleName_);
00437                 return;
00438             }
00439         } else {
00440             logger_>warn("Can't find firstSep. Invalid module name pattern: {}", this->moduleName_);
00441             return;
00442         }
00443     } else {
00444         logger_>warn("Module name is empty. Can't extract critical input port.");
00445         return;

```

```

00446 }
00447
00448 // Add additional wires for intermediate outputs that aren't part of the chain
00449 if (this->outputPins_.size() > 1) { // Only add if there are multiple output pins
00450     for (int i = 0; i < instance_count_ - 1; i++) { // Skip the first and last instances
00451         for (const auto &outputPin : this->outputPins_) {
00452             if (outputPin != criticalOutputPort) { // Found an intermediate output pin
00453                 auto &newNetNode = parse("\n wire P_" + std::to_string(i) + "__" + outputPin + ";");
00454                 insertAtBack(module.members, newNetNode);
00455                 logger->debug("Added intermediate wire: {}", newNetNode.toString());
00456             }
00457         }
00458     }
00459 }
00460
00461 std::string portList_str = module.header->ports->toString();
00462 logger->debug("Port list: {}", portList_str);
00463
00464 std::vector<std::string> allPorts;
00465 auto ansiPortList = module.header->ports->as_if<slang::syntax::AnsiPortListSyntax>();
00466 if (!ansiPortList) {
00467     logger->warn("Port list is not ANSI style.");
00468     return;
00469 }
00470 portInfoMap_.clear(); // Clear the port info map
00471 for (const auto portMember : ansiPortList->ports) {
00472     if (portMember->kind == slang::syntax::SyntaxKind::ImplicitAnsiPort) {
00473         const auto implicitPort = portMember->as_if<slang::syntax::ImplicitAnsiPortSyntax>();
00474         const auto &directionToken =
00475             implicitPort->header->as_if<slang::syntax::VariablePortHeaderSyntax>()->direction;
00476         const auto &nameToken = implicitPort->declarator->name;
00477
00478         std::string_view portName = nameToken.valueText();
00479         std::string_view direction = directionToken.valueText();
00480         portInfoMap_[std::string(portName)] = std::string(direction); // Store port info in the map
00481         logger->debug("Port Name: {}, Direction: {}", portName, direction);
00482     } else {
00483         logger->warn("Port member is not ImplicitAnsiPort.");
00484     }
00485 }
00486
00487 for (int i = 0; i < instance_count_; i++) {
00488     std::string instanceName = "I_" + this->cellName_ + "__X" + std::to_string(i) + "__" +
00489         criticalInputPort + "__" + criticalOutputPort;
00490     std::string instanceCode = "\n " + this->cellName_ + " " + instanceName + " (";
00491     std::vector<std::string> portConnections;
00492
00493     for (const auto &pair : portInfoMap_) {
00494         std::string portName = pair.first;
00495         std::string direction = pair.second;
00496         std::string connectionName;
00497
00498         if (portName == criticalInputPort) {
00499             if (i == 0) {
00500                 connectionName = portName; // First instance: connect to module input port
00501             } else {
00502                 connectionName =
00503                     "OP_" + std::to_string(i - 1); // Intermediate instances: connect to previous OP wire
00504             }
00505         } else if (portName == criticalOutputPort) {
00506             if (i == instance_count_ - 1) {
00507                 connectionName = portName; // Last instance: connect to module output port
00508             } else {
00509                 connectionName = "OP_" + std::to_string(i); // Intermediate instances: connect to OP wire
00510             }
00511         } else if (direction == "output") { // Handle other output ports
00512             if (i < instance_count_ - 1) {
00513                 connectionName = "P_" + std::to_string(i) + "__" +
00514                     portName; // Connect to intermediate P_i__portName wire

```

```

00515         } else {
00516             connectionName = portName; // Last instance: connect to module output port
00517         }
00518     } else { // For input ports (and inout?), connect directly to module port
00519         connectionName = portName;
00520     }
00521     portConnections.push_back("." + portName + "(" + connectionName + ")");
00522 }
00523
00524 // Connect all ports with comma and space
00525 if (!portConnections.empty()) {
00526     instanceCode += portConnections[0];
00527     for (size_t i = 1; i < portConnections.size(); ++i) {
00528         instanceCode += ", " + portConnections[i];
00529     }
00530 }
00531 instanceCode += ");";
00532
00533 // Blank line before the first instance
00534 if (i == 0) {
00535     instanceCode = "\n" + instanceCode;
00536 }
00537
00538 // Insert the instance code
00539 auto &instanceNode = parse(instanceCode);
00540 insertAtBack(module.members, instanceNode);
00541 }
00542 }
00543
00544 void getAST(const std::string &verilog_file, const std::string &cell) {
00545     try {
00546         spdlog::info("Starting get AST from Verilog file: '{}'", verilog_file);
00547         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00548         if (result) {
00549             spdlog::info("Successfully parsed Verilog file.");
00550
00551             try {
00552                 // First, use the visitor to print basic information
00553                 VerilogVisitor visitor(cell);
00554                 visitor.visit(result.value()->root());
00555
00556                 // If a target cell is specified, use the rewriter to extract the relevant code
00557                 if (!cell.empty()) {
00558                     // Create a rewriter to only keep the target cell related code
00559                     CellExtractor extractor(cell);
00560                     auto extractedTree = extractor.transform(result.value());
00561
00562                     if (extractor.foundTargetCell()) {
00563                         // Use SyntaxPrinter to print the extracted code
00564                         std::string extractedCode = slang::syntax::SyntaxPrinter::printFile(*extractedTree);
00565
00566                         // Save to a file named after the cell name
00567                         std::string outputFile = cell + ".v";
00568                         std::ofstream cellOut(outputFile);
00569                         if (cellOut) {
00570                             cellOut << extractedCode;
00571                             cellOut.close();
00572                             spdlog::info("Extracted '{}' cell code to '{}'", cell, outputFile);
00573                         } else {
00574                             spdlog::error("Failed to write extracted cell code to '{}'", outputFile);
00575                         }
00576                     } else {
00577                         spdlog::warn("Target cell '{}' not found in the Verilog file", cell);
00578                     }
00579                 }
00580
00581                 // spdlog::info("Print full source code to 'full_source_code.v'");
00582                 // // Optionally save the entire syntax tree
00583                 // std::string fullOutput = slang::syntax::SyntaxPrinter::printFile(*result.value());

```

```
00584         // std::ofstream out("full_source_code.v");
00585         // out << fullOutput;
00586         // out.close();
00587
00588         spdlog::info("Print target cell code to 'cell_code.v using toString()");
00589         // Use CellPrinter to print the target cell's code
00590         std::ofstream cellOut("cell_code.v");
00591         CellPrinter cellPrinter(cell, cellOut);
00592         cellPrinter.visit(result.value()->root());
00593         cellOut.close();
00594
00595     } catch (const std::exception &e) {
00596         spdlog::error("Exception during AST traversal: {}", e.what());
00597     } catch (...) {
00598         spdlog::error("Unknown exception during AST traversal");
00599     }
00600 } else {
00601     spdlog::error("Error parsing Verilog file.");
00602 }
00603 } catch (const std::exception &e) {
00604     spdlog::error("Exception during Verilog parsing: {}", e.what());
00605 } catch (...) {
00606     spdlog::error("Unknown exception during Verilog parsing");
00607 }
00608 }
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